
INTRODUCTION TO BUSINESS



*Trade, Enterprise and Organisation,
with Cases from Bangladesh*

A TEXTBOOK FOR FIRST-YEAR STUDENTS

BIJOY?!

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with Cases from Bangladesh*



BIJOY ? !

First edition, in British spelling. Amounts are in Bangladeshi taka (Tk). Laws are summarised in plain words as they stood when the text was written; laws and rules change, so readers should consult the current text of any Act before relying on it.

The running business, Padma Bakery, and its owners are *illustrative*: they are invented to show ideas at work and are not a real firm. Historical cases are real, but each is told in brief.

Plate engravings shown are placeholders, to be replaced by public-domain plates before printing. Set in Libre Caslon Text and IM Fell English.

TO THE READER

This book assumes that you know nothing about business. Every new word is explained when it first appears, and the explanation is in simple English.

How a page works. The wide column holds the main text. The narrow column on the left holds notes. A note in *italic* sums up the paragraph beside it. A note that starts with a word in bold gives the meaning of that word. A bold word in the main text means that the word appears for the first time; it is explained in the note and again in the glossary at the back. Small numbers in the text point to footnotes at the foot of the page. Bold type is never used for ordinary emphasis.

A note marked **GO SLOWLY HERE** shows an idea that many readers find hard. Read it twice, and say its main point aloud in your own words.

How a chapter works. Each chapter opens with a question. It then teaches ideas one layer at a time: the idea, an example, the rule, and the cases where the rule fails. It follows the story of Padma Bakery, a small illustrative business, and ends with a summary, a problem to think through, common mistakes, a true case from history, and review questions. Short answers are at the back.

A word on numbers. Where a chapter uses figures, every step is shown. Begin with the small example, then read the second.

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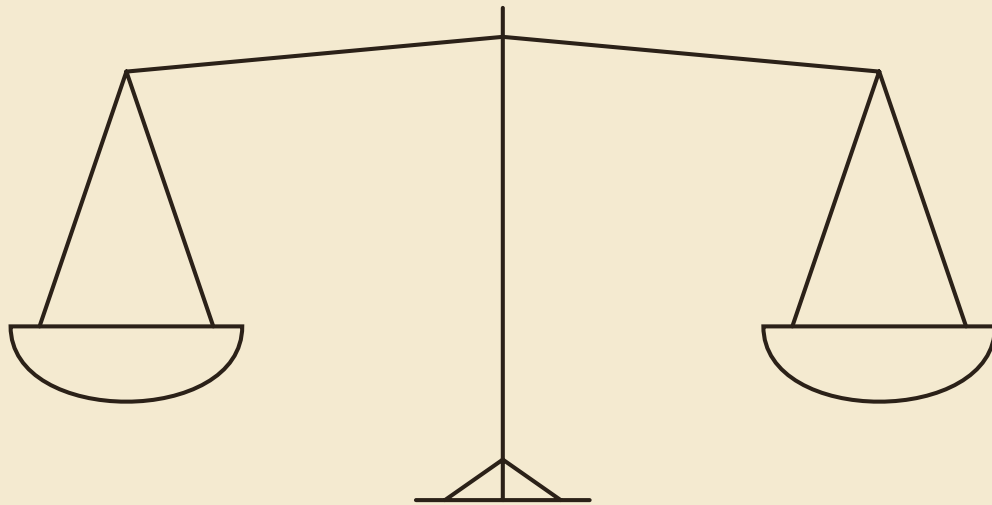


Plate I. [An engraving will be placed here.]

CHAPTER I

The Nature of Business

Is every sale a business, and who decides?



This chapter explains what a business is, how its parts are named, and what it aims to achieve. It ends with the qualities that help a business to last.

A neighbour asks to buy a cake.

A business sells goods or services to earn profit.

Business. An activity that provides goods or services to customers to earn a profit.

Goods. Things you can touch and keep, such as a cake or a chair.

Services. Useful work done for others, such as teaching or transport.

Customer. A person or firm that pays for goods or services.

Profit is what remains after costs are paid.

Profit. The money left from sales after all costs are paid.

Sales. The total money received from customers.

The test: regular selling, for profit.

GO SLOWLY HERE.

Five features mark every business.

Risk. The chance of loss, because the future cannot be known.

Utility. The usefulness that makes something worth buying.
Profit is the reward for taking risk.

THE FIRST QUESTION

Salma Akter bakes cakes at home in Mirpur, Dhaka. One day a neighbour tastes one and asks to buy another. Salma agrees, and the next week three more neighbours ask. Is Salma now in business? Before a single taka is spent on a shop, this question needs an answer, because the way we name an activity decides what we expect of it.¹

The word **business** means an activity that provides goods or services to customers in order to earn a profit. **Goods** are things you can touch, such as a cake. **Services** are useful work done for others, such as transport. A **customer** is a person who pays for them.

Profit means the money left after all costs are paid. Business people use several words for the money that customers pay: *revenue*, *sales* and *turnover*. This book uses one word, **sales**, for the total money received from customers.

Now we can answer the opening question. One sale to a neighbour is a favour. Regular selling to many customers, with the aim of earning profit, is business. By that test, Salma crossed the line in her second week.

WHAT MAKES A BUSINESS A BUSINESS

Five features appear in almost every business. First, it involves *exchange*: goods or services pass to a customer, and money passes back. Second, it is *regular*; dealings repeat. Third, it aims at profit. Fourth, it carries **risk**, which means that the future is not known and a loss may happen. Fifth, it creates **utility**, which means usefulness. Flour has some use; a baked cake is more useful to a hungry customer. The baker has added usefulness, and the customer pays for it.

Risk and profit belong together. Salma buys flour before she knows how many cakes she will sell. If customers do not come, the flour is wasted. The profit she earns is, in part, her reward for accepting this chance of loss.

¹ Taka is the money of Bangladesh. Its short sign in this book is Tk.

Exceptions: some bodies sell but do not seek profit.

There are exceptions. A school sells education, and a charity may sell goods, yet neither aims at profit for its owners. Such bodies are *not-for-profit*. They use business methods but not the profit aim. A hospital may do both: charge fees and treat some patients free.

You should now be able to say: a business sells goods or services, regularly, to earn profit, and it accepts risk.

INDUSTRY, COMMERCE AND TRADE

Three words that students often confuse.

Industry. All activities that produce or change goods, or extract them from nature.

Trade. The buying and selling of goods.

Commerce. Trade together with the services that help trade happen.

Three words are often mixed up. Each has its own meaning. **Industry** means the activities that produce goods or take them from nature, such as farming, mining and making cakes. **Trade** means buying and selling goods. **Commerce** means trade together with all the services that help trade to happen.

TABLE 1. INDUSTRY, TRADE AND COMMERCE COMPARED

Word	What it does	Example
Industry	Produces or extracts goods	Baking cakes in an oven
Trade	Buys and sells goods	Selling cakes to a shop
Commerce	Trade plus its helpers	Selling, banking, transport

Commerce is trade plus its helpers.

Aids to trade. Services that help goods move from maker to buyer, such as banking.

Trade may be local, or between countries.

Wholesale. Selling goods in large amounts to other businesses.

Retail. Selling goods in small amounts to the final user.

The helpers have a name: **aids to trade**. These include banking, insurance, transport, warehousing, advertising and communication. A bank lends money to Salma; a van delivers her cakes; an advertisement tells customers where to find them. None of these makes a cake, but without them selling becomes slow and costly.

Trade has two main kinds. *Home trade* takes place within one country. It may be **wholesale**, which means selling in large amounts to other businesses, or **retail**, which means selling in small amounts to the final user. *Foreign trade* crosses a national border; Chapter VII treats it.

Selling over the internet is commerce too.

E-commerce. Buying and selling through the internet or other electronic networks.

The internet has added a newer form. **E-commerce** means buying and selling through electronic networks. A customer orders a cake on a phone, pays through a mobile wallet, and the cake arrives by rider. Nothing about the idea is new. Only the place of meeting has changed, from a shop to a screen.

The *scope* of business, then, covers all of this: industry that makes goods, trade that sells them, and the aids that connect the two, including newer electronic means. Industry itself has branches. Extractive industry takes goods from nature, as mining and fishing do. Manufacturing changes materials into finished goods. Construction

builds. Service industries, such as banking and teaching, sell work rather than goods.

WHAT A BUSINESS AIMS AT

An objective is a goal the business works toward.

Objective. A goal that a business tries to reach.

Profit keeps the business alive.

Planning, honesty and care for customers.

An **objective** means a goal that a business tries to reach. Most firms have several, and they fall into four groups. *Economic objectives* include earning profit, surviving through bad years, and growing larger. *Social objectives* include fair prices, good goods, and jobs for the community. *Human objectives* include fair pay and safe work for employees. *National objectives* include paying tax and helping the country to produce more.

Profit comes first because it pays the other bills. A business that loses money for long enough must close, and then it can serve no one. Yet profit is the means by which a business stays alive, and not the only purpose. Chapter III returns to how economic and social aims fit together.

What helps a business to last? Experience points to a few pre-requisites. There must be a clear idea that customers want, enough money to begin, and a good location (Chapter II). There must be skill in managing people, honest dealing, and the willingness to change when customers change. A successful owner is usually *alert* to chances, *careful* with money, *honest* with customers, *patient* in hard months and *ready to learn*.

A WORKED CASE

SITUATION. Salma sells 20 cakes a week at Tk 400 each. Flour, sugar, eggs and butter cost Tk 250 per cake. This is the first entry in the case file of Padma Bakery, an illustrative business.

QUESTION. How much profit does Salma earn in one week?

REASONING. Sales equal 20 cakes times Tk 400, which is Tk 8,000. Costs equal 20 cakes times Tk 250, which is Tk 5,000. Profit is sales minus costs, so Tk 8,000 less Tk 5,000.

CONCLUSION. Salma earns Tk 3,000 a week.

CHECK. Each cake earns Tk 400 less Tk 250, which is Tk 150. Twenty cakes at Tk 150 each give Tk 3,000, the same answer.



SUMMARY

A business sells goods or services regularly, for profit, and carries risk. Industry makes or extracts goods, trade sells them, and commerce is trade plus its helpers. Objectives are economic, social, human and national.

THINK IT THROUGH. A tailor sews uniforms for one school, once a year, and charges only the cost of cloth. Is this a business? Which feature is missing?

WHERE PEOPLE GO WRONG. Students often say that *commerce* and *trade* are the same. They are not: commerce is wider, because it

includes the aids. Another mistake is to say that profit is the only objective. It is the first, not the only.

IN HISTORY. In 1776 Adam Smith described a small pin workshop. One worker alone might make hardly one pin a day, he wrote, but ten workers dividing the tasks could make many thousands. The story showed why businesses organise work into specialised parts, an idea Chapter II uses.

SOURCES

Smith (1776), Book I, Chapter 1.

REVIEW QUESTIONS

1. What are the five features of a business?
2. Explain the difference between trade and commerce.
3. Name the four groups of business objectives.

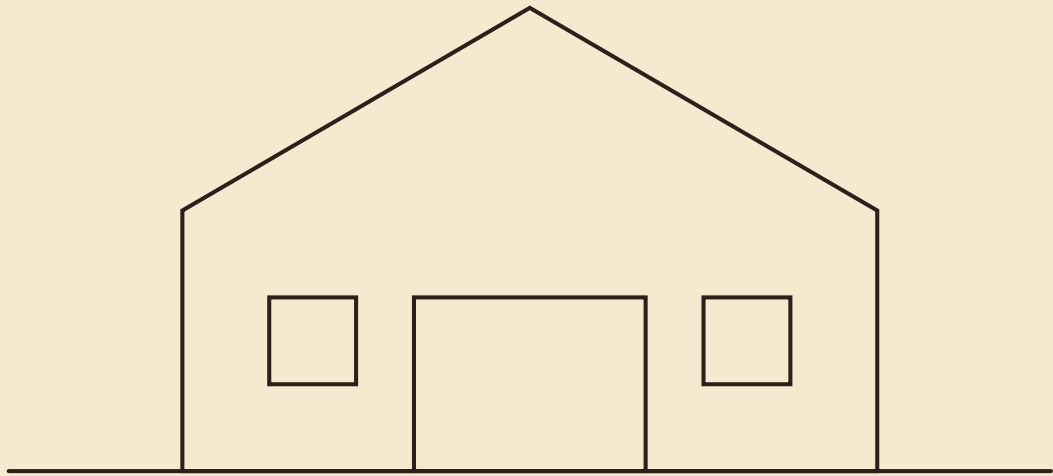


Plate II. [An engraving will be placed here.]

CHAPTER II

Resources and Location

What must Salma gather before the first cake is sold, and where should she bake?



This chapter names the four resources every business needs, explains what each one earns, and sets out the factors that guide the choice of a business site.

THE PUZZLE OF THE EMPTY SHOP

Salma has rented nothing yet. She has a recipe and a wish. To open a shop she must find a room, bakers, an oven, money for flour, and someone to organise it all. Which of these is the most important? The question is harder than it looks, because without any one of them the shop cannot open.

THE FOUR RESOURCES

Economists group business inputs into four.

Factors of production. The resources used to make goods and services: land, labour, capital and organisation.

Land means all gifts of nature.

Land. All natural resources, including soil, water, minerals and the space a building stands on.

Rent. The payment made to the owner of land for its use.

Labour is human work, of mind or body.

Labour. The physical and mental work of people in production.

Wages. The payment made to workers for their labour.

Division of labour. Splitting a job into parts, with each worker doing one part.

The things a business uses to produce goods or services are called **factors of production**. There are four: land, labour, capital and organisation. Each one earns its owner a different reward.

Land means more than soil. It includes water, minerals, forests and the ground under a shop. The reward for the use of land is **rent**. Salma pays rent for her shop room.

Labour means the work of people, whether of hands or of mind. A baker kneading dough and an accountant keeping records both give labour. The reward for labour is **wages**. Labour is more productive when tasks are divided. Adam Smith's pin workshop (Chapter I) showed this. **Division of labour** means splitting a job into parts so that each worker repeats one part and becomes quick at it.

Capital needs care, because the word has two meanings. In daily life it often means money. In this chapter it means the oven, the

GO SLOWLY HERE.

Capital here means goods used to make goods.

Capital. Man-made goods, such as machines and tools, that are used to produce other goods.

Interest. The payment made for the use of borrowed money or capital.

The entrepreneur organises and takes the risk.

Entrepreneur. A person who organises production and takes the risk of loss.

Each factor can be hard to replace.

Location means the place where a business operates.

Location. The place where a business sets up its premises.

Distance to customers matters most for a shop.

Power, water, rent and rules also count.

Weigh all factors, and add up the costs.

tables and the tools used to make cakes. Money used to buy them is called *finance*. The reward for providing capital is **interest**.

The fourth factor is organisation. The person who brings the other three together, makes the decisions and bears the risk is the **entrepreneur**. Salma is the entrepreneur of Padma Bakery. The reward of the entrepreneur is profit (Chapter I), which may be large or may be a loss.

TABLE 2. THE FOUR FACTORS AND THEIR REWARDS

Factor	Meaning	Reward
Land	Natural resources and space	Rent
Labour	Human work	Wages
Capital	Tools, machines, buildings	Interest
Organisation	Planning and risk-taking	Profit

These factors have limits. Land cannot be moved, and good land is scarce. Skilled labour takes time to train. Capital wears out and must be replaced. An entrepreneur who is careless with one factor will find that the others cannot make up for it. This is why a business plan lists all four.

WHERE TO LOCATE

Location means the place where a business sets up its shop, office or factory. A poor site is costly to change, because moving means new rent, new fittings and lost customers. The choice therefore deserves care.

Several factors guide the choice. *Nearness to customers* matters most for a shop selling cakes, because cakes go stale and buyers will not travel far. *Nearness to raw materials* matters more for a mill or a mine, where the materials are heavy. *Labour* matters, since a site must offer workers with the right skills at fair wages. *Transport* matters, as roads, ports and railways carry goods in and out.

Other factors add to these. A bakery needs steady electricity and clean water. Rent and the price of land differ from place to place. Government rules may forbid certain businesses in certain areas, or offer lower taxes in others. Finally, the *climate*, the risk of flooding, and the safety of the area affect daily work.

No site is best on every count. The owner must weigh the factors and, where possible, add up the costs of each choice. The worked case shows how.

A WORKED CASE

SITUATION. Salma compares two sites. A shop near the Mirpur market rents for Tk 15,000 a month, and delivery to customers costs Tk 2,000. A plot in an industrial area rents for Tk 6,000, but delivery over the longer distance costs Tk 14,000. Both sites are otherwise the same.

QUESTION. Which site has the lower total monthly cost?

REASONING. Market site: Tk 15,000 plus Tk 2,000 equals Tk 17,000. Industrial site: Tk 6,000 plus Tk 14,000 equals Tk 20,000.

CONCLUSION. The market site costs Tk 3,000 a month less, so Salma takes it.

CHECK. The industrial site saves Tk 9,000 in rent but adds Tk 12,000 in delivery. The net difference is Tk 12,000 less Tk 9,000, which is Tk 3,000, the same answer. The case file now records the shop, the oven at Tk 120,000, and two helpers paid Tk 10,000 a month each.



SUMMARY

A business uses four factors of production: land earns rent, labour earns wages, capital earns interest, and organisation earns profit. The site should be chosen by weighing nearness to customers, materials, labour, transport, power, rules and cost.

THINK IT THROUGH. A brick factory is choosing between a site beside a clay field and a site beside a city. Which factors should weigh most? Would your answer change for a shop selling bricks?

WHERE PEOPLE GO WRONG. Students often use *capital* to mean cash only. Here it means goods used to make goods. Another mistake is to pick the cheapest rent without adding the other costs, as the worked case shows.

IN HISTORY. In 1776 Adam Smith counted the benefits of dividing labour in a pin workshop. His idea explains why factories gather skilled workers in one place, and why cities grew around them.

REVIEW QUESTIONS

1. Name the four factors of production and the reward of each.
2. Why should an owner compare total costs and not only rent?
3. List three factors that guide the choice of a location.

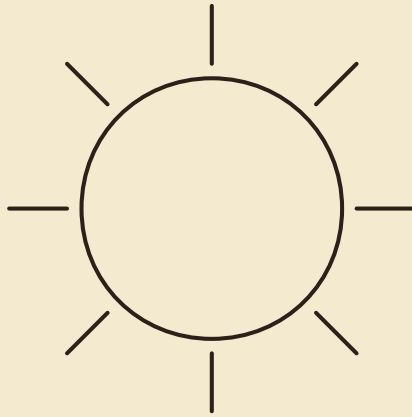


Plate III. [An engraving will be placed here.]

CHAPTER III

The Business and Its World

Can a shop ignore the street, the law and the neighbours?



This chapter explains the forces inside and outside a business, and then asks what a business owes to the people it touches, from workers to investors.

THE PUZZLE OF THE FLOUR PRICE

One morning the price of flour rises by a fifth. Salma did nothing to cause it. A harvest failed in another country, a ship was late, and a price changed in a market she has never seen. Yet her profit falls. A business is therefore not a closed box. It lives among forces that it cannot control, and it also affects the people around it.

THE BUSINESS ENVIRONMENT

The environment is everything around the business.

Business environment. All the forces, inside and outside a business, that affect how it works.

Six outside forces shape most businesses.

Business and environment act on each other.

Responsibility means duties beyond the law.

Social responsibility. The duty of a business to act in ways that benefit society, beyond what the law demands.

Four layers, from profit to giving.

The **business environment** means all the forces that affect how a business works. It has two parts. The *internal environment* is what the owner can control: the owner's aims, the staff, the machines, and the way work is organised. The *external environment* is what the owner cannot control.

Six outside forces matter most. *Economic* forces include prices, incomes and interest rates. *Political and legal* forces include laws, taxes and licences; a bakery needs a food-safety licence. *Social and cultural* forces include tastes and customs; demand for sweets rises before festivals. *Technological* forces include new machines and on-line selling. *Natural* forces include weather, floods and heat. *Competitors, suppliers and customers* form a group often called the market environment.

The relation runs both ways. The environment acts on the business: a new tax raises costs. The business acts on the environment: a large factory changes local jobs, traffic and air. An owner who watches the external environment can prepare, as a farmer who watches the sky prepares for rain.

RESPONSIBILITY TO SOCIETY

Social responsibility means the duty of a business to act in ways that help society, and not only itself. It is often shortened to CSR, for *corporate social responsibility*. The law sets a minimum. Responsibility begins where the minimum ends.

Archie Carroll (1991) pictured four layers of duty. A business must be *profitable* to survive. It must *obey the law*. It should be *ethical*, doing what is fair even where no law demands it. It may be a *good citizen*, giving back to the community. Each layer rests on the one below.

This leads to a useful word. A **stakeholder** means any person or group who is affected by a business. Owners are stakeholders. So are

GO SLOWLY HERE.

A stakeholder is anyone affected by the business.

Stakeholder. Any person or group affected by what a business does.

Customers: safe goods and honest prices.

Workers: fair pay and safe work.

Supplier. A person or firm that sells materials or goods to a business.

How to tell a responsible business.

Milton Friedman held that profit is the duty.

Economic and social aims can support each other.

workers, customers, suppliers, lenders and the local community. The duties of a business differ for each group.

To *customers*, the business owes safe goods, true weights, honest advertising and fair prices. A baker who uses banned colouring harms customers who cannot see the danger.

To *workers*, it owes fair wages, safe conditions and respect. To *suppliers*, a **supplier**, it owes prompt payment and fair dealing. A late payer may ruin a small flour merchant. To *investors*, who put in money, it owes honest accounts and fair use of their funds. To the *community*, it owes clean surroundings and care for the environment.

How can we recognise a responsible business? Look for signs: honest records, safe workplaces, fair pay, goods that match their descriptions, and care for waste and pollution. Such firms keep promises even when no one is checking.

PROFIT AND DUTY TOGETHER

Not everyone agrees on how far duties extend. Milton Friedman (1970) argued that the one responsibility of a business is to raise its profits within the rules of the law. Others reply that a firm which harms its workers or its town will in time lose customers and trust. Their view is that responsibility protects profit in the long run.

The two aims often agree. Safe work reduces accidents, which saves money. Honest dealing keeps customers. Where they conflict, as when safety costs money today, the owner must choose, and the choice shows what the business values. Fair profit and fair dealing together are the usual answer.

A WORKED CASE

SITUATION. Flour for one cake costs Tk 30 at Tk 60 a kilogram, because a cake uses half a kilogram. The price of flour rises by one fifth. All other costs stay the same. Salma sells 20 cakes a week at Tk 400, with a cost of Tk 250 each before the rise.

QUESTION. What is the new weekly profit?

REASONING. One fifth of Tk 60 is Tk 12, so flour becomes Tk 72 a kilogram. Flour per cake is then half of Tk 72, which is Tk 36, a rise of Tk 6. Cost per cake becomes Tk 256. Profit per cake is Tk 400 less Tk 256, which is Tk 144. For 20 cakes, profit is Tk 2,880.

CONCLUSION. Weekly profit falls from Tk 3,000 to Tk 2,880, a fall of Tk 120.

CHECK. Extra cost is Tk 6 a cake. Twenty cakes times Tk 6 equals Tk 120, the same fall.



SUMMARY

The environment of a business has an internal part that the owner controls and an external part that the owner cannot control. Social responsibility asks a firm to do more than the law requires. Each stakeholder has a claim on the business.

THINK IT THROUGH. A factory pours waste water into a river. It breaks no law, because the rule is not yet written. Is it acting responsibly? Which stakeholders are harmed?

WHERE PEOPLE GO WRONG. Many students think social responsibility is only gifts to charity. It is wider; it includes how workers, suppliers and customers are treated every day. Another mistake is to think that the owner alone is a stakeholder.

IN HISTORY. On 24 April 2013 the Rana Plaza building in Savar, near Dhaka, collapsed, and more than 1,100 garment workers died. Cracks had been seen the day before. The disaster led to the Accord on Fire and Building Safety in Bangladesh, signed in May 2013. It showed that a firm's duty to workers is real even when the risk is hidden.

SOURCES

Carroll (1991); Friedman (1970).

REVIEW QUESTIONS

1. Name the six outside forces in the business environment.
2. What is a stakeholder? Give two examples.
3. State one reason why a firm may find profit and duty to agree.

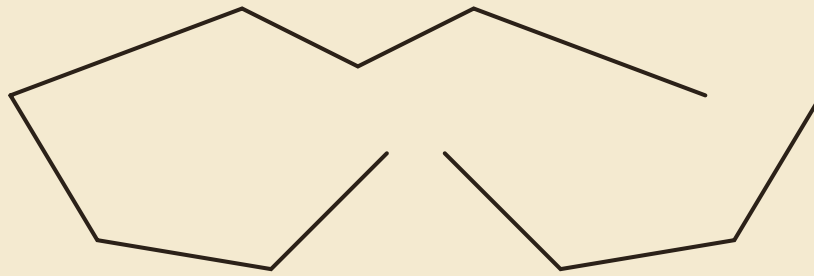


Plate IV. [An engraving will be placed here.]

CHAPTER IV

The Owner and the Partner

Two people, one undertaking, and the question of who answers for what.



This chapter explains the oldest forms of ownership: the business of one person and the business of several. It sets out what each gains and risks, and how a partnership may end.

THE BUSINESS OF ONE

One owner holds all the control.

Sole proprietorship. A business owned, run and financed by one person.

Strengths: speed, control and a full share of profit.

Weaknesses: small capital, unlimited risk, no continuity.

Unlimited liability. Duty to pay all the business's debts, even from personal property.

Best suited to small, personal work.

Two people, each lacking something, decide to join forces.

Salma's shop began as a business that belongs to one person. Many shops, workshops and farms are of this kind. A **sole proprietorship** means a business owned and controlled by one person, who takes all the profit and bears all the loss. The law does not treat the business and its owner as separate.

This form has real strengths. It is quick to start, since few formalities are needed. The owner decides alone, so decisions are fast. All the profit belongs to the owner, which is a strong reason to work hard. The owner knows each customer, and personal care often wins loyalty. Business affairs can also be kept private.

The weaknesses are equally real. Capital is limited to what the owner can save or borrow. The owner has **unlimited liability**, which means that if the business cannot pay its debts, the owner's house and savings may be used. One person has limited skills and cannot be everywhere. And the business is bound to the life and health of its owner.

The form suits small work that depends on personal skill: a tailor, a barber, a small shop, a repairer, a village farm. These activities need little capital and gain from personal attention. It is for this reason that sole proprietors survive beside large firms. Customers who want a custom garment or a neighbour's shop, and not a standard product, remain with the small owner.

THE IDEA OF PARTNERSHIP

Salma's shop is growing. She needs a larger oven and more money than she has. Anwar Hossain, who has savings and an oven, wishes to join. Neither can do it alone. Before a single sack of flour is ordered, a question stands between them: who owns what, and who answers for what?

Law calls it a partnership when people agree to share profit.

Partnership. A business owned by two or more people who agree to run it together and share its profits.

Partners. The individual owners of a partnership.

Firm. The partners taken together.

A partnership begins with agreement, not family or ownership.

The deed is the written agreement.

Partnership deed. The written agreement setting out each partner's rights and duties.

An illustrative case, not a real firm.

Unless agreed otherwise, partners share equally.

GO SLOWLY HERE.

Debts fall on all partners, without limit.

The law gives the name **partnership** to such an arrangement. The people are the **partners**, and together they are the **firm**.¹ In Bangladesh, the Partnership Act of 1932 governs the matter.²

The Act says that a partnership arises from agreement, and not from status. Brothers who inherit a shop are not partners for that reason alone; they become partners if they agree to run it together for profit. More than twenty persons (ten for banking) cannot form a partnership. Beyond that number, the law requires them to form a company.

THE PARTNERSHIP DEED

The agreement may be spoken, but a written one is wiser. Such a document is a **partnership deed**. It records each partner's contribution, the share of profit, the duties, and what happens when a partner leaves. Without it, every disagreement is settled from memory, and memory is a poor witness.

Case file, illustrative. Anwar contributes the oven and Tk 300,000. Salma contributes Tk 100,000, her recipes and her customers. They agree to divide profit equally, because each believes the other's part is worth the same. Whether that is fair is a question the worked case below tests.

SHARING PROFIT, LOSS AND DEBT

Unless the deed says otherwise, the law treats partners as sharing profit and loss equally, whatever each has put in.

There is a second duty, and beginners often overlook it. Each partner is responsible, without limit, for the debts of the firm, and each can be made to pay all of them. If the firm cannot pay, its creditors may look to the personal property of any partner, and not only to what that partner invested. This is unlimited liability again, and it is the heaviest risk of partnership.

The law also adds *implied duties*, which apply even if no one writes them down. Partners must deal with each other in utmost good faith. They must give true accounts, work for the common benefit, and not make a secret profit from the firm's business. They

¹ The firm is not a legal person apart from its partners. Chapter V shows how a company differs.

² The Partnership Act, 1932 (Act IX of 1932), sections 4 and 5.

Partners owe each other honesty and fair play.

Partners differ in work, visibility and risk.

A partnership may be for a task, a period, or open-ended.

Advantages: more money, skills and shared load.

Dissolution means the firm ends.

Dissolution. The ending of a partnership, so that the firm no longer exists.

must use the firm's property for the firm only. Each partner is also an *agent* of the firm: a promise made by one in the course of business binds all.

KINDS OF PARTNER AND OF PARTNERSHIP

Not every partner plays the same role. An *active partner* manages the business. A *sleeping partner* invests money but takes no part in running it. A *nominal partner* only lends a name. A *partner by holding out* is a person who lets others believe that he or she is a partner; the law then makes that person liable like one. The Act of 1932 does not provide for partners whose liability is limited.

Partnerships also differ in how long they last. A *partnership at will* has no fixed end and may be ended by notice. A *particular partnership* is made for one task, such as building a single block. A *partnership for a fixed term* ends when the term does.

GAINS, RISKS AND COMPARISON

A partnership brings more capital than one person can raise. It combines different skills, as Salma's recipes and Anwar's money. It spreads the work and the risk among several people, and its formation needs few formalities. The disadvantages are the unlimited liability, the risk of quarrels, the fact that one partner's mistake binds all, and the difficulty of raising very large sums.

TABLE 3. PARTNERSHIP AND SOLE PROPRIETORSHIP COMPARED

Point	Sole proprietorship	Partnership
Owners	One	Two or more
Capital	One person's means	Pooled by partners
Decisions	By the owner alone	Shared, may cause delay
Liability	Unlimited, one person	Unlimited, each partner for all
Legal basis	No special Act	Partnership Act, 1932
Continuity	Ends with the owner	May end on a partner's death

HOW A PARTNERSHIP ENDS

Dissolution means the ending of a partnership. It may happen in several ways. By *agreement*, when all partners consent. By *notice*, when a partnership at will is ended by one partner telling the others. *Compulsorily*, when all but one partner become insolvent, or when the business becomes unlawful. By *contingency*, such as the end of a fixed term, the completion of a task, or the death of a partner, unless the deed provides otherwise. By *order of a court*, if a partner has become of unsound mind, is permanently unable to act, is guilty of misconduct, or if the business can only run at a loss.

A WORKED CASE

SITUATION. In its first year Padma Bakery makes a profit of Tk 240,000. Anwar contributed Tk 300,000 and Salma Tk 100,000. The deed says only that profit is shared equally.

QUESTION. What does each partner receive? What would each receive if profit followed capital?

REASONING. Equal sharing gives Tk 240,000 divided by 2, which is Tk 120,000 each. By capital, the ratio is 300,000 to 100,000, which is 3 to 1, making 4 parts. One part is Tk 60,000. Anwar receives 3 parts, Tk 180,000; Salma receives 1 part, Tk 60,000.

CONCLUSION. Under the deed, each partner receives Tk 120,000. Sharing by capital would give Anwar Tk 180,000 and Salma Tk 60,000.

CHECK. Tk 120,000 plus Tk 120,000, and Tk 180,000 plus Tk 60,000, each equal Tk 240,000. Equal sharing favours Salma, who gave her skills, and capital sharing favours Anwar, who gave money. The deed should say which is intended.

SUMMARY

A sole proprietorship has one owner who has full control and full risk. A partnership arises from an agreement among two or more persons to share profit; its deed records the terms. Partners have unlimited liability and implied duties. A partnership ends by agreement, notice, compulsion, contingency or court order.

THINK IT THROUGH. Two partners share profits equally, but one works full-time and the other only invests. Is that fair? What would you add to the deed?

WHERE PEOPLE GO WRONG. Students often think a sleeping partner is free of the firm's debts. The sleeping partner is liable too. Another mistake is to think that partners are always equal in profit; the law gives equality only when the deed is silent.

IN HISTORY. The rules on partnership in British India were once part of the Indian Contract Act of 1872. In 1932 they were taken out and made into a separate Act, because partnership had grown into a subject of its own. That Act still governs partnership in Bangladesh.

SOURCES

The Partnership Act, 1932.

REVIEW QUESTIONS

1. What does the law mean by a partnership, and what is its base?
2. Name three kinds of partner.
3. State two differences between a partnership and a sole proprietorship.

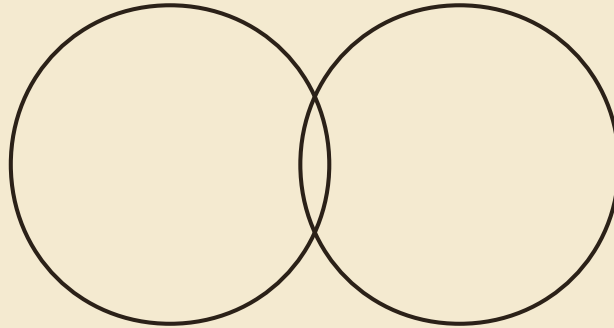


Plate V. [An engraving will be placed here.]

CHAPTER V

The Company

Can a business live on after its owners are gone?



This chapter explains the joint stock company: why it exists, how it is formed, and how private and public companies differ. It weighs its strengths against its weaknesses.

THE PUZZLE OF THE LARGE SUM

The bakery now earns a steady profit. Anwar and Salma wish to build a factory that will cost Tk 5,000,000. They do not have it, and no bank will lend it to partners whose personal wealth is small. Meanwhile, many people in Dhaka have small savings and would gladly join a profitable business if their risk were limited. How can many small sums be gathered into one large one, safely?

THE IDEA OF A JOINT STOCK COMPANY

A company is owned by many through shares.

Joint stock company. A business whose capital is divided into shares and owned by shareholders.

Share. One equal unit of a company's capital, showing part ownership.

Shareholder. A person who owns one or more shares in a company.

A **joint stock company**, usually called a company, means a business whose capital is divided into small equal parts called shares. A **share** means one such part. A person who owns shares is a **shareholder**. The shareholders together own the company.

GO SLOWLY HERE.

The company is a person in law.

Legal personality. The status of being treated by law as a person separate from the owners.

A company has a special feature that a partnership lacks. The law treats it as a *separate legal person*. It can own property, make contracts, sue and be sued in its own name, and it continues to exist when shareholders die or sell their shares. This is called **legal personality**. In Bangladesh the Companies Act, 1994 governs companies.

Limited liability caps the loss at the unpaid share.

Limited liability. A shareholder's duty to pay company debts only up to the unpaid value of his or her shares.

The second feature is **limited liability**. It means that a shareholder who has paid for all of his or her shares cannot be made to pay the company's debts from personal property. The most such a person can lose is the money put into the shares. This is the opposite of the partner's position in Chapter IV, and it explains why people dare to invest.

Shares can be sold; the company stays.

Directors. People elected by shareholders to manage the company.

The third feature is the ease of transfer. A shareholder who wants money can sell shares to another person, and the company carries on unchanged. A fourth feature is management by representatives: the shareholders elect **directors**, who run the company for them.

PRIVATE AND PUBLIC COMPANIES

The Companies Act recognises two chief types. A **private company** has at least two and no more than fifty members, restricts the transfer of its shares, and may not invite the public to buy them. A **public**

Two types, divided by size and by the public.

Private company. A company that limits the number of members and cannot invite the public to buy its shares.

Public company. A company that may invite the public to buy its shares and has no upper limit on members.

A private company suits a family firm.

Formation goes through four stages.

Two documents are filed at registration.

Memorandum of association. The document stating a company's name, office, objects, liability and capital.

Articles of association. The document of internal rules for managing the company.

A prospectus invites the public to invest.

Prospectus. A document inviting the public to buy a company's shares, with full details.

company needs at least seven members, may invite the public to subscribe, and allows free transfer of its shares.

TABLE 4. PRIVATE AND PUBLIC LIMITED COMPANIES COMPARED

Point	Private company	Public company
Members	2 to 50	At least 7, no limit
Shares	Transfer restricted	Freely transferable
Public invitation	Not allowed	Allowed, by prospectus
Start of business	After incorporation	Needs certificate of commencement
Size of capital	Usually smaller	Usually larger

A private company suits a family or a small group that wants limited liability yet wishes to stay in control. A public company suits a business that needs large sums from many people.

HOW A COMPANY IS FORMED

Forming a company follows stages. First comes *promotion*: someone has the idea, studies it and gathers the founders. Second comes *incorporation*, which means registration with the Registrar of Joint Stock Companies and Firms. Third, for a public company, comes the raising of capital. Fourth comes the start of business.

At registration two documents are essential. The **memorandum of association** states the company's name, its registered office, the objects for which it exists, the limit on its members' liability, and its capital. It fixes what the company may do. The **articles of association** set the internal rules: how directors are chosen, how meetings are held, and how profits are paid. In short, the memorandum is the charter and the articles are the rulebook. When the Registrar is satisfied, a *certificate of incorporation* is issued, and the company is born as a legal person.

A public company that wants money from the public issues a **prospectus**. It describes the company's name and objects, its directors, the capital, the number and price of shares offered, the money it must raise at minimum, how the money will be used, and the risks. Its purpose is honest information, so that investors know what they buy. A public company must also obtain a *certificate of commencement of business* before it begins to trade.

STRENGTHS AND WEAKNESSES

A company can raise large sums from many people. Limited liability lowers each investor's risk. It has a continuing life and can hire

*Large capital, limited risk,
long life.*

*Weaknesses: costs, rules and
distance of owners.*

Is the company the best form?

Dividend. The share of
profit paid to shareholders.

skilled managers. Shares are easy to transfer. A large company can use large machines and gain from large-scale production.

There are weaknesses. Formation is costly and slow, and the law imposes many rules, reports and audits. The owners are far from the work, so directors may serve themselves. A few big shareholders may control a company against the small ones. Secrets are harder to keep, since accounts must be published.

Is the company the best form of business? The answer depends on the task. For a large factory needing much capital, it is hard to match. For a small tailor, it would be too heavy. The best form is the one that fits the size, the risk and the aim.

A WORKED CASE

SITUATION. Padma Bakery becomes Padma Bakery Private Limited. Capital is Tk 1,000,000 in 100,000 shares of Tk 10 each. Anwar holds 60,000 shares and Salma 40,000. In the first year the company decides to pay Tk 120,000 of its profit to shareholders as **dividend**.

QUESTION. What dividend does each share earn, and how much does each person receive?

REASONING. Dividend per share is Tk 120,000 divided by 100,000 shares, which is Tk 1.20. Anwar receives 60,000 times Tk 1.20, which is Tk 72,000. Salma receives 40,000 times Tk 1.20, which is Tk 48,000.

CONCLUSION. Each share earns Tk 1.20. Anwar receives Tk 72,000 and Salma Tk 48,000.

CHECK. Tk 72,000 plus Tk 48,000 equals Tk 120,000, the whole dividend. Note too that if the company later fails owing more than it owns, Anwar and Salma lose at most the value of their shares.



SUMMARY

A company is owned by shareholders through shares, is a separate legal person, and gives limited liability. A private company has 2 to 50 members and cannot invite the public; a public company has at least 7 members and may do so. Formation needs a memorandum, articles, and incorporation.

THINK IT THROUGH. A shop owner wants limited liability but wishes to keep the shop within the family. Which type of company would you advise, and why?

WHERE PEOPLE GO WRONG. Limited liability does not mean that the company cannot be sued. The company can, and the shareholder is protected from its debts. Another mistake is to mix up the memorandum, which fixes the company's objects, with the articles, which govern its daily rules.

IN HISTORY. In the case of *Salomon v A Salomon & Co Ltd* (House of Lords, 1896), a shoemaker turned his business into a company and held nearly all its shares. When the company failed, creditors asked to claim from him personally. The court held that the

company was a separate person, and Mr Salomon was not liable for its debts. The decision still guides the law of companies.

SOURCES

The Companies Act, 1994; *Salomon v A Salomon & Co Ltd* (1896).

REVIEW QUESTIONS

1. Name the main features of a joint stock company.
2. How do private and public companies differ?
3. What do the memorandum and articles of association contain?

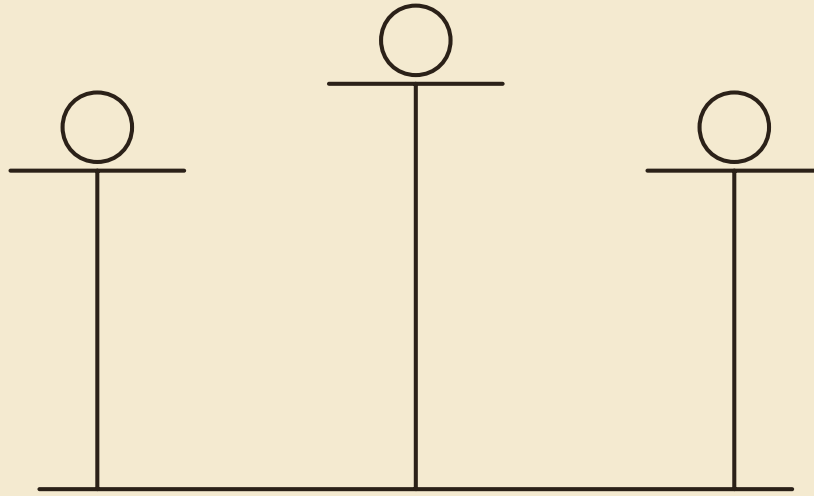


Plate VI. [An engraving will be placed here.]

CHAPTER VI

Many Hands: Cooperatives and the State

Who should own a business when profit is not the first aim?



This chapter explains two forms of ownership whose first aim is service: the cooperative society, owned by its members, and the state enterprise, owned by the public.

THE PUZZLE OF THE FLOUR BAG

Salma buys flour by the sack, and she pays the full shop price. So do nine other bakers in Mirpur. A merchant who sold to all ten at once could offer a lower price, but each baker alone is too small to ask for one. What if the ten bakers bought together, and shared the savings? This question leads to a form of business owned by those who use it.

THE COOPERATIVE SOCIETY

A cooperative is owned by its members for their own use.

Cooperative society. A business owned and run by its members to serve their common needs, not to earn profit for outsiders.

Seven principles guide cooperatives worldwide.

GO SLOWLY HERE.

One member, one vote: the key difference from a company.

How a society works, step by step.

Cooperatives come in many types.

Cooperatives help farmers sell their produce.

A **cooperative society** means a business formed by people who join voluntarily to meet a common need, and who own and control it together. Its first aim is service to its members. In Bangladesh, the Cooperative Societies Act of 2001 governs such societies, which must be registered with the Registrar of Cooperative Societies.

Cooperatives follow shared principles, set out by the International Cooperative Alliance (1995). Membership is *voluntary and open* to all who can use the service. Control is *democratic*: each member has one vote, whatever the shares held. Members contribute *capital* fairly and share the surplus. The society is *independent* of outside control. It provides *education* for members. Cooperatives *work together*. And they show *concern for the community*.

This differs sharply from a company. In a company, power follows shares: a person with more shares has more votes. In a cooperative, power follows people: one member has one vote. This rule keeps a rich member from ruling the poor.

A society works in this way. Members, who usually number at least a set minimum, apply for registration with a draft of rules called *by-laws*. Each member buys shares, and the liability of each is usually limited to the amount of the shares. A general meeting of all members elects a managing committee. The committee runs the society and answers to the members. At the year's end, the *surplus* is divided: a part goes to a reserve fund, a limited part is paid as a dividend on shares, and the rest is returned to members in proportion to their use of the society.

In Bangladesh, there are many types: *agricultural* societies that supply seeds and tools, *credit* societies that lend small sums, *consumers'* societies that sell goods at fair prices, *marketing* societies, *weavers'* and *fishermen's* societies, *housing* societies, and *dairy* societies.

The cooperative has a particular value in marketing farm produce. A single farmer who sells alone to a trader is often paid a low price. A society can collect the produce of many farmers, grade it, store it, and sell in bulk at a better price. It cuts out some middlemen, gives price information, and may lend money until the sale is made.

Cooperatives also have weaknesses. Members may lack business skill. Funds may be small. Poor management or political pressure

Limits: poor management and weak funds.

can damage trust. A cooperative is only as good as the care of its members.

THE STATE ENTERPRISE

State enterprise is owned and run by the government.

State enterprise. A business owned and controlled by the government.

A **state enterprise** means a business owned and controlled by the government on behalf of the public. It may be fully owned or partly owned by the state. Governments run enterprises for several reasons.

Four reasons for public ownership.

First, some goods are too important or too costly to leave to private owners, such as railways and defence supplies. Second, some activities have a *natural monopoly*, where one supplier serves best, as with a gas pipeline. Third, a government may wish to hold prices down for the poor. Fourth, new industries may need public money before private investors will enter. The fields best suited to state ownership therefore include defence, railways, power, water, and basic industries.

TABLE 5. FORMS OF STATE ENTERPRISE

Form	Meaning	Example
Departmental undertaking	Run as a government department	Bangladesh Railway
Public corporation	Set up by a special law, with its own board	Bangladesh Petroleum Corporation
Government company	A company under the Companies Act, with the government holding most shares	Titas Gas Transmission and Distribution Company Ltd

Each form gives different freedom.

A departmental undertaking is closest to government and has least freedom. A public corporation has a board and more freedom of action. A government company works like a private company, and the government is its main shareholder.

A fair debate: for and against.

Is it wise to support state enterprise? The arguments differ. Supporters say that it serves the public, protects strategic industries, and can hold prices down. Critics say that many state enterprises in Bangladesh and elsewhere have run at a loss. The causes commonly named are weak management, too many employees, political interference, prices fixed below cost, old machines, and little reward for efficiency. A reader should weigh both sides, and decide for which tasks, and under what controls, state ownership serves the public best.

A WORKED CASE

SITUATION. Ten Mirpur bakers form a flour-buying society. A 50-kilogram bag costs Tk 3,000 in a shop and Tk 2,850 in bulk. Padma Bakery buys 20 bags a month.

QUESTION. What does Padma save each month by buying through the society?

REASONING. The saving per bag is Tk 3,000 less Tk 2,850, which is Tk 150. For 20 bags the saving is 20 times Tk 150.

CONCLUSION. Padma saves Tk 3,000 a month, or Tk 36,000 in a year.

CHECK. Shop cost is 20 times Tk 3,000, which is Tk 60,000. Bulk cost is 20 times Tk 2,850, which is Tk 57,000. The difference is Tk 3,000, the same answer.



SUMMARY

A cooperative society is owned by its members, who each have one vote; its first aim is service. A state enterprise is owned by the government and takes three forms: departmental, public corporation and government company. Each has strengths and limits.

THINK IT THROUGH. A village has one doctor and no clinic. Should the clinic be a cooperative, a state enterprise or a private company? What would each choice cost and give?

WHERE PEOPLE GO WRONG. Students often say that a cooperative earns no profit. It earns a surplus, but returns it to members. Another mistake is to assume that state enterprises always lose money; some earn profit, and results differ widely.

IN HISTORY. In 1844 a group of 28 workers in Rochdale, England, many of them weavers, opened a small shop to sell pure goods at fair prices. They gave each member one vote and shared the surplus by purchases. Their rules became the model for cooperatives around the world. In Bangladesh, the Academy for Rural Development at Comilla, set up in 1959, later developed a two-tier cooperative scheme for villages.

SOURCES

The Cooperative Societies Act, 2001; International Cooperative Alliance (1995).

REVIEW QUESTIONS

1. State the principles on which a cooperative society is built.
2. Name the three forms of state enterprise.
3. Give two reasons for state enterprise.

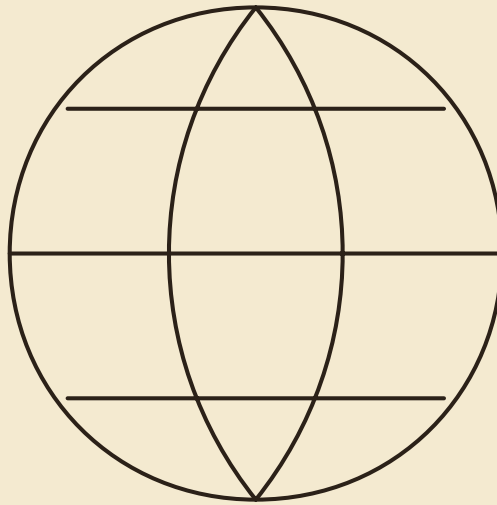


Plate VII. [An engraving will be placed here.]

CHAPTER VII

Across the Border

Why does a country that grows rice buy wheat from abroad?



This chapter explains why nations trade, how trade is measured, how goods are exported and imported, and what barriers stand in the way. It ends with the firm that works in many countries.

THE PUZZLE OF THE FOREIGN OVEN

Padma Bakery needs a larger oven. The best ones are built in another country. To buy one, Anwar must pay in foreign money, deal with a seller he has never met, and bring the machine through a customs office. At the same time, a bakery in another land might want to buy Bangladeshi jute bags. Why do people cross borders to trade, and how is it done safely?

WHY NATIONS TRADE

No country has everything.

Foreign trade. Buying and selling of goods and services between people in different countries.

Exports. Goods or services sold to another country.

Imports. Goods or services bought from another country.

Foreign trade means buying and selling across national borders. Goods sold to another country are **exports**. Goods bought from another country are **imports**. Nations trade first because they differ: in climate, soil, minerals, skills and machines.

GO SLOWLY HERE.

Each country should make what costs least to give up.

Comparative advantage.

The ability to make something at a lower opportunity cost than others.

A deeper reason was given by David Ricardo (1817). A country gains by making what it makes at the lowest *opportunity cost*, and buying the rest, even if it could make everything. Suppose that in a day a baker can make 40 loaves or mend 4 ovens, while a mechanic can make 10 loaves or mend 5 ovens. The mechanic is better at mending, and the baker is better at loaves compared with mending. Each should specialise, then trade. This is **comparative advantage**.

Trade brings foreign money, jobs and choice.

Foreign trade matters for several reasons. Exports earn foreign money, which pays for needed imports. Export industries create jobs. Imports bring goods that the country cannot make, and bring new machines and ideas. Competition from abroad pushes local firms to improve, and larger markets allow large-scale production.

MEASURING TRADE BETWEEN NATIONS

Nations measure their trade in two ways. The **balance of trade** means the difference between the value of goods exported and goods imported. If exports are larger, the balance is a surplus; if imports are larger, a deficit. The **balance of payments** is wider. It records *all* payments between a country and the rest of the world in a period: goods, services such as shipping and tourism, transfers such

Two measures: of goods only, and of everything.

Balance of trade. The difference between the value of a country's goods exported and goods imported.

Balance of payments. A record of all money flowing into and out of a country in a period.

Export follows a chain of steps.

Letter of credit. A bank's promise to pay the seller if the agreed documents are shown.

Import follows the reverse chain.

The documents carry the proof.

Bill of lading. A receipt for goods shipped, which also acts as proof of ownership.

Bill of lading and charter party differ.

Charter party. A contract hiring a whole ship, or part of it, for a voyage.

Barriers: narrow range, slow ports, costly funds.

as money sent home by workers abroad, and capital flows such as loans and investment.

The balance of trade can show a deficit while the wider balance of payments is in surplus, because money sent home by citizens working abroad can fill the gap.

EXPORTING AND IMPORTING

The *export procedure* runs in steps. The exporter first registers as an exporter with the government's import and export office. An enquiry arrives from a buyer, and the exporter sends a quotation, often as a *proforma invoice*. The buyer places an order and opens a **letter of credit** through a bank. This promise reassures the seller. The exporter then makes or collects the goods, arranges inspection, packs them, and books a ship or plane through a clearing agent. Customs inspect the goods and permit export. The carrier issues a transport document. The exporter hands the documents to the bank, which collects payment.

The *import procedure* is the reverse. The importer registers, asks the seller for a proforma invoice, and opens a letter of credit. When the goods arrive, the importer files a *bill of entry* with customs, pays duty, and takes delivery with the shipping documents.

The documents used include the *commercial invoice* (the bill), the *packing list*, the *certificate of origin* (where the goods were made), the *inspection certificate*, the *bill of exchange* (an order to pay) and the transport document. For sea freight the transport document is the **bill of lading**.

Do not confuse it with a **charter party**. The bill of lading is a receipt for goods and a title to them, issued by the shipping company to each shipper. The charter party is the contract by which a merchant hires the whole ship or a space on it. One ship may carry goods under many bills of lading, but be hired under one charter party.

BARRIERS AND SUPPORT

Bangladeshi exporters meet barriers. Exports depend heavily on a narrow range of goods, chiefly ready-made garments. Port and road delays and a shortage of power raise costs. Loans may be costly, and customs work may be slow. Foreign buyers set strict standards of quality and safety. Other countries may raise import taxes, and competitors abroad are strong.

Governments both support and regulate trade. A **tariff** raises the price of imports. A **quota** caps the amount. A government may give exporters cash support, cheap loans or tax relief, and runs bodies to help them: the Export Promotion Bureau, and the export processing zones (Chapter VIII). At the world level, the World Trade Organiza-

Governments tax, limit and support trade.

Tariff. A tax on imported goods.

Quota. A limit on the quantity of a good that may be imported.
A multinational works in several countries.

Multinational corporation. A large firm that owns and runs business operations in more than one country.

Home country. The country where a multinational has its headquarters.

Host countries. The countries in which a multinational runs operations away from its home.

Staff come from three places.

PCN. A parent country national: an employee from the firm's home country.

HCN. A host country national: an employee from the country where the branch operates.

TCN. A third country national: an employee from a country that is neither home nor host.

What hosts gain.

What hosts may lose.

tion, formed in 1995, sets the rules by which its member countries trade and settles disputes between them.

THE FIRM IN MANY COUNTRIES

A **multinational corporation** (MNC) is a firm that owns and runs operations in more than one country. Its first base is its **home country**. The countries where it opens branches are **host countries**. Bangladesh is a host to many.

The firm's staff are grouped in three. A **PCN** comes from the home country, a **HCN** from the host country, and a **TCN** from some other country.

Host countries gain from MNCs. They bring capital, modern machines and skills. They create jobs, pay taxes, and open foreign markets for local products. Local firms learn new methods.

Host countries may also suffer. Profits may be sent abroad. Large MNCs may crowd out small local firms. Some MNCs pay low wages, harm the environment or influence politics. Technology may be kept within the parent firm. Whether the gain outweighs the loss depends on the host's laws and its ability to bargain.

A WORKED CASE

SITUATION. An illustrative country, in billions of dollars. Goods exports are 40 and goods imports are 52. Services earn 6 and cost 9. Workers abroad send home 22. Take the current account to be the sum of these items.

QUESTION. What are the balance of trade and the current account?

REASONING. Balance of trade is 40 less 52, which is a deficit of 12. Services net is 6 less 9, which is minus 3. Add the transfers of plus 22. The current account is minus 12, minus 3, plus 22.

CONCLUSION. The trade balance is a deficit of 12 billion. The current account is a surplus of 7 billion.

CHECK. Receipts are 40 plus 6 plus 22, which is 68. Payments are 52 plus 9, which is 61. The difference, 68 less 61, is 7, the same answer.

SUMMARY

Nations trade because they differ and because specialising lowers cost. The balance of trade counts goods only; the balance of payments counts all payments. Export and import follow set steps and documents. MNCs bring gains and risks to host countries.

THINK IT THROUGH. A country sells only one product abroad. What risks does it run? How might it reduce them?

WHERE PEOPLE GO WRONG. Students often use balance of trade and balance of payments as if they meant the same. They do not. A trade deficit is also not always bad; imports of machines can build future output.

IN HISTORY. The English East India Company received a royal charter on 31 December 1600 to trade with Asia. After the battle of Plassey in 1757 and the grant of the *diwani* in 1765, it began to collect revenue in Bengal. It was an early multinational, and it shows how trade, power and local welfare may become tangled.

SOURCES

Ricardo (1817), chapter on foreign trade.

REVIEW QUESTIONS

1. What is the difference between the balance of trade and the balance of payments?
2. Name four documents used in export.
3. Explain PCN, HCN and TCN.

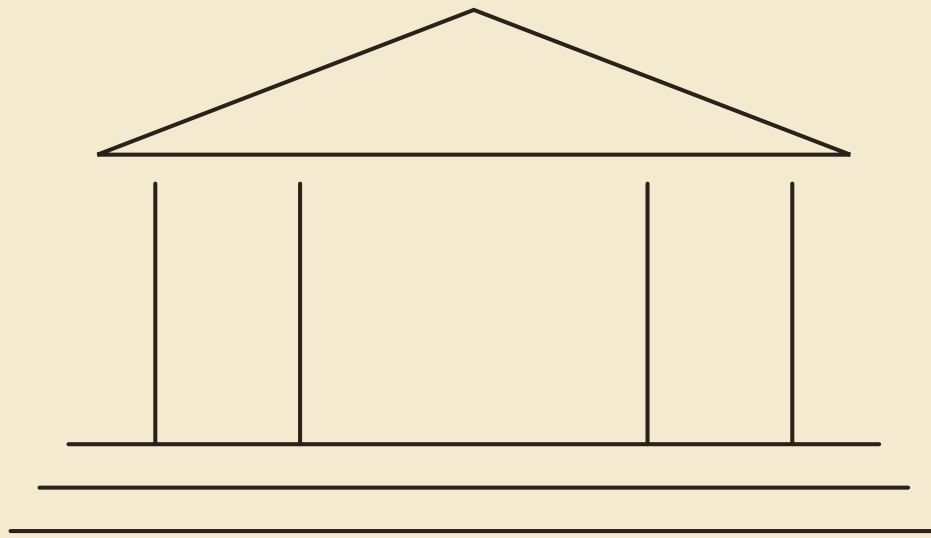


Plate VIII. [An engraving will be placed here.]

CHAPTER VIII

Shares, Markets and Helping Institutions

How does a small baker find strangers willing to share her risk?



This chapter explains the share market, the Dhaka Stock Exchange and the IPO, and then names the bodies that regulate, promote and support business in Bangladesh.

THE PUZZLE OF THE STRANGER'S MONEY

Padma Bakery now wishes to raise Tk 5,000,000 from the public. A stranger in Chattogram has savings and might buy shares, but how would she know the company is honest? And if she needed her money back next year, who would buy her shares? Two problems must be solved: trust, and the chance to sell. The institutions in this chapter exist to solve them.

THE SHARE MARKET

The share market is where shares are bought and sold.

Share market. The market where shares of companies are issued, bought and sold.

WORD IN USE.

Stock. Daily life: goods held for sale. Finance: shares of a company.

An IPO is a company's first public offer of shares.

Initial public offering. The first sale of a company's shares to the public, called IPO for short.

Two types of share, with different aims.

Ordinary share. A share whose holder votes and receives a dividend that depends on profit.

Preference share. A share that receives its fixed dividend before ordinary shares.

The **share market** is the market in which shares of companies are issued, bought and sold. It has two parts. In the *primary market*, a company sells new shares to investors for the first time and receives the money. In the *secondary market*, investors sell shares to each other, and the company receives nothing. A person who buys shares hopes for dividends and for a rise in the share's price.

The word *stock* has two meanings, and finance uses the second. In a shop, stock means the goods held for sale. In the share market, it means the shares of companies, so *stock market* and *share market* mean the same place.

When a company first offers its shares to the public, the offer is an **initial public offering**, known as an IPO. The company publishes a prospectus (Chapter V), and the regulator must approve the offer. Investors apply for shares, and if applications exceed the shares offered, the shares are shared out by a set rule. After the IPO, the shares are listed on an exchange so that they can be traded.

Shares are of two main types. An **ordinary share** gives a vote and a dividend that rises and falls with profit. A **preference share** receives a fixed dividend before ordinary shareholders, but usually has no vote. Which is best for an investor? It depends on the aim. An investor who wants steady income and lower risk may prefer preference shares. An investor who accepts more risk for possible growth may prefer ordinary shares.

THE DHAKA STOCK EXCHANGE

An exchange is the organised place for trading.

Stock exchange. An organised market where listed shares are bought and sold under set rules.

A **stock exchange** is an organised market in which listed shares are traded under set rules. Dhaka Stock Exchange (DSE) began in 1954. Its functions are these. It provides a fair, orderly place for trading. It lists companies that meet its rules. It publishes prices so that all can see them. It gives investors the chance to sell, so that savers are willing to invest. It supervises its member brokers and settles trades. It thus supports companies in raising capital.

HELPING INSTITUTIONS

A regulator protects investors.

Bangladesh Securities and Exchange Commission.

The state regulator of the share market, set up in 1993.

Trade bodies speak for business.

FBCCI. The Federation of Bangladesh Chambers of Commerce and Industry, the main body of business chambers.

Exports are helped by three bodies.

Export processing zone.

An area where firms making goods for export enjoy special tax and service benefits.

Other bodies support the whole system.

The **Bangladesh Securities and Exchange Commission (BSEC)** was set up under the Securities and Exchange Commission Act of 1993. It regulates the share market. It approves public offers, licenses brokers and exchanges, watches for unfair practice, and protects investors.

The **FBCCI** is the national federation of chambers of commerce and industry. It speaks for business to the government, offers advice and organises trade fairs.

The Export Promotion Bureau (EPB) is a government body that helps exporters with information, fairs and approvals. An **export processing zone (EPZ)** is an area where firms producing for export enjoy special benefits such as duty-free import of machines and materials, tax relief, and one office for services. The Bangladesh Export Processing Zones Authority, created by a law of 1980, runs them; the first zone, at Chattogram, began work in 1983. The Trading Corporation of Bangladesh (TCB, sometimes TCCB) is a state body that buys and sells essential goods, such as edible oil and pulses, to hold prices fair.

Other bodies help too. The central bank, Bangladesh Bank, regulates money and banks. The Bangladesh Investment Development Authority helps investors, both local and foreign, to start and register business.

A WORKED CASE

SITUATION. Padma Bakery Public Limited offers 500,000 shares at Tk 10 each. Investors apply for 1,000,000 shares in all.

QUESTION. How much money is raised? How many times is the offer subscribed? What share of what each applicant asked for can be given, if shares are divided in proportion?

REASONING. Money raised is 500,000 times Tk 10, which is Tk 5,000,000. Subscription is 1,000,000 divided by 500,000, which is 2 times. The share given is 500,000 divided by 1,000,000, which is one half.

CONCLUSION. The company raises Tk 5,000,000. The offer is subscribed twice, and each applicant would receive about half of the shares asked for. In practice, the regulator's rules fix how the shares are divided.

CHECK. Half of 1,000,000 is 500,000, which equals the shares offered.



SUMMARY

The share market has a primary part, where companies raise money, and a secondary part, where investors trade. An IPO is the first public offer. The BSEC regulates the market; the DSE runs trading; the FBCCI, EPB, EPZ authority and TCB each help a different part of business.

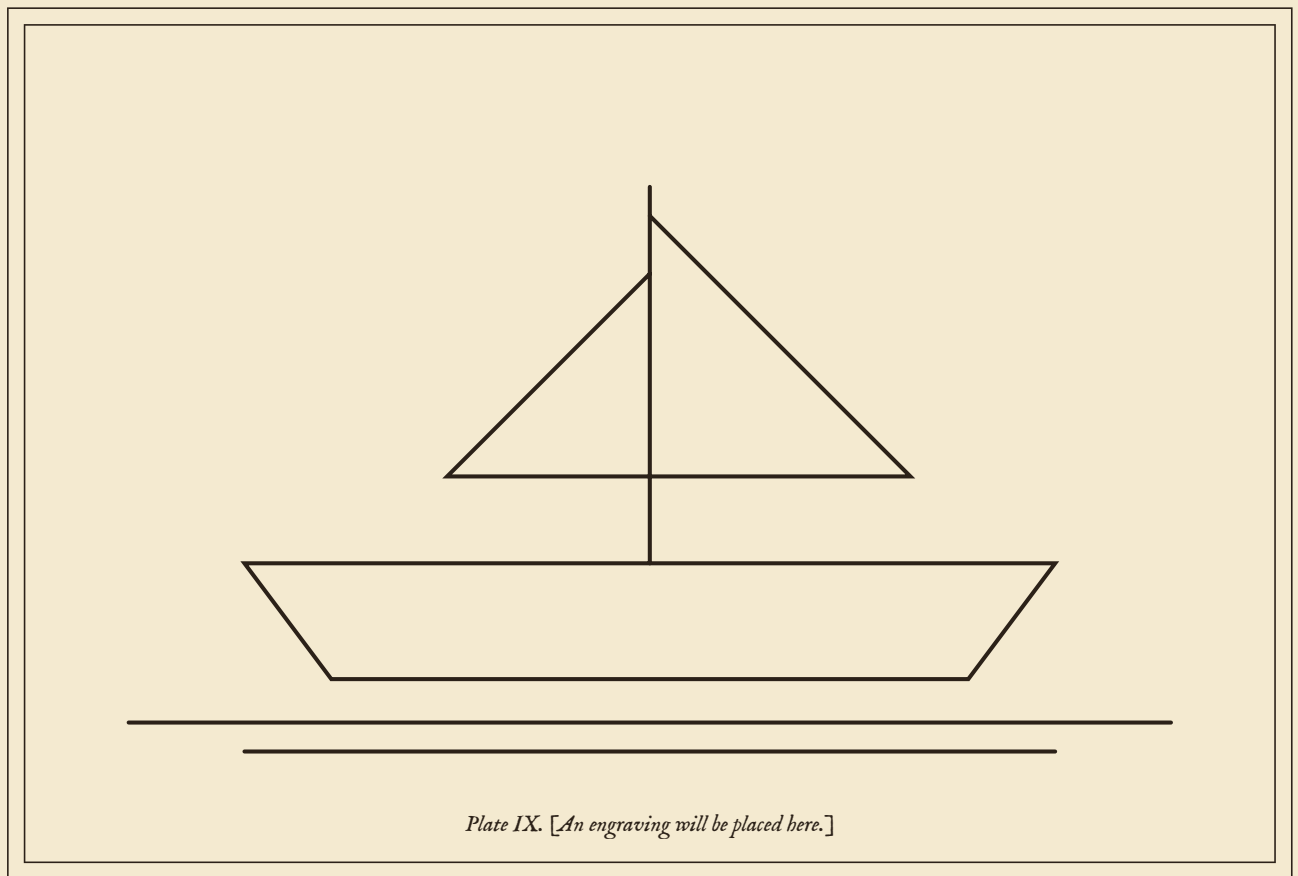
THINK IT THROUGH. A retired teacher wants steady income and fears losing her savings. Which kind of share might suit her better? What might she lose by choosing it?

WHERE PEOPLE GO WRONG. Students often think the company receives money whenever its shares are traded. It receives money only when it issues new shares. Another mistake is to assume that a high share price means a good company.

IN HISTORY. In 1602 the Dutch East India Company sold shares to the public, and soon people traded them on the Amsterdam exchange. It is often called the first company to have shares traded in this way, and it created the pattern on which exchanges later grew.

REVIEW QUESTIONS

1. What is an IPO, and who approves it in Bangladesh?
2. State three functions of the Dhaka Stock Exchange.
3. Distinguish ordinary from preference shares.



CHAPTER IX

Joining Together

Why would two rivals choose to become one?



This chapter explains why businesses combine, what forms combination takes, and what it offers and threatens.

THE PUZZLE OF THE RIVAL SHOP

Across the road from Padma Bakery stands Mirpur Sweets. Each shop pays rent, buys flour, and competes for the same customers. If they joined, rent could be shared, flour bought in bulk, and price wars ended. Customers, though, might pay more. Why do firms combine, and should the public welcome it?

WHY AND HOW FIRMS COMBINE

A combination links independent businesses.

Business combination. The joining of separate businesses into a single group or firm.

Three directions of joining.

Four common forms.

Cartel. An agreement among firms to fix prices or share markets while staying separate.

Trust. A combination in which firms hand control to a board that runs them as one.

Holding company. A company that owns enough shares in other companies to control them.

Merger. The joining of one firm into another, so that only one remains.

A **business combination** means the joining of separate businesses so that they act together or as one. Firms combine for several reasons. They want to cut costs by sharing and by buying in bulk. They want to reduce competition. They wish to be sure of raw materials or of outlets for their goods. They want larger markets, stronger funds, and a way to spread risk.

Combinations are grouped by direction. A *horizontal* combination joins firms at the same stage, such as two bakeries. A *vertical* combination joins firms at different stages of one chain, such as a flour mill and a bakery. A *conglomerate* joins firms in unrelated lines of business.

Combinations take several forms. A **cartel**, also called a pool, is an agreement among independent firms to fix prices or divide markets. A **trust** is a combination in which owners hand control to a single board. A **holding company** owns enough shares in other companies, called its subsidiaries, to control them. A **merger** joins one firm into another, so that only one remains; in an *amalgamation* two or more firms join to form a new one.

TABLE 6. FORMS OF COMBINATION COMPARED

Form	Firms remain separate?	Control
Cartel	Yes	Agreement only
Trust	Not in practice	One board
Holding company	Yes in law	Share ownership
Merger	No	One firm remains

OPPORTUNITY AND THREAT

Gains: lower cost, stronger firms, wider reach.

GO SLOWLY HERE.

Threats: high prices, lost jobs, less choice.

Combination offers real gains. Large firms enjoy lower cost per unit, can pay for research, deal better with suppliers, and can sell in new places. A weak firm may be saved by joining a strong one.

Combination also carries threats. If competition ends, a combined firm may raise prices and give poorer service. Jobs may be cut when duplicate work ends. Small suppliers may lose bargaining power. Very large firms may influence government. For these reasons, many countries have competition laws. Bangladesh passed the Competition Act of 2012 to watch for unfair agreements and abuse of strong positions.

A WORKED CASE

SITUATION. Padma Bakery's monthly costs are Tk 140,000 and Mirpur Sweets' are Tk 120,000. After a merger, shared rent, bulk flour and one office reduce the combined monthly costs to Tk 235,000.

QUESTION. What is the saving, as an amount and as a percentage of the old combined cost?

REASONING. Old combined cost is Tk 140,000 plus Tk 120,000, which is Tk 260,000. Saving is Tk 260,000 less Tk 235,000, which is Tk 25,000. As a percentage, 25,000 divided by 260,000 is about 0.096, or 9.6 per cent.

CONCLUSION. The merger saves Tk 25,000 a month, nearly 10 per cent of old costs.

CHECK. Ten per cent of Tk 260,000 is Tk 26,000, a little more than Tk 25,000, so the answer of 9.6 per cent is sensible.

SUMMARY

Firms combine to cut costs, reduce competition, secure supplies and grow. Combinations may be horizontal, vertical or conglomerate, and take the form of cartels, trusts, holding companies or mergers. They bring gains and threats, so laws watch them.

THINK IT THROUGH. Two of the three bus companies in a town propose to merge. Name one gain and one threat for passengers.

WHERE PEOPLE GO WRONG. Students often think that every combination ends competition. A merger of two small firms may create a better rival to a large one. Another mistake is to call every large firm a trust; a trust is a particular form.

IN HISTORY. In 1882 John D. Rockefeller's oil firms were joined in the Standard Oil Trust in the United States. It came to control most of the country's oil refining. In 1911 the US Supreme Court ordered the company to be broken into separate companies. The case shaped modern competition law.

REVIEW QUESTIONS

1. Why do business firms combine?
2. Name the four common forms of combination.

3. State one gain and one threat of combination.

ANSWERS TO REVIEW QUESTIONS

- I.** (1) Exchange, regularity, profit aim, risk, and creation of utility. (2) Trade is buying and selling; commerce adds the aids such as banking and transport. (3) Economic, social, human and national.
- II.** (1) Land: rent; labour: wages; capital: interest; organisation: profit. (2) Other costs, such as delivery, may change the total. (3) For example, customers, raw materials, labour, transport or power.
- III.** (1) Economic, political and legal, social and cultural, technological, natural, and the market. (2) A person or group affected by a business, such as workers and customers. (3) Safe work reduces accidents and costs, and honesty keeps customers.
- IV.** (1) A business of two or more persons who agree to share profit; its base is agreement. (2) Active, sleeping and nominal (or by holding out). (3) Number of owners, and pooled capital against one person's means.
- V.** (1) Shares, separate legal personality, limited liability, transferable shares and elected directors. (2) In members, transfer of shares, and the right to invite the public. (3) The memorandum states name, office, objects, liability and capital; the articles give internal rules.
- VI.** (1) Open voluntary membership, one member one vote, fair capital, independence, education, cooperation, community concern. (2) Departmental undertaking, public corporation and government company. (3) For example, strategic industries and natural monopolies.
- VII.** (1) Trade counts goods only; payments count goods, services, transfers and capital. (2) For example, commercial invoice, packing list, bill of lading and certificate of origin. (3) Employees from the home, host and a third country.
- VIII.** (1) The first public sale of a company's shares; the BSEC approves it. (2) For example, providing a trading place, listing companies and publishing prices. (3) Ordinary shares vote and earn variable dividend; preference shares earn fixed dividend first.
- IX.** (1) To cut costs, reduce competition, secure supplies, grow and spread risk. (2) Cartel (pool), trust, holding company and merger or amalgamation. (3) Lower cost per unit; higher prices if competition ends.

GLOSSARY

Aids to trade. Services that help goods move from maker to buyer, such as banking. Ch. I, p. 7

Articles of association. The document of internal rules for managing the company. Ch. V, p. 27

Balance of payments. A record of all money flowing into and out of a country in a period. Ch. VII, p. 36

Balance of trade. The difference between the value of a country's goods exported and goods imported. Ch. VII, p. 36

Bangladesh Securities and Exchange Commission. The state regulator of the share market, set up in 1993. Ch. VIII, p. 43

Bill of lading. A receipt for goods shipped, which also acts as proof of ownership. Ch. VII, p. 37

Business. An activity that provides goods or services to customers to earn a profit. Ch. I, p. 6

Business combination. The joining of separate businesses into a single group or firm. Ch. IX, p. 46

Business environment. All the forces, inside and outside a business, that affect how it works. Ch. III, p. 16

Capital. Man-made goods, such as machines and tools, that are used to produce other goods. Ch. II, p. 12

Cartel. An agreement among firms to fix prices or share markets while staying separate. Ch. IX, p. 46

Charter party. A contract hiring a whole ship, or part of it, for a voyage. Ch. VII, p. 37

Commerce. Trade together with the services that help trade happen. Ch. I, p. 7

Comparative advantage. The ability to make something at a lower opportunity cost than others. Ch. VII, p. 36

Cooperative society. A business owned and run by its members to serve their common needs, not to earn profit for outsiders. Ch. VI, p. 32

Customer. A person or firm that pays for goods or services. Ch. I, p. 6

Directors. People elected by shareholders to manage the company. Ch. V, p. 26

Dissolution. The ending of a partnership, so that the firm no longer exists. Ch. IV, p. 22

Dividend. The share of profit paid to shareholders. Ch. V, p. 28

Division of labour. Splitting a job into parts, with each worker doing one part. Ch. II, p. 12

E-commerce. Buying and selling through the internet or other electronic networks. Ch. I, p. 7

Entrepreneur. A person who organises production and takes the risk of loss. Ch. II, p. 13

Export processing zone. An area where firms making goods for export enjoy special tax and service benefits. Ch. VIII, p. 43

Exports. Goods or services sold to another country. Ch. VII, p. 36

Factors of production. The resources used to make goods and services: land, labour, capital and organisation. Ch. II, p. 12

FBCCI. The Federation of Bangladesh Chambers of Commerce and Industry, the main body of business chambers. Ch. VIII, p. 43

Firm. The partners taken together. Ch. IV, p. 21

Foreign trade. Buying and selling of goods and services between people in different countries. Ch. VII, p. 36

Goods. Things you can touch and keep, such as a cake or a chair. Ch. I, p. 6

HCN. A host country national: an employee from the country where the branch operates. Ch. VII, p. 38

Holding company. A company that owns enough shares in other companies to control them. Ch. IX, p. 46

Home country. The country where a multinational has its headquarters. Ch. VII, p. 38

Host countries. The countries in which a multinational runs operations away from its home. Ch. VII, p. 38

Imports. Goods or services bought from another country. Ch. VII, p. 36

Industry. All activities that produce or change goods, or extract them from nature. Ch. I, p. 7

Initial public offering. The first sale of a company's shares to the public, called IPO for short. Ch. VIII, p. 42

Interest. The payment made for the use of borrowed money or capital. Ch. II, p. 13

Joint stock company. A business whose capital is divided into shares and owned by shareholders. Ch. V, p. 26

Labour. The physical and mental work of people in production. Ch. II, p. 12

Land. All natural resources, including soil, water, minerals and the space a building stands on. Ch. II, p. 12

Legal personality. The status of being treated by law as a person separate from the owners. Ch. V, p. 26

Letter of credit. A bank's promise to pay the seller if the agreed documents are shown. Ch. VII, p. 37

Limited liability. A shareholder's duty to pay company debts only up to the unpaid value of his or her shares. Ch. V, p. 26

Location. The place where a business sets up its premises. Ch. II, p. 13

Memorandum of association. The document stating a company's name, office, objects, liability and capital. Ch. V, p. 27

Merger. The joining of one firm into another, so that only one remains. Ch. IX, p. 46

Multinational corporation. A large firm that owns and runs business operations in more than one country. Ch. VII, p. 38

Objective. A goal that a business tries to reach. Ch. I, p. 8

Ordinary share. A share whose holder votes and receives a dividend that depends on profit. Ch. VIII, p. 42

Partners. The individual owners of a partnership. Ch. IV, p. 21

Partnership. A business owned by two or more people who agree to run it together and share its profits. Ch. IV, p. 21

Partnership deed. The written agreement setting out each partner's rights and duties. Ch. IV, p. 21

PCN. A parent country national: an employee from the firm's home country. Ch. VII, p. 38

Preference share. A share that receives its fixed dividend before ordinary shares. Ch. VIII, p. 42

Private company. A company that limits the number of members and cannot invite the public to buy its shares. Ch. V, p. 26

Profit. The money left from sales after all costs are paid. Ch. I, p. 6

Prospectus. A document inviting the public to buy a company's shares, with full details. Ch. V, p. 27

Public company. A company that may invite the public to buy its shares and has no upper limit on members. Ch. V, p. 26

Quota. A limit on the quantity of a good that may be imported. Ch. VII, p. 37

Rent. The payment made to the owner of land for its use. Ch. II, p. 12

Retail. Selling goods in small amounts to the final user. Ch. I, p. 7

Risk. The chance of loss, because the future cannot be known. Ch. I, p. 6

Sales. The total money received from customers. Ch. I, p. 6

Services. Useful work done for others, such as teaching or transport. Ch. I, p. 6

Share. One equal unit of a company's capital, showing part ownership. Ch. V, p. 26

Share market. The market where shares of companies are issued, bought and sold. Ch. VIII, p. 42

Shareholder. A person who owns one or more shares in a company. Ch. V, p. 26

Social responsibility. The duty of a business to act in ways that benefit society, beyond what the law demands. Ch. III, p. 16

Sole proprietorship. A business owned, run and financed by one person. Ch. IV, p. 20

Stakeholder. Any person or group affected by what a business does. Ch. III, p. 16

State enterprise. A business owned and controlled by the government. Ch. VI, p. 33

Stock exchange. An organised market where listed shares are bought and sold under set rules. Ch. VIII, p. 42

Supplier. A person or firm that sells materials or goods to a business. Ch. III, p. 17

Tariff. A tax on imported goods. Ch. VII, p. 37

TCN. A third country national: an employee from a country that is neither home nor host. Ch. VII, p. 38

Trade. The buying and selling of goods. Ch. I, p. 7

Trust. A combination in which firms hand control to a board that runs them as one. Ch. IX, p. 46

Unlimited liability. Duty to pay all the business's debts, even from personal property. Ch. IV, p. 20

Utility. The usefulness that makes something worth buying. Ch. I, p. 6

Wages. The payment made to workers for their labour. Ch. II, p. 12

Wholesale. Selling goods in large amounts to other businesses. Ch. I, p. 7

FURTHER READING

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INTRODUCTION TO BUSINESS
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INTRODUCTION TO MANAGEMENT



*The Principles of Management
for the First-Year Student*

A TEXTBOOK FOR FIRST-YEAR STUDENTS

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INTRODUCTION TO MANAGEMENT



*The principles of planning, organizing, leading and controlling,
explained from first principles for readers new to business*

B I J O Y ? !

First edition. Set in Libre Caslon Text and IM Fell English.
Illustrative cases are invented and are not real firms.

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TO THE READER

This book assumes that you know nothing about management. It begins with ordinary life, adds one idea at a time, and defines each new word in plain English before it is used.

Look at a page. The wide column holds the main text. The narrow column on its left holds *side notes*. A note in italics summarises the paragraph beside it. A note that begins with a word in bold gives the plain meaning of a **key term**. The same word appears in bold in the main text at its first use only, and nowhere else; italics are used for emphasis.

The words **GO SLOWLY HERE** mark an idea that is new and abstract. Read it twice. Small numbers in the text point to footnotes at the foot of the page. An imaginary bakery, set in Bangladesh and priced in taka (Tk), grows through the chapters as a *case file*. It is illustrative, not real.

Each chapter opens with a puzzle, teaches its ideas in layers (the idea, an example, the rule, and where the rule fails), and closes with a summary, a problem to think through, common mistakes, and a true story from history. Review questions are gathered at the back, followed by a glossary and an index.

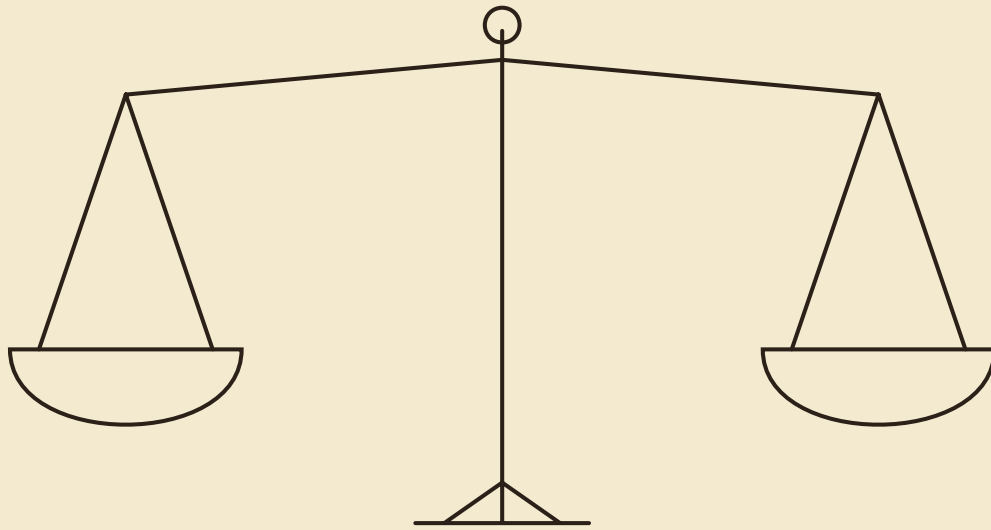


Plate I. [An engraving will be placed here.]

CHAPTER I

What Management Is

Who makes a group of people more than a crowd?



This chapter defines management, shows what managers do and what skills they need, and asks whether management is a science, an art or a profession.

THE PUZZLE OF THE TWO BAKERIES

*Same oven, same flour,
different results.*

Two bakeries stand on one street in Rajshahi. Both bought the same oven and the same flour. After a year, one earns a steady profit and the other is closing. Nothing in the ovens explains the difference. The difference lies in how the people were guided: who decided what to bake, who was told to do what, and who checked the result. That guiding work has a name, and this chapter explains it.

THE MEANING OF MANAGEMENT

Management. Getting work done through other people, to reach goals.

Organization. A group of people working together for a shared goal.

We begin with two words. An **organization** is a group of people who work together to reach a shared goal; a bakery, a school and a bank are all organizations. **Management** means the process of using people and other resources to reach the goals of the organization, by planning, organizing, leading and controlling. Notice that the manager's result comes through the work of others.

A manager's job, then, is not to bake the bread. It is to see that the right bread is baked, at the right time, at a cost the shop can bear.¹

WHY MANAGEMENT MATTERS

*Management turns scarce
resources into useful
results.*

Every organization has more wants than resources. Managers decide how money, people, materials and information are used. Good management reduces waste, keeps people working toward one goal, and helps the organization survive change. It also creates the conditions in which workers find their work worth doing. This is why the subject is studied by every future owner, employee and public official.

NATURE AND PURPOSE

Purpose. The reason an activity exists.

The **nature** of management means its main features. Management is *purposeful*: it aims at goals. It is *universal*: a shop, a hospital and a ministry all need it. It is *continuous*, because work never ends. It is a *group* activity, since managers need others. It works at *every level*, and it depends on *judgement*. Its **purpose** is to reach the organization's goals with the least waste of effort and money.

A second purpose is social. Managers also try to serve customers, give fair work to employees, and respect the community. These purposes sometimes pull apart, and balancing them is part of the manager's task. Chapter III returns to this.

GO SLOWLY HERE.

Function. A main kind of work that every manager performs.

THE FUNCTIONS OF MANAGEMENT

Slowly, now. Management is made of four **functions**, and together they form the **management process**. *Planning* chooses goals and the way to reach them. *Organizing* arranges people and resources to carry out the plan. *Leading* guides and motivates people. *Controlling* compares results with the plan and corrects the difference. In daily life you perform all four when you organise a journey: you choose a destination, pack, guide your family and check the time.

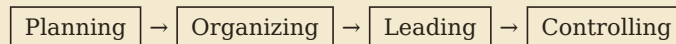


FIG. 1. The management process: each function feeds the next, and controlling leads back to planning.

These functions overlap in practice. A manager may plan in the morning, correct a mistake at noon and encourage a tired worker in the afternoon. Still, the four names help a beginner to sort a confusing day into clear kinds of work. *You should now be able to say*: management is planning, organizing, leading and controlling.

MANAGERS AND THEIR RESOURCES

Top managers. Those who set direction for the whole organization.

Middle managers. Those who carry out top managers' plans through a department.

First-line managers. Those who supervise workers directly.

Resources. Things an organization uses to reach its goals.

Not all managers are alike. **Top managers** set the direction of the whole organization. **Middle managers** turn that direction into plans for a department. **First-line managers** supervise the workers who do the actual work. By area, a *functional* manager is responsible for one activity such as sales, and a *general* manager is responsible for all of them.

Managers use four kinds of **resources**. *Human resources* are the people and their skills. *Financial resources* are the money. *Physical resources* are buildings, machines and materials. *Information resources* are the facts needed for decisions. The skill of a manager lies in combining all four, since a shortage of any one holds back the rest. Each manager is also responsible for results: managerial responsibility means accepting the duty to account for the work of one's group.

¹ Henri Fayol, a French mining engineer, wrote in 1916 that to manage is to forecast, organize, command, coordinate and control. The wording has changed since; the idea has lasted.

Technical skill. Knowing how to do a specific kind of work.

Human skill. Ability to work with and understand people.

Conceptual skill. Ability to see the organization as a whole.

SKILLS OF MANAGERS

The researcher Robert Katz wrote in 1955 that managers need three skills. **Technical skill** is knowing how to do a particular kind of work, such as baking or accounting. **Human skill** is the ability to understand, motivate and work with people. **Conceptual skill** is the ability to see the whole organization and how its parts fit together.

SKILLS NEEDED AT EACH LEVEL

Level	Technical	Human	Conceptual
First-line	High	High	Low
Middle	Medium	High	Medium
Top	Low	High	High

Read each row as a manager's mix of skills. A supervisor needs much technical skill, because he or she shows workers how to do the job. A top manager needs little of it but much conceptual skill. Human skill matters equally at every level, since every manager works through people. The *classic view* treats skills as fixed abilities of a role. The *contemporary view* adds that skills such as communication, flexibility and learning can be developed throughout a career.

PRODUCTIVITY, EFFICIENCY AND EFFECTIVENESS

Productivity. Output produced for each unit of input used.

Efficiency. Doing the thing right, with little waste.

Effectiveness. Doing the right thing, so goals are reached.

These three words are often confused. **Productivity** is output divided by input. If one baker makes 120 loaves in 8 hours, productivity is $120 \div 8 = 15$ loaves per hour. **Efficiency** means using the fewest resources for a given result. **Effectiveness** means reaching the goal that was set. A shop can be efficient and yet ineffective: it may bake perfect loaves that nobody wants to buy.

SCIENCE, ART OR PROFESSION?

Profession. An occupation that requires long training and follows a code of conduct.

Management is *science* in that it has organised knowledge and tested principles. It is *art* in that applying them to a living situation takes personal skill and judgement. Most scholars call it both: a science in its knowledge and an art in its practice. It is only partly a **profession**. It has bodies of knowledge and training, but a person may manage without a licence, and no single code binds every manager.



Case file, illustrative. Golden Crust Bakery opens in Rajshahi in January 2025. Two partners invest Tk 600,000 in an oven, tools and rent. In month one they sell 3,000 loaves at Tk 30 each, a

revenue of Tk 90,000. One partner will plan and buy; the other will bake and serve. This is the first sign of managerial work: dividing the tasks.

Summary

Management is getting results through people by planning, organizing, leading and controlling. It uses human, financial, physical and information resources, needs technical, human and conceptual skills, and aims at efficiency and effectiveness together. It is both science and art, and only partly a profession.

Think it through

Golden Crust bakes 150 loaves in 10 hours with two workers. Next month it bakes 180 loaves in 10 hours with the same workers. Calculate the productivity each month and state whether efficiency or effectiveness is shown by the change.

Where people go wrong

Students often say that a manager is anyone with a senior title. In fact, a person who supervises a team of one is a manager. Others think efficient means effective; the bakery that makes unwanted bread shows the difference.

In history

In 1911 the American engineer Frederick W. Taylor published *The Principles of Scientific Management*. He argued that work could be studied and improved as a subject, instead of being left to habit. This idea gave management its first claim to be a field of study, and the next chapter tells the story.

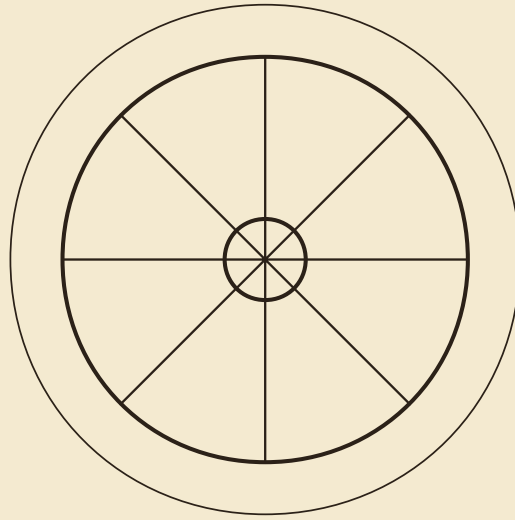


Plate II. [An engraving will be placed here.]

CHAPTER II

The Growth of Management Thought

Is there one best way to run an organization?



This chapter traces how ideas about management developed, from scientific management and Fayol's principles to the human, quantitative and situational approaches.

Many people have asked whether one right method exists.

Industrial Revolution.
The change, from the late 1700s, from hand work at home to machine work in factories.

Scientific management.
Studying work by observation and measurement to find the best way to do it.

Taylor studied the job; Fayol studied the whole organization.

Principle. A general guide to action, to be applied with judgement.

THE PUZZLE OF THE BEST WAY

Imagine ten bakers who each shape a loaf in a different way. One method is faster, and another wastes less dough. Should the shop's owner find the best method and require everyone to use it? Or do people work better when they choose their own way? For more than a century, thinkers have argued about such questions, and each answer produced a school of thought that still shapes how organizations work.

MANAGEMENT BEFORE THE FACTORY

Management has been practised for thousands of years, in building pyramids and running armies. As a subject of study, however, it is young. The **Industrial Revolution**, which began in Britain in the late 1700s, put hundreds of workers and costly machines under one roof. Owners now needed methods to organise people, control costs and keep output steady. The theories below answered that need.

SCIENTIFIC MANAGEMENT

Scientific management means using careful observation and measurement, instead of habit, to find the best way to do a job. Its founder was Frederick W. Taylor (1856-1915), an American engineer who began as a worker and later became a manager in steel works. He published *The Principles of Scientific Management* in 1911.¹

Taylor proposed four principles. First, replace rules of thumb with a science of each task. Second, select workers by fit and train them carefully. Third, cooperate with workers so that the work follows the science. Fourth, divide the work so that managers plan and workers perform. He also urged fair pay linked to output, to reward higher productivity.

ADMINISTRATIVE MANAGEMENT: FAYOL

Taylor looked at the workshop. Henri Fayol (1841-1925), a French mining engineer who became managing director of a coal and iron company, looked at the whole organization from the top. In 1916 he listed five duties of a manager (now four functions, as Chapter I showed) and fourteen **principles of management**, which are general guides for managing, not fixed laws.

Fayol insisted that principles are flexible. The table gives each principle with its plain meaning.

FAYOL'S FOURTEEN PRINCIPLES

Principle	Plain meaning
1. Division of work	Split work into specialised tasks to raise skill and output
2. Authority and responsibility	The right to give orders goes with the duty to answer for results
3. Discipline	Obedience and respect for agreed rules, from managers and workers
4. Unity of command	Each worker has one superior only
5. Unity of direction	One plan and one head for each group of related activities
6. Subordination of individual interest	Group goals come before personal ones
7. Remuneration	Pay is fair to both worker and firm
8. Centralization	The balance between top control and local freedom is set to suit the case
9. Scalar chain	A line of authority runs from top to bottom
10. Order	A place for each person and thing, and each in its place
11. Equity	Kindness joined with justice toward workers
12. Stability of tenure	Workers need time to learn, so turnover is kept low
13. Initiative	Allow people to plan and carry out their ideas
14. Esprit de corps	Team spirit, which comes from harmony and unity

Read a row as a pair: the name, then what it asks of the manager. Take unity of command. If a baker receives conflicting orders from two partners, she cannot satisfy both, and tension follows. The principle prevents this by naming one superior for each worker.

BUREAUCRACY AND THE HUMAN SIDE

Bureaucracy. An organization run by fixed rules, clear ranks and written records.

Human relations. The view that workers' feelings and social needs affect output.

Max Weber (1864-1920), a German sociologist, described the **bureaucracy**: clear division of labour, a hierarchy of authority, written rules, impersonal treatment, and careers based on merit. He saw it as fair and efficient for large tasks. Critics later found that it can become slow and rigid.

From 1924 to 1932 researchers studied workers at the Hawthorne Works of the Western Electric Company near Chicago. Elton Mayo of Harvard joined the work in 1927. They found that attention, group bonds and a sense of being valued changed output as much as pay or lighting did. This was the start of the **human relations** movement, which turned managers' eyes toward people.

QUANTITATIVE, SYSTEMS AND SITUATIONAL VIEWS

Management science. Using mathematical models and computers to solve management problems.

System. A set of connected parts working toward one purpose.

Contingency. Depending on the situation.

During the Second World War, teams of scientists used mathematics to solve problems of supply and defence. After the war, business adopted these methods, called **management science** (or operations research). Do not confuse it with scientific management. Scientific management studies *how a job is done*, by observation of work. Management science builds *mathematical models* to help managers decide, for example, how much stock to hold.

The **systems approach** sees an organization as a **system**: parts such as departments that depend on each other and on the outside world. The **contingency approach** teaches that no single method suits every case; the right action depends on the situation. Modern managers use all the schools together, choosing what the situation demands.

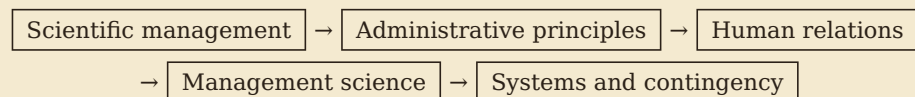


FIG. 2. Approaches to managing, in order of appearance (about 1900 to the present).

Case file, illustrative. Golden Crust's partners time each step of making a loaf and cut the shaping time from 4 minutes to 3 (scientific management). They then give each worker one supervisor (unity of command) and ask workers for ideas on new flavours (initiative).

Summary

Scientific management (Taylor) improved jobs by measurement. Fayol set out fourteen general principles for the whole organization. Weber described bureaucracy, and the Hawthorne studies showed the importance of people. Management science adds mathematics; systems and contingency views teach that the best method depends on the situation.

Think it through

A new worker at Golden Crust is told to bake by the head baker and also to sell by the manager. Which of Fayol's principles is broken, and how would you fix it?

Where people go wrong

Students confuse scientific management with management science; remember that one studies tasks and the other builds models. Others treat Fayol's principles as laws; Fayol himself called them flexible.

¹ Taylor measured tasks with a stopwatch. This is called time study, and it remains in use in factories.

In history

In 1888 Fayol became managing director of a French mining company that was in serious difficulty. He led it for thirty years, and by his retirement in 1918 it was financially sound. He credited the practice of the principles that he later wrote down.

Sources

Taylor, F. W. (1911), *The Principles of Scientific Management*.
Fayol, H. (1916), *Administration industrielle et générale*.
Weber, M., writings on bureaucracy, translated into English in 1947. Fayol's work is the landmark of administrative management.

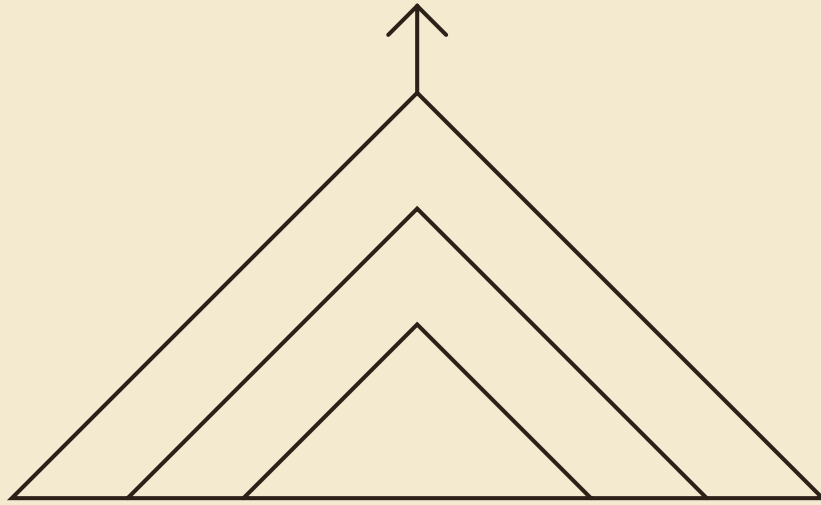


Plate III. [An engraving will be placed here.]

CHAPTER III

The Environment and Social Responsibility

*What happens outside the shop door still
reaches the oven.*



*This chapter shows how forces inside and outside an
organization shape its work, how it can respond, and
what duties it owes to society.*

A flood far away changes a baker's costs.

Environment. Everything inside and outside an organization that affects its work.

Internal environment. Forces inside: owners, employees, culture.

General environment. Outside forces that affect all organizations.

Task environment. Outside groups that deal directly with one organization.

GO SLOWLY HERE.

Boundary spanning. Work of staff who deal with outsiders for the firm.

THE PUZZLE OF THE FLOODED FIELDS

Heavy floods destroy wheat fields in a distant region. Within weeks, the price of flour rises in Rajshahi, and a bakery that never saw the flood must raise the price of bread. The shop did nothing wrong, and yet it is affected. Managers cannot control every force around them, but they can learn to watch these forces and prepare for them.

INSIDE AND OUTSIDE

The **environment** of an organization is everything that affects its work. The **internal environment** is made of owners, the board, employees and the shared habits and values called *culture*. Managers can change much of it directly. The **external environment** lies outside and is much harder to change.

Experts divide the outside world in two. The **general environment** has five parts: *economic* (prices, income, interest rates), *technological* (new machines and methods), *sociocultural* (tastes, values, population), *political-legal* (laws, taxes, government), and *international* (foreign trade and competition). The **task environment** holds the groups that deal directly with the firm: customers, competitors, suppliers, and regulators.

HOW THE OUTSIDE SHAPES THE INSIDE

Consider the effect of the outside on the inside. A new law on food safety (political-legal) forces the bakery to buy new equipment, which changes its budget and its staff training. A new rival nearby (competitor) forces a change in price, which changes its profit and its plans. A wave of interest in healthy food (sociocultural) changes what it bakes. Each external change travels inside as a change in plans, people or money. *You should now be able to say:* outside forces become inside decisions.

ADAPTING TO CHANGE

Organizations have several ways to respond. They can *forecast* changes and plan for them. They can *adapt*, by keeping structures flexible. They can *influence* the outside, by advertising or by presenting a case to government. They can *buffer*, by holding spare stock against shortages. They can use **boundary spanning**: staff such as buyers and public relations officers who deal with outsiders and bring back news. They can join with others through partnerships or mergers.

A baker who holds three weeks of flour has buffered against price shocks; one who signs yearly contracts with several mills has joined with suppliers. No single response is best. The

choice depends on cost and on how fast the environment moves.

SOCIAL RESPONSIBILITY

Corporate social responsibility. A firm's duty to act for the good of society, beyond profit and the law.

Davis model. Keith Davis's five propositions on business and society.

Business ethics. Standards of right and wrong that guide conduct in business.

Sustainability. Meeting present needs without harming the chances of future generations.

Corporate social responsibility (CSR) means the duty of a business to act in ways that benefit society, beyond making a profit and obeying the law. Supporters say that business uses society's resources and gains from its stability, so it should give something back, and that responsible firms earn trust. Opponents, such as the economist Milton Friedman (1970), argued that managers should serve owners and that social causes belong to government.

The **Davis model**, from the American scholar Keith Davis (1960), states five propositions. Social responsibility comes from *social power*. Business should operate openly and listen to society. Social costs and benefits should be weighed before an activity is begun. Social costs should be added to the price of the product. And business should help solve social problems beyond its normal area when it has the skill. Davis added an *iron law*: those who do not use social power responsibly will lose it.

Business ethics means the standards of right and wrong that guide managers. The law sets a minimum; ethics asks for more. **Sustainability** means meeting present needs without damaging the ability of future generations to meet theirs, a definition from the 1987 report of the Brundtland Commission. In business it asks managers to balance profit, people and the natural environment.

Managers carry social duties to several groups: to *owners* (a fair return), *employees* (safe work, fair pay), *customers* (honest goods), the *community*, the *natural environment*, and *government* (obeying law). Responding to these influences means listening, forecasting, and sometimes changing policy before the law forces a change.



Case file, illustrative. After a rise in flour prices, Golden Crust buys from two mills, uses sealed bins for food safety, and gives its unsold bread to a local shelter each Friday. Each act answers one outside force or one social duty.

Summary

The environment has an internal part and an external part, itself divided into general and task parts. Outside changes become inside decisions. Organizations adapt by forecasting, flexibility, influence, buffering, boundary spanning and alliances. Social responsibility, ethics and sustainability are duties to society; Davis set out the main model.

Think it through

A new tax raises the price of imported flour at Golden Crust by 10 per cent. Name the part of the environment involved, and propose two responses from this chapter.

Where people go wrong

Some think ethics means obeying the law. Law sets the minimum, ethics goes further. Others think CSR is charity; Davis's model shows that it is a duty tied to power, not a gift.

In history

On 24 April 2013 an eight-storey factory building, Rana Plaza, collapsed at Savar near Dhaka, and more than 1,100 garment workers died. Afterwards, many firms and unions signed the Accord on Fire and Building Safety in Bangladesh, and inspections of factories began. The tragedy made the duty of managers toward workers a matter of public concern.

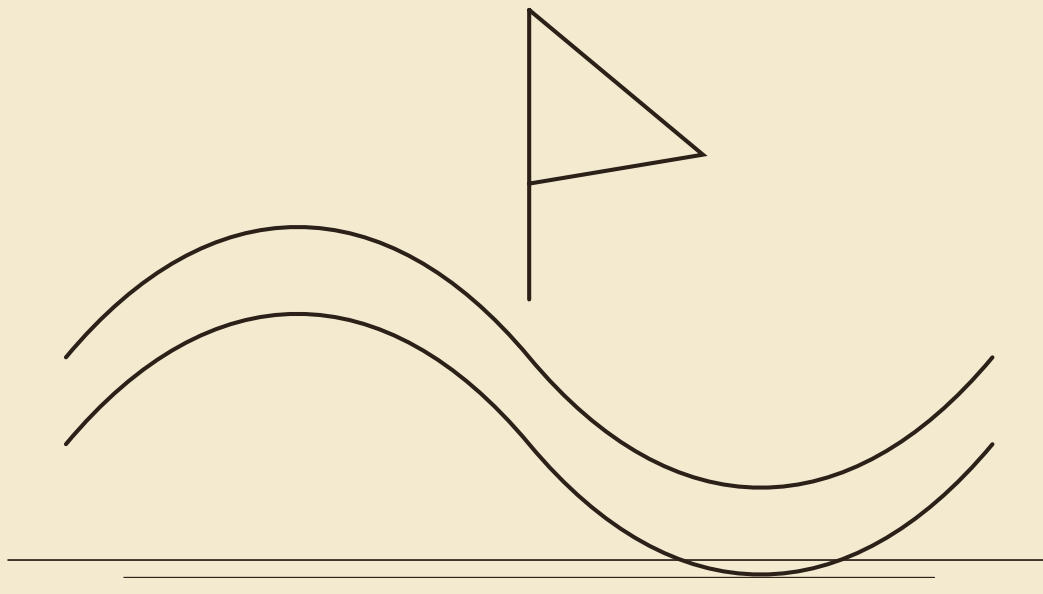


Plate IV. [An engraving will be placed here.]

CHAPTER IV

Planning

*How does a traveller reach a place that has
not yet been chosen?*



*This chapter explains what planning is, why it comes
first among the functions, the kinds of plan, the six
steps of planning, and why plans fail.*

A journey needs an end before a route.

Planning. Choosing goals and deciding in advance how to reach them.

Plan. A written or spoken statement of what will be done, when, and by whom.

Strategic plan. Long-range plan for the whole organization.

Tactical plan. Plan for a part of the organization.

Operational plan. Daily plan for a specific task.

Standing plan. A plan made once for situations that repeat.

Single-use plan. A plan made for one situation that will not repeat.

THE PUZZLE OF THE UNCHOSEN DESTINATION

A traveller leaves Dhaka by bus. Asked where she is going, she answers that she will decide on the road. By evening she is tired, far from any useful place, and unsure whether she has arrived. A journey with no destination cannot succeed or fail. Organizations are the same: until they decide where they are going, nobody can say whether they are on the right road.

MEANING, NATURE AND PURPOSE

Planning means choosing goals and deciding in advance the actions needed to reach them. The result is a **plan**. Planning looks forward: it asks what we want, where we stand now, and what we must do to close the gap.

Planning has several features. It is *primary*, because the other functions depend on it. It is *pervasive*, found at every level. It is *continuous*, since plans are revised. It is *forward-looking*, and it is *a choice among alternatives*. Its purposes are to give direction, to reduce risk by looking ahead, to cut waste, to set standards for control, and to help coordination because everyone works to one plan.

TYPES OF PLANS

Plans can be sorted in two ways. By *level*, a **strategic plan** covers the whole organization for several years and is set by top managers. A **tactical plan** turns it into steps for a department, and an **operational plan** guides daily tasks.

By *use*, a **standing plan** is made once and used again and again; a **single-use plan** is made for one occasion and then discarded. Standing plans include *policies* (general guides to decisions), *procedures* (steps in order) and *rules* (actions allowed or forbidden). Single-use plans include *programs* (a set of linked projects), *projects* (one task with a start and end) and *budgets* (a plan stated in money).

STANDING AND SINGLE-USE PLANS

Kind	Used	Examples
Standing	Many times	Hiring policy, order procedure, no-smoking rule
Single-use	Once	Opening of a new branch, this year's budget

Read a row from left to right: the kind, how often it is used, and an example. A bakery's rule that bread is baked fresh every morning is standing. Its plan to open a second shop in March is single-use.

GO SLOWLY HERE.

Planning premise. An assumption about the future on which a plan is built.

THE SIX STEPS OF PLANNING

Slowly, now. Six steps make a plan. *Step one:* be aware of an opportunity or problem, by studying the present position. *Step two:* set objectives, which are the results wanted. *Step three:* develop **planning premises**, which are assumptions about the future, such as expected prices or demand. *Step four:* find alternative ways to reach the objectives. *Step five:* evaluate them and choose the best. *Step six:* put the plan into action and prepare the supporting plans, such as budgets.

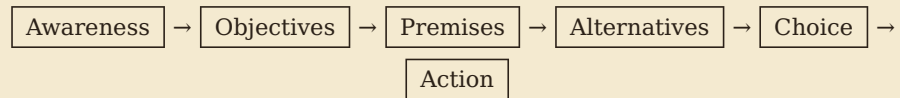


FIG. 3. The six steps of the planning process.

Consider a family planning a wedding. They ask what they can afford (awareness), fix the guest list (objective), assume the date and the season (premise), compare halls (alternatives), pick one (choice) and book it (action). The business plan follows the same order. *You should now be able to say:* a plan starts from a look at the present and ends in action.

WHY PLANS FAIL AND HOW TO IMPROVE THEM

Plans fail for causes inside and outside the plan.

Plans fail for many reasons. The objectives may be unclear. The premises may be wrong. The plan may be too rigid to change, or too costly to make. Employees who were not consulted may not support it. Top managers may lack commitment, or fail to check results. Outside changes may be too sudden to forecast. These are also the *limits of planning*: the future is uncertain, planning takes time and money, and it can make people slow to act.

To make planning effective, state clear objectives; use realistic premises; involve those who will carry out the plan; keep it flexible with a revision date; link each plan to a budget; and compare results against the plan. This last point joins planning to controlling: a plan gives control its standards, and control tells planning what to repair. Without plans there is nothing to control, and without control plans are only hopes.



Case file, illustrative. In January 2025 Golden Crust plans: objective, sell 3,600 loaves a month by June; premise, flour stays at Tk 55 per kilogram; alternatives, advertise or add a second counter; choice, second counter at Tk 40,000; action, build it in February.

Summary

Planning chooses goals and the way to reach them. Plans are strategic, tactical or operational by level, and standing or single-use by repetition. The six steps run from awareness to action. Plans fail from unclear aims, wrong premises, rigidity and lack of support, and improve when people take part and results are checked.

Think it through

Golden Crust assumed flour would cost Tk 55 a kilogram, but it rises to Tk 66. Name the step of planning that was weak, and say what the plan should have included.

Where people go wrong

Students confuse policy with procedure: a policy guides thought, a procedure lists steps. Others think a plan must never change; a good plan has a date for revision.

In history

Before the Normandy landings in June 1944, Allied commanders planned for more than a year and prepared alternative dates and weather forecasts. When storms delayed the landing by one day, the plan allowed it. The story shows planning with premises and flexibility.

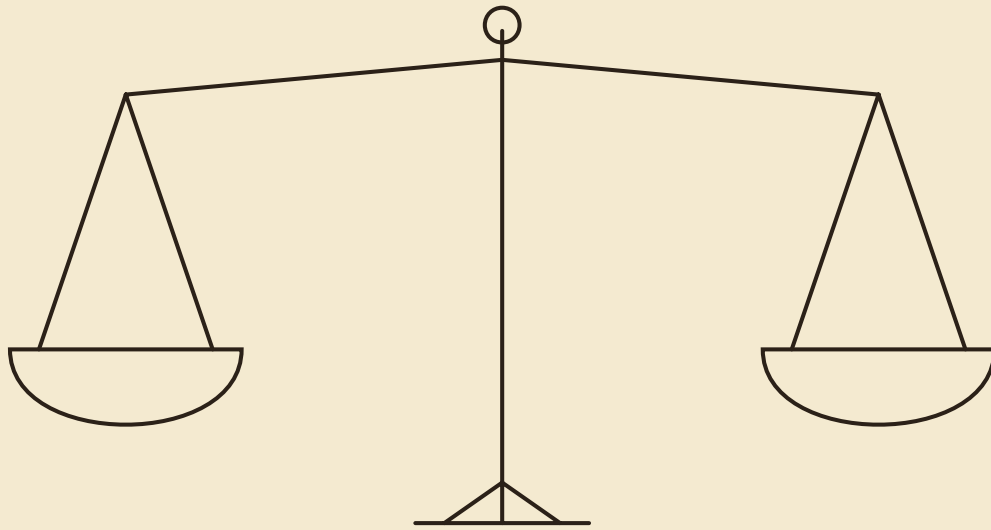


Plate V. [An engraving will be placed here.]

CHAPTER V

Objectives and Management by Objectives

How can a whole shop pull in one direction?



This chapter explains what objectives are, how to set them, the eight areas where they matter, and how management by objectives works, with its strengths and weaknesses.

A boat needs all oars to follow one beat.

Objective. A result that an organization or person wants to reach.

Mission. The basic reason an organization exists.

Good objectives pass five tests.

Key result area. A field in which results decide the survival of the organization.

Management by objectives. A cycle in which managers and employees agree on goals, then review results.

GO SLOWLY HERE.

THE PUZZLE OF THE ROWERS

Eight rowers sit in a boat. Each rows hard, but each rows toward a different bank. The boat spins and goes nowhere. Hard work was not missing; shared direction was. Objectives give a group a single bank to row toward, and management by objectives is a method for making each rower's target fit the boat's.

THE NATURE OF OBJECTIVES

An **objective** is a result that an organization or person wants to reach. It differs from the **mission**, which states why the organization exists; objectives are the steps that carry out the mission. Objectives are *specific and measurable, set for a time, arranged in a hierarchy* (from top goals to daily targets), and *linked*, so that each supports the next level.

SETTING OBJECTIVES

A well-set objective is specific, measurable, achievable, relevant and time-bound; this checklist is credited to George Doran (1981). "Sell more bread" fails the tests. "Sell 3,600 loaves a month by June 2025" passes them. People also work harder when they share in setting the objective and when they know how progress will be measured.

Peter Drucker, in 1954, named eight **key result areas** where objectives should be set: *market standing* (share of customers), *innovation* (new ideas), *productivity* (output per input), *physical and financial resources*, *profitability*, *manager performance and development*, *worker performance and attitude*, and *public responsibility*. An organization that sets objectives in all eight is less likely to succeed in one area by neglecting another.

MANAGEMENT BY OBJECTIVES

Management by objectives (MBO) is a method in which managers and workers set goals together, agree on how they will be measured, and review the results at a fixed time. Drucker described it in *The Practice of Management* (1954). Its core idea is that people work better when they help to choose their targets.

The process is a cycle of steps. *First*, top managers set the organization's objectives. *Second*, these are passed down, so each department sets its own, matched to the top ones. *Third*, each manager and employee agree on personal objectives together. *Fourth*, each person plans the actions needed. *Fifth*, progress is checked at agreed times. *Sixth*, performance is appraised and rewarded, and the findings begin the next cycle.

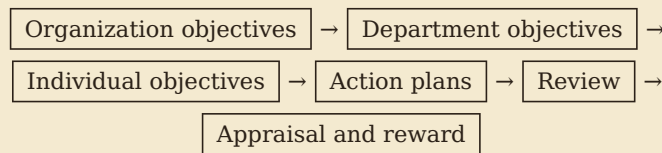


FIG. 4. The MBO cycle: the appraisal feeds the setting of the next objectives.

Read the figure from left to right and then back to the start. Imagine a class: the school sets a pass rate (organization), each teacher sets a target for her class (department), each student agrees to a score (individual), and results are reviewed each term. *You should now be able to say: MBO links every person's target to the organization's goal, and repeats.*

BENEFITS AND WEAKNESSES

MBO has clear strengths and clear limits.

Its *benefits* are better planning, clearer roles, higher motivation because people join in choosing goals, easier control since targets exist, and better communication between levels. Its *weaknesses* are as real. It takes much paperwork and time. It may push people toward goals that are easy to measure and away from those that matter. It can fail if top managers do not support it, if goals are forced downward without discussion, or if the environment changes faster than goals can be revised. Rewards tied only to targets may encourage short-term thinking.



Case file, illustrative. Golden Crust's partners set the shop objective: profit of Tk 30,000 a month by December 2025. The baker's target is to waste under 5 per cent of dough, and the seller's is 3,600 loaves a month. They review both targets on the last day of each month.

Summary

Objectives are specific, measurable and time-bound results. Drucker named eight areas where they matter. MBO turns organization goals into personal ones through joint setting, action plans, review and reward. It improves motivation and control but costs time and can neglect goals that are hard to measure.

Think it through

Rewrite the objective "improve customer service" so that it meets the five tests, and name one key result area it belongs to.

Where people go wrong

Students mix objectives with plans. An objective is the result wanted; a plan is how to get it. Others think MBO means orders from the top; it depends on joint setting.

In history

Peter Drucker studied the management of General Motors and published *Concept of the Corporation* in 1946. In 1954 he described management by objectives in *The Practice of Management*. Many firms adopted the method in the following decades, which made joint goal setting a common practice.

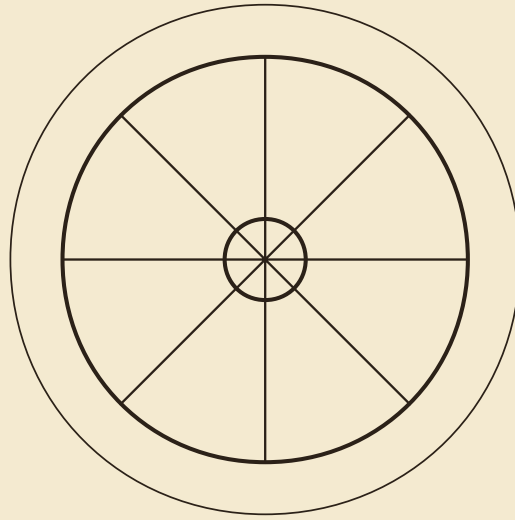


Plate VI. [An engraving will be placed here.]

CHAPTER VI

Decision Making

When the choice is costly and the future is hidden, how should a manager choose?



This chapter sets out the elements of a decision, the steps of the rational process, the conditions of certainty, risk and uncertainty, the tools of probability, and group decisions.

A costly choice with an uncertain result.

Decision. A choice among alternatives.

Decision maker. The person who makes the choice.

Rational decision making. Choosing by steps that weigh facts and alternatives logically.

Certainty. The outcome of each alternative is known.

Risk. Outcomes are uncertain, but their chances are known.

Uncertainty. Chances are unknown.

THE PUZZLE OF THE SECOND OVEN

The partners of Golden Crust can buy a second oven for Tk 300,000. If demand grows, the oven pays for itself. If demand stays flat, it sits idle. Nobody can know the future. Yet a choice must be made, and each choice leads to a different result. How should they decide? This is the central question of the chapter, and many writers say it is the heart of management, because every function ends in a choice.

THE ELEMENTS OF A DECISION

A **decision** is a choice among alternatives. Every decision situation has the same *elements*. There is a **decision maker**, the person who chooses. There are *objectives* that the choice should serve. There are *alternatives*, the possible courses of action. There are *states of nature*, the future events that the decision maker cannot control. There are *outcomes*, the results of each alternative under each state. And there is a *preference*, the value placed on each outcome.

THE DECISION-MAKING PROCESS

The **rational decision-making process** moves in steps. *First*, identify and define the problem. *Second*, collect the facts. *Third*, develop alternatives. *Fourth*, evaluate each alternative against the objectives. *Fifth*, choose the best one. *Sixth*, put it into action. *Seventh*, follow up and check the result, and begin again if the problem remains.

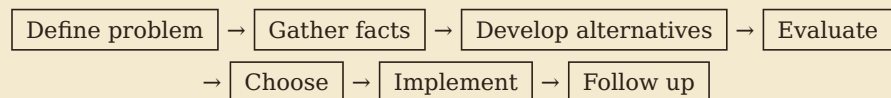


FIG. 5. The rational decision-making process; follow-up may return the manager to the first step.

Managers often ask which process is best. No process guarantees a good result, but the rational one is the best guide because it forces a clear problem, a search for alternatives, and a check after the action. Right decisions do not drive management alone, but wrong ones undo all the other functions.

CONDITIONS OF DECISION MAKING

Managers decide under three **conditions**. Under **certainty**, the outcome of each alternative is known: buying flour at a fixed price. Under **risk**, outcomes are uncertain but their chances can be estimated, as when past sales suggest a 40 per cent chance of high demand. Under **uncertainty**, the chances cannot even be estimated, as with a new product with no past record. Most important decisions are made under risk or uncertainty.

TOOLS: PROBABILITY AND DECISION TREES

GO SLOWLY HERE.

Expected value. The average result if a choice were repeated many times.

Under risk, managers use **probability theory**: each possible outcome receives a chance between 0 and 1, and the chances add to 1. The **expected value** of an alternative is each outcome multiplied by its chance, then summed. A **decision tree** draws the choice as branches: first the alternatives, then the states of nature, then the outcomes. Slowly, here is a first example. A coin pays Tk 100 on heads and Tk 0 on tails. Expected value = $0.5 \times 100 + 0.5 \times 0 = \text{Tk } 50$.

Worked example. Given: the second oven costs Tk 300,000. Net payoff after cost: high demand (chance 0.4) Tk 500,000; medium demand (0.4) Tk 200,000; low demand (0.2) a loss of Tk 100,000. Not buying pays Tk 0. *Find:* the expected value of buying. *Formula:* expected value equals the sum of chance times payoff. *Steps:* $0.4 \times 500,000 = 200,000$; $0.4 \times 200,000 = 80,000$; $0.2 \times (-100,000) = -20,000$. Sum: $200,000 + 80,000 - 20,000 = 260,000$. *Answer:* the expected value of buying is Tk 260,000, against Tk 0 for not buying. *Check:* the result lies between the worst (-100,000) and the best (500,000), and the chances add to 1.

Notice a *common slip*: using a loss as a positive number. Also note the limit: the expected value is an average, and the partners will earn one of the three outcomes, not the average. A careful manager asks whether the firm could survive the worst case.

GROUP DECISIONS AND OTHER FACTORS

Groups bring knowledge, but they take time.

Bounded rationality. People choose with limited information and time.

Decision support system. Computer system that gives managers data for decisions.

A group has *advantages*: more knowledge, more alternatives, wider acceptance of the decision, and better understanding. Its *disadvantages* are slowness, pressure to agree, domination by a few, and divided responsibility. Groups suit complex decisions where support matters; individuals suit urgent or simple ones.

Real decisions face *constraints*: limited time, money and information, and the decision maker's own feelings and habits. Herbert Simon (1947) called this **bounded rationality**; managers often choose an option that is good enough, rather than the best. Managers also meet *programmed decisions*, which repeat and follow rules, and *non-programmed decisions*, which are new and need judgement. A **decision support system** helps with the second kind by letting managers test questions on data.



Case file, illustrative. Golden Crust's partners compute the expected value of the oven (Tk 260,000), judge that they can bear the worst loss of Tk 100,000, and decide in March 2025 to buy.

Summary

A decision is a choice among alternatives, with decision maker, objectives, alternatives, states of nature, outcomes and preferences. The rational process runs from defining the problem to follow-up. Decisions are made under certainty, risk or uncertainty. Expected value and decision trees guide choice under risk; groups add knowledge but cost time.

Think it through

If the chance of high demand falls to 0.2 and medium demand rises to 0.6, with low demand still 0.2, recalculate the expected value of the oven and say whether the decision changes.

Where people go wrong

Students mix risk and uncertainty; under risk the chances are known. Others think the highest expected value is always the right choice, forgetting that the firm must survive the worst case.

In history

Herbert Simon, whose book *Administrative Behavior* appeared in 1947, received the Nobel Memorial Prize in Economic Sciences in 1978 for his research on decision making in organizations. His idea of bounded rationality changed how teachers describe managers' choices.

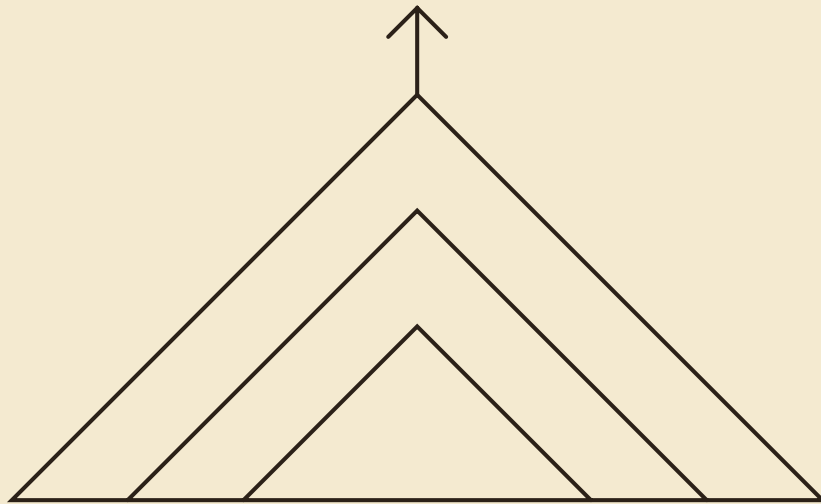


Plate VII. [An engraving will be placed here.]

CHAPTER VII

Organizing

Two partners can share a counter; how do two hundred share a company?



This chapter explains what organizing is, the steps of the process, the forms and structure of organization, span of management, authority and delegation, centralization, coordination, and how an organization is designed.

Growth forces a shop to divide its work.

Organizing. Arranging people, tasks and resources so that plans can be carried out.

Organization structure. The formal pattern of jobs and reporting lines.

Steps. Five steps turn a plan into a structure.

Formal organization. The planned structure with defined jobs and rules.

Informal organization. The unplanned network of personal relations.

THE PUZZLE OF THE GROWING SHOP

Golden Crust began with two partners who did everything. A year later it has ten workers. Who orders flour? Who decides the price? To whom does a new baker report? A shop of two can answer such questions in conversation. A shop of ten cannot, and a company of two hundred certainly cannot. The work of arranging people, tasks and authority so that a plan is carried out is called organizing.

MEANING, NATURE AND PURPOSE

Organizing means arranging people, tasks and resources so that plans can be carried out. Organizing is the main way in which managers activate plans: planning only says what to do, and organizing gives people the means to do it. The resulting pattern of jobs and reporting lines is the **organization structure**. Organizing is continuous, needs people, and depends on the plan and the environment.

Its purposes are to divide work among people, to define authority and duties, to avoid overlapping, to use skills well, and to make coordination possible. A good structure also lets each person know what is expected, and to whom he or she answers.

THE ORGANIZING PROCESS

The process has five steps. *First*, review the plan and the objectives. *Second*, list the work to be done and divide it into tasks. *Third*, group related tasks into jobs and departments. *Fourth*, assign each job to a person and give the authority needed. *Fifth*, join the parts by lines of reporting and coordination, then review the result and adjust it.

FORMAL AND INFORMAL ORGANIZATION

The **formal organization** is the planned structure shown on the organization chart, with defined jobs, rules and reporting lines. The **informal organization** grows by itself from friendship, shared interests and daily contact. It carries gossip but also help and advice, and managers who ignore it lose useful knowledge.

FORMAL AND INFORMAL ORGANIZATION

Feature	Formal	Informal
Origin	Planned by management	Grows from personal relations
Authority	From position	From personal influence
Purpose	Reach organization goals	Meet social needs
Rules	Written	Unwritten
Shown on chart	Yes	No

FORMS OF ORGANIZATION

Line organization.

Authority flows straight down, one superior for each person.

Functional organization.

Specialists direct work in their field.

Line-and-staff

organization. Line managers decide; staff specialists advise them.

The *line organization* is the simplest: orders go down one chain from owner to worker, as unity of command asks. Its strength is speed and clarity; its weakness is that one head cannot know everything. The *functional organization* groups workers by specialty, and each specialist directs all workers in that field. It uses expert knowledge but may break unity of command.

The *line-and-staff organization* adds specialists who advise but do not command, such as a legal adviser or accountant. It joins unity of command with expert advice, though friction may arise between line and staff. A *committee* is a group given a task, used for hard problems that need many views. A *matrix organization* gives a worker two bosses, one for the function and one for the project; it suits complex projects but needs careful coordination.

DEPARTMENTALIZATION

Departmentalization.

Grouping jobs into departments.

Departmentalization is the grouping of jobs into departments. Common bases are *function* (baking, sales, finance), *product* (bread, cakes), *geography* (Rajshahi branch, Dhaka branch), *customer* (retail, wholesale) and *process* (mixing, baking, packing). Each has a strength. The functional form builds expertise; the geographic form serves local needs. Large organizations often combine several bases.

SPAN OF MANAGEMENT

GO SLOWLY HERE.**Span of management.**

The number of subordinates one manager supervises.

Slowly, now. The **span of management** (also called span of control) is the number of people who report directly to one manager. If a head baker supervises 5 workers, her span is 5. A *narrow* span means close supervision but many levels; a *wide* span means freedom but less supervision. More workers need more levels, so the span and the number of levels are linked: with a narrow span, the organization grows *tall*; with a wide span, it stays *flat*.

Factors that decide the best span: the *skill and training* of workers (high skill allows a wide span); the *similarity* of tasks; the *clarity* of plans; the *need for personal contact*; the *quality of communication*; the *distance* between workers; and the manager's own ability and time. Span is therefore an important concept, for it sets the shape and cost of the structure. *You should now be able to say:* the more people a manager supervises, the fewer levels the organization needs.

AUTHORITY, RESPONSIBILITY AND DELEGATION

Authority. The right to give orders and make decisions.

Responsibility. The duty to do an assigned task.

Accountability. The duty to answer for results.

Line authority. Right to command in the chain.

Staff authority. Right to advise.

Functional authority. Limited right to direct others in one specialty.

Delegation. Giving part of one's authority and tasks to a subordinate.

Authority is the right to give orders and to make decisions. **Responsibility** is the duty to perform a task. **Accountability** is the duty to answer to a superior for the result. The three move together: responsibility without matching authority leaves the person unable to act, and authority without accountability invites abuse. This is the relationship between responsibility and authority.

Kinds of authority. **Line authority** is the right to command subordinates in the chain; **staff authority** is the right to advise; **functional authority** is a limited right of a specialist to direct people outside his own department in his own field, as a finance officer may require every department to follow the same accounting rules. It is delegated by the line manager, in writing and for a defined area. Authority differs from *power*: authority is a right that comes from position, and power is the ability to influence, which may come from skill or personality.

Delegation means giving a subordinate part of your tasks, with the authority to do them, and holding him accountable. It has three steps: assign the duty, grant the authority, and create the obligation to report. Delegation is the means by which authority is *distributed* through the organization, from the top to the lower levels. Obstacles come from the manager (fear of losing control, belief that "I can do it better," lack of trust), from the subordinate (fear of mistakes, lack of confidence) and from the organization (unclear duties, no culture of trust).

CENTRALIZATION AND DECENTRALIZATION

Centralization. Keeping decision power at the top.

Decentralization. Spreading decision power to lower levels.

In a **centralized** organization, important decisions are made at the top; in a **decentralized** one, they are made at lower levels. Centralization gives *uniform policy, close control and lower duplication*, but is slow and uses little local knowledge. Decentralization gives *speed, local knowledge, motivation and training of managers*, but risks inconsistency and higher cost. Most firms balance the two. Decentralization is related to departmentalization: when a firm splits into product or geographic departments, it must decide how much power each department receives.

COORDINATION

Coordination. Joining the work of different people and parts toward one goal.

Coordination is the joining of separate activities toward one goal. Division of labour raises the need for it: the more the work is divided, the more the parts must be fitted together. Its basic activities are *setting common objectives, clear communication, scheduling, and sharing information*.

Structural techniques build coordination into the design. They are the *hierarchy* (the chain of command), *rules and procedures*, *plans and schedules*, *liaison persons*, *task forces and committees*, *integrating managers* (such as a project head), and *shared information systems*. Used together, they let ten departments work as one company.

ORGANIZATIONAL DESIGN

Bureaucratic model.

Design by fixed rules, ranks and procedures.

Behavioral model. Design that stresses people, teams and flexibility.

The **bureaucratic model** designs the organization by rules, ranks and clear duties; it brings fairness and stability, but may be slow. The **behavioral model** stresses people, flat structures and teamwork; it brings motivation and speed, but may lack clear lines and is hard to run at large scale. Compare them: the first suits stable work such as a tax office; the second suits changing work such as a design studio.

What decides the design? The basic *situational factors* are the *environment* (stable or changing), the *technology* (routine or complex), the *size* of the firm, the *strategy*, and the *people* who work in it. These same forces, with the owner's wishes and the law, influence the formal structure.



Case file, illustrative. By 2026, Golden Crust has ten workers and two shops. The partners group workers by function (baking, sales), keep a span of five, and delegate daily purchasing to the head baker, who must report weekly to the partners.

Summary

Organizing arranges people and tasks so plans can be carried out. Its steps run from listing work to joining the parts. Forms include line, functional, line-and-staff, committee and matrix. Span of management shapes the number of levels. Authority, responsibility and accountability go together; delegation spreads authority; coordination holds the parts together; design depends on the situation.

Think it through

A manager supervises 8 workers and finds that mistakes are rising. Should he widen or narrow his span? Name two factors from the chapter that would help you decide.

Where people go wrong

Students treat authority and power as the same thing. They differ: authority is a right of position, power is influence. Others believe a manager may delegate accountability; he may delegate the task, but he remains accountable.

In history

In 1920 Alfred P. Sloan, later the president of General Motors, proposed an organization plan giving the car divisions control of daily operations while the centre kept control of policy and finance. This balance of decentralization with coordination helped General Motors to grow into a leading car maker.

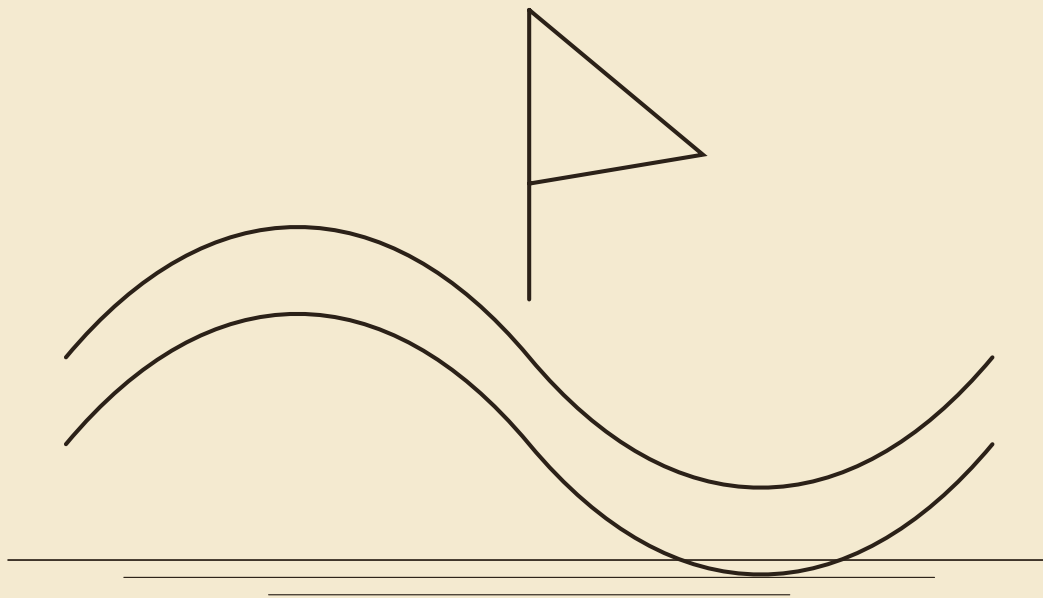


Plate VIII. [An engraving will be placed here.]

CHAPTER VIII

Leading

*Why does one worker give her best and
another give only her time?*



*This chapter examines the human side of
management: motivation and its theories, groups,
creativity, and the nature, power, styles and models
of leadership.*

*Equal pay and equal skill,
but unequal effort.*

Human factors. The needs, attitudes and relationships of people at work.

Theory X. The view that people dislike work and must be controlled.

Theory Y. The view that people enjoy work and can direct themselves.

Motivation. The inner force that makes a person act toward a goal.

GO SLOWLY HERE.

Need. Something a person feels the lack of.

THE PUZZLE OF THE TWO COUNTERS

Two sellers work at the same bakery for the same pay. One arrives early, remembers the customers' names, and suggests new products. The other watches the clock. The manager cannot explain the difference by skill or pay. The difference lies in what moves each person to act, and in how well the manager guides them. This guiding of people is the function of leading.

THE HUMAN FACTORS IN MANAGEMENT

Human factors are the needs, attitudes, abilities and relationships of people at work. Workers differ in what they want, how they see events and how they act in groups. A manager who studies these differences can match tasks to people and can predict how they will respond to change. The ideas below build on this study.

In 1960 Douglas McGregor described two sets of assumptions managers hold. **Theory X** assumes that people dislike work, avoid responsibility, and must be directed and threatened. **Theory Y** assumes that work is natural, that people can direct themselves toward goals they accept, and that many seek responsibility. A manager with Theory X assumptions supervises closely; one with Theory Y assumptions gives freedom. McGregor argued that Theory Y suits most modern workers, but each manager must judge the situation.

MOTIVATION

Motivation is the inner force that makes a person act to reach a goal. Managers must understand it for three reasons: a motivated worker produces more, stays longer, and needs less supervision. Motivation theory rests on three assumptions: behaviour has a cause; behaviour is directed at a goal; and people act to meet needs.

Slowly, now. The *needs-goal model* connects these assumptions. A **need** is something a person feels the lack of, such as hunger. The need creates tension, called a *drive*. The drive leads to *goal-directed behaviour*: the person looks for food. If the goal is reached, the need is met and tension falls. The result is *satisfaction*, which is *fed back* to the person, who adjusts future behaviour. If the goal is not reached, new behaviour is tried.

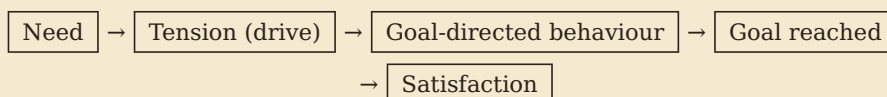


FIG. 6. The needs-goal model of motivation; feedback from satisfaction returns to the person's needs.

A worker needs more income. The tension leads her to work overtime. She earns the money and feels satisfied, and the lesson of that overtime guides her next choice. *You should now be*

Hierarchy of needs. Maslow's ordered list of five human needs.

able to say: motivation begins with a need and ends in satisfaction.

Abraham Maslow set out his **need hierarchy** in 1943. He proposed five needs, from lowest to highest: *physiological* (food, rest), *safety* (secure job, health), *social* (belonging, friendship), *esteem* (respect, status) and *self-actualization* (using one's full ability). A lower need must be mostly met before the next one grows strong, and a met need no longer motivates. A manager should therefore ask which need each worker feels now.



FIG. 7. Maslow's hierarchy of needs; the highest need sits at the top.

Read the figure from the base up. A worker without a steady wage (safety) will not respond to praise (esteem). Critics note that the order is not fixed for every person or culture. Still, the theory gives managers a practical question to ask.

GROUPS, CREATIVITY AND INNOVATION

Informal group. A group formed from personal ties, not by management.

People join **informal groups** and teams for many reasons: to meet social needs, to feel safe in numbers, to gain status, to share knowledge and help one another, and to reach goals they cannot reach alone. Such groups can raise morale or resist change, so a manager works with them rather than against them.

Creativity. Producing new and useful ideas.

Innovation. Putting a new idea into use.

Creativity is the making of new and useful ideas, and **innovation** is the use of such an idea in a product or method. A creative idea is only an idea until the firm acts on it. Managers encourage both by allowing time for ideas, accepting honest failure, and rewarding good suggestions.

LEADERSHIP

Leadership. Influencing people to work willingly toward a goal.

Leadership is the process of influencing people to work willingly toward group goals. A leader need not hold a post; a respected worker may lead. A *manager* is appointed and handles planning, organizing and controlling; a *leader* gains followers by influence. The two roles overlap: managers are most effective when they also lead, and a leader without management skill cannot keep results steady. In short, management is about handling the work; leadership is about inspiring the people.

Power. The ability to influence others.

John French and Bertram Raven (1959) named five bases of **power**. *Legitimate power* comes from position. *Reward power* comes from the ability to give pay or praise. *Coercive power* comes from the ability to punish. *Expert power* comes from special knowledge. *Referent power* comes from admiration for

Emotional intelligence. Skill in understanding and managing one's own and others' feelings.

Trait approach. Leaders are born with special qualities.

Situational approach. Best leadership depends on the situation.

Autocratic style. The leader decides alone.

Democratic style. The leader shares decisions.

Laissez-faire style. The leader leaves decisions to the group.

Transformational leadership. Inspiring followers to look beyond self-interest to a shared vision.

Super leadership. Leading others to lead themselves.

the leader's person. The first three come from the organization; the last two come from the leader.

Daniel Goleman (1995) described **emotional intelligence** as five skills: *self-awareness*, *self-regulation*, *motivation*, *empathy* and *social skill*. A leader with these skills stays calm under pressure, understands the worries of the team and builds trust. It is necessary because leaders work through the feelings of others, and a leader who cannot manage his own anger loses the willing effort of the group.

APPROACHES AND STYLES OF LEADERSHIP

The **trait approach** holds that leaders have special qualities, such as intelligence, confidence and drive. Research found that no list of traits guarantees success. The **situational approach** holds that the best behaviour depends on the leader, the followers and the task. It is now widely accepted, in line with the contingency view of Chapter II.

Kurt Lewin and colleagues (1939) described three *styles*. The **autocratic** leader decides alone and gives orders; it is quick, but breeds fear. The **democratic** leader shares decisions with the group; it raises commitment but is slower. The **laissez-faire** leader leaves decisions to the group; it suits skilled, self-driven people and fails with others. No style is best; the manager chooses by situation.

Robert Tannenbaum and Warren Schmidt (1958) showed styles as a *continuum*: the manager chooses a point along a line, according to how much freedom is given to the group.

Manager decides and announces	Manager sells decision	Manager presents ideas, invites questions	Manager proposes tentative decision	Manager presents problem, takes suggestions	Manager sets limits, group decides	Group decides within limits
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FIG. 8. The Tannenbaum and Schmidt continuum: boss-centred behaviour on the left, subordinate-centred on the right.

Read the figure from left to right: the manager's authority falls and the group's freedom rises. The right point depends on three forces: the *manager* (values, confidence in workers), the *subordinates* (readiness, desire for freedom) and the *situation* (urgency, type of organization).

Three modern forms are noted. **Transformational leadership** (Burns, 1978) inspires people to rise above self-interest and work toward a vision. *Coaching* develops people by guidance, feedback and encouragement. **Super leadership** (Manz and Sims, 1989) leads others to lead themselves, by teaching self-direction.



Case file, illustrative. Golden Crust's partners notice that the seller who watches the clock lacks recognition. They add a

monthly award (esteem need), ask for her ideas (Theory Y) and let her lead the Saturday counter (delegation and leadership).

Summary

Leading guides people. McGregor's Theories X and Y describe managers' assumptions. Motivation begins with a need and follows the needs-goal model; Maslow ranks needs in five levels. Leaders use five bases of power, need emotional intelligence, and choose from styles that range from autocratic to democratic, guided by the situation.

Think it through

A worker who has a steady job and good friends keeps working slowly. Using Maslow, which need might now matter most, and which two actions could the manager take?

Where people go wrong

Students think a leader and a manager are the same person. They differ in how authority is gained. Others read Theory X and Theory Y as two kinds of worker; they are two sets of assumptions held by managers.

In history

From 1800, Robert Owen managed the cotton mills at New Lanark in Scotland. He improved housing for workers, opened a school for their children, and raised the age at which children could work. The mills remained profitable, which showed that care for workers and sound business could go together.

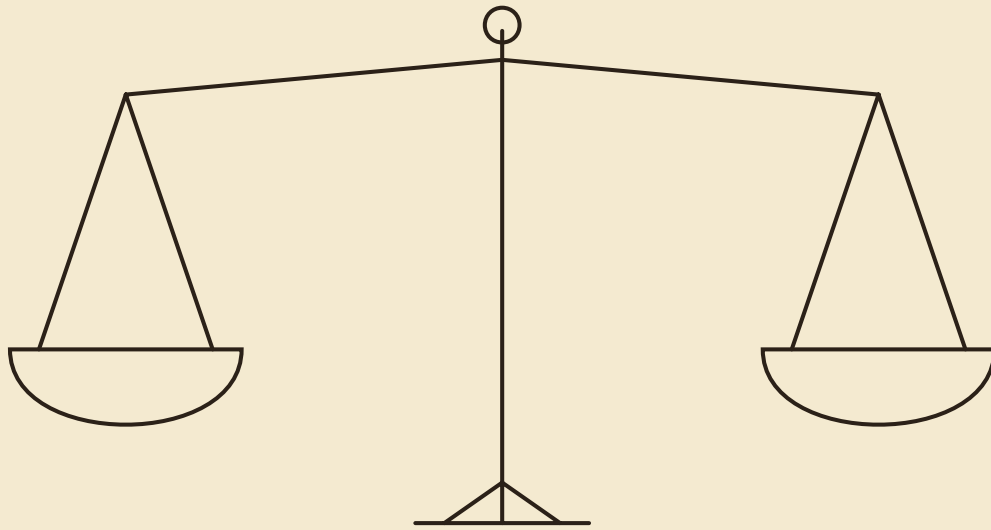


Plate IX. [An engraving will be placed here.]

CHAPTER IX

Controlling

How does a manager know that the plan is still working?



This chapter explains what controlling is, how the control process runs, the types of control, what makes control effective, and what stands in its way.

THE PUZZLE OF THE LIGHTER LOAF

A small drift grows unseen.

A customer complains that her loaf of bread feels lighter than last week's. The baker had not changed the recipe, but a worn scale had slowly drifted. By the time anybody noticed, hundreds of loaves had gone out under weight. A check made each morning would have caught the error on the first day. Controlling is the function that makes such checks regular and useful.

MEANING AND NATURE

Controlling. Comparing actual results with the plan and correcting the difference.

Monitoring. Watching progress to see what is happening.

Controller. A staff manager who designs and runs the control system.

Controlling means comparing actual results with planned results and taking action to correct any difference. **Monitoring** is the watching of progress; it provides the facts, whereas controlling adds the decision to act. Control is *continuous, forward-looking, linked to planning, a duty of every manager*, and aimed at *action*, not blame. Its basic principle is that control should match the plan: a measure has meaning only against an objective.

The relationship between controlling and the **controller** is this: the controller is an officer, often in finance, who builds and runs the system and reports results, but the duty to control remains with every line manager. The controller supplies the instruments; the managers use them.

THE CONTROL PROCESS

GO SLOWLY HERE.

Standard. A target or measure against which results are judged.

Slowly, now. The control process has three steps. *First*, set **standards**: a standard is a target, such as "each loaf weighs 400 grams." *Second*, measure actual performance: weigh a sample of loaves. *Third*, compare, and correct: if the difference is large, find the cause and act. The cycle then repeats.

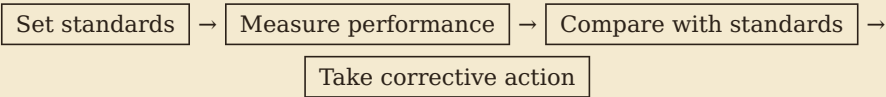


FIG. 9. The control process; corrective action sometimes changes the standard or the plan itself.

Deciding *what* to measure matters: measure what shows success (loaf weight, sales, waste), not only what is easy to count. Deciding *how* to measure matters too: by sample, by report or by observation, with enough accuracy and at a reasonable cost. A wrong measure leads people to chase the wrong result. *You should now be able to say*: control compares results with standards, then acts.

Worked example. Given: standard loaf weight 400 g; ten sampled loaves weigh 4,000 g in total. Find: the average weight and the variance. Formula: average equals total weight divided by number of loaves; variance equals actual minus standard. Steps: $4,000 \div 10 = 400$ g, so the variance is $400 - 400 = 0$. If the total were 3,800 g, the average is $3,800 \div 10 =$

380 g, and the variance is $380 - 400 = -20$ g. *Answer:* the second batch is 20 g (5 per cent) under standard. *Check:* $20 \div 400 = 0.05$, which is 5 per cent. A common slip is to ignore the sign of the variance.

TYPES OF CONTROL

- Feedforward control.** Action before work starts, to prevent problems.
- Concurrent control.** Action during the work.
- Feedback control.** Action after the work, from results.

There are three basic types. **Feedforward** (or preliminary) control acts *before* work begins, such as inspecting flour before it is used. **Concurrent** control acts *during* the work, such as a baker checking dough as it is mixed. **Feedback** control acts *after* the work, from the result, such as a monthly profit report. Feedforward prevents errors; concurrent corrects them at once; feedback teaches for the future but cannot undo the past.

THREE TYPES OF CONTROL		
Type	When	Example
Feedforward	Before	Test flour quality
Concurrent	During	Supervise baking
Feedback	After	Monthly sales report

Common *methods* include budgets, reports, audits, inspection, and direct observation. Each is a means of one of the three types, as a budget can warn before spending (feedforward) and compare after it (feedback).

EFFECTIVE CONTROL AND ITS BARRIERS

Control must suit the task and the people.

Effective control is *accurate, timely, economical* (it must cost less than the loss it prevents), *flexible, simple to understand, focused on critical points* and *accepted* by those controlled. A control that arrives late or that nobody understands has no value.

Barriers to successful controlling include *resistance* from workers who feel watched, *overcontrol* that kills initiative, *wrong standards, difficulty in measuring* some work such as creativity, *delay* in reporting, a *short-term view* that ignores long-term results, and *cost*. A manager overcomes them by involving people in setting standards, explaining the purpose, and controlling only the critical points.

A control process succeeds when its standards come from clear objectives, measures are accurate and prompt, those affected understand and accept it, and managers act on the results. Control also makes a manager more efficient: it exposes waste early, saves time through exceptions (acting only on large deviations), and frees attention for planning. As Chapter IV showed, planning and controlling are twins: planning sets the standard, and controlling tells planning where to repair. Power and control are also related, for the right to control is a form of power, and those who control resources gain influence.



Case file, illustrative. Golden Crust weighs ten loaves each morning against a 400 g standard. In June 2025 the average is 380 g. The head baker recalibrates the scale, and the next day the average returns to 400 g. This is concurrent control; the monthly profit report is feedback control.

Summary

Controlling compares results with standards and corrects the difference. Its three steps are to set standards, measure performance and act. Control may be feedforward, concurrent or feedback. It works when it is accurate, timely, economical, understood and accepted, and it is hindered by resistance, over-control and poor measures. Planning and controlling are linked.

Think it through

Golden Crust's ten sampled loaves weigh 3,900 g in total against a 400 g standard. Calculate the average weight and the variance, and say which type of control the weighing is.

Where people go wrong

Students think controlling means punishing. Its aim is correction. Others confuse monitoring with controlling; monitoring only gathers facts, and controlling includes action.

In history

In 1924, Walter Shewhart of Bell Telephone Laboratories drew the first control chart, a graph that shows when variation in a process is too large to be chance. The chart became a basic tool of quality control in factories across the world.

REVIEW QUESTIONS

Think each question through in your own words. If you cannot answer one, return to the chapter named beside it.

CHAPTER I: WHAT MANAGEMENT IS

1. Define management and name its four functions.
2. What is the difference between top, middle and first-line managers?
3. Describe the three skills of Katz and say which matters most at each level.
4. Distinguish between productivity, efficiency and effectiveness, with an example.
5. Is management a science, an art or a profession? Explain.
6. Why is management important to an organization?

CHAPTER II: THE GROWTH OF MANAGEMENT THOUGHT

1. Define scientific management and state Taylor's four principles.
2. Explain Fayol's fourteen principles of management.
3. Describe the evolution of management thought in your own words.
4. Differentiate between management science and scientific management.
5. What did the Hawthorne studies show?
6. List the different approaches to managing.

CHAPTER III: THE ENVIRONMENT AND SOCIAL RESPONSIBILITY

1. How does the external environment influence the internal environment of an organization?
2. How can an organization adapt to changes in the external environment?
3. What effect do social, political and legal forces have on an enterprise?
4. Define corporate social responsibility and explain the Davis model.
5. Define business ethics and sustainability.
6. What are the major social responsibilities of managers?

CHAPTER IV: PLANNING

1. What is planning, and what is its nature?
2. What are the purposes of planning?
3. Describe the six steps in the planning process.
4. What is a planning premise?
5. Distinguish between standing plans and single-use plans.
6. Why do plans fail, and how can planning be made effective?
7. Discuss the limits of planning.

CHAPTER V: OBJECTIVES AND MANAGEMENT BY OBJECTIVES

1. What is meant by an organizational objective?
2. Discuss the nature of objectives.
3. Describe the eight key areas in which objectives should be set.
4. What is management by objectives, and how would you describe it?
5. Discuss the MBO process with the help of a diagram.
6. Mention the benefits and the weaknesses of MBO.

CHAPTER VI: DECISION MAKING

1. What do you understand by decision making?
2. Elaborate the steps in the decision-making process, and show the model.
3. Mention the elements in a decision situation and describe them briefly.
4. What are the conditions of decision making?
5. Discuss probability theory and decision trees as decision-making tools.

6. Discuss the advantages and disadvantages of group decisions.
7. What is the best decision-making process?

CHAPTER VII: ORGANIZING

1. Define organizing, and explain why organizing is the primary mechanism by which managers activate plans.
2. List and describe the steps in the organizing process.
3. Distinguish between formal and informal organization.
4. Describe various forms of management organization.
5. What is span of management, and what factors decide it?
6. Is there a relationship between organizational levels and span of management?
7. Distinguish line, staff and functional authority.
8. Discuss the relationship between responsibility, authority and delegation, and the obstacles to delegation.
9. Discuss the advantages and disadvantages of centralization and decentralization.
10. Illustrate the basic coordinating activities and structural coordination techniques.
11. Compare the bureaucratic and behavioral models, and state the situational factors affecting design.

CHAPTER VIII: LEADING

1. Define motivation and explain why managers must understand it.
2. Draw and explain a model that illustrates the needs-goal theory of motivation.
3. Describe Maslow's need hierarchy theory.
4. What are the assumptions of Theory X and Theory Y?
5. How would you describe the human factors that affect management?
6. What is leadership, and how does it differ from management?
7. Illustrate the different types of leadership power.
8. What is emotional intelligence, and why does a leader need it?
9. Why do people join informal groups and teams?
10. Distinguish the trait approach from the situational approach.
11. Explain different styles of leadership; draw and explain the Tannenbaum and Schmidt model.
12. Describe transformational leadership, coaching and super leadership.

CHAPTER IX: CONTROLLING

1. What are meant by controlling and monitoring, and what is the nature of control?
2. What is the relationship between controlling and the controller?
3. What are the main steps of the controlling process?
4. Explain the significance of how and what to measure, and say what a standard is.
5. List and define the three basic types of control.
6. What are the requirements of effective controls, and how may the process succeed?
7. Describe several potential barriers to successful controlling.
8. Discuss the relationship between planning and controlling, and between power and control.
9. How can controlling help a manager to become more efficient?

GLOSSARY

Accountability. The duty to answer for results. (*Ch. VII*)

Authority. The right to give orders and make decisions. (*Ch. VII*)

Autocratic style. The leader decides alone. (*Ch. VIII*)

Behavioral model. Design that stresses people, teams and flexibility. (*Ch. VII*)

Boundary spanning. Work of staff who deal with outsiders for the firm. (*Ch. III*)

Bounded rationality. People choose with limited information and time. (*Ch. VI*)

Bureaucracy. An organization run by fixed rules, clear ranks and written records. (*Ch. II*)

Bureaucratic model. Design by fixed rules, ranks and procedures. (*Ch. VII*)

Business ethics. Standards of right and wrong that guide conduct in business. (*Ch. III*)

Centralization. Keeping decision power at the top. (*Ch. VII*)

Certainty. The outcome of each alternative is known. (*Ch. VI*)

Conceptual skill. Ability to see the organization as a whole. (*Ch. I*)

Concurrent control. Action during the work. (*Ch. IX*)

Contingency. Depending on the situation. (*Ch. II*)

Controller. A staff manager who designs and runs the control system. (*Ch. IX*)

Controlling. Comparing actual results with the plan and correcting the difference. (*Ch. IX*)

Coordination. Joining the work of different people and parts toward one goal. (*Ch. VII*)

Corporate social responsibility. A firm's duty to act for the good of society, beyond profit and the law. (*Ch. III*)

Creativity. Producing new and useful ideas. (*Ch. VIII*)

Davis model. Keith Davis's five propositions on business and society. (*Ch. III*)

Decentralization. Spreading decision power to lower levels. (*Ch. VII*)

Decision. A choice among alternatives. (*Ch. VI*)

Decision maker. The person who makes the choice. (*Ch. VI*)

Decision support system. Computer system that gives managers data for decisions. (*Ch. VI*)

Delegation. Giving part of one's authority and tasks to a subordinate. (*Ch. VII*)

Democratic style. The leader shares decisions. (*Ch. VIII*)

Departmentalization. Grouping jobs into departments. (*Ch. VII*)

Effectiveness. Doing the right thing, so goals are reached. (*Ch. I*)

Efficiency. Doing the thing right, with little waste. (*Ch. I*)

Emotional intelligence. Skill in understanding and managing one's own and others' feelings. (*Ch. VIII*)

Environment. Everything inside and outside an organization that affects its work. (*Ch. III*)

Expected value. The average result if a choice were repeated many times. (*Ch. VI*)

Feedback control. Action after the work, from results. (*Ch. IX*)

Feedforward control. Action before work starts, to prevent problems. (*Ch. IX*)

First-line managers. Those who supervise workers directly. (*Ch. I*)

Formal organization. The planned structure with defined jobs and rules. (*Ch. VII*)

Function. A main kind of work that every manager performs. (*Ch. I*)

Functional authority. Limited right to direct others in one specialty. (*Ch. VII*)

Functional organization. Specialists direct work in their field. (*Ch. VII*)

General environment. Outside forces that affect all organizations. (*Ch. III*)

Hierarchy of needs. Maslow's ordered list of five human needs. (*Ch. VIII*)

Human factors. The needs, attitudes and relationships of people at work. (*Ch. VIII*)

Human relations. The view that workers' feelings and social needs affect output. (*Ch. II*)

Human skill. Ability to work with and understand people. (*Ch. I*)

Industrial Revolution. The change, from the late 1700s, from hand work at home to machine work in factories. (*Ch. II*)

Informal group. A group formed from personal ties, not by management. (*Ch. VIII*)

Informal organization. The unplanned network of personal relations. (*Ch. VII*)

Innovation. Putting a new idea into use. (*Ch. VIII*)

Internal environment. Forces inside: owners, employees, culture. (*Ch. III*)

Key result area. A field in which results decide the survival of the organization. (*Ch. V*)

Laissez-faire style. The leader leaves decisions to the group. (*Ch. VIII*)

Leadership. Influencing people to work willingly toward a goal. (*Ch. VIII*)

Line authority. Right to command in the chain. (*Ch. VII*)

Line organization. Authority flows straight down, one superior for each person. (*Ch. VII*)

Line-and-staff organization. Line managers decide; staff specialists advise them. (*Ch. VII*)

Management. Getting work done through other people, to reach goals. (*Ch. I*)

Management by objectives. A cycle in which managers and employees agree on goals, then review results. (*Ch. V*)

Management science. Using mathematical models and computers to solve management problems. (*Ch. II*)

Middle managers. Those who carry out top managers' plans through a department. (*Ch. I*)

Mission. The basic reason an organization exists. (*Ch. V*)

Monitoring. Watching progress to see what is happening. (*Ch. IX*)

Motivation. The inner force that makes a person act toward a goal. (*Ch. VIII*)

Need. Something a person feels the lack of. (*Ch. VIII*)

Objective. A result that an organization or person wants to reach. (*Ch. V*)

Operational plan. Daily plan for a specific task. (*Ch. IV*)

Organization. A group of people working together for a shared goal. (*Ch. I*)

Organization structure. The formal pattern of jobs and reporting lines. (*Ch. VII*)

Organizing. Arranging people, tasks and resources so that plans can be carried out. (*Ch. VII*)

Plan. A written or spoken statement of what will be done, when, and by whom. (*Ch. IV*)

Planning. Choosing goals and deciding in advance how to reach them. (*Ch. IV*)

Planning premise. An assumption about the future on which a plan is built. (*Ch. IV*)

Power. The ability to influence others. (*Ch. VIII*)

Principle. A general guide to action, to be applied with judgement. (*Ch. II*)

Productivity. Output produced for each unit of input used. (*Ch. I*)

Profession. An occupation that requires long training and follows a code of conduct. (*Ch. I*)

Purpose. The reason an activity exists. (*Ch. I*)

Rational decision making. Choosing by steps that weigh facts and alternatives logically. (*Ch. VI*)

Resources. Things an organization uses to reach its goals. (*Ch. I*)

Responsibility. The duty to do an assigned task. (*Ch. VII*)

Risk. Outcomes are uncertain, but their chances are known. (*Ch. VI*)

Scientific management. Studying work by observation and measurement to find the best way to do it. (*Ch. II*)

Single-use plan. A plan made for one situation that will not repeat. (*Ch. IV*)

Situational approach. Best leadership depends on the situation. (*Ch. VIII*)

Span of management. The number of subordinates one manager supervises. (*Ch. VII*)

Staff authority. Right to advise. (*Ch. VII*)

Standard. A target or measure against which results are judged. (*Ch. IX*)

Standing plan. A plan made once for situations that repeat. (*Ch. IV*)

Steps. Five steps turn a plan into a structure. (*Ch. VII*)

Strategic plan. Long-range plan for the whole organization. (*Ch. IV*)

Super leadership. Leading others to lead themselves. (*Ch. VIII*)

Sustainability. Meeting present needs without harming the chances of future generations. (*Ch. III*)

System. A set of connected parts working toward one purpose. (*Ch. II*)

Tactical plan. Plan for a part of the organization. (*Ch. IV*)

Task environment. Outside groups that deal directly with one organization. (*Ch. III*)

Technical skill. Knowing how to do a specific kind of work. (*Ch. I*)

Theory X. The view that people dislike work and must be controlled. (*Ch. VIII*)

Theory Y. The view that people enjoy work and can direct themselves. (*Ch. VIII*)

Top managers. Those who set direction for the whole organization. (*Ch. I*)

Trait approach. Leaders are born with special qualities. (*Ch. VIII*)

Transformational leadership. Inspiring followers to look beyond self-interest to a shared vision. (*Ch. VIII*)

Uncertainty. Chances are unknown. (*Ch. VI*)

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INTRODUCTION TO MANAGEMENT
BIJOY?!

PRINCIPLES AND PRACTICES OF ACCOUNTING



*An Introduction for Readers
Who Know Nothing Yet*

A TEXTBOOK FOR FIRST-YEAR STUDENTS

BIJOY?!

PRINCIPLES AND PRACTICES OF ACCOUNTING

An Introduction for Readers Who Know Nothing Yet



BUSINESS, MANAGEMENT AND ACCOUNTING

BIJOY?!

TO THE READER

This book teaches accounting from the beginning. You need no earlier knowledge of business. You need only arithmetic and basic algebra. Every new word is explained in plain English when it first appears.

Bold type marks a new term at its first appearance. Bold is never used for any other purpose. **Italic type** is used for emphasis, for titles and for words from other languages.

The **narrow column** beside the text carries notes. Some notes are short summaries of the paragraph beside them, set in italic. Others give a new term in bold with a one-line meaning. A note marked **Go slowly here** tells you that the passage is new and deserves care. A note marked **Word in use** warns you that an everyday word has a special meaning in accounting.

Small raised numbers in the text refer to notes. Because this edition is set electronically, the notes are collected at the end of each chapter.

Each chapter opens with a puzzle, builds an idea in layers (the idea, an example, the rule, the exceptions), and closes with a summary, a problem, a list of common mistakes, a true story from history, and review questions. Short answers to the review questions are at the back.

Work every example with a pencil. Accounting is learned by doing. When a calculation appears, follow the same order each time: what is given, what is asked, the formula, the steps, the answer, and a check.

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A NOTE ON THE EXAMPLES

The running business in this book is **Padma Stationers**, a small shop in Dhaka owned by Nadia Rahman. It is **illustrative**: invented to show ideas at work, and not a real firm. Its figures appear as a case file: dates, amounts and decisions only.

All amounts are in **taka**, written Tk. Dates are written day, month, year. Spelling is British. Where law or standards are mentioned, the country is Bangladesh unless the text says otherwise. Laws and standards change, so confirm the current rules before relying on them.

Historical accounts in the book are true to the best of the author's knowledge. Where a date or figure could not be verified, it has been left out.

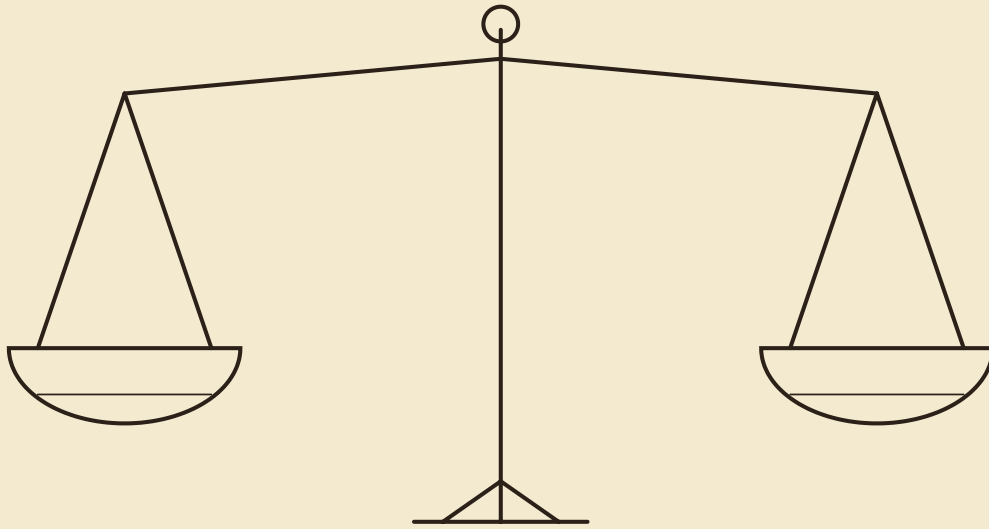


Plate I. [An engraving will be placed here.]

CHAPTER I

Business and Accounting

A shopkeeper is asked, "Are you making money?" What does she say?



This chapter explains what a business does, what accounting is and who needs it, the rules accountants follow, and the one equation on which all of accounting rests. It ends with the statements that report a business to the world.

Nadia Rahman plans to open a stationery shop in Dhaka. She has Tk 500,000 saved. A cousin asks her, a year before she opens, a plain question: "How will you know whether the shop is making money?" Nadia cannot answer. She can count the cash in the till. But cash alone does not show what she owes suppliers, what customers owe her, or what her goods are worth.

A business needs a method for answering such questions. The method has a name, **accounting**, and this chapter introduces it. Later chapters build on every idea here, so read slowly.

YOU WILL NEED: nothing earlier. This is where the book begins. Chapter II uses the equation of this chapter at once.

WHAT A BUSINESS DOES

A business sells something to earn a profit.

Business. An organisation that sells goods or services to customers in order to earn a profit.

Businesses fall into three kinds.

Service company. A business that earns money by doing work for customers, such as a barber or a school.

Each kind has its own accounts.

Merchandising company. A business that buys finished goods and sells them again without making any change.

Manufacturing company. A business that turns materials into new goods and then sells those goods.

Three common forms of ownership.

Sole trader. A business owned and run by one person, who is personally responsible for its debts.

A **business** means an organisation that sells goods or services to customers in order to earn a profit. A **profit** means the money left after all costs of the business are paid. If costs are larger than the money earned, the result is a **loss**. Nadia's shop is a business. It buys pens and notebooks and sells them for more than they cost.

Businesses are grouped by what they sell. A **service company** sells work, such as a barber, a clinic or a photocopy bureau. A **merchandising company** buys finished goods and sells them again without change, as a stationery shop does. A **manufacturing company** makes goods from materials and sells them, as a factory making notebooks does.

The three kinds are different in one important way. A service company holds no goods for sale. A merchandising company holds goods bought from others. A manufacturing company holds materials, goods half-made, and goods finished. Each extra kind of goods needs extra records, which is why this book moves from service to merchandising, and touches manufacturing in Chapter VI.

A business also has a form of ownership. A **sole trader** is owned by one person. A partnership is owned by two or more people (Chapter IX). A company is a separate legal body owned by shareholders (Chapter X). Padma Stationers begins as a sole trader, and later chapters show how its owner might change the form.

WHAT ACCOUNTING IS

Accounting records, sorts and reports money facts.

Accounting. The process of recording, sorting and reporting the money facts of a business so that people can make decisions.

The idea is old. People called it the language of business.

Accounting means recording, sorting and reporting the money facts of a business so that people can make decisions. A shop sells forty notebooks on a Monday. Accounting records the sale, adds it to the week's sales, and reports the total to the owner. The record is of no use unless someone acts on it.

Accounting has often been called the language of business. The description is fair in three ways. Accounting turns many different events into one common unit, money. It is understood by owners, bankers and tax officers alike. And, like any language, it has rules of grammar that must be followed. The description fails in one way: a language can say anything, but accounting reports only what can be measured in money.

Accounting has several branches.

Financial accounting. The branch that reports a business's results to people outside it, such as lenders and tax officers.

Management accounting. The branch that gives managers the figures they need to plan and control the business.

Bookkeeping is only the first step.

Accounting has several branches. **Financial accounting** reports results to people outside the business. **Management accounting** gives managers figures to plan and control work. **Cost accounting** finds what each product or service costs to make. **Auditing** means an independent check that the records are fair. **Tax accounting** prepares the figures for tax returns. This book teaches mainly financial accounting.

A word often confused with accounting is *bookkeeping*. Bookkeeping means the daily recording of transactions. Accounting includes bookkeeping and then goes further: it classifies the records, summarises them, and explains what they mean.

WHO USES ACCOUNTING INFORMATION

Users sit inside and outside the business.

User. A person who reads accounting information in order to make a decision.

Each user asks a different question.

A **user** means a person who reads accounting information in order to decide something. Users of two kinds exist. *Internal users* work inside the business: the owner and the managers. *External users* sit outside it: lenders, suppliers, customers, tax officers, employees and possible investors.

The same figures serve different questions. The owner asks, "Did we earn a profit?" A bank asks, "Can this business repay a loan?" A supplier asks, "Will we be paid on time?" The tax officer asks, "Is the tax correct?" An employee asks, "Is the business secure enough for my job?" A viewpoint changes the question, but the record is the same.

THE RULES OF THE GAME

Standards make reports comparable.

Accounting standards. Written rules that tell accountants how to record and report items, so that reports can be compared.

Principles and concepts sit beneath the standards.

GAAP. Generally Accepted Accounting Principles: the common rules and ideas that accountants agree to follow.

Some basic concepts follow.

Accounting standards means written rules that tell accountants how to record and report items. They are needed for one reason. Without common rules, two shops with the same facts could report different profits, and no reader could compare them. In Bangladesh, companies follow International Financial Reporting Standards (IFRS) as adopted locally.¹

The ideas beneath the standards are called **Generally Accepted Accounting Principles**, shortened to GAAP. They ensure consistency, which means that the same method is used from one year to the next. They also ensure fairness, because every user reads figures prepared by the same rules.

Several basic concepts are used throughout this book. The *business entity* concept says the business is kept separate from its owner's private affairs. The *money measurement* concept says only facts that can be stated in money are recorded. The *cost* concept says an asset is recorded at what was paid for it. The *going concern* concept assumes the business will continue. The *periodicity* concept divides life into periods, such as a year. The *accrual* concept records an item when it happens, not when cash moves (Chapter III). Prudence asks that losses are not hidden and profits are not overstated.

GO SLOWLY HERE.

Everything in accounting begins with a simple balance.

GO SLOWLY HERE.

Asset. Something of value that a business owns and can use in future, such as cash or equipment.

Liability. An amount the business owes to someone outside it.

Equity. The owner's claim on the business: what is left after liabilities are taken from assets.

GO SLOWLY HERE.

Accounting equation. Assets equal liabilities plus equity. It holds after every transaction.

GO SLOWLY HERE.

Equity changes by four items.

Revenue. Money earned from selling goods or services. It raises equity.

Expense. A cost of running the business, such as rent or wages. It lowers equity.

A transaction is a money event that changes the equation.

Transaction. A business event stated in money that changes the assets, liabilities or equity.

Given and find: the first worked example.

Each line balances.

WORD IN USE. Capital. In daily life, money. In accounting, the owner's investment in the business.

THE ACCOUNTING EQUATION

Think of a household. Your family owns a house worth Tk 1,000,000. A bank lent you Tk 400,000 of the price. The part you truly own is the rest, Tk 600,000. What you own equals what you owe plus what is yours. That is the whole idea of this section.

An **asset** means something of value that a business owns, such as cash, goods and equipment. A **liability** means an amount the business owes to others, such as a bank loan. **Equity** means the owner's claim: what remains after liabilities are subtracted from assets. Equity is also called owner's equity, or stockholders' equity in a company. Equity and liability are alike because both are claims on the assets. They differ because a liability must be repaid by a fixed date, but equity is repaid only after all debts.

The relation is the **accounting equation**. In words: assets equal liabilities plus equity. In symbols: $A = L + E$. Here A stands for assets, L for liabilities and E for equity. The equation holds at all times, because every asset was paid for either by borrowing or by the owner.

Equity is changed by four items. The owner's investment raises it, and **drawings**, meaning money the owner takes out for private use, lower it. **Revenue** means money earned by selling goods or services; it raises equity. An **expense** means a cost of running the business, such as rent; it lowers equity. Revenue minus expenses is the profit. We write the full form: $A = L + (\text{Capital} + \text{Revenue} - \text{Expenses} - \text{Drawings})$.

A **transaction** means an event, stated in money, that changes assets, liabilities or equity. Buying equipment is one. Agreeing to hire a worker next month is not, because no money value has changed yet. A transaction is also supported by a document, such as a receipt, and it affects the business itself.

GIVEN. During January 2025 Padma Stationers, then a photocopy service, has seven transactions: (1) Nadia invests Tk 500,000 cash; (2) buys equipment for Tk 150,000 cash; (3) buys supplies on credit for Tk 20,000; (4) earns fees of Tk 40,000 in cash; (5) pays rent Tk 12,000; (6) pays Tk 10,000 to the supplier; (7) draws Tk 15,000 for private use. **FIND.** The totals of A, L and E after each transaction. **FORMULA.** $A = L + E$. **STEPS.** See the table below.

PADMA STATIONERS: EQUATION AFTER EACH TRANSACTION (TK)

Transaction	Assets	Liabilities	Equity
1 Investment	500,000	0	500,000
2 Equipment, cash	500,000	0	500,000
3 Supplies on credit	520,000	20,000	500,000
4 Fees earned	560,000	20,000	540,000
5 Rent paid	548,000	20,000	528,000
6 Supplier paid	538,000	10,000	528,000
7 Drawings	523,000	10,000	513,000

How to read the table and check it.

ANSWER. After transaction 7 the business has assets of Tk 523,000, owes Tk 10,000, and equity is Tk 513,000. CHECK. Equity should equal capital 500,000 plus revenue 40,000, less rent 12,000, less drawings 15,000. That is 513,000, which agrees. Each row is a transaction, and in every row the left total equals the right total.

Common slips.

A common slip is to treat drawings as an expense. Drawings reduce equity but are not a cost of running the business. Another slip is to forget that paying a supplier lowers both an asset and a liability, leaving equity unchanged.

REPORTING TO THE WORLD

Four statements report the business.

Financial statement. A formal report that sums up the results or position of a business.

A **financial statement** means a formal report that sums up the results or position of a business. A profit-seeking business prepares four: the *income statement*, which shows revenue, expenses and profit for a period; the *statement of changes in equity*; the *balance sheet*, which lists assets, liabilities and equity at one date; and the *cash flow statement*, which shows cash received and paid. Notes explain the figures. The first three are learned in Chapters IV and V.

Periods are fixed in advance.

The income statement covers a period. A **fiscal year** means any twelve-month period a business chooses for its reports, such as 1 July to 30 June. A *calendar year* runs from 1 January to 31 December. *Interim periods* are shorter, such as a quarter, used for reports between year-ends.

SUMMARY

Business sells goods or services for profit. Accounting records, sorts and reports money facts for internal and external users. Standards and concepts keep reports fair and comparable. All records rest on $A = L + E$, and every transaction leaves it in balance.

THINK IT THROUGH

Nadia buys a printer for Tk 30,000, paying Tk 10,000 now and owing the rest. Which of A, L and E change, and by how much? (Assets rise by 30,000 minus 10,000 cash, so net 20,000; liabilities rise by 20,000.)

WHERE PEOPLE GO WRONG

Many students think profit and cash are the same. They are not: a sale on credit raises profit before it raises cash. Many also think equity means money in the bank. Equity is a claim, not a sum of cash.

IN HISTORY

Clay tablets from Mesopotamia, dating from about 3000 BC, record quantities of grain and animals owed or delivered. People needed a record because memory failed once an exchange spread over time and many people. Accounting began where trust needed a witness.

REVIEW QUESTIONS

1. Name the three kinds of company and give one example of each.
2. State the accounting equation and say how a liability differs from equity.
3. A business has assets Tk 80,000 and liabilities Tk 30,000. Find equity.
4. Why are accounting standards needed?

¹ Standards are revised. A reader should check the current requirements of the Financial Reporting Council and the Institute of Chartered Accountants of Bangladesh.

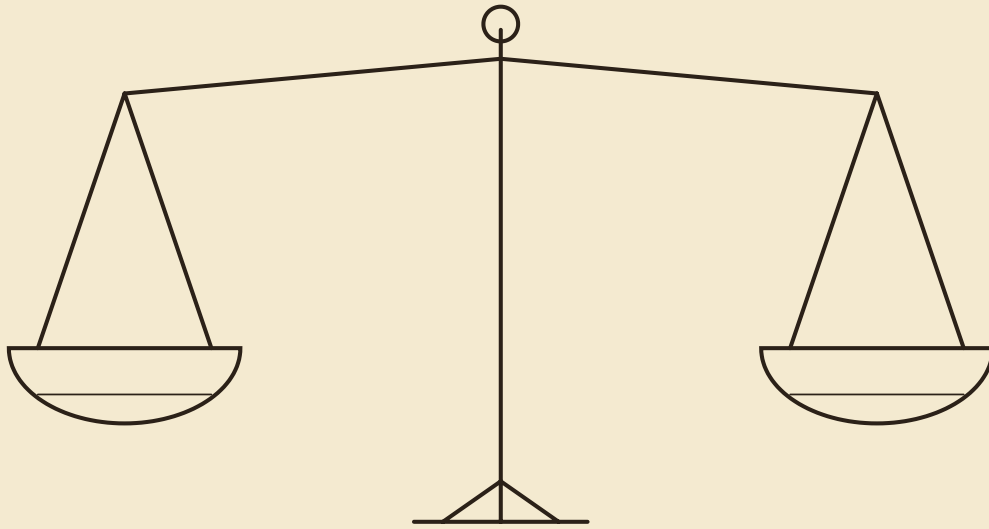


Plate II. [An engraving will be placed here.]

CHAPTER II

Recording the Transactions

One transaction must be written twice. Why?



This chapter explains accounts, debits and credits, the journal and the ledger, and the trial balance. It follows one month of Padma Stationers from first entry to a proved balance.

Nadia buys a copier for Tk 150,000 and pays cash. The records must say two things: the business now has a copier, and it now has less cash. If only one is written, the books no longer balance. Over a year a shop has thousands of such events. How can they be written so that mistakes reveal themselves?

The answer was published in Venice more than five hundred years ago. It is called double entry, and it is the subject of this chapter.

YOU WILL NEED: the accounting equation, assets, liabilities, equity, revenue, expense and transaction (Chapter I).

THE ACCOUNTING CYCLE

The cycle is a fixed list of steps.

Accounting cycle. The steps repeated every period, from recording transactions to closing the books.

Each step guards accuracy.

The **accounting cycle** means the steps an accountant repeats every period. First, analyse each transaction. Second, record it in the journal. Third, post it to the ledger. Fourth, prepare the unadjusted trial balance. Fifth, make adjusting entries. Sixth, prepare the adjusted trial balance. Seventh, prepare the financial statements. Eighth, close the temporary accounts. This chapter teaches steps one to four.

Each step checks the one before it. Analysis ensures that the event is a transaction. The journal records it in order of date. The ledger sorts it by account. The trial balance tests whether debits equal credits. Later steps update and report. Skipping any step lets errors pass unseen.

ACCOUNTS, DEBITS AND CREDITS

GO SLOWLY HERE.

An account is a named record.

Account. A record of all changes to one kind of item, such as Cash or Rent Expense.

WORD IN USE. Account. In daily life, a bank record. In accounting, a named record of one kind of item.

GO SLOWLY HERE.

The T-account has two sides and a name.

T-account. A simple picture of an account with the name on top, debit on the left and credit on the right.

GO SLOWLY HERE.

The sign of increase depends on the account type.

An **account** means a record of all changes to one kind of item, such as Cash or Rent Expense. Think of a notebook with a page headed *Cash*. Every time cash arrives, a line is added on the left. Every time cash leaves, a line is added on the right. That page is the idea of an account.

The page is drawn as a **T-account**, named for its shape. It has three parts: the *title* at the top, the *debit side* on the left, and the *credit side* on the right. **Debit** means the left side of an account; **credit** means the right side. The words do not mean good or bad. They mean left and right.

Six types of account exist. Assets, drawings and expenses are *increased* by a debit and decreased by a credit. Liabilities, equity (capital) and revenue are *increased* by a credit and decreased by a debit. The rule follows from the equation: $A = L + E$ has assets on the left, so they grow on the left.

Learn this table. It is the whole of the system.

SIX ACCOUNT TYPES

Type	Normal balance	Increase	Decrease
Asset	Debit	Debit	Credit
Expense	Debit	Debit	Credit
Drawings	Debit	Debit	Credit
Liability	Credit	Credit	Debit
Capital	Credit	Credit	Debit
Revenue	Credit	Credit	Debit

GO SLOWLY HERE.

Double entry: every debit has an equal credit.

Double-entry system. A method in which every transaction is recorded as equal debits and credits in at least two accounts.

Pause and check.

The **double-entry system** means a method in which every transaction is recorded with debits equal to credits, in at least two accounts. Buying the copier increases an asset, Equipment, so Equipment is debited Tk 150,000. It decreases another asset, Cash, so Cash is credited Tk 150,000. The totals match, and the equation stays in balance.

PAUSE AND CHECK. The shop pays Tk 12,000 rent. Which account is debited, and which is credited? (Debit Rent Expense, since expenses rise by debit. Credit Cash, since assets fall by credit.)

THE JOURNAL

The journal is the book of first entry.

Journal. The book in which transactions are first recorded, in order of date, with debits and credits.

The **journal** means the book in which transactions are first recorded, in date order, with debits and credits and a short explanation. Each record is a *journal entry*. It names the date, the account debited, the account credited, the amounts, and a narration. The debit is written first, and the credit is indented.

JOURNAL OF PADMA STATIONERS, JANUARY 2025 (TK)

Date	Particulars	Debit	Credit
1 Jan	Cash	500,000	
	Capital		500,000
3 Jan	Equipment	150,000	
	Cash		150,000
5 Jan	Supplies	20,000	
	Accounts Payable		20,000
10 Jan	Cash	40,000	
	Fees Revenue		40,000
15 Jan	Rent Expense	12,000	
	Cash		12,000
20 Jan	Accounts Payable	10,000	
	Cash		10,000
25 Jan	Drawings	15,000	
	Cash		15,000

Accounts payable and receivable.

Accounts payable. Money the business owes to suppliers for goods or services bought on credit.

Accounts receivable. Money customers owe the business for goods or services sold on credit.

Two names appear often. **Accounts payable** means money the business owes to suppliers; it is a liability. **Accounts receivable** means money customers owe the business; it is an asset. They are opposites: one is a debt of the business, the other is a debt to the business.

The ledger sorts by account.

Ledger. The complete set of accounts of a business, in which journal entries are sorted by account.

Posting the Cash account.

THE LEDGER

The **ledger** means the complete set of accounts. Posting means copying each journal line to its account. The journal tells the story by date; the ledger gathers every change to Cash on one page. After posting, the *balance* of an account is found by subtracting the smaller side from the larger.

GIVEN. The seven journal entries above. **FIND.** The balance of Cash. **STEPS.** Debit side: $500,000 + 40,000 = 540,000$. Credit side: $150,000 + 12,000 + 10,000 + 15,000 = 187,000$. Balance: $540,000 - 187,000 = 353,000$, a debit balance. **ANSWER.** Cash is Tk 353,000. **CHECK.** Assets after transaction 7 in Chapter I were 523,000, and Cash 353,000 + Supplies 20,000 + Equipment 150,000 = 523,000.

The trial balance tests the books.

Trial balance. A list of all account balances at one date, used to test that total debits equal total credits.

Debits equal credits: the test passes.

THE TRIAL BALANCE

A **trial balance** means a list of all ledger balances at one date, in two columns, to test that total debits equal total credits. It is not a financial statement. The balance sheet reports position to outsiders and divides assets, liabilities and equity; the trial balance is an inside check that lists every account, including revenue and expenses.

TRIAL BALANCE, 31 JANUARY 2025 (TK)

Account	Debit	Credit
Cash	353,000	
Supplies	20,000	
Equipment	150,000	
Accounts Payable		10,000
Capital		500,000
Drawings	15,000	
Fees Revenue		40,000
Rent Expense	12,000	
Totals	550,000	550,000

Limits of the test.

A balanced trial balance does not prove the books are right. It misses an entry left out, an entry posted to the wrong account, an error repeated on both sides, and two errors that cancel. It does catch a one-sided posting and most errors of addition.

Finding an error.

When the totals disagree, work in order. Re-add both columns. Find the difference. If it is divisible by 9, suspect a *transposition*, such as 1,520 written as 1,250 (difference 270, and $270 \div 9 = 30$). If the difference equals one item, suspect that item was left out; if it is twice an item, suspect that item was placed in the wrong column. Then check that every ledger balance was copied. Correct an error by a new journal entry, not by erasing.

SUMMARY

Every transaction is analysed, journalised, posted and tested. Debits equal credits. Assets, expenses and drawings rise by debit. Liabilities, capital and revenue rise by credit. The trial balance tests arithmetic but not judgement.

THINK IT THROUGH

Record in T-accounts: (a) owner invests Tk 60,000; (b) buys goods on credit Tk 8,000; (c) pays Tk 3,000 of the debt. Find the Cash and Accounts Payable balances. (Cash 57,000 debit; Accounts Payable 5,000 credit.)

WHERE PEOPLE GO WRONG

Beginners believe debit always means increase. It does for assets, but not for liabilities. Another error is to think the trial balance is the balance sheet. A third is to record only one side of a transaction.

IN HISTORY

In 1494 Luca Pacioli, a friar and teacher of mathematics, published *Summa de arithmetica* in Venice. One section described the merchant's methods of bookkeeping, including journal, ledger and balancing by equal debits and credits. Pacioli did not invent them: Italian merchants used the method earlier. He wrote it down, and so it spread.

SOURCES

Pacioli, L. (1494), *Summa de arithmetica, geometria, proportioni et proportionalita*. It matters because it is the earliest printed account of double entry.

REVIEW QUESTIONS

1. What are the six types of account and which are increased by debit?
2. Record: owner invests Tk 50,000 cash; buys equipment Tk 20,000 cash.
3. A trial balance differs by Tk 360. What is the first check?
4. How does a trial balance differ from a balance sheet?
5. Name the parts of a T-account.

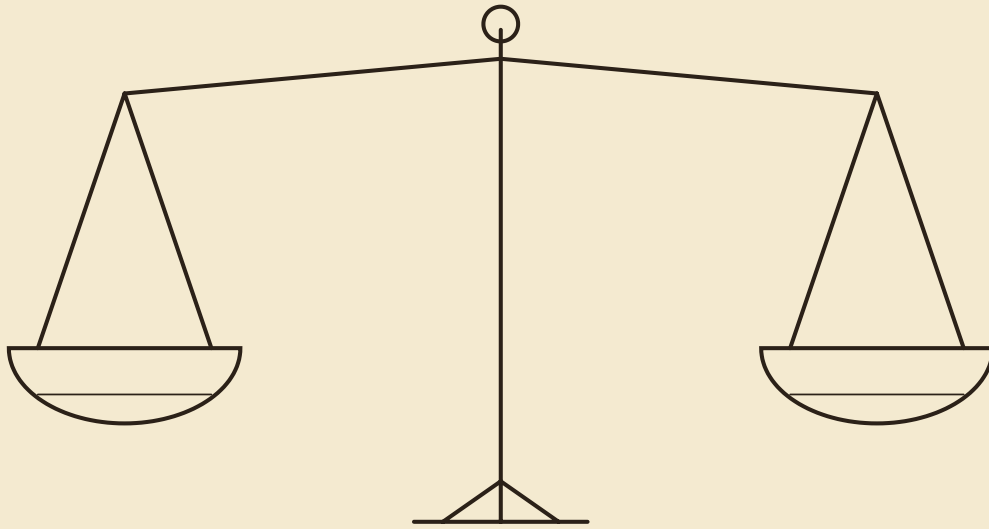


Plate III. [An engraving will be placed here.]

CHAPTER III

Adjusting the Accounts

On 31 January the shop's books say one thing, and the truth is another.



This chapter explains why the books must be corrected at the end of each period, the kinds of adjustment, and how to record them, using Padma Stationers at the end of January 2025.

It is 31 January. The unadjusted trial balance of Padma Stationers is balanced. Yet three facts are missing from it. Part of the supplies has been used. Workers have worked five days for which they have not been paid. A customer paid in advance for work not yet done. If Nadia reads her profit from the books as they stand, she reads a wrong number.

YOU WILL NEED: the six account types, journal entries and the trial balance (Chapter II); revenue and expense (Chapter I).

WHY ADJUSTMENTS ARE NEEDED

Profit belongs to the period that earned it.

Accrual basis. A method that records revenue when it is earned and expenses when they are incurred, whether or not cash moves.

The matching idea.

Reasons for adjusting.

There are two ways to measure profit. The *cash basis* records an item when cash moves. The **accrual basis** records revenue when it is earned and an expense when it is used, whatever the date of payment. Accrual gives a truer picture, so standards require it. Adjusting entries are what make the accrual basis work.

Accrual rests on the **matching** idea: the expenses of a period are set against the revenue they helped to earn. Wages for the last week of January belong to January, although they will be paid in February. The books, which record only what has happened, must be adjusted to show what belongs.

There are three reasons. Some payments cover several periods and must be divided. Some costs and revenues have arisen but not yet been recorded. And some assets lose value gradually, which the daily records do not show. For these reasons an **adjusting entry** is made at the end of every period. It changes an account balance to its correct amount. It never involves Cash.

TYPES OF ADJUSTMENT

GO SLOWLY HERE.

Four types, in two families.

Prepaid expense. A cost paid in advance that is recorded as an asset until it is used.

Unearned revenue. Money received before the work is done; it is a liability until the work is finished.

GO SLOWLY HERE.

Accruals record what cash has not yet shown.

Accrued expense. A cost already used but not yet paid or recorded, shown as a liability.

Accrued revenue. Revenue already earned but not yet billed or received, shown as an asset.

Adjustments come in two families. *Deferrals* hold back a cash payment or receipt that belongs partly to the future. A **prepaid expense** is a cost paid before it is used, such as insurance, and it is an asset until used. **Unearned revenue** is money received before work is done, and it is a liability until the work is finished.

Accruals record items that exist before cash moves. An **accrued expense** is a cost already used but not yet paid, such as wages for days worked. An **accrued revenue** is revenue earned but not yet received, such as fees for work done and not yet billed. A fifth case needs an estimate: **depreciation**, the spreading of an asset's cost over its life (Chapter VIII).

ADJUSTING THE BOOKS OF PADMA STATIONERS

The starting point.

UNADJUSTED TRIAL BALANCE, 31 JANUARY 2025 (TK)

Account	Debit	Credit
Cash		360,000
Supplies		20,000
Prepaid Insurance		6,000
Equipment		150,000
Accounts Payable		10,000
Unearned Fees		9,000
Capital		500,000
Drawings	15,000	
Fees Revenue		64,000
Rent Expense	12,000	
Wages Expense	20,000	
Totals	583,000	583,000

The six facts found at month end.

GIVEN. (a) Supplies on hand are Tk 8,000. (b) The Tk 6,000 insurance covers six months from 1 January. (c) The equipment cost Tk 150,000, will last 10 years (120 months) and be worth Tk 30,000 at the end. (d) Wages of Tk 3,000 are owed to workers for 27 to 31 January. (e) Half of the Tk 9,000 advance has now been earned. (f) Fees of Tk 6,000 are earned but not yet billed.

GO SLOWLY HERE.

Method: ask what the balance should be, then record the difference.

FIND. The six adjusting entries. METHOD. For each account ask two questions: what is the balance now, and what should it be? The entry is the difference. STEPS. (a) Supplies used = $20,000 - 8,000 = 12,000$. (b) Insurance used = $6,000 \div 6 = 1,000$. (c) Depreciation = $(150,000 - 30,000) \div 120 = 1,000$. (d) Given. (e) $9,000 \div 2 = 4,500$. (f) Given.

ADJUSTING ENTRIES, 31 JANUARY 2025 (TK)

Particulars	Debit	Credit
(a) Supplies Expense	12,000	
Supplies		12,000
(b) Insurance Expense	1,000	
Prepaid Insurance		1,000
(c) Depreciation Expense	1,000	
Accumulated Depreciation		1,000
(d) Wages Expense	3,000	
Wages Payable		3,000
(e) Unearned Fees	4,500	
Fees Revenue		4,500
(f) Accounts Receivable	6,000	
Fees Revenue		6,000

Accumulated depreciation is a contra account.

Accumulated depreciation.

The total depreciation charged on an asset so far. It is subtracted from the asset's cost.

Accumulated depreciation means the total depreciation charged so far. It is a *contra account*: it has a credit balance and is subtracted from the asset it belongs to. The asset itself stays at its cost. The balance sheet shows the cost, the accumulated depreciation, and the difference, called the book value.

Answer and check.

ANSWER. The entries add expenses of $12,000 + 1,000 + 1,000 + 3,000 = 17,000$ and revenue of $4,500 + 6,000 = 10,500$, so net income falls by 6,500, from 32,000 to 25,500. CHECK. Every entry has equal debits and credits, and the total debits (27,500) equal the total credits (27,500). The adjusted trial balance then totals Tk 593,000 on each side.

WHERE ADJUSTMENTS GO WRONG

Common slips.

The usual slips are three. A student debits Cash, though adjustments never touch cash. A student adjusts only one side. Or a student uses the wrong direction, such as crediting an expense account for an accrued wage. Ask each time: is it an asset or an expense, a liability or a revenue?

Relation to the statements.

The adjusted trial balance is the bridge. The unadjusted trial balance shows the books as recorded. The adjusting entries correct them. The adjusted trial balance lists every account at its right amount, and from it the income statement and balance sheet are drawn without further change.

SUMMARY

Accrual accounting matches expenses to the revenue they earn. At the end of each period, deferrals and accruals are adjusted, along with depreciation. Adjusting entries never touch Cash. The adjusted trial balance feeds the statements.

THINK IT THROUGH

A shop pays Tk 24,000 on 1 October for a year's insurance. How much is expense by 31 December and how much is still an asset? (6,000 expense; 18,000 prepaid.)

WHERE PEOPLE GO WRONG

Many students think an adjusting entry corrects a mistake. It does not. It brings correct records up to date. Others record accrued wages only when paid, which puts the cost in the wrong month.

IN HISTORY

In 1752 Britain changed from the Julian to the Gregorian calendar and lost eleven days. The tax year had begun on 25 March. To keep the full year of 365 days for taxpayers, the start moved to 5 April, and later, from 1800, to 6 April. The tax year of Britain still begins on that date. A period is a human choice, and the books must follow it.

REVIEW QUESTIONS

1. "Adjusting entries are essential for accrual accounting." Explain with an example.
2. Name the types of adjustment.
3. Insurance of Tk 12,000 was paid for 12 months from 1 March. Adjust at 31 December.
4. Why does an adjusting entry never include Cash?

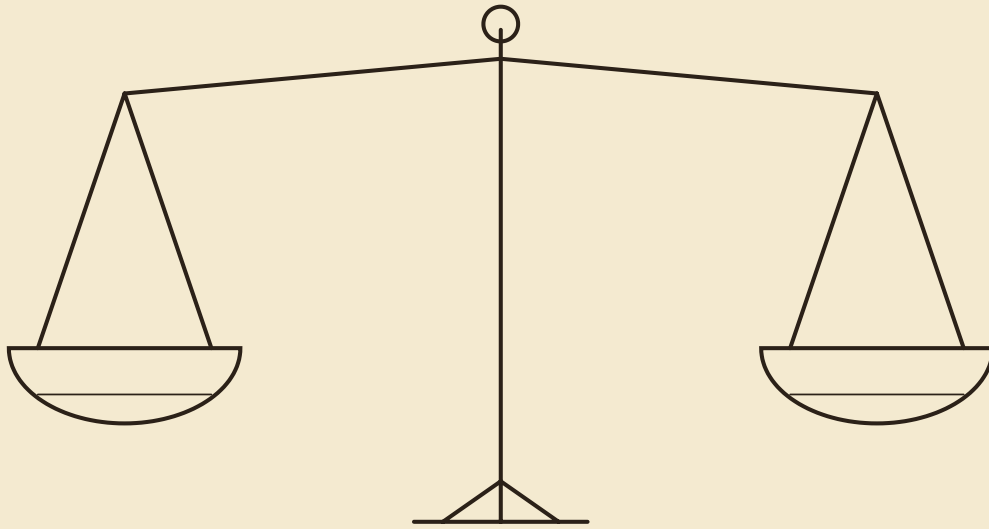


Plate IV. [An engraving will be placed here.]

CHAPTER IV

The Worksheet and the Closing of the Books

*How can an accountant see the whole year on one sheet
before writing a single statement?*



*This chapter explains the ten-column worksheet, the financial
statements drawn from it, and the closing and reversing entries
that prepare the books for the next period.*

After the adjusting entries, an accountant has three lists: the unadjusted trial balance, the adjustments, and the amounts they produce. The statements still have to be written. If one error remains, it may be found only after the statements are printed. Would it not be safer to see everything on one sheet, and test it, first?

YOU WILL NEED: the trial balance (Chapter II) and adjusting entries (Chapter III). We continue with Padma Stationers at 31 January 2025.

THE WORKSHEET

The worksheet is a working paper.

Worksheet. A large sheet on which the accountant sorts the trial balance, adjustments and statement figures before preparing the statements.

Why and when it is prepared.

GO SLOWLY HERE.

The sheet has ten columns in five pairs.

GO SLOWLY HERE.

Method in four steps.

A **worksheet** means a large sheet on which the trial balance, the adjustments and the statement figures are sorted side by side. It is a working paper for the accountant, and it is not a formal document. It is not published, and it is not part of the books. The ledger is changed only by the journal entries.

A worksheet is prepared at the end of a period, after the trial balance and before the statements. It gives three benefits: it shows the effect of adjustments at once, it tests the arithmetic before any statement is written, and it makes the statements easy to prepare. It is prepared by the bookkeeper or accountant, for the owner or a senior accountant who reviews the work.

The ten-column sheet has five pairs of columns, each pair a debit and a credit. The pairs are: the *unadjusted trial balance*, the *adjustments*, the *adjusted trial balance*, the *income statement* and the *balance sheet*. Work from left to right. Add the pair across, then move to the next.

STEPS. (1) Copy the trial balance into the first pair, and total it. (2) Enter the adjustments in the second pair, with letters, and check that they balance. (3) Add or subtract across to make the third pair. (4) Carry each account to the income statement pair, or to the balance sheet pair. Revenue and expenses go to the income statement; assets, liabilities, capital and drawings go to the balance sheet.

PADMA STATIONERS: TEN-COLUMN WORKSHEET, 31 JANUARY 2025 (TK)

Account	Unadj Dr	Unadj Cr	Adj Dr	Adj Cr	Adj TB Dr	Adj TB Cr	Income Dr	Income Cr	Bal Sheet Dr	Bal Sheet Cr
Cash	360,000				360,000				360,000	
Accounts Receivable			(f) 6,000		6,000				6,000	
Supplies	20,000			(a) 12,000	8,000				8,000	
Prepaid Insurance	6,000			(b) 1,000	5,000				5,000	
Equipment	150,000				150,000				150,000	
Accumulated Depreciation				(c) 1,000		1,000				1,000
Accounts Payable		10,000				10,000				10,000
Wages Payable				(d) 3,000		3,000				3,000
Unearned Fees		9,000	(e) 4,500			4,500				4,500
Capital		500,000				500,000				500,000
Drawings	15,000				15,000				15,000	
Fees Revenue		64,000		(e,f) 10,500		74,500		74,500		
Rent Expense	12,000				12,000		12,000			
Wages Expense	20,000		(d) 3,000		23,000		23,000			
Supplies Expense			(a) 12,000		12,000		12,000			
Insurance Expense			(b) 1,000		1,000		1,000			
Depreciation Expense			(c) 1,000		1,000		1,000			
Totals	583,000	583,000	27,500	27,500	593,000	593,000	49,000	74,500	544,000	518,500
Net income							25,500			25,500
Totals							74,500	74,500	544,000	544,000

How to read the sheet and check it.

Read a row from left to right. Take Supplies: the trial balance shows a debit of 20,000; adjustment (a) credits 12,000; the adjusted debit is 8,000; it is an asset, so it goes to the balance sheet debit column. The first check is that each pair totals equally. The last check is that net income is the same number in the income statement and balance sheet columns. Here both are Tk 25,500.

Net income balances two pairs.

The income statement columns show credits of 74,500 and debits of 49,000. The difference of 25,500 is net income. It is added to the debit column to make the pair equal. In the balance sheet columns, the same figure is added to the credit side, because net income increases capital.

The statements come straight from the sheet.

Income statement. A report of revenue, expenses and net income for a period of time.

From the income columns.

THE STATEMENTS

The **income statement** reports revenue, expenses and net income for a period. The balance sheet reports the position at one date. They are linked: net income from the first is added to equity in the second. This is how they are connected.

INCOME STATEMENT, MONTH ENDED 31 JANUARY 2025 (TK)

Item	Tk
Fees revenue	74,500
Less expenses:	
Wages	23,000
Rent	12,000
Supplies used	12,000
Insurance	1,000
Depreciation	1,000
Total expenses	49,000
Net income	25,500

Equity statement, then the balance sheet.

Balance sheet. A report of the assets, liabilities and equity of a business at one date.

STATEMENT OF OWNER'S EQUITY AND BALANCE SHEET, 31 JANUARY 2025 (TK)

Item	Tk
Capital, 1 January	500,000
Add net income	25,500
Less drawings	(15,000)
Capital, 31 January	510,500
Assets: cash, receivable, supplies, prepaid	379,000
Equipment less accumulated depreciation (150,000 – 1,000)	149,000
Total assets	528,000
Liabilities (10,000 + 3,000 + 4,500)	17,500
Capital	510,500
Total liabilities and capital	528,000

CLOSING AND REVERSING

GO SLOWLY HERE.

Closing returns the temporary accounts to zero.

Closing entries. Entries made at the end of a period to bring revenue, expense and drawings accounts to zero and move the result to capital.

GO SLOWLY HERE.

Four closing entries.

Revenue, expense and drawings accounts measure one period only. They are called *temporary* accounts. **Closing entries** bring them to zero so the next period starts clean, and move the profit to capital. Asset, liability and capital accounts are *permanent* and are carried forward.

STEPS. (1) Close revenue: Dr Fees Revenue 74,500, Cr Income Summary 74,500. (2) Close expenses: Dr Income Summary 49,000, Cr each expense account (Wages 23,000; Rent 12,000; Supplies 12,000; Insurance 1,000; Depreciation 1,000). (3) Close the summary to capital: Dr Income Summary 25,500, Cr Capital 25,500. (4) Close drawings: Dr Capital 15,000, Cr Drawings 15,000. **CHECK.** Capital is now 500,000 + 25,500 – 15,000 = 510,500, equal to the statement.

A **reversing entry** is an optional entry on the first day of the next period that cancels an earlier accrual. On 1 February: Dr Wages Payable 3,000, Cr Wages Expense 3,000. Suppose Padma pays Tk 6,000 of wages on 5 February: Dr Wages Expense 6,000, Cr Cash 6,000. The February expense then shows 6,000 – 3,000 = 3,000, which is right, and no one needs to remember the accrual.

SUMMARY

The worksheet tests the work before the statements are written. It sorts accounts into the income statement and balance sheet. Closing entries zero the temporary accounts. Reversing entries are optional.

THINK IT THROUGH

On a worksheet, an account has an adjusted debit balance of Tk 8,000 and is a supply asset. In which two columns does it appear? (Adjusted trial balance debit; balance sheet debit.)

WHERE PEOPLE GO WRONG

A common error is to treat the worksheet as a book of record. It is not. Another is to place net income in the wrong column of the balance sheet pair. Net income goes on the credit side.

Reversing entries are optional.

Reversing entry. An optional entry on the first day of a period that cancels an earlier accrual so that routine payments can be recorded simply.

IN HISTORY

Francesco Datini, a merchant of Prato in Italy, died in 1410. His business archive survives in Prato, with some five hundred account books and about 150,000 letters. It shows a merchant who balanced his books at set dates and compared branch with branch.

REVIEW QUESTIONS

1. What is a worksheet? Is it a formal document?
2. Why and when is it prepared?
3. In the worksheet above, what is the sum of the adjusted trial balance debits?
4. Why are revenue and expense accounts closed?
5. Write the closing entry for drawings of Tk 15,000.

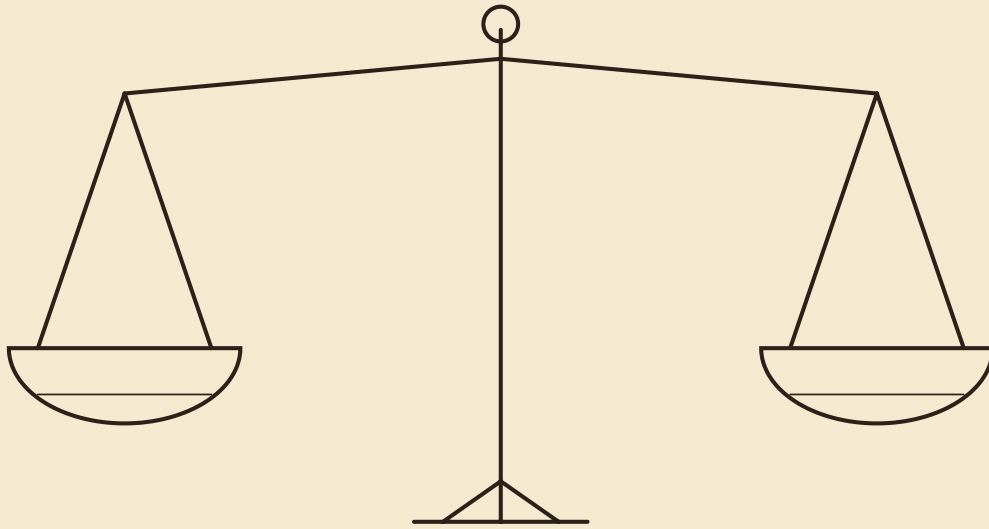


Plate V. [An engraving will be placed here.]

CHAPTER V

The Merchandising Company

*When a shop buys at one price and sells at another, where
does the profit hide?*



*This chapter shows how the accounts change when a business buys
goods and resells them. It teaches the classified income statement
and balance sheet, the difference between product and period costs,
and the correction of errors.*

Padma Stationers now buys notebooks and pens from wholesalers and sells them to students. At the end of the year Nadia asks two questions. How much did the goods she sold cost her? And how much did she earn above that cost? A service business has no such questions. A merchandising business must answer them first.

YOU WILL NEED: revenue, expense, the income statement and the balance sheet (Chapters I and IV). The figures below are a fresh illustrative set for the year 2025.

BUYING AND SELLING GOODS

GO SLOWLY HERE.

Goods bought are an asset until sold.

Inventory. Goods a business holds for sale in the ordinary course of trade.

Cost of goods sold. The cost of the goods that were sold during the period.

GO SLOWLY HERE.

Four accounts record the buying.

WORD IN USE. Goods. In daily life, any things. In accounting, items bought for resale.

GO SLOWLY HERE.

The cost of goods sold is found by a short sum.

Inventory means goods held for sale. When goods are bought they are an asset. When they are sold, their cost becomes an expense called **cost of goods sold** (COGS). **Gross profit** means sales minus cost of goods sold. It shows what the goods earned before the cost of running the shop.

Four accounts record the buying. *Purchases* holds the cost of goods bought. *Purchase returns* holds goods sent back to the supplier, and lowers cost. *Freight-in* holds the cost of carrying the goods to the shop, and raises cost. *Sales returns* holds goods customers send back, and lowers sales. Under the *periodic system* used here, goods on hand are counted at the end of the period.

FORMULA. In words: cost of goods sold equals opening inventory, plus net purchases, minus closing inventory. Net purchases are purchases, less returns, plus freight-in. In symbols: $\text{COGS} = \text{OI} + (\text{P} - \text{R} + \text{F}) - \text{CI}$. The opening inventory is what was on hand at the start, and the closing inventory is what is counted at the end.

THE CLASSIFIED INCOME STATEMENT

Classification groups like items.

Classified statement. A statement in which items of the same kind are grouped under headings and subtotals.

A **classified statement** groups items of the same kind under headings, with subtotals. It serves readers: a banker can find the gross profit at a glance, and an owner can compare selling costs with administrative costs. A list of twenty items in no order tells the reader little. The classified income statement has these parts: net sales, cost of goods sold, gross profit, operating expenses (selling and administrative), operating income, other income and expenses, and net income.

PADMA STATIONERS: CLASSIFIED INCOME STATEMENT, YEAR ENDED 31
DECEMBER 2025 (TK)

Item	Tk	Tk
Sales	1,200,000	
Less sales returns	(30,000)	
Net sales		1,170,000
Opening inventory	90,000	
Purchases	700,000	
Less purchase returns	(20,000)	
Add freight-in	15,000	
Goods available for sale	785,000	
Less closing inventory	(110,000)	
Cost of goods sold		(675,000)
Gross profit		495,000
Selling expenses (salaries 120,000, advertising 30,000, delivery 25,000)	175,000	
Administrative expenses (rent 96,000, salaries 80,000, depreciation 12,000, insurance 6,000)	194,000	
Total operating expenses		(369,000)
Operating income		126,000
Interest income	4,000	
Interest expense	(10,000)	
Net other expense		(6,000)
Net income		120,000

Check the statement.

CHECK. Gross profit should be about 42 per cent of net sales: $495,000 \div 1,170,000 = 0.423$. A shop selling at twice cost would show 50 per cent, and one with thin margins far less, so the figure is sensible for stationery. Net income of 120,000 equals 126,000 less 6,000.

THE CLASSIFIED BALANCE SHEET

Assets and liabilities are grouped by time.

Current asset. An asset expected to be turned into cash or used up within one year.

Current liability. A debt that must be paid within one year.

GO SLOWLY HERE.

The grouping shows safety.

WORD IN USE. Current. In daily life, of now. In accounting, due or usable within one year.

On the balance sheet, assets are grouped by how soon they turn to cash. A **current asset** is one expected to become cash or be used within one year: cash, receivables, inventory, supplies and prepaid items, listed in that order of nearness to cash. A **non-current asset** lasts longer, such as equipment. A **current liability** is a debt due within one year. A *non-current liability* is due later, such as a five-year loan.

The classified balance sheet serves a clear purpose. A lender compares current assets to current liabilities to see if short debts can be paid from short assets. Here that is 588,000 against 66,000, a comfortable margin. Without the grouping, the reader cannot judge.

PADMA STATIONERS: CLASSIFIED BALANCE SHEET, 31 DECEMBER 2025 (TK)

Item	Tk	Tk
Current assets		
Cash	380,000	
Accounts receivable	85,000	
Inventory	110,000	
Supplies	10,000	
Prepaid insurance	3,000	
Total current assets		588,000
Non-current assets		
Equipment (cost 150,000 less accumulated depreciation 12,000)		138,000
Total assets		726,000
Current liabilities		
Accounts payable	60,000	
Wages payable	6,000	
Total current liabilities		66,000
Bank loan (non-current)		100,000
Total liabilities		166,000
Capital 1 January 500,000 + net income 120,000 – drawings 60,000		560,000
Total liabilities and capital		726,000

PRODUCT COSTS AND PERIOD COSTS

Costs are sorted by what they buy.

Product cost. A cost of acquiring or making goods for sale; it stays in inventory until the goods are sold.

Period cost. A cost of running the business in a period, such as rent or advertising; it is expensed at once.

The statements are linked.

A **product cost** is a cost of acquiring or making goods for sale. It is held in inventory as an asset until the goods are sold, and then it becomes COGS. A **period cost** is a cost of running the business, such as rent or advertising. It is expensed in the period in which it occurs, whether or not goods are sold. Freight-in is a product cost; delivery to customers is a period cost.

The income statement and balance sheet are connected at one point: net income. The 120,000 earned increases capital in the balance sheet. A trial balance, by contrast, is a test list. The balance sheet, shown above, is a report of position that groups the accounts for readers.

CORRECTING ERRORS

Errors are of known kinds.

Errors are sorted by kind. An *error of omission* leaves out a whole transaction. An *error of commission* records an amount in the wrong account of the same class. An *error of principle* puts an item in the wrong class, such as a repair debited to an asset. An *error of original entry* records the wrong amount on both sides. *Compensating errors* cancel one another. A *reversal error* debits what should be credited.

GO SLOWLY HERE.

Correct by a new entry that fixes the effect.

GIVEN. Padma paid Tk 5,000 for repairs to the shop but debited Equipment. **FIND.** The correcting entry. **STEPS.** The wrong entry was Dr Equipment 5,000, Cr Cash 5,000. The right entry is Dr Repairs Expense 5,000, Cr Cash 5,000. Remove the wrong debit and make the right one: Dr Repairs Expense 5,000, Cr Equipment 5,000. **CHECK.** Equipment falls by 5,000, expense rises by 5,000, and Cash is correct.

SUMMARY

Cost of goods sold = opening inventory + net purchases – closing inventory. A classified income statement shows gross profit, then operating income, then net income. A classified balance sheet separates current from non-current items. Product costs wait in inventory; period costs are expensed at once.

THINK IT THROUGH

Opening inventory is Tk 40,000, purchases 300,000, purchase returns 10,000, freight-in 5,000, closing inventory 50,000. Find COGS. (40,000 + 300,000 – 10,000 + 5,000 – 50,000 = 285,000.)

WHERE PEOPLE GO WRONG

Students often subtract purchase returns from sales. They belong in the cost of goods. Others place freight-in with selling expenses, though it is a cost of getting the goods.

IN HISTORY

In the early 1770s, during a trade depression, the potter Josiah Wedgwood studied his costs and found that he had been underestimating them. He used what he learned to adjust output and prices. The episode is often cited as an early use of cost analysis in manufacturing.

REVIEW QUESTIONS

1. What are a classified income statement and a classified balance sheet, and what do they serve?
2. Compute gross profit: net sales Tk 500,000, COGS Tk 320,000.
3. Differentiate product cost from period cost.
4. A purchase of goods for Tk 8,000 was debited to Equipment. Give the correction.
5. What is the difference between accounts payable and accounts receivable?

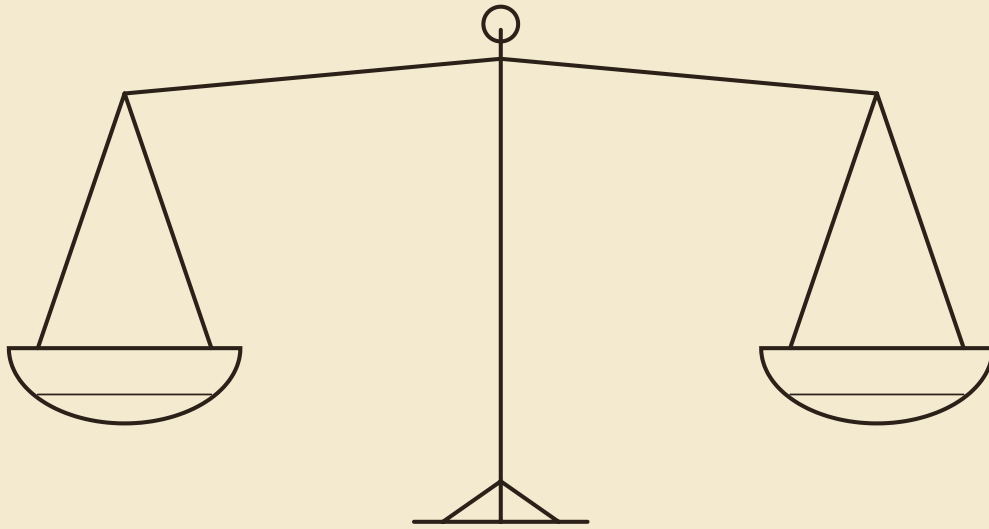


Plate VI. [An engraving will be placed here.]

CHAPTER VI

Inventory

*Three deliveries of the same notebook arrive at three prices.
Which one did she sell?*



This chapter explains what inventory is and why a business holds it, then teaches how to value it by the main methods and how the choice changes profit.

Padma Stationers starts the quarter with 100 notebooks that cost Tk 40 each. It then buys 200 more at Tk 44, and later 100 at Tk 48. Over the quarter, 250 notebooks are sold. The notebooks are identical, and no one can say which were sold. Yet the cost of the 250 must be stated, and the cost of the 150 left on the shelf. The answer depends on a rule.

YOU WILL NEED: cost of goods sold, gross profit and the periodic system (Chapter V).

Inventory is a current asset.

Inventory (types). Goods held for sale, goods being made, and materials waiting to be used.

Work in process and finished goods differ.

WHAT INVENTORY IS

Inventory is a current asset, because it is expected to become cash within a year. Its type depends on the business. A merchandising company holds *merchandise inventory*, goods bought for resale. A manufacturing company holds three kinds: *raw materials* (cloth, paper), *work in process* (goods partly made), and *finished goods* (goods complete and unsold). Supplies used in the office are not for sale and are kept apart.

The difference between the last two kinds is one of completion. A notebook factory's paper and card are raw materials. Sheets cut and printed but not yet bound are work in process. Bound notebooks in the warehouse are finished goods. Each kind carries its own accumulated cost.

WHY INVENTORY IS HELD, AND THE RISK OF HOLDING IT

Reasons for holding stock.

Too little and too much both cost money.

Organisations hold inventory for four reasons. Customers must be served at once, or they go elsewhere. Buying in larger lots often lowers the price per unit. Stock protects against delays by suppliers. And a manufacturer needs materials ready so production does not stop.

Too little inventory leads to lost sales, unhappy customers and stopped work. Too much ties up cash, needs storage, and risks theft, spoilage and goods that go out of fashion. Management therefore aims at the amount that serves customers without waste.

VALUING INVENTORY

GO SLOWLY HERE.

Cost flows need a rule.

Cost flow assumption. A rule stating which unit costs are treated as sold first, used to divide the cost of goods between sold and unsold.

GO SLOWLY HERE.

FIFO sells the oldest costs first.

FIFO. First in, first out: the oldest costs are treated as sold first, so inventory carries the newest costs.

Weighted average. A method that gives each unit the average cost of all units available.

When unit costs differ, the cost of goods available must be divided between goods sold and goods unsold. A **cost flow assumption** is a rule for doing so. The rule need not match the physical movement of goods. Four methods are in common use. *Specific identification* tracks the real cost of each unit, and suits cars or jewels. The three that follow suit identical goods.

FIFO means first in, first out: the oldest costs are treated as sold first. **LIFO** means last in, first out: the newest costs are treated as sold first. **Weighted average** gives every unit the same average cost, found by dividing total cost by total units.

GOODS AVAILABLE FOR SALE, PADMA STATIONERS (PERIODIC SYSTEM)

Lot	Units	Unit cost (Tk)	Total (Tk)
Opening inventory	100	40	4,000
Purchase 1	200	44	8,800
Purchase 2	100	48	4,800
Goods available	400		17,600

GO SLOWLY HERE.

Worked example, step by step.

GIVEN. The table above. Sales are 250 units at Tk 70, a total of Tk 17,500. Units remaining: $400 - 250 = 150$. **FIND.** COGS and closing inventory by each method. **STEPS.** FIFO: COGS is $100 \times 40 + 150 \times 44 = 4,000 + 6,600 = 10,600$; closing is $50 \times 44 + 100 \times 48 = 2,200 + 4,800 = 7,000$. LIFO: COGS is $100 \times 48 + 150 \times 44 = 4,800 + 6,600 = 11,400$; closing is $100 \times 40 + 50 \times 44 = 4,000 + 2,200 = 6,200$. Average: $17,600 \div 400 = 44$ per unit; COGS is $250 \times 44 = 11,000$; closing is $150 \times 44 = 6,600$.

RESULTS OF THE THREE METHODS (TK)

Method	COGS	Closing inventory	Gross profit
FIFO	10,600	7,000	6,900
Weighted average	11,000	6,600	6,500
LIFO	11,400	6,200	6,100

Check and meaning.

CHECK. For each method COGS plus closing inventory equals goods available: $10,600 + 7,000 = 17,600$; $11,400 + 6,200 = 17,600$; $11,000 + 6,600 = 17,600$. When costs rise, as here, FIFO gives the highest profit and the highest closing inventory. LIFO gives the lowest profit. The average lies between.

Choosing a method for tax.

If tax were charged on profit and the owner could choose freely, LIFO would show less profit when prices rise and so delay tax. But IFRS (IAS 2) does not allow LIFO, and tax rules may state their own requirements. Among permitted methods, FIFO shows the highest profit and the highest tax, and weighted average lies between. The rules in force must be checked. Whatever method is chosen must be kept from year to year, so that results can be compared.

Two limits.

When prices fall, the order of results reverses. And inventory is shown at the lower of cost and net realisable value, which means the expected selling price less selling costs. Damaged or obsolete stock is written down to that lower figure.

SUMMARY

Inventory is held for sale, made, or used up. The cost of goods available is divided by FIFO, LIFO or weighted average. In rising prices FIFO gives the highest profit. Under IFRS, LIFO is not allowed.

THINK IT THROUGH

Goods available: 50 units at Tk 10, then 50 at Tk 12. Sell 60. FIFO COGS? ($50 \times 10 + 10 \times 12 = 620$.) Closing inventory? ($40 \times 12 = 480$.)

WHERE PEOPLE GO WRONG

Many students reverse the LIFO order and sell the oldest costs first. Others forget to count the opening inventory as part of goods available. A third error is to compute the average from unit costs alone instead of total cost over total units.

IN HISTORY

The Revenue Act of 1939 in the United States is generally given as the law that allowed LIFO for tax purposes to businesses in general. Since firms had to use it in their books if they used it for tax, accounting for stock became a matter of tax planning as well as measurement.

REVIEW QUESTIONS

1. Define and classify inventory.
2. Why do organisations maintain inventory?
3. What are the consequences of too little and too much inventory?
4. Using the data in this chapter, find FIFO COGS if 300 units were sold.
5. Which method gives the highest profit when prices rise?

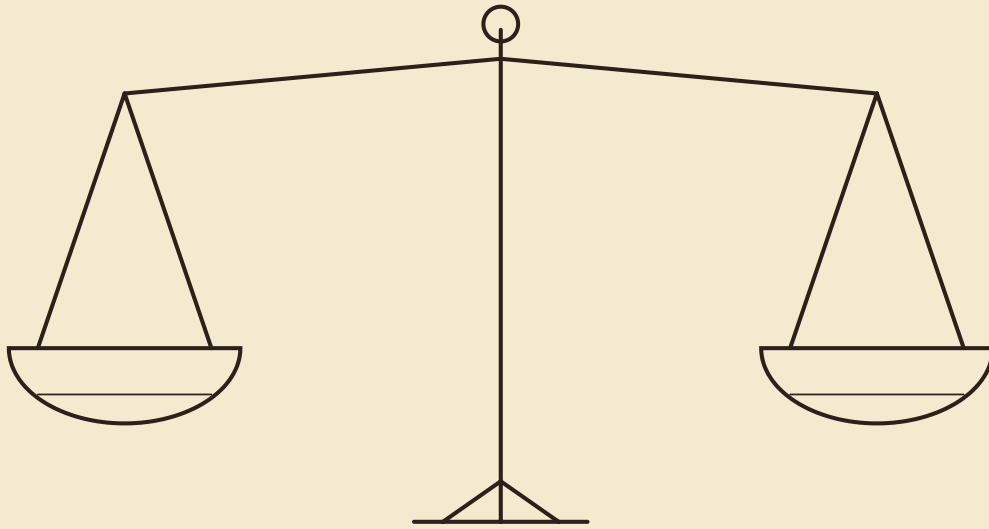


Plate VII. [An engraving will be placed here.]

CHAPTER VII

Special Journals, Cash and the Bank

A shop makes two hundred sales in a day. Must each one be written in the same book?



This chapter explains the special journals that save labour, the cash book, and the bank reconciliation statement that compares the books with the bank's.

Padma Stationers grows. It makes dozens of credit purchases and sales a week. If every one is written in the general journal, with two lines each, the book fills fast and the same words are copied again and again. A sensible accountant sorts the events by kind and gives each kind its own book.

YOU WILL NEED: the journal, ledger, accounts payable and receivable (Chapter II), and the periodic system (Chapter V).

Each journal handles one kind of event.

Special journal. A journal used for one repeated kind of transaction, such as credit sales.

Subsidiary ledgers and control accounts.

Control account. A general ledger account whose balance equals the total of many individual customer or supplier accounts.

SPECIAL JOURNALS

A **special journal** is a journal used for one repeated kind of transaction. The *sales journal* records credit sales of goods. The *purchases journal* records credit purchases of goods. The *cash receipts* and *cash payments journals* record cash events. Anything else goes to the *general journal*. The main gain is time: only the total of a column is posted, once a month.

The buyers on credit are many, so a **subsidiary ledger** keeps one account for each customer, and another for each supplier. The general ledger holds a single **control account**, Accounts Receivable or Accounts Payable, whose balance must equal the total of the subsidiary accounts. Each posting is made to both, and the agreement is a check.

PURCHASES JOURNAL, PADMA STATIONERS, MAY 2025 (TK)

Date	Supplier	Invoice	Amount
3 May	Dhaka Paper Co	101	40,000
8 May	Star Pens	102	25,000
Total (Dr Purchases, Cr Accounts Payable)			65,000

SALES JOURNAL, PADMA STATIONERS, MAY 2025 (TK)

Date	Customer	Invoice	Amount
4 May	Alam Stores	201	30,000
14 May	Rahim Traders	202	18,000
Total (Dr Accounts Receivable, Cr Sales)			48,000

THE CASH BOOK

The cash book combines journal and ledger.

Cash book. A book that records all cash received and paid and also serves as the Cash account.

The **cash book** is a book that records all cash received and paid, and also serves as the Cash account itself, so it is posted to no separate Cash account. It has a debit side for receipts and a credit side for payments. A discount allowed is shown on the debit side, and a discount received on the credit side.

CASH PAYMENTS JOURNAL, PADMA STATIONERS, MAY 2025 (TK)

Date	Paid to	Payables Dr	Other Dr	Discount Cr	Cash Cr
12 May	Dhaka Paper Co	40,000		800	39,200
18 May	Rent expense		8,000		8,000
Totals	40,000	8,000	800	47,200	

Worked check.

GIVEN. On 12 May Padma pays Dhaka Paper Co the full invoice of Tk 40,000 less a 2 per cent discount for early payment; on 18 May it pays rent of Tk 8,000. FIND. The cash paid. STEPS. Discount = $40,000 \times 2\% = 800$; cash paid = $40,000 - 800 = 39,200$; rent 8,000; total cash 47,200. CHECK. Debits $40,000 + 8,000 = 48,000$ equal credits $800 + 47,200 = 48,000$.

THE BANK RECONCILIATION STATEMENT

GO SLOWLY HERE.

The bank's record and the owner's record rarely agree.

Bank reconciliation statement.

A statement that explains the difference between the cash book balance and the bank statement balance at the same date.

GO SLOWLY HERE.

Causes of difference.

Padma's cash book says the balance at the bank is Tk 48,500. The bank statement says Tk 52,300. Neither is wrong. A **bank reconciliation statement** is a statement that explains the difference between the two balances at one date. It is prepared by the business, usually monthly, to find errors and to bring the cash book up to date.

There are two groups of cause. *Timing differences* mean the bank has not yet seen something the business has recorded: a **deposit in transit**, which is cash banked but not yet credited, and **outstanding cheques**, which are cheques issued but not yet paid by the bank. *Unrecorded items* mean the bank has done something the business has not recorded: bank charges, interest earned, direct deposits by customers, and cheques returned unpaid. The bank sends notices: a **credit memo** adds to the account, and a **debit memo** subtracts from it.

GIVEN. Cash book balance Tk 48,500; bank statement Tk 52,300. Deposit in transit Tk 4,000; outstanding cheques Tk 8,400; bank charges Tk 300; interest credited Tk 500; customer deposit direct to bank (credit memo) Tk 1,200; customer cheque returned unpaid (debit memo) Tk 2,000. FIND. The true cash balance. STEPS. Correct each side as shown.

BANK RECONCILIATION, 31 DECEMBER 2025 (TK)

Item	Tk
Bank statement balance	52,300
Add deposit in transit	4,000
Less outstanding cheques	(8,400)
Adjusted bank balance	47,900
Cash book balance	48,500
Add interest 500 and customer deposit 1,200	1,700
Less bank charges 300 and returned cheque 2,000	(2,300)
Adjusted cash book balance	47,900

Answer, entries and check.

ANSWER. The true cash balance is Tk 47,900. CHECK. Both adjusted balances agree. Only the cash book side needs journal entries: Dr Cash 500, Cr Interest Income 500; Dr Cash 1,200, Cr Accounts Receivable 1,200; Dr Bank Charges 300, Cr Cash 300; Dr Accounts Receivable 2,000, Cr Cash 2,000. Timing items need no entry: the bank will catch up.

SUMMARY

Special journals record repeated events by kind and are posted by totals. The cash book is both journal and ledger. A bank reconciliation explains why the two balances differ and gives the true figure. Only unrecorded items need entries.

THINK IT THROUGH

Cash book Tk 20,000; bank statement Tk 21,500. Outstanding cheques are Tk 3,000, and the bank charged Tk 500 not yet recorded. Is there a deposit in transit? (Adjusted cash book 19,500. Bank side $21,500 - 3,000 = 18,500$, so a deposit in transit of 1,000.)

WHERE PEOPLE GO WRONG

A frequent error is to make a journal entry for an outstanding cheque. It needs none. Another is to add an item to the wrong side: bank charges lower the cash book, while outstanding cheques lower the bank balance.

IN HISTORY

The Bank of England was founded in 1694 by royal charter. As cheques spread in later centuries, customers began to keep their own record of cheques written and to compare it with the bank's statement, which is the practice this chapter teaches.

REVIEW QUESTIONS

1. What is a bank reconciliation statement?
2. What are the common causes of differences?
3. What are credit and debit memos?
4. Cash book Tk 30,000, bank statement Tk 33,000; outstanding cheques Tk 5,000, deposit in transit Tk 2,000. Prove the difference.
5. Prepare a purchases journal entry: bought Tk 12,000 goods on credit from Star Pens on 3 June.

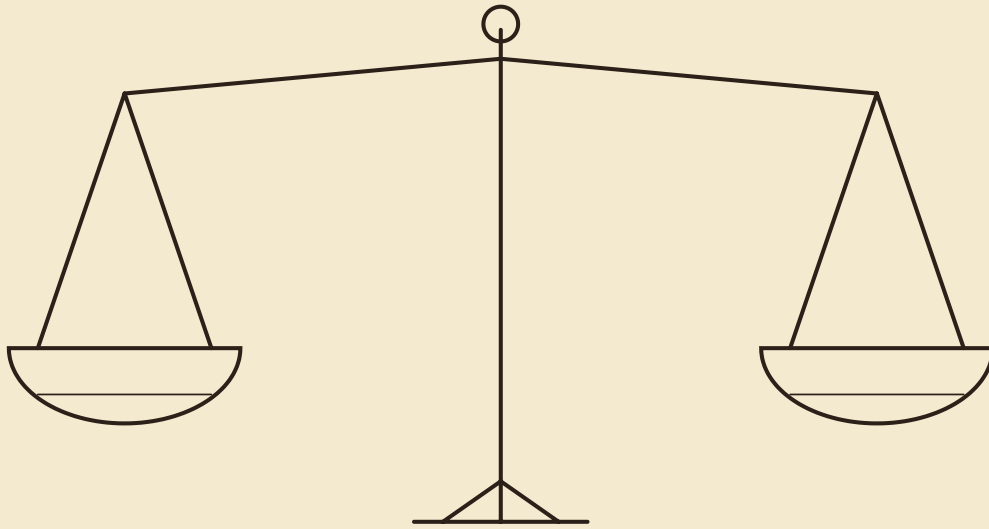


Plate VIII. [An engraving will be placed here.]

CHAPTER VIII

Plant Assets and Depreciation

A copier lasts ten years. Why is its whole cost not an expense of the first?



This chapter explains what plant assets are, why their cost is spread over their lives, and how the main methods of depreciation work.

Nadia buys a copier for Tk 440,000. It will serve the shop for about ten years. If the whole price were treated as an expense in year one, that year would show a heavy loss and the next nine a false gain. The copier helps earn revenue in every year it works. Its cost should therefore be shared among those years.

YOU WILL NEED: asset, expense, adjusting entries and accumulated depreciation (Chapters I and III).

Plant assets are long-lasting tools.

Plant asset. A physical asset used for many years in running the business, not held for sale, such as equipment, buildings and land.

Cost includes all costs of readiness.

GO SLOWLY HERE.

Depreciation spreads cost over use.

Depreciation. The spreading of the cost of a plant asset, less its residual value, over the years it is expected to be used.

GO SLOWLY HERE.

Depreciation is not a valuation, and not a saving of cash.

Reasons for charging depreciation.

Three factors fix the charge.

Residual value. The amount a business expects to recover when it disposes of a plant asset at the end of its life.

Useful life. The period over which a business expects to use an asset.

PLANT ASSETS

A **plant asset** means a physical asset used for many years to run the business and not held for sale. Examples are a copier, shelves, a van, a building and land. A van bought by a van dealer to sell is inventory. The same van bought by the shop to deliver goods is a plant asset. Use decides the class.

The recorded cost of a plant asset is everything paid to get it ready for use: the price, taxes that cannot be recovered, freight, and installation. Later repairs that only keep it working are expenses, not part of cost. For a machine bought at Tk 600,000, with freight of Tk 20,000 and installation of Tk 30,000, the cost is Tk 650,000.

DEPRECIATION

Depreciation means spreading the cost of a plant asset, less any amount expected to be recovered at the end, over the years in which it is used. It is charged on assets that are *physical*, that have a *limited life*, and that are *used in the business*. Land is not depreciated, because it does not wear out. Inventory is not depreciated, because it is sold.

Two misunderstandings are common. Depreciation does not measure what the asset could be sold for today. It is a way of dividing cost. And it does not set cash aside: the entry Dr Depreciation Expense, Cr Accumulated Depreciation moves no cash at all.

There are three reasons. It matches the cost of the asset to the revenue it helps to earn. It gives a fairer profit each year. And it shows the asset on the balance sheet at a sensible figure, its book value, which is cost less accumulated depreciation.

Three estimates fix the yearly charge. The first is *cost*. The second is **residual value**, the amount expected from sale or scrap at the end of life. The third is **useful life**, the period of expected use, measured in years or in units of output. The last two are judgements, so depreciation is an estimate. Wear, age and new inventions, called obsolescence, all shorten life.

METHODS OF DEPRECIATION

GO SLOWLY HERE.

Straight-line gives the same charge each year.

Straight-line method. A method that charges the same depreciation every year: cost less residual value divided by useful life.

GO SLOWLY HERE.

Worked example: straight-line.

The **straight-line method** charges equal amounts each year. **FORMULA.** In words: annual depreciation equals cost, less residual value, divided by useful life in years. In symbols: $D = (C - S) \div N$. Here C is cost, S is residual value, and N is life.

GIVEN. Padma buys equipment for Tk 440,000. Useful life is 10 years. Residual value is Tk 40,000. **FIND.** Annual depreciation and the journal entry. **STEPS.** $C - S = 440,000 - 40,000 = 400,000$. $D = 400,000 \div 10 = 40,000$. **ANSWER.** Each year Dr Depreciation Expense 40,000, Cr Accumulated Depreciation 40,000. **CHECK.** Ten years at 40,000 total 400,000, which is cost less residual value; the book value then equals the residual 40,000.

STRAIGHT-LINE SCHEDULE (TK)

Year	Expense	Accumulated depreciation	Book value
At purchase			440,000
1	40,000	40,000	400,000
2	40,000	80,000	360,000
3	40,000	120,000	320,000

The difference between expense and accumulated depreciation.

Part of a year.

GO SLOWLY HERE.

Units of production follows use.

Units-of-production method. A method that charges depreciation by the amount of work the asset does in each period.

GO SLOWLY HERE.

Declining balance gives larger charges early.

Declining-balance method. A method that charges a fixed percentage each year on the book value at the start of the year, so charges fall as the asset ages.

Depreciation expense is the charge for one year, and it appears on the income statement. *Accumulated depreciation* is the total charged since purchase, and it appears on the balance sheet. In year 3 the expense is 40,000 and the accumulated depreciation is 120,000.

If the equipment is bought on 1 April, the first year's charge covers nine months of a twelve-month year: $40,000 \times 9 \div 12 = 30,000$.

The **units-of-production method** charges depreciation by use. **FORMULA.** Rate per unit equals (cost – residual value) divided by total units expected. **GIVEN.** The machine costing Tk 650,000 will give 600,000 copies and then be worth Tk 50,000. This year it makes 90,000 copies. **STEPS.** Rate = $(650,000 - 50,000) \div 600,000 = 1.00$ per copy. Depreciation = $90,000 \times 1.00 = 90,000$. **CHECK.** At 600,000 copies the total charge would be 600,000, equal to cost less residual value.

The **declining-balance method** charges a fixed percentage of the opening book value, so the charge is largest in the early years, when the asset is most useful. The *double-declining-balance* rate is twice the straight-line rate. **GIVEN.** Cost Tk 100,000, life 5 years, residual value Tk 5,000. **STEPS.** Straight-line rate = $1 \div 5 = 20$ per cent; double rate = 40 per cent. Year 1: $40\% \times 100,000 = 40,000$; book value 60,000. Year 2: $40\% \times 60,000 = 24,000$; book value 36,000. Year 3: $40\% \times 36,000 = 14,400$; book value 21,600. The residual value is ignored in the rate, but the book value must never fall below it.

The method should suit how the asset gives its service. A machine that wears with output suits units of production. A building that gives steady service suits straight-line. Once chosen, the method is kept from year to year.

Choosing a method.

SUMMARY

A plant asset is physical, long-lasting and used in the business. Its cost is spread over its useful life by depreciation. Straight-line, units-of-production and declining-balance are the main methods. Accumulated depreciation is the total charged to date.

THINK IT THROUGH

A van costs Tk 800,000, with residual value Tk 80,000 and life 6 years. Find the straight-line charge and book value after 2 years. ($720,000 \div 6 = 120,000$; book value $800,000 - 240,000 = 560,000$.)

WHERE PEOPLE GO WRONG

Students often subtract nothing for residual value, or charge depreciation on land. Others record accumulated depreciation as an expense. It is a deduction from the asset.

IN HISTORY

In the nineteenth century, British railway companies owned vast tracks, bridges and engines. Many paid dividends without providing for wear and replacement. When the assets decayed, the money to renew them was not there. The lesson helped bring depreciation into ordinary practice.

REVIEW QUESTIONS

1. What is depreciation, and on which assets is it charged?
2. Point out the reasons for charging depreciation.
3. Cost Tk 330,000, residual Tk 30,000, life 10 years. Straight-line depreciation and journal entry?
4. How does depreciation expense differ from accumulated depreciation?
5. A machine of 64,000 expected copies and cost Tk 128,000 with residual Tk 0 makes 8,000 copies. Depreciation?

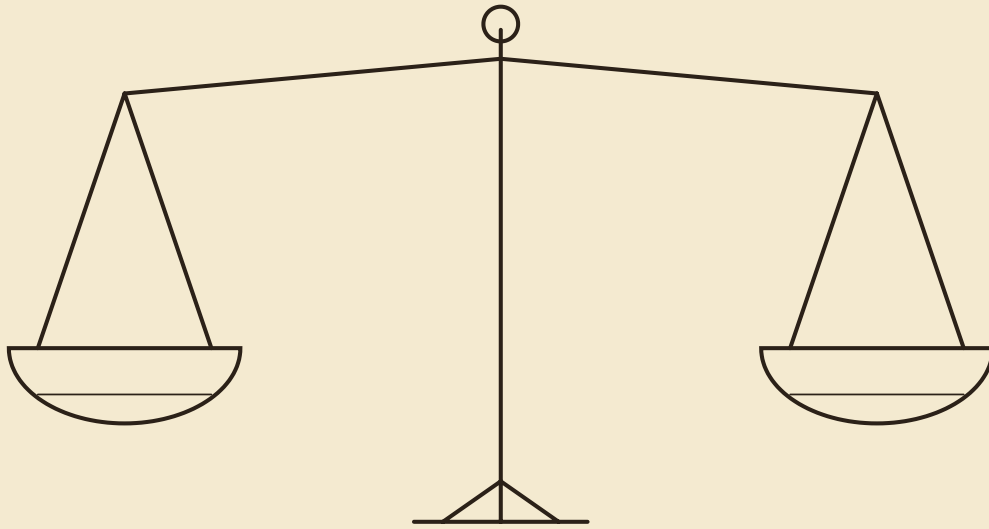


Plate IX. [An engraving will be placed here.]

CHAPTER IX

Different Kinds of Organisation

Some groups keep accounts but do not seek profit. Some firms share profit among friends. How do their books differ?



This chapter shows how a club keeps its accounts, then follows Padma Stationers as it becomes a partnership: the agreement, the sharing of profit, and the ending of the firm.

A cricket club collects fees, pays a groundsman and buys balls. It makes no profit, yet its members want to know whether it is spending wisely. Meanwhile Nadia has invited a friend, Imran, to join her shop. They must decide how to share what the shop earns. Each case needs accounts suited to its purpose.

YOU WILL NEED: revenue, expense and accrual (Chapters I and III), and the statements (Chapters IV and V).

THE NON-TRADING CONCERN

A club exists to serve its members.

Non-trading concern. An organisation, such as a club or charity, that provides service to members or the public and does not exist to earn profit.

Two statements are prepared.

Receipts and payments account. A summary of the cash received and paid during a period, with no regard to the period to which items belong.

GO SLOWLY HERE.

Worked example: converting cash to accrual.

A **non-trading concern** is an organisation that exists to serve its members or the public, and not to earn profit. Clubs, schools and charities are examples. They have no sales and no profit. Their summary of the year is a *surplus* if income is greater than expense, or a *deficit* if it is less.

A **receipts and payments account** summarises the cash received and paid, whatever period the items belong to. It is a cash statement. An **income and expenditure account** is the equivalent of the income statement. It uses the accrual basis: it includes only the income and expenses that belong to the year, and leaves out capital items, such as the purchase of equipment.

GIVEN. The club received subscriptions of Tk 60,000 in the year. At the start, members owed Tk 2,000 from the previous year; at the end, they owed Tk 5,000. The club paid salaries of Tk 24,000 and still owed Tk 2,000 at year end. It paid rent of Tk 18,000 and bought sports equipment of Tk 15,000. **FIND.** The surplus. **STEPS.** Subscriptions earned = 60,000 – 2,000 + 5,000 = 63,000. Salaries incurred = 24,000 + 2,000 = 26,000. Rent = 18,000. The equipment is a capital item and is left out. Surplus = 63,000 – 26,000 – 18,000 = 19,000. **CHECK.** The adjustments are the same as accrual adjustments in Chapter III.

THE PARTNERSHIP

Partnership begins with an agreement.

Partnership. A business owned by two or more people who have agreed to run it together and to share its profits.

Partner. A person who is an owner of a partnership.

Characteristics of a partnership.

A **partnership** means a business owned by two or more people who have agreed to run it together and share its profits. The people are called **partners**, and together they are the *firm*. In Bangladesh the matter is governed by the Partnership Act of 1932. It holds that a partnership arises from an agreement, and not from family status or the mere joint ownership of property.¹

A partnership has these features. There are at least two partners. There is an agreement, which may be spoken or written. The business is carried on for profit. Each partner can bind the others in the ordinary work of the business, a feature called *mutual agency*. Each partner is responsible, without limit, for the debts of the firm, which is called **unlimited liability**: creditors may look to the private property of any partner.

The deed records the terms.

Partnership deed. The written agreement setting out each partner's rights and duties.

GO SLOWLY HERE.

Profit is shared in steps.

A written agreement is called a **partnership deed**. It states the name and purpose of the firm, the capital each partner brings, the ratio in which profits and losses are shared, any interest on capital or salary to partners, the rules for drawings, the rights to join or leave, the rules for settling accounts, and the way the firm may end. Without a deed, the Act supplies a default rule: profits and losses are shared equally.

The deed fixes how profit is divided. This is done in an *appropriation account*. First, interest on capital is credited to each partner. Second, salaries are given to partners who work in the business. Third, the remaining profit is shared in the agreed ratio. **GIVEN.** Net profit Tk 300,000. Capital: Nadia Tk 600,000 and Imran Tk 400,000. Interest on capital 10 per cent. Imran's salary Tk 60,000. Remaining profit shared 3 to 2.

APPROPRIATION OF PROFIT (TK)

Item	Nadia	Imran	Total
Interest on capital (10%)	60,000	40,000	100,000
Salary		60,000	60,000
Share of remainder (3:2)	84,000	56,000	140,000
Total	144,000	156,000	300,000

Check and entries.

STEPS. Remainder = 300,000 – 100,000 – 60,000 = 140,000. Nadia: $3 \div 5 \times 140,000 = 84,000$. Imran: $2 \div 5 \times 140,000 = 56,000$. **CHECK.** 144,000 + 156,000 = 300,000, equal to net profit. The formation entry is Dr Cash 1,000,000, Cr Nadia, Capital 600,000, Cr Imran, Capital 400,000.

WHEN THE FIRM ENDS

Reasons for liquidation.

Liquidation. The process of ending a business by selling its assets, paying its debts and sharing what is left among the owners.

GO SLOWLY HERE.

Order of payment.

GO SLOWLY HERE.

Worked example of liquidation.

Liquidation means ending a business by selling the assets, paying the debts, and giving the owners what is left. A partnership may end for several reasons: the agreed term expires, the venture is complete, the partners agree, a partner dies, retires or becomes insolvent and the deed gives no right to continue, a court orders it, or the business becomes unlawful. These reasons are set by the deed and the Act.

The order is fixed. First, the non-cash assets are sold, and the gain or loss on realisation is shared among the partners in the profit ratio. Second, outside debts are paid. Third, what remains is paid to the partners according to their capital balances.

GIVEN. Cash Tk 50,000; other assets Tk 450,000; liabilities Tk 150,000; capital Nadia Tk 200,000 and Imran Tk 150,000. Profit ratio 3 to 2. The other assets sell for Tk 360,000. **STEPS.** (1) Dr Cash 360,000, Dr Loss on Realisation 90,000, Cr Other Assets 450,000. (2) Share the loss: Dr Nadia, Capital 54,000, Dr Imran, Capital 36,000, Cr Loss on Realisation 90,000. (3) Pay debts: Dr Liabilities 150,000, Cr Cash 150,000. (4) Pay partners: Cash is now 50,000 + 360,000 – 150,000 = 260,000. Capital is Nadia 146,000 and Imran 114,000. Dr Nadia 146,000, Dr Imran 114,000, Cr Cash 260,000. **CHECK.** Capital paid (260,000) equals cash left, and every account is zero.

SUMMARY

A non-trading concern reports a surplus or deficit, on an accrual basis. A partnership rests on an agreement, written in a deed. Profit is shared by interest, salary and ratio. On liquidation, assets are sold, debts paid, and the rest goes to partners.

THINK IT THROUGH

Two partners share profits equally. A has capital Tk 100,000, B Tk 50,000, and profit is Tk 40,000. If there is no deed, how is it divided? (Tk 20,000 each. Without agreement, the law gives equal shares regardless of capital.)

WHERE PEOPLE GO WRONG

Students forget that unlimited liability reaches private property. Others pay out the partners before paying creditors. Debts always come first.

IN HISTORY

Before 1932 the law of partnership in this region was contained in Chapter XI of the Indian Contract Act of 1872. The Indian Partnership Act of 1932 gave it a separate statute. In Bangladesh it continues as the Partnership Act, 1932.

REVIEW QUESTIONS

1. Describe the elements of a partnership deed.
2. What are the characteristics of a partnership?
3. What are the common reasons for liquidation?
4. Net profit Tk 90,000, ratio 2 to 1. Share the profit.
5. What is a receipts and payments account?

¹ The Partnership Act, 1932 (Act IX of 1932), sections 4 and 5. The firm is not a legal person separate from its partners.

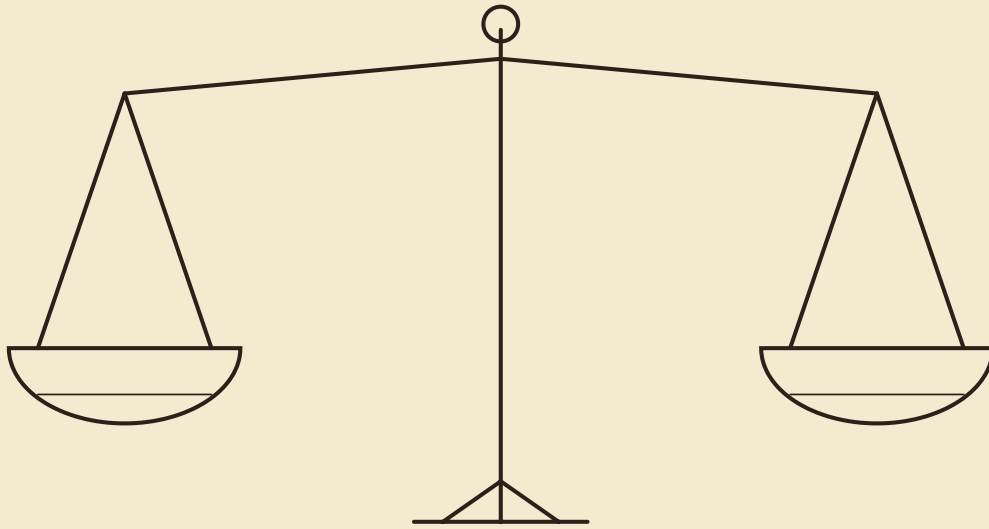


Plate X. [An engraving will be placed here.]

CHAPTER X

The Company and Its Shares

*How can a hundred people own one business without any
of them being responsible for its debts?*



*This chapter explains the company form, the kinds of shares, and
how shares are issued and recorded.*

Padma Stationers has done well, and Nadia wants to open ten shops. She needs Tk 100 million, more than she and her partner can supply. She could ask a hundred friends to invest. But who would invest if, should the business fail, their homes could be taken? The company form answers that fear.

YOU WILL NEED: equity and capital (Chapter I), and partnership and unlimited liability (Chapter IX).

WHAT A COMPANY IS

A company is a legal person.

Company. A business that the law treats as a person separate from its owners, with its capital divided into shares.

Owners risk only what they invest.

Limited liability. The rule that owners lose at most the amount they agreed to pay for their shares.

A **company** means a business that the law treats as a person separate from its owners, with its capital divided into **shares**. In Bangladesh companies are formed under the Companies Act, 1994. The company owns property, makes contracts and can be sued, in its own name. It does not die when an owner does.

The owners are called *shareholders*. Their **limited liability** means they can lose at most the sum they agreed to pay for their shares. This differs from a partner's unlimited liability. Companies are *private* when they may not offer shares to the public, and *public* when they may. Equity in a company is called *shareholders' equity* or stockholders' equity.

KINDS OF SHARES

Ordinary and preference shares.

Ordinary share. A share that gives a vote and a share of profits after all other claims, with no fixed dividend.

Preference share. A share that receives a fixed dividend before any dividend on ordinary shares.

Capital has several names.

An **ordinary share** gives its owner a vote and a share in profits, but it receives a dividend only after others are paid. A **preference share** has a fixed dividend, paid before ordinary shares. A **dividend** means a share of profit paid to shareholders. Preference shareholders usually have no vote.

Several terms describe the capital. *Authorised capital* is the largest amount the company may issue, set in its founding documents. *Issued capital* is the part offered and taken. *Paid-up capital* is the part for which shareholders have actually paid. The **par value**, or face value, is the amount printed on a share, often Tk 10. A share sold above par is sold at a *premium*.

ISSUING SHARES

GO SLOWLY HERE.

Payment may be in stages.

Shares are often paid for in stages. The buyer pays *application money* when applying, *allotment money* when shares are given, and may pay one or more *calls* later. A separate account holds each stage until the shares are allotted.

GO SLOWLY HERE.

Worked example: shares issued at a premium.

GIVEN. Padma Ltd issues 10,000 ordinary shares of Tk 10 at Tk 12 each. Payment: Tk 3 on application, Tk 5 on allotment (including the premium of Tk 2), and Tk 4 on the first and final call. All money is received. FIND. The journal entries and the totals. STEPS. Application money: $10,000 \times 3 = 30,000$. Allotment: $10,000 \times 5 = 50,000$. Call: $10,000 \times 4 = 40,000$.

JOURNAL ENTRIES, ISSUE OF SHARES (TK)

Particulars	Debit	Credit
Bank	30,000	
Share Application		30,000
Share Application	30,000	
Share Capital		30,000
Share Allotment	50,000	
Share Capital		30,000
Share Premium		20,000
Bank	50,000	
Share Allotment		50,000
Share Call	40,000	
Share Capital		40,000
Bank	40,000	
Share Call		40,000

Answer and check.

ANSWER. Share capital is $30,000 + 30,000 + 40,000 = 100,000$, which is 10,000 shares at Tk 10. Share premium is 20,000. Bank receives 120,000. CHECK. 10,000 shares at Tk 12 give 120,000, which equals $100,000 + 20,000$. Share premium is not profit; it is part of equity, shown apart from share capital.

SUMMARY

A company is a legal person owned by shareholders with limited liability. Ordinary shares carry votes and residual profit; preference shares carry a fixed dividend first. Shares are issued at par or premium, often in stages, and recorded in capital and premium accounts.

THINK IT THROUGH

A company issues 5,000 shares of Tk 10 at Tk 15, all payable at once. Find share capital and premium. (50,000 and 25,000; bank 75,000.)

WHERE PEOPLE GO WRONG

Many students enter the premium as revenue. It is not. Others confuse authorised capital with issued capital: the first is a limit, the second is what has been sold.

IN HISTORY

In 1602 the Dutch East India Company was founded in the Netherlands. Its shares could be bought and sold in Amsterdam, and its owners were not personally responsible for its debts beyond their investment. It is often called the first company with freely traded shares.

REVIEW QUESTIONS

1. What is a company and how does it differ from a partnership?
2. Differentiate ordinary and preference shares.
3. Distinguish authorised, issued and paid-up capital.
4. Journalise: 2,000 shares of Tk 10 issued at par, all paid in cash.
5. What is share premium?

ANSWERS TO REVIEW QUESTIONS

I.1 Service (clinic), merchandising (stationery shop), manufacturing (notebook factory).

I.2 $A = L + E$. A liability is owed to outsiders by a fixed date; equity is the owner's claim, repaid last.

I.3 Tk 50,000 (80,000 – 30,000).

I.4 Common rules make reports from different businesses fair and comparable.

II.1 Asset, expense, drawings (debit); liability, capital, revenue (credit).

II.2 Dr Cash 50,000, Cr Capital 50,000; Dr Equipment 20,000, Cr Cash 20,000.

II.3 Divide by 9: $360 \div 9 = 40$ suggests a transposition. Re-add the columns first.

II.4 The trial balance lists all accounts as an internal test; the balance sheet reports assets, liabilities and equity to outsiders.

II.5 Title, debit (left) side, credit (right) side.

III.1 Accrual matches expenses to the revenue they earn. Example: wages for the last days of January are expensed in January by Dr Wages Expense, Cr Wages Payable.

III.2 Prepaid expense, unearned revenue, accrued expense, accrued revenue, and depreciation.

III.3 10 months used: Dr Insurance Expense 10,000, Cr Prepaid Insurance 10,000.

III.4 It records items that belong to a period but where cash has moved at another time, or not at all.

IV.1 A working paper used to prepare the statements; it is not formal and not part of the books.

IV.2 At period end, to test the arithmetic and sort accounts before the statements.

IV.3 Tk 593,000.

IV.4 They measure one period only; closing brings them to zero for the next period and moves profit to capital.

IV.5 Dr Capital 15,000, Cr Drawings 15,000.

V.1 They group like items under headings and subtotals so that readers can find gross profit, operating income, and current against non-current items.

V.2 Tk 180,000.

V.3 Product costs are held in inventory until sale; period costs are expensed when incurred.

V.4 Dr Purchases 8,000, Cr Equipment 8,000.

V.5 Payable is owed by the business to suppliers; receivable is owed to the business by customers.

VI.1 Goods held for sale, goods in the making, and materials awaiting use: merchandise, raw materials, work in process, finished goods.

VI.2 To serve customers at once, to buy in lots at lower prices, to guard against delay, and to keep production running.

VI.3 Too little: lost sales and stopped production. Too much: tied-up cash, storage costs, spoilage and obsolescence.

VI.4 $COGS = 100 \times 40 + 200 \times 44 = 12,800$.

VI.5 FIFO.

VII.1 A statement explaining the difference between the cash book and bank statement balances at one date.

VII.2 Deposits in transit, outstanding cheques, bank charges, interest, direct deposits and returned cheques, and errors.

VII.3 Bank notices that add to (credit memo) or subtract from (debit memo) the account.

VII.4 $33,000 + 2,000 - 5,000 = 30,000$. The statements reconcile.

VII.5 Purchases journal: 3 June, Star Pens, 12,000; Dr Purchases, Cr Accounts Payable.

VIII.1 The spreading of cost less residual value over useful life; on physical assets with a limited life used in the business.

VIII.2 To match cost with revenue, to state profit fairly, and to show book value on the balance sheet.

VIII.3 $(330,000 - 30,000) \div 10 = 30,000$. Dr Depreciation Expense 30,000, Cr Accumulated Depreciation 30,000.

VIII.4 Expense is one year's charge (income statement); accumulated depreciation is the total to date (balance sheet).

VIII.5 Rate 2 per copy; $8,000 \times 2 = 16,000$.

IX.1 Name and purpose, capitals, profit ratio, interest and salaries, drawings, admission and retirement, settlement of accounts, and ending of the firm.

IX.2 Two or more persons, an agreement, a business for profit, mutual agency and unlimited liability.

IX.3 Expiry of term, completion of venture, agreement, death or insolvency, court order, illegality.

IX.4 60,000 and 30,000.

IX.5 A summary of cash received and paid in a period, with no regard to the period to which items belong.

X.1 A legal person separate from owners, with shares and limited liability; a partnership is not separate and has unlimited liability.

X.2 Ordinary: vote, residual profit. Preference: fixed dividend first, usually no vote.

X.3 Authorised is the limit; issued is what is offered and taken; paid-up is what has been paid.

X.4 Dr Bank 20,000, Cr Share Capital 20,000.

X.5 The amount received above par value. It is part of equity, not profit.

GLOSSARY

Account. A record of all changes to one kind of item, such as Cash or Rent Expense. (*Ch. II*)

Accounting. The process of recording, sorting and reporting the money facts of a business so that people can make decisions. (*Ch. I*)

Accounting cycle. The steps repeated every period, from recording transactions to closing the books. (*Ch. II*)

Accounting equation. Assets equal liabilities plus equity. It holds after every transaction. (*Ch. I*)

Accounting standards. Written rules that tell accountants how to record and report items, so that reports can be compared. (*Ch. I*)

Accounts payable. Money the business owes to suppliers for goods or services bought on credit. (*Ch. II*)

Accounts receivable. Money customers owe the business for goods or services sold on credit. (*Ch. II*)

Accrual basis. A method that records revenue when it is earned and expenses when they are incurred, whether or not cash moves. (*Ch. III*)

Accrued expense. A cost already used but not yet paid or recorded, shown as a liability. (*Ch. III*)

Accrued revenue. Revenue already earned but not yet billed or received, shown as an asset. (*Ch. III*)

Accumulated depreciation. The total depreciation charged on an asset so far. It is subtracted from the asset's cost. (*Ch. III*)

Asset. Something of value that a business owns and can use in future, such as cash or equipment. (*Ch. I*)

Balance sheet. A report of the assets, liabilities and equity of a business at one date. (*Ch. IV*)

Bank reconciliation statement. A statement that explains the difference between the cash book balance and the bank statement balance at the same date. (*Ch. VII*)

Business. An organisation that sells goods or services to customers in order to earn a profit. (*Ch. I*)

Cash book. A book that records all cash received and paid and also serves as the Cash account. (*Ch. VII*)

Classified statement. A statement in which items of the same kind are grouped under headings and subtotals. (*Ch. V*)

Closing entries. Entries made at the end of a period to bring revenue, expense and drawings accounts to zero and move the result to capital. (*Ch. IV*)

Company. A business that the law treats as a person separate from its owners, with its capital divided into shares. (*Ch. X*)

Control account. A general ledger account whose balance equals the total of many individual customer or supplier accounts. (*Ch. VII*)

Cost flow assumption. A rule stating which unit costs are treated as sold first, used to divide the cost of goods between sold and unsold. (*Ch. VI*)

Cost of goods sold. The cost of the goods that were sold during the period. (*Ch. V*)

Current asset. An asset expected to be turned into cash or used up within one year. (*Ch. V*)

Current liability. A debt that must be paid within one year. (*Ch. V*)

Declining-balance method. A method that charges a fixed percentage each year on the book value at the start of the year, so charges fall as the asset ages. (*Ch. VIII*)

Depreciation. The spreading of the cost of a plant asset, less its residual value, over the years it is expected to be used. (*Ch. VIII*)

Double-entry system. A method in which every transaction is recorded as equal debits and credits in at least two accounts. (*Ch. II*)

Equity. The owner's claim on the business: what is left after liabilities are taken from assets. (*Ch. I*)

Expense. A cost of running the business, such as rent or wages. It lowers equity. (*Ch. I*)

FIFO. First in, first out: the oldest costs are treated as sold first, so inventory carries the newest costs. (*Ch. VI*)

Financial accounting. The branch that reports a business's results to people outside it, such as lenders and tax officers. (*Ch. I*)

Financial statement. A formal report that sums up the results or position of a business. (*Ch. I*)

GAAP. Generally Accepted Accounting Principles: the common rules and ideas that accountants agree to follow. (*Ch. I*)

Income statement. A report of revenue, expenses and net income for a period of time. (*Ch. IV*)

Inventory. Goods a business holds for sale in the ordinary course of trade. (*Ch. V*)

Inventory (types). Goods held for sale, goods being made, and materials waiting to be used. (*Ch. VI*)

Journal. The book in which transactions are first recorded, in order of date, with debits and credits. (*Ch. II*)

Ledger. The complete set of accounts of a business, in which journal entries are sorted by account. (*Ch. II*)

Liability. An amount the business owes to someone outside it. (*Ch. I*)

Limited liability. The rule that owners lose at most the amount they agreed to pay for their shares. (*Ch. X*)

Liquidation. The process of ending a business by selling its assets, paying its debts and sharing what is left among the owners. (*Ch. IX*)

Management accounting. The branch that gives managers the figures they need to plan and control the business. (*Ch. I*)

Manufacturing company. A business that turns materials into new goods and then sells those goods. (*Ch. I*)

Merchandising company. A business that buys finished goods and sells them again without making any change. (*Ch. I*)

Non-trading concern. An organisation, such as a club or charity, that provides service to members or the public and does not exist to earn profit. (*Ch. IX*)

Ordinary share. A share that gives a vote and a share of profits after all other claims, with no fixed dividend. (*Ch. X*)

Partner. A person who is an owner of a partnership. (*Ch. IX*)

Partnership. A business owned by two or more people who have agreed to run it together and to share its profits. (*Ch. IX*)

Partnership deed. The written agreement setting out each partner's rights and duties. (*Ch. IX*)

Period cost. A cost of running the business in a period, such as rent or advertising; it is expensed at once. (*Ch. V*)

Plant asset. A physical asset used for many years in running the business, not held for sale, such as equipment, buildings and land. (*Ch. VIII*)

Preference share. A share that receives a fixed dividend before any dividend on ordinary shares. (*Ch. X*)

Prepaid expense. A cost paid in advance that is recorded as an asset until it is used. (*Ch. III*)

Product cost. A cost of acquiring or making goods for sale; it stays in inventory until the goods are sold. (*Ch. V*)

Receipts and payments account. A summary of the cash received and paid during a period, with no regard to the period to which items belong. (*Ch. IX*)

Residual value. The amount a business expects to recover when it disposes of a plant asset at the end of its life. (*Ch. VIII*)

Revenue. Money earned from selling goods or services. It raises equity. (*Ch. I*)

Reversing entry. An optional entry on the first day of a period that cancels an earlier accrual so that routine payments can be recorded simply. (*Ch. IV*)

Service company. A business that earns money by doing work for customers, such as a barber or a school. (*Ch. I*)

Sole trader. A business owned and run by one person, who is personally responsible for its debts. (*Ch. I*)

Special journal. A journal used for one repeated kind of transaction, such as credit sales. (*Ch. VII*)

Straight-line method. A method that charges the same depreciation every year: cost less residual value divided by useful life. (*Ch. VIII*)

T-account. A simple picture of an account with the name on top, debit on the left and credit on the right. (*Ch. II*)

Transaction. A business event stated in money that changes the assets, liabilities or equity. (*Ch. I*)

Trial balance. A list of all account balances at one date, used to test that total debits equal total credits. (*Ch. II*)

Unearned revenue. Money received before the work is done; it is a liability until the work is finished. (*Ch. III*)

Units-of-production method. A method that charges depreciation by the amount of work the asset does in each period. (*Ch. VIII*)

Useful life. The period over which a business expects to use an asset. (*Ch. VIII*)

User. A person who reads accounting information in order to make a decision. (*Ch. I*)

Weighted average. A method that gives each unit the average cost of all units available. (*Ch. VI*)

Worksheet. A large sheet on which the accountant sorts the trial balance, adjustments and statement figures before preparing the statements. (*Ch. IV*)

MATHS YOU WILL USE

Percentage. A percentage is a part of 100. To find 2 per cent of 40,000, multiply 40,000 by $2 \div 100$: the answer is 800.

Percentage change. Change divided by the starting value, times 100. A price rising from 40 to 44 changes by $4 \div 40 \times 100 = 10$ per cent.

Ratio. A ratio of 3 to 2 means five equal parts in all: three for the first share, two for the second. To split 140,000, one part is $140,000 \div 5 = 28,000$.

Average. Add the amounts and divide by their number. For weighted average, divide the total cost by the total units, not by the number of lots.

Rearranging an equation. From $A = L + E$, equity is $E = A - L$, and liabilities are $L = A - E$. Whatever is done to one side is done to the other.

Rounding. Carry full figures through a calculation and round only the final answer, to the nearest taka unless told otherwise.

Reading a graph. The horizontal axis shows the item measured, often time. The vertical axis shows the amount. A point shows the amount at that moment.

FURTHER READING

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PRINCIPLES AND PRACTICES OF ACCOUNTING
BIJOY?!

MICROECONOMY



*Prices, Markets, Costs and Rewards,
explained from first principles*

A TEXTBOOK FOR FIRST-YEAR STUDENTS

BIJOY?!

MICROECONOMY

Prices, Markets, Costs and Rewards, explained from first principles

A Primer for First-Year Students of Business Studies

B I J O Y ? !

British spelling. Currency: the Bangladeshi taka (Tk). The bakery used in the examples is *illustrative*: it is not a real business. All figures in the examples are invented to teach method.

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TO THE READER

This book teaches **microeconomics**, the study of how people, households and firms make choices and how those choices meet in markets. You need no earlier knowledge of business. You need only arithmetic and basic algebra.

How a page is built. The wide column holds the main text. The narrow column at the outer edge holds notes. An *italic* note gives the one-line point of the paragraph beside it. A note that begins with a word in **bold type** gives the plain meaning of a key term.

Bold type in the main text marks a key term at its first use only. Italics are used for emphasis, for titles of books and for foreign words. Each key term is explained in simple English, then shown in an example. A full list appears at the back, in the Glossary and Index, with the page of first use.

How a chapter is built. Each chapter opens with a question. Each idea then follows four steps: the idea, an example, the rule, and the exceptions. A **worked example** shows every step of a calculation in the same order: Given, Find, Formula, Steps, Answer, Check. A **case file** follows one small bakery through the book. The bakery is invented for teaching.

How to study. Read slowly. Where a passage ends with “You should now be able to say”, test yourself aloud before you move on. Do the review questions at the end of each chapter, then check your answers at the back.

The mathematics. Percentage change means the change divided by the starting value, multiplied by 100. A ratio compares two numbers by division. To read a graph, find a point on the horizontal axis, go up to the curve, then read across to the vertical axis. Nothing else is needed.



CHAPTER THE FIRST

WHAT ECONOMICS IS ABOUT

Why can nobody have everything?

The chapter begins with one oven and two choices. It shows why choice is unavoidable, defines economics and its two branches, and explains what an economic law is.

On 1 January 2024 the owner of a small bakery in Dhaka opens for the first time. She has one oven and eight hours a day to use it. In one day the oven can bake 400 loaves of bread or 80 cakes. It cannot bake both at full strength. Which should she bake? Any answer gives something up. Baking cakes means fewer loaves, and baking loaves means no cakes. This small problem is the whole of economics in miniature.

Start with a question you can almost answer.

CASE FILE: RUPALI BAKERY (ILLUSTRATIVE)

1 Jan 2024: opens in Mirpur, Dhaka. One oven, eight oven hours a day. Output choice per day: 400 loaves or 80 cakes, or a mixture in the same ratio (1 cake = 5 loaves). The owner is the only decision-maker.

THE MEANING OF ECONOMICS

Economics means the study of how people choose to use limited things to satisfy their wants. A **want** is anything a person would like to have, such as food, a phone or a holiday. A **resource** is anything used to make goods and services, such as land, labour, machines and money. The baker's oven hours are a resource. Her customers' wish for bread and cake is a want.

The central fact is **scarcity**. Scarcity means that resources are limited while wants are without end. Scarcity does not mean poverty. A rich country and a poor country both face it. Even a person with great wealth has only twenty-four hours in a day. Because of scarcity, every choice has a price, and that price is what we give up.

Writers have defined economics in several ways. Adam Smith (1776) described it as an inquiry into the nature and causes of the wealth of nations. Alfred Marshall (1890) called it a study of mankind in the ordinary business of life, and linked it to human welfare. Lionel Robbins (1932) wrote that it studies human behaviour as a relation between ends and scarce means that have alternative uses. Most modern teachers prefer the last, because it applies to every choice, not only to money.

The three definitions differ in their focus. Smith's focuses on wealth, Marshall's on welfare, and Robbins's on scarcity and choice. Critics said Smith

Economics the study of how people choose to use limited resources to satisfy wants.

The idea: choice under limits.

Scarcity resources are limited while wants are not.

Scarcity is the root of all economic problems.

Three famous definitions.

Each definition looked at a different part.

Opportunity cost the value of the best alternative given up when a choice is made.

The real cost of a choice is what is given up.

ignored well-being, and that Marshall's definition left out services that do not add to welfare. Robbins's definition covers both, but it says little about growth over time. For a first course, remember this: economics is about choice made necessary by scarcity.

The baker's problem shows a key idea. If she bakes 80 cakes, she gives up 400 loaves. The loaves given up are the **opportunity cost** of the cakes. Opportunity cost means the value of the best alternative that you give up when you choose. We study it fully in Chapter 9. For now, hold the idea: choosing one thing means not choosing another.

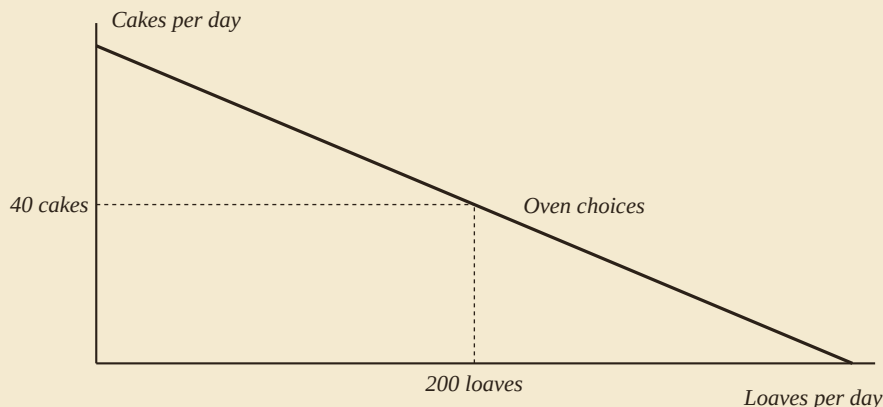


FIG. 1. Every point on the line is one possible day's baking. The point shown is a mixture.

Reading the figure.

Read the figure in three steps. The horizontal axis counts loaves, and the vertical axis counts cakes. The line joins the two extreme choices: all cakes at the top, all loaves at the right. A point on the line, such as 200 loaves and 40 cakes, shows a mixture. To gain more cakes, you must move left, and that means fewer loaves.

You should now be able to say: scarcity forces choice, and every choice gives something up.

You should now be able to say: "Resources are scarce, so a choice for one thing is a choice against another." Test yourself before you read on.

THE BASIC PROBLEMS OF AN ECONOMY

Economic system the way a society organises the choices of what, how and for whom to produce.

Every society answers three questions.

Two further problems follow from these three.

Every society, rich or poor, must answer three questions. **What** goods shall be produced, and in what amounts? **How** shall they be produced, by hand or by machine, in small shops or large factories? **For whom** shall they be produced, that is, how shall the output be shared? The baker answers all three for her shop. A country answers them for millions of people, through its **economic system**.

Two more problems follow. Society must use its resources fully and without waste, and it must decide how much to save for the future so that output can grow. Unused ovens and idle workers are a waste. Both problems still rest on scarcity. Markets, governments or a mixture of both give the answers. Bangladesh is a mixed economy: private firms and the state both take part.

MICRO AND MACRO

Microeconomics the study of the choices of single households, firms and markets.

Micro looks at the parts.

Macroeconomics the study of the whole economy, such as total output, prices and jobs.

Macro looks at the whole.

Microeconomics means the study of the choices made by single households and firms, and of the single markets where they meet. The word comes from the Greek *mikros*, "small". It asks why a loaf costs Tk 30, why a baker hires four workers, or why bread sells more than cake.

Macroeconomics means the study of the whole economy: total output, the general level of prices, employment and the money supply. The word comes

from *makros*, “large”. It asks why prices rise across the country, or why total output grows slowly.

MICRO AND MACRO COMPARED

Point of difference	Microeconomics	Macroeconomics
Unit studied	One household, firm or market	The whole economy
Main question	How is price set?	What decides total output?
Main tools	Demand, supply, cost	Income, spending, money
Example	The price of bread	The rate of inflation

The two branches are not rivals. Total output is the sum of many single choices, and each firm’s choices depend on the state of the whole economy. A fall in national income reduces bread sales. Still, you must learn the parts before the whole, and that is why this book starts with micro.

The two are linked.

THE SCOPE AND USE OF MICROECONOMICS

The **scope** of a subject means the range of topics it covers. Microeconomics covers the theory of demand, which explains buyers; the theory of supply and production, which explains sellers; the theory of the firm and of market structures, which explains competition and monopoly; and the theory of distribution, which explains rent, wages, interest and profit. Welfare economics, which asks whether the result is good for society, is a further part.

Scope means what the subject covers.

Its objectives are practical. It explains how prices are set, so that firms can decide what to produce and at what price. It shows how resources are shared among uses. It helps governments judge taxes, subsidies and price controls. It also helps a consumer plan spending. A firm that knows its costs and its customers decides better.

Why the study is useful.

In Bangladesh the subject has particular value. The country has many small farms, workshops and shops, so market prices matter to millions of people. Planners need to judge the effect of taxes and price rules on a developing economy. Support for firms comes from institutions such as Bangladesh Bank, the Bangladesh Small and Cottage Industries Corporation (BSCIC) and the Bangladesh Competition Commission. These bodies rely on microeconomic ideas when they set rules for credit, small enterprise and fair competition.

Prospects in Bangladesh.

ECONOMIC LAWS, SCIENCE AND ART

An **economic law** means a statement of how people usually behave in economic matters. An example: when the price of a good rises, buyers tend to buy less. Such laws differ from the laws of physics in four ways. They state tendencies, not certainties, because human beings choose. They hold only if other things stay the same, a condition known by the Latin phrase *ceteris paribus*. They are less exact, as they cannot be tested in a laboratory. And they apply to groups of people more than to one person.

Economic law a statement of the usual relation between economic events, true when other things stay equal.

A law states a tendency, not a certainty.

Economics contains two kinds of statement. A **positive** statement says what is, and facts can test it: “A higher bread price reduces bread sales.” A **normative** statement says what ought to be, and it rests on values: “The government should fix the price of bread.” Positive economics explains. Normative economics judges. Good reasoning keeps the two apart, because facts can settle the first but not the second.

Positive statement a statement about what is, which facts can test.

Positive: what is. Normative: what ought to be.

Is economics a science or an art?

A **science** is an organised body of knowledge that links causes to effects. An **art** is the skilful use of knowledge to reach a goal. Economics is both. It is a science because it builds laws from observation and reasons from cause to effect. It is an art because a baker, a banker or a minister applies those laws to reach a goal. Positive economics is the science, and normative economics guides the art.

CASE FILE: DECISION ON 1 JANUARY 2024

The owner chooses 300 loaves and 16 cakes for the first day. Check against the oven: 300 loaves use 6 of 8 oven hours; 16 cakes use 1.6 hours; total 7.6 hours. The remaining 0.4 hour is unused, which is waste in the language of this chapter.

SUMMARY. Scarcity forces choice. Economics studies how people choose when resources are limited. Micro studies single decision-makers and markets; macro studies the whole economy. Economic laws state tendencies that hold when other things are equal. Positive statements describe; normative statements judge.

THINK IT THROUGH. A student can spend Saturday working for Tk 600 or studying for a test. She chooses to study. State what she gave up and why this is a choice caused by scarcity.

WHERE PEOPLE GO WRONG. “Scarcity means poverty.” It does not. Even the richest country cannot satisfy every want. Also, a normative opinion is not wrong because it is an opinion; it is simply not a positive statement.

IN HISTORY. In 1776 Adam Smith described a pin factory in which ten workers, each doing one task, made about 48,000 pins a day. Working alone, each could hardly make twenty. He used the case to show that dividing tasks raises output from the same scarce labour.

REVIEW QUESTIONS

1. Why does scarcity not mean poverty?
2. Give one micro and one macro question.
3. Classify: “Bread prices should be capped.” Positive or normative? Why?



CHAPTER THE SECOND

UTILITY AND SATISFACTION

Why does the fifth bun taste less sweet?

The chapter begins with a puzzle about water and diamonds. It defines utility, builds the law of diminishing marginal utility from a hungry student's buns, and shows where the law holds and where it fails.

Water is cheap, though without it we die. Diamonds are costly, though we can live without them. Why? Why does the more useful good cost less? The puzzle puzzled thinkers for a century. The answer needs a new idea: the value we place on a good depends on the extra satisfaction we gain from one more unit of it. We build that idea step by step.

Start with a question you can almost answer.

UTILITY, TOTAL AND MARGINAL

Utility means the satisfaction a person gets from using a good or service. The word does not mean usefulness in a moral sense. A cigarette has utility for a smoker, though it harms health. Utility depends on the person, the time and the place. A cold drink gives more utility on a hot day than on a cold one.

Utility the satisfaction a person gets from using a good.

Utility is satisfaction.

Two measures matter. **Total utility** (TU) means the whole satisfaction from all the units consumed. **Marginal utility** (MU) means the extra satisfaction from one more unit. In words, marginal utility equals the change in total utility divided by the change in the number of units. The word *marginal* means “at the edge”, that is, the last or the next unit.

Total utility the whole satisfaction from all the units consumed.

Marginal utility the extra satisfaction from one more unit.

Two measures: total and marginal.

WORKED EXAMPLE. Total and marginal utility

GIVEN	A hungry student eats buns. TU after 1, 2, 3, 4, 5, 6 buns is 20, 36, 48, 56, 60, 60 utils.
FIND	The marginal utility of each bun.
FORMUL	MU of the nth bun = TU(n) minus TU(n-1).
A	
STEPS	MU1 = 20 - 0 = 20. MU2 = 36 - 20 = 16. MU3 = 48 - 36 = 12. MU4 = 56 - 48 = 8. MU5 = 60 - 56 = 4. MU6 = 60 - 60 = 0.
ANSWER	The marginal utilities are 20, 16, 12, 8, 4, 0 utils.
CHECK	Add them: 20+16+12+8+4+0 = 60, the total utility. The check holds.

A **util** is an imaginary unit used to count satisfaction. No machine measures it. Is satisfaction measurable at all? Early economists, such as Jevons and Marshall, said yes: a person can state how much they would pay, and that measures it in money. This is the **cardinal** view. Later writers, such as Hicks,

A unit of measure.

said a person can only rank satisfactions as first, second and third. This is the **ordinal** view, and we use it in Chapter 5.

THE LAW OF DIMINISHING MARGINAL UTILITY

Law of diminishing marginal utility as a person consumes more units, the extra satisfaction from each further unit falls.

The more we have, the less the next one adds.

Look at the example again. The marginal utility falls with each bun: 20, 16, 12, 8, 4, 0. This is the **law of diminishing marginal utility**. It states that, as a person consumes more units of a good in a given period, the marginal utility of each extra unit falls. Total utility still rises while marginal utility is positive. It reaches its highest point when marginal utility is zero, here after the fifth bun.

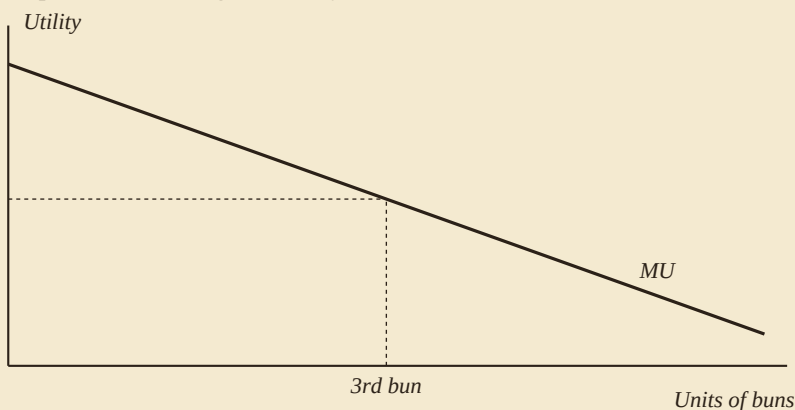


FIG. 2. Marginal utility falls as units rise (the schedule of buns, drawn smoothly).

Reading the figure.

To read the figure, find a number of buns on the horizontal axis and go up to the line. The height shows the extra satisfaction from that bun. The line slopes down, which is the law in picture form. Where the line meets the horizontal axis, the extra satisfaction is zero, and the student should stop eating.

The assumptions behind the law.

The law holds under conditions. The units must be of the same kind and of reasonable size. Consumption must be continuous, with no long breaks. The consumer's taste, income and the prices of related goods must stay the same. The consumer must act sensibly. If a condition fails, the law may appear to fail too.

Limits and exceptions.

Some cases seem to break the law. The urge for a hobby, such as collecting stamps, may grow with each stamp. A person who loves money may want more money as the pile grows. A drug user may want ever more of the drug. Knowledge may bring more interest in further knowledge. Most economists answer that in these cases a condition fails: the units are not of the same kind, the taste is changing, or the consumption is not rational. Even then, at some point the satisfaction from the next unit falls.

Back to the puzzle.

Now return to water and diamonds. Water is so plentiful that its marginal utility is low. Diamonds are rare, so the marginal utility of one more diamond is high. Price follows marginal utility, not total utility. Water has a very high total utility, but its price is low. In 1871 Jevons and Menger, and in 1874 Walras, each gave this answer separately.

CASE FILE: CUSTOMER SURVEY, 20 JAN 2024

Rupali Bakery asks one customer how much she would pay for each bun in a day. Answers: Tk 20 for the first, 16 for the second, 12 for the third, 8 for the fourth, 4 for the fifth. These are her marginal utilities in money. At a price of Tk 10 she buys three buns.

SUMMARY. Utility is satisfaction. Total utility is the whole; marginal utility is the extra from one more unit. As consumption rises, marginal utility falls. The law needs stated conditions, and the apparent exceptions break those conditions. Value follows marginal utility.

THINK IT THROUGH. Using the survey, how many buns would the customer buy at Tk 5? Explain by comparing price with marginal utility.

WHERE PEOPLE GO WRONG. “Diminishing utility means total utility falls.” It does not. Total utility rises more slowly, and it falls only when marginal utility becomes negative.

IN HISTORY. In 1738 Daniel Bernoulli argued that an extra ducat means less to a rich man than to a poor man. He used it to explain risky choices. The marginal revolution of 1871-74 turned this insight into the basis of value theory.

REVIEW QUESTIONS

1. Define marginal utility in one sentence.
2. If TU is 30, 50, 60, 62, what is MU of the third unit?
3. Name two conditions the law needs.



CHAPTER THE THIRD

THE THEORY OF DEMAND

Why do we buy less when prices rise?

The chapter begins with a bakery price change. It defines demand, states the law, builds the demand curve from a schedule, and examines the exceptions and the forces that shift the curve.

On 1 March 2024 the bakery raises the price of a loaf from Tk 25 to Tk 30. By the end of the week, fewer loaves are sold. Was this a surprise? Why does a higher price so often meet lower sales? The answer is the oldest regularity in economics, and it is the first half of the story of every market.

Start with a question you can almost answer.

DEMAND AND ITS LAW

Demand means the quantity of a good that buyers are willing and able to buy at each possible price in a given period. Both parts matter. A wish without money is not demand. A **demand schedule** is a table of prices and quantities. A **demand curve** is the same information drawn as a line.

Demand the quantity of a good that buyers are willing and able to buy at each price in a given period.

Demand needs both wish and ability to pay.

DAILY DEMAND FOR LOAVES IN THE NEIGHBOURHOOD

Price per loaf (Tk)	20	25	30	35	40
Quantity demanded	180	150	120	90	60

The table shows the **law of demand**. It states that, when the price of a good rises, the quantity demanded falls, and when the price falls, the quantity rises, if other things stay the same. Three reasons lie behind it. As price rises, we buy cheaper substitutes (the substitution effect). A higher price also reduces our real buying power (the income effect). And by the law of diminishing marginal utility, we will pay less for each extra unit.

Law of demand when the price of a good rises, the quantity demanded falls, other things equal.

Price up, quantity down.

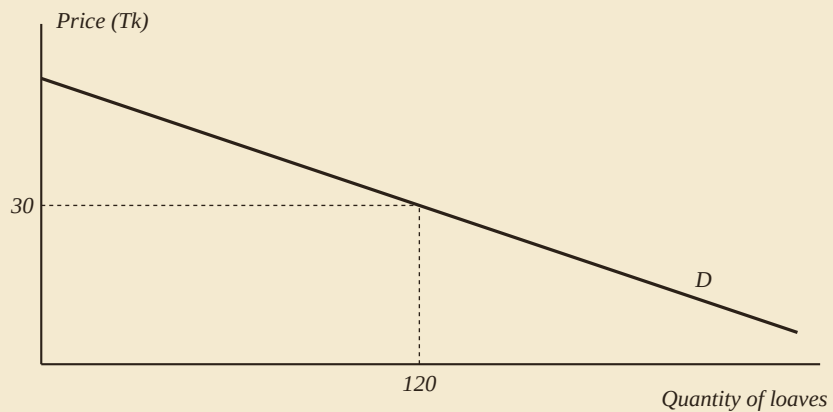


FIG. 3. The demand curve for loaves. It slopes downward from left to right.

Reading the figure.

To read the curve, pick a price on the vertical axis and go across to the line, then down to the quantity. At Tk 30 the quantity is 120 loaves. Moving along the curve, because of a change in price, is a **change in quantity demanded**. This is different from a shift of the whole curve.

FROM ONE BUYER TO THE MARKET

Adding up buyers.

A **market demand** curve is the sum of the demands of all buyers at each price. If two customers each buy 60 loaves at Tk 30, market demand at Tk 30 is 120 loaves. A price rise therefore reduces quantity both because each buyer buys less and because some buyers drop out. Throughout this book, “demand” means market demand unless we say otherwise.

Substitute goods goods that can replace each other, such as tea and coffee.

Complementary goods goods used together, such as bread and butter.

Related goods change demand.

Goods can be related. **Substitute goods** can replace each other, so a rise in the price of one raises demand for the other. **Complementary goods** are used together, so a rise in the price of one lowers demand for the other. Suppose the price of petrol rises. People drive less, and demand for cars, which are used with petrol, probably falls. Demand for bicycles, a substitute, probably rises.

SHIFTS OF THE CURVE

Other things that move demand.

The demand curve is drawn for given values of other things: the buyer’s income, tastes, the prices of related goods, expectations and the number of buyers. When one of these changes, the whole curve shifts. This is a **change in demand**. A rise in income usually moves the curve to the right: buyers want more at every price. A fall moves it to the left. Remember the difference: price moves us along the curve; every other cause shifts it.

EXCEPTIONS TO THE LAW

Giffen good a very basic good for which demand rises when its price rises.

Rare, but a real exception.

The law has a few exceptions. A **Giffen good** is a basic food for which quantity demanded rises when the price rises. Marshall (1895) credited Robert Giffen with the idea and linked it to the potato in nineteenth-century Ireland, though modern historians dispute the evidence. The reasoning is this: for a very poor family, a basic food takes most of its income. When its price rises, the family cannot afford better food, and it buys more of the basic food to stay fed.

Other exceptions.

Two other cases concern pride and crowds. The **snob effect** is the wish to own a good because few others have it: a rise in price may make a rare watch more desirable. The **bandwagon effect** is the wish to own what others own:

demand rises when many people are seen buying a popular phone. Expectations also matter. If buyers expect the price to rise further, they may buy more today. In each case the demand curve may slope upward over part of its range.

CASE FILE: DEMAND SCHEDULE, 1 MARCH 2024

Daily quantity demanded in the neighbourhood: Tk 20 gives 180 loaves, Tk 25 gives 150, Tk 30 gives 120, Tk 35 gives 90, Tk 40 gives 60. In equation form, $Q = 300 - 6P$. Check: $P = 30$ gives $Q = 300 - 180 = 120$.

SUMMARY. Demand is the quantity buyers are willing and able to buy at each price. The law says price and quantity move in opposite directions. A price change moves us along the curve; a change in income, taste, related prices or number of buyers shifts it. Giffen goods, snob and bandwagon effects are exceptions.

THINK IT THROUGH. From $Q = 300 - 6P$, find the quantity demanded at Tk 22. Then say whether a rise in the price of flour shifts this demand curve.

WHERE PEOPLE GO WRONG. “A fall in price shifts the curve to the right.” It does not. A price change moves us along the same curve; only other causes shift it.

IN HISTORY. Alfred Marshall’s *Principles* (1890) drew demand with price on the vertical axis and quantity on the horizontal. Economists have kept that convention for more than a century.

REVIEW QUESTIONS

1. State the law of demand.
2. Name two causes of a shift in demand.
3. What is a Giffen good?



CHAPTER THE FOURTH

ELASTICITY OF DEMAND

How much does demand react to a price change?

The chapter begins with two price rises that bring different results. It defines elasticity, measures price, income and cross elasticity, names the five forms, and lists the factors behind each.

Two shops each raise their price by 10 per cent. The bread shop loses almost no customers. The cake shop loses a third of them. Why do the same price rises give such different results? The law of demand tells us the direction of the change. It does not tell us the size. To measure size, we need a new tool.

Start with a question you can almost answer.

PRICE ELASTICITY OF DEMAND

Price elasticity of demand means the percentage change in quantity demanded divided by the percentage change in price. In symbols, $E_d = (\% \text{change in } Q) / (\% \text{change in } P)$. Percentage change means the change divided by the starting value, times 100. Because price and quantity move in opposite directions, the answer is negative. We usually drop the minus sign and speak of the size.

Price elasticity of demand the percentage change in quantity demanded divided by the percentage change in price.

Elasticity measures how strongly buyers react.

WORKED EXAMPLE. Price elasticity from a schedule

GIVEN	From the demand schedule of Chapter 3: at Tk 30 quantity is 120; at Tk 35 quantity is 90.
FIND	The price elasticity of demand for the rise from Tk 30 to Tk 35.
FORMUL A	$E_d = (\% \text{change in } Q) / (\% \text{change in } P)$.
STEPS	Change in $Q = 90 - 120 = -30$. $\% \text{change in } Q = -30/120 \times 100 = -25\%$. Change in $P = 35 - 30 = 5$. $\% \text{change in } P = 5/30 \times 100 = 16.7\%$. $E_d = -25 / 16.7 = -1.5$.
ANSWER	The size of elasticity is 1.5. Demand is elastic.
CHECK	Check: the percentage fall in quantity (25) is larger than the percentage rise in price (16.7), so the ratio must exceed 1. It does.

Now take a fall in price from Tk 25 to Tk 20. Quantity rises from 150 to 180. The change in Q is +30, which is 20 per cent of 150. The change in P is -5, which is 20 per cent of 25. So $E_d = 20 / (-20) = -1$. Take care: a common slip is to use the new value instead of the starting value, which gives a different answer. Always divide by the starting value, or use the midpoint if your teacher asks.

A second example, with messier numbers.

There are five forms. If E_d is greater than 1, demand is **elastic**: quantity changes by a larger percentage than price. If E_d is less than 1, demand is **inelastic**: quantity changes by a smaller percentage. If E_d equals 1, demand is **unitary elastic**. At the two extremes, demand is **perfectly inelastic** ($E_d = 0$,

Elastic demand quantity changes by a larger percentage than price.

Inelastic demand quantity changes by a smaller percentage than price.

Five forms, from zero to infinity.

quantity does not change at all) or **perfectly elastic** (E_d is infinite, buyers will buy any amount at one price and none above it).

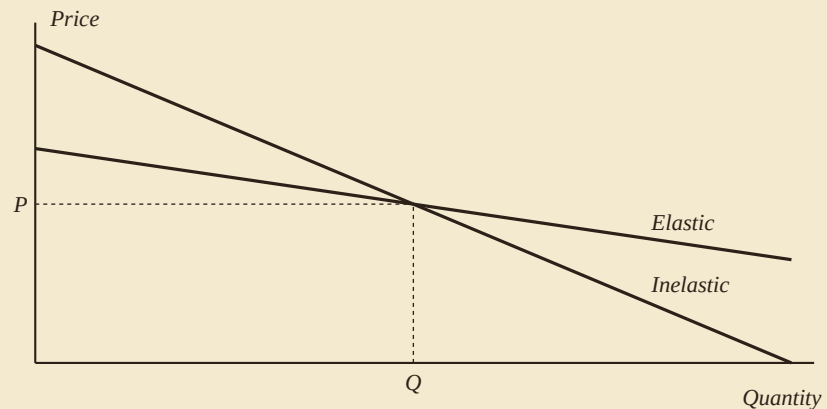


FIG. 4. Two demand curves through the same point. The flatter one is more elastic.

What decides elasticity?

Four factors matter. The more substitutes a good has, the more elastic its demand: if tea is dear, buyers switch to coffee. Necessities, such as salt, have inelastic demand; luxuries have elastic demand. A good that takes a large share of income is more elastic than one that takes a small share. And demand is more elastic over a long period, because buyers have time to find other goods.

INCOME AND CROSS ELASTICITY

Income elasticity of demand the percentage change in quantity demanded divided by the percentage change in income.

Income elasticity links buying to income.

Cross elasticity of demand the percentage change in quantity of one good divided by the percentage change in the price of another good.

Cross elasticity shows how goods are related.

Income elasticity of demand means the percentage change in quantity demanded divided by the percentage change in income. If income rises 10 per cent and cake purchases rise 20 per cent, the value is 2. A value above 1 marks a luxury; between 0 and 1 marks a necessity; below 0 marks an **inferior good**, one that people buy less of as they grow richer, such as very cheap coarse rice.

Cross elasticity of demand means the percentage change in the quantity of one good divided by the percentage change in the price of another. If a rival's bread price rises 10 per cent and our sales rise 5 per cent, the value is 0.5. A positive value shows that the goods are substitutes. A negative value shows that they are complements. A value near zero shows that the goods are unrelated.

WHY ELASTICITY MATTERS

Revenue and the price decision.

Total revenue means price multiplied by quantity sold. With elastic demand, a price cut raises revenue, because quantity rises by more than price falls. With inelastic demand, a price cut lowers revenue. A firm that knows its elasticity can choose the right price. So a seller of an inelastic good, such as salt, can raise its price with little loss of sales; a seller of an elastic good, such as cake, cannot.

CASE FILE: PRICE CHANGE TEST, 1 APRIL 2024

The bakery moves the loaf price from Tk 30 to Tk 35. Revenue before: $30 \times 120 =$ Tk 3,600. Revenue after: $35 \times 90 =$ Tk 3,150. Revenue falls, as the elasticity of 1.5 predicts. The bakery returns to Tk 30.

SUMMARY. Elasticity measures how much quantity responds to a change in price (or income, or another price). Demand is elastic above 1, inelastic below 1. Substitutes, necessity, share of income and time decide its size. With elastic demand, a price cut raises revenue.

THINK IT THROUGH. A firm sells 200 units at Tk 10. It raises the price to Tk 11 and sells 150. Find the elasticity and say what happens to revenue.

WHERE PEOPLE GO WRONG. “A steep curve and an inelastic curve are the same.” Slope and elasticity are different, because elasticity uses percentages and slope uses units. A straight line has one slope but a different elasticity at each point.

IN HISTORY. Marshall introduced the measure of price elasticity of demand in 1890. He wanted a number that did not depend on units, so that the demand for wheat and the demand for cloth could be compared.

REVIEW QUESTIONS

1. Define price elasticity of demand.
2. $Q = 200$, $P = 10$; Q falls by 50 when P rises by 1. Find E_d .
3. What does a cross elasticity of -0.8 show?



CHAPTER THE FIFTH

CONSUMER SURPLUS AND INDIFFERENCE CURVE

Why does a bargain feel like a gift?

The chapter begins with a customer who pays less than she was willing to pay. It measures this gain as consumer surplus, then shows how indifference curves describe choice without counting utils.

In Chapter 2 a customer was ready to pay Tk 20 for her first bun. The price is only Tk 10. She walks away pleased. Where did the pleasure come from, and can we count it in taka? The answer gives economists a way to judge whether a market makes people better off.

Start with a question you can almost answer.

CONSUMER'S SURPLUS

Consumer's surplus means the difference between the most a buyer is willing to pay and the price actually paid. Dupuit (1844) first described the idea, and Marshall (1890) developed it. It rests on diminishing marginal utility: because each extra unit is worth less, only the last unit is worth exactly the price.

Consumer's surplus the difference between what a buyer is willing to pay and what the buyer actually pays.

The gain from paying less than you would.

WORKED EXAMPLE. Consumer's surplus from the survey

GIVEN	The customer values the buns at Tk 20, 16, 12, 8 and 4. The price is Tk 10.
FIND	Her consumer's surplus.
FORMULA	Surplus on a unit = value minus price. Buy a unit only if value is at least the price.
STEPS	She buys the first three buns, valued at 20, 16 and 12. Surplus: $(20 - 10) + (16 - 10) + (12 - 10) = 10 + 6 + 2 = 18$.
ANSWER	Her consumer's surplus is Tk 18.
CHECK	Check: she paid $3 \times 10 = \text{Tk } 30$ for buns she valued at $20 + 16 + 12 = \text{Tk } 48$. The gap is $48 - 30 = 18$. It agrees.

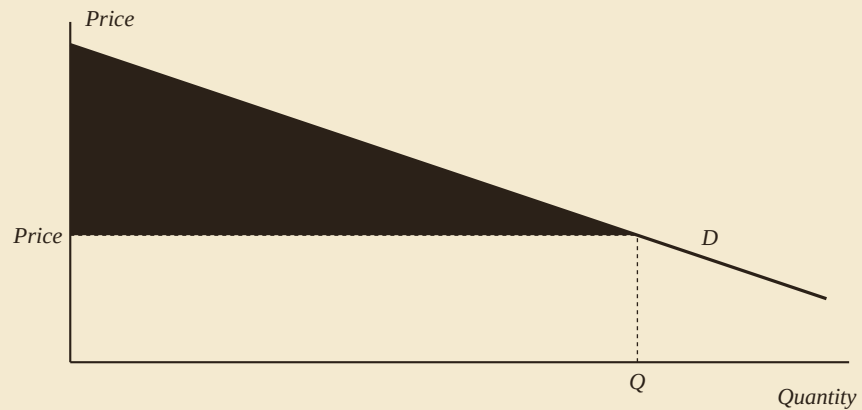


FIG. 5. Consumer's surplus is the shaded triangle between the demand curve and the price line.

Reading the figure.

The demand curve shows the most buyers would pay for each unit. The price line shows what they pay. The gap between the two, summed over all units bought, is the shaded area. If the price falls, the triangle grows, and buyers gain more. This is why a fall in price is a gain for buyers as a group.

Limits of the idea.

The measure has limits. It assumes that money has the same marginal utility for everybody, which is not exactly true. It cannot be applied to necessities, where the first unit is worth more than any sum. And it assumes there is no close substitute. Even so, it is a useful guide, and the next paragraph shows where.

Producer's surplus the difference between the price a seller receives and the lowest price the seller would accept.

The sellers' gain, as the mirror image.

Producer's surplus is the sellers' counterpart: the difference between the price received and the lowest price the seller would have accepted. Together, consumer's and producer's surplus measure the gain from trade. Governments use them to compare policies: a tax, a price ceiling or a monopoly reduces the sum, and the lost part is called the welfare loss. Public bodies use consumer's surplus to price water and power, and to weigh a new bridge against its cost.

CONSUMPTION AND CHOICE

Consumption the use of goods and services to satisfy wants.

Consumption ends the chain of production.

Consumption means the use of goods and services to satisfy wants. A consumer's budget is limited, so the consumer must choose among goods. Utility counting, with its utils, is hard to defend. The next method needs only ranking.

INDIFFERENCE CURVE ANALYSIS

Indifference curve a line showing all combinations of two goods that give a consumer equal satisfaction.

Every point on one curve is equally good.

Four properties.

An **indifference curve** means a line that joins all combinations of two goods that give the consumer the same satisfaction. If you are equally happy with 4 buns and 2 cups of tea, or 2 buns and 5 cups of tea, both points lie on one curve. The consumer is "indifferent" between them. Edgeworth (1881), Pareto (1906) and Hicks and Allen (1934) developed the tool.

Indifference curves have four properties. They slope downward, because more of one good must replace less of the other. They are convex to the origin, because of a falling **marginal rate of substitution** (MRS), the amount of one good given up for one more of the other. They never cross, since crossing would make one bundle both better and equal. And a curve higher and further from the origin shows greater satisfaction.

Budget line all combinations of two goods that the consumer's income can buy at given prices.

Income and prices draw the budget line.

A consumer cannot choose any point she likes. The **budget line** shows all combinations she can buy with her income at given prices. Suppose income is Tk 120, a bun costs Tk 10 and a cup of tea costs Tk 20. She can buy 12 buns and no

tea, or 6 cups and no buns, or any point between, such as 6 buns and 3 cups. In words: spending on buns plus spending on tea equals income.

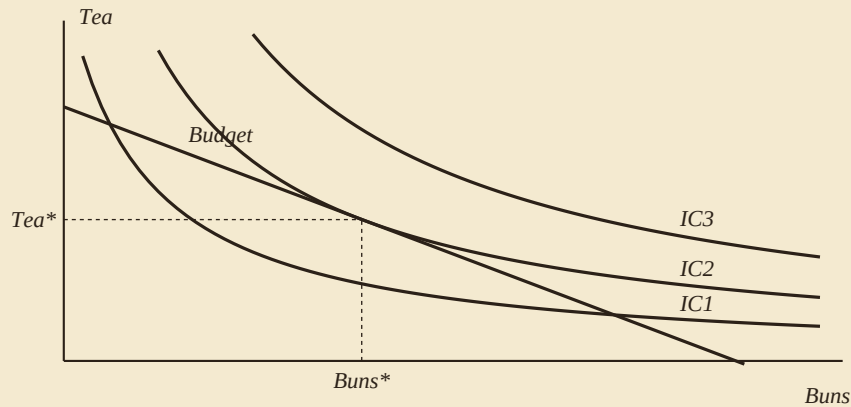


FIG. 6. The consumer chooses the point where the budget line just touches the highest curve she can reach.

The consumer reaches **equilibrium**, the point of best choice, where the budget line touches the highest indifference curve. At that point the two lines have the same slope. So the rate at which she is willing to swap goods (MRS) equals the rate at which the market lets her swap them (the price ratio). If the price of one good falls, the budget line turns outward, and she reaches a higher curve. The extra satisfaction can be valued in money, which links this method back to consumer's surplus.

Hicks's method measures the same surplus without utils. After a price fall, the consumer reaches a higher indifference curve. Now ask how much money we must take from her income to push her back to her old curve. That sum is her gain from the price fall, and so her consumer's surplus in money. The method needs only a ranking of bundles, not a count of satisfaction.

SUMMARY. Consumer's surplus is value minus price, summed over the units bought. Producer's surplus is its mirror. Indifference curves rank bundles without counting utils; the budget line shows what can be bought; the best choice is where the two touch.

THINK IT THROUGH. With income Tk 120, a bun at Tk 10 and tea at Tk 20, can the consumer buy 4 buns and 4 cups? Show your working.

WHERE PEOPLE GO WRONG. "Indifference curves can cross if the goods are similar." They cannot. If they crossed, one bundle would be equal to two others that are not equal to each other.

IN HISTORY. In 1844 the French engineer Jules Dupuit asked how to value a public bridge when it charged a toll. He saw that users would have paid more than the toll, and he measured the difference. This became consumer's surplus.

REVIEW QUESTIONS

1. Define consumer's surplus.
2. Give two properties of indifference curves.
3. With income Tk 100, bun Tk 10, how many buns if nothing else is bought?

Equilibrium of the consumer.

Measuring surplus with indifference curves.



CHAPTER THE SIXTH

THE THEORY OF SUPPLY

Why do sellers offer more at a higher price?

The chapter begins with a baker who bakes more when bread sells for more. It states the law of supply, lists what shifts supply, notes the exceptions, and measures the response by elasticity of supply.

In the spring of 2024 the price of bread in the neighbourhood rises. Rupali Bakery bakes more, and two other bakeries do the same. A higher price gives sellers a reason to offer more. This is the second half of every market story, and it mirrors the story of buyers.

Start with a question you can almost answer.

SUPPLY AND ITS LAW

Supply means the quantity of a good that sellers are willing and able to offer for sale at each price in a given period. A **supply schedule** is a table of prices and quantities. A **supply curve** is the same table drawn as a line. A schedule gives separate numbers; a curve joins them into one picture.

Supply the quantity of a good that sellers are willing and able to offer at each price in a given period.

Supply is the sellers' side.

DAILY SUPPLY OF LOAVES IN THE NEIGHBOURHOOD

Price per loaf (Tk)	20	25	30	35	40
Quantity supplied	60	90	120	150	180

The table shows the **law of supply**. It states that, when the price of a good rises, the quantity supplied rises, and when the price falls, the quantity falls, if other things stay the same. Two reasons lie behind it. A higher price makes it more profitable to produce. And producers face rising cost per unit when they expand, so they need a higher price to cover it. In equation form, $Q_s = 6P - 60$.

Law of supply when the price of a good rises, the quantity supplied rises, other things equal.

Price up, quantity up.

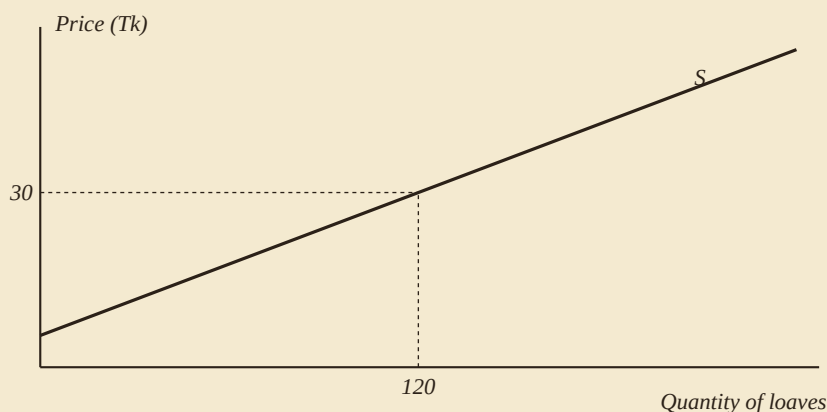


FIG. 7. The supply curve slopes upward from left to right.

Determinants of supply.

Exceptional supply curve a supply curve that does not slope upward in the usual way.

A few cases break the rule.

Price elasticity of supply the percentage change in quantity supplied divided by the percentage change in price.

How strongly sellers respond.

WHAT MOVES THE SUPPLY CURVE

A change in price moves us along the curve. Other causes shift the whole curve. The main ones are: the cost of inputs, such as flour and labour; technology, which lowers cost; taxes and subsidies; the prices of other goods the seller could make instead; the sellers' expectations of future prices; and the number of sellers. Cheaper flour moves the curve to the right: sellers offer more at every price. A tax on bread moves it to the left.

EXCEPTIONS TO THE LAW

An **exceptional supply curve** does not rise in the usual way. There are several cases. The supply of labour may bend backward: at very high wages, a worker may choose more leisure and work fewer hours. Goods with fixed supply, such as old paintings or land in a city, cannot rise with price. Perishable goods, such as fish at the end of the day, may be sold at any price rather than wasted. If sellers expect the price to rise further, they may hold stock back and offer less now.

ELASTICITY OF SUPPLY

Price elasticity of supply means the percentage change in quantity supplied divided by the percentage change in price. In symbols, $E_s = (\% \text{change in } Q_s) / (\% \text{change in } P)$. The answer is positive, because price and quantity move together. If E_s is above 1, supply is **elastic**; below 1, **inelastic**; at 1, unitary. If E_s is zero, supply is **perfectly inelastic**; if infinite, **perfectly elastic**.

WORKED EXAMPLE. Elasticity of supply

GIVEN	From the supply schedule: at Tk 30 supply is 120; at Tk 35 supply is 150.
FIND	E_s for the rise from Tk 30 to Tk 35.
FORMUL	$E_s = (\% \text{change in } Q_s) / (\% \text{change in } P)$.
A	
STEPS	Change in $Q_s = 150 - 120 = 30$. $\% \text{change} = 30/120 \times 100 = 25\%$. Change in $P = 5$. $\% \text{change} = 5/30 \times 100 = 16.7\%$. $E_s = 25 / 16.7 = 1.5$.
ANSWER	$E_s = 1.5$. Supply is elastic.
CHECK	Check: quantity rose by a larger percentage than price, so E_s is above 1. It is.

A neat fact: any straight supply line that starts at the point of origin has an elasticity of exactly 1 at every point. For such a line, quantity is a fixed multiple of price, so a 10 per cent rise in price gives a 10 per cent rise in quantity. A line that cuts the vertical axis (as our schedule does) has elasticity above 1. A line that cuts the horizontal axis has elasticity below 1.

A line from the origin.

Three factors matter. The first is spare capacity: a bakery with an idle oven can expand fast. The second is the ability to store output: goods that can be stored, such as rice, have elastic supply, and goods that spoil do not. The third is time: supply is inelastic in the very short run, when inputs are fixed, and more elastic in the long run, when firms can build and hire.

What decides the size?

CASE FILE: CAPACITY, 1 JUNE 2024

The bakery adds a second shift and can raise output from 120 to 150 loaves a day when the price rises from Tk 30 to Tk 35, as in the schedule. Its own elasticity of supply over that range is 1.5.

SUMMARY. Supply is the quantity sellers will offer at each price. The law says price and quantity rise together. Costs, technology, taxes, related prices, expectations and number of sellers shift the curve. Elasticity of supply measures the response; spare capacity, storability and time decide its size.

THINK IT THROUGH. Using $Q_s = 6P - 60$, find supply at Tk 25. Then say what happens to the curve if flour becomes cheaper.

WHERE PEOPLE GO WRONG. “A higher price shifts the supply curve to the right.” It does not; it moves us along the curve. Only a change in cost, technology and the like shifts it.

IN HISTORY. In 1890 Marshall compared supply and demand to the two blades of a pair of scissors. He said that it is as useless to ask which blade cuts as to ask whether price is set by supply or demand alone.

REVIEW QUESTIONS

1. State the law of supply.
2. Name two causes that shift supply to the right.
3. What is E_s of a straight supply line through the origin?



CHAPTER THE SEVENTH

MARKETS AND EQUILIBRIUM

How does a price find itself?

The chapter begins with a bakery that tries a price too high and a price too low. It defines a market, shows how supply and demand meet at an equilibrium price, describes the forces that bring it about, and asks whether lower prices are always good.

Suppose Rupali Bakery tries a loaf price of Tk 40 on Monday. At that price buyers want 60 loaves and sellers offer 180. On Tuesday it tries Tk 20. Buyers want 180 but sellers offer 60. Neither price lasts. On Wednesday the price settles at Tk 30, and buyers and sellers are both content. Why does Tk 30 work, and who set it?

Start with a question you can almost answer.

WHAT A MARKET IS

A **market** means any arrangement that brings buyers and sellers of a good together so that they can trade. It need not be a place. A phone call, a website or a bazaar can be a market. Markets differ by area (local, national, world), by time (very short, short and long period) and by the amount of competition. A **competitive market** has many buyers and many sellers, so that no one of them can set the price. Other kinds, monopoly and monopolistic competition, are studied in Chapter 10.

Market any arrangement that brings buyers and sellers of a good together to trade.

A market is a meeting, not necessarily a place.

EQUILIBRIUM

Equilibrium means a state in which nothing pushes the situation to change. In a market, the equilibrium price is the price at which quantity demanded equals quantity supplied. At that price, every buyer who wishes to buy finds a seller, and every seller finds a buyer.

Equilibrium a state in which quantity demanded equals quantity supplied, so there is no pressure for change.

Equilibrium is a point of rest.

WORKED EXAMPLE. Finding the equilibrium

GIVEN	Demand: $Q_d = 300 - 6P$. Supply: $Q_s = 6P - 60$.
FIND	The equilibrium price and quantity.
FORMULA	Set $Q_d = Q_s$.
STEPS	$300 - 6P = 6P - 60$. Add $6P$ to both sides and add 60 : $360 = 12P$. So $P = 360 / 12 = 30$. Put $P = 30$ in demand: $Q = 300 - 180 = 120$.
ANSWER	Equilibrium price is Tk 30 and quantity is 120 loaves.
CHECK	Check in supply: $Q_s = 6 \times 30 - 60 = 120$. Both sides agree.

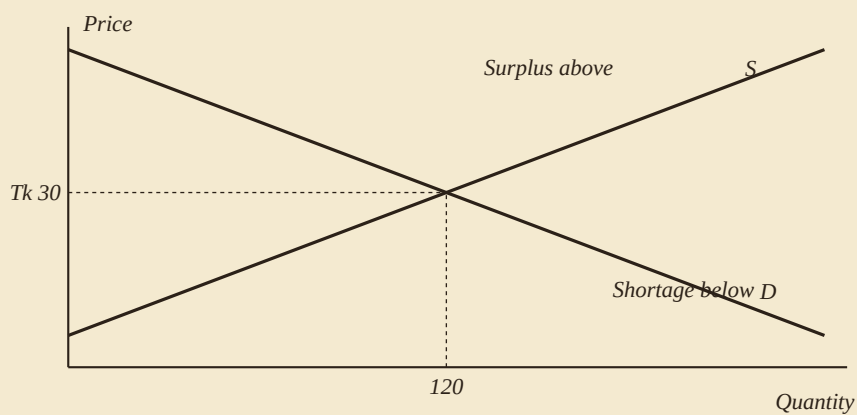


FIG. 8. The curves cross at the equilibrium. Above the price there is a surplus; below it, a shortage.

The forces at work.

Now see why other prices fail. At Tk 40, quantity supplied is 180 and quantity demanded is 60, so there is a **surplus** of 120 loaves. Sellers have unsold bread, and they cut the price. At Tk 20, quantity demanded is 180 and quantity supplied is 60, so there is a **shortage** of 120. Buyers compete, and the price rises. In both cases the price moves toward Tk 30. No one sets it. It emerges from many separate choices.

When a curve shifts.

Suppose a change in tastes adds 60 loaves to demand at every price, so $Q_d = 360 - 6P$. Then $360 - 6P = 6P - 60$, so $420 = 12P$ and $P = 35$. Quantity is $Q = 6 \times 35 - 60 = 150$. A rise in demand raises both price and quantity. A rise in supply does the opposite to price: it lowers price and raises quantity. Always find which curve moved, in which direction, and then read the new crossing.

PRICE IN DIFFERENT PERIODS

Time matters.

Marshall (1890) distinguished three periods. In the **market period**, supply is fixed, as with the morning's baked loaves, so demand alone sets the price. In the **short run**, firms can change output using their existing ovens, but cannot build new ones. In the **long run**, firms can enter and leave, and all inputs can change. Prices in the long run follow cost more closely, and prices in the short run follow demand more closely. Under other market types, as Chapter 10 shows, a firm may influence the price itself.

Is a lower price always good?

Not always. A lower price helps buyers, and consumer's surplus rises. But if the price falls below the cost of production, firms leave, supply shrinks and shortages follow. A government price ceiling set below equilibrium creates exactly the shortage described above. A very low price may also signal poor quality, or may be used by a large firm to drive out smaller rivals. A lower price is good for buyers only when it reflects lower cost and when the sellers remain.

CASE FILE: MARKET TEST, 4-6 MARCH 2024

Monday Tk 40: demand 60, supply 180, surplus 120. Tuesday Tk 20: demand 180, supply 60, shortage 120. Wednesday Tk 30: demand 120, supply 120. Revenue at the equilibrium: $30 \times 120 = \text{Tk } 3,600$ a day.

SUMMARY. A market brings buyers and sellers together. The equilibrium price makes quantity demanded equal to quantity supplied. A surplus pulls price down and a shortage pushes it up. A shift in demand or supply creates a new equilibrium. Lower prices are not always good.

THINK IT THROUGH. Supply rises by 30 at every price, so $Q_s = 6P - 30$. Using $Q_d = 300 - 6P$, find the new price and quantity.

WHERE PEOPLE GO WRONG. “Equilibrium means everyone is happy.” It means only that plans match. Some buyers still cannot afford the good.

IN HISTORY. In the 1890 edition of his book, Marshall used diagrams where demand and supply cross, and called the crossing point the equilibrium. Before him, writers argued which one set the price.

REVIEW QUESTIONS

1. Define equilibrium price.
2. At a price above equilibrium, what exists and what happens to price?
3. $Q_d = 100 - 2P$, $Q_s = 3P - 20$. Find P .



CHAPTER THE EIGHTH

PRODUCTION AND RETURNS

What happens when we add one more baker?

The chapter begins with a bakery that hires workers one by one. It defines the production function, shows why extra workers add less and less, and distinguishes this from returns to scale.

The bakery has one oven. With one baker it makes 40 loaves a day. A second baker makes it possible to bake 100. A third raises output to 180. A seventh baker, however, makes output fall. Why should a helper ever reduce output? The answer explains the shape of every cost curve in the next chapter.

Start with a question you can almost answer.

THE PRODUCTION FUNCTION

Production means the making of goods and services that satisfy wants. Economists group the resources used into four **factors of production**: land (all natural resources), labour (human effort), capital (machines, buildings and tools) and enterprise (the skill of organising and taking risk).

The **production function** means the relationship between the inputs used and the greatest output they can make. In words: output depends on land, labour, capital and enterprise. In symbols, $Q = f(L, K)$, where Q is output, L is labour and K is capital. In the **short run**, at least one input is fixed. In the **long run**, all inputs can change.

Production the creation of goods and services that satisfy wants.

Production creates utility.

Production function the relation between the inputs used and the maximum output obtained.

Inputs in, output out.

THE LAW OF DIMINISHING RETURNS

Take the short run, with the oven fixed and labour variable. **Total product** (TP) is the total output. **Average product** (AP) is output per worker, TP divided by workers. **Marginal product** (MP) is the extra output from one more worker.

Three measures.

OUTPUT OF THE BAKERY WITH ONE OVEN

Bakers	Total product	Average product	Marginal product
1	40	40	40
2	100	50	60
3	180	60	80
4	240	60	60
5	280	56	40
6	300	50	20
7	290	41	-10

Law of diminishing returns when one input is added to fixed inputs, the extra output from each further unit eventually falls.

Eventually each extra worker adds less.

The table shows the **law of diminishing returns**. It states that, when more of one input is added to fixed amounts of the others, the marginal product of the added input eventually falls. Here it begins after the third baker, whose MP is 80. Turgot noted the effect in 1767, and Malthus, West and Ricardo stated it in 1815. Ricardo applied it to farmland.

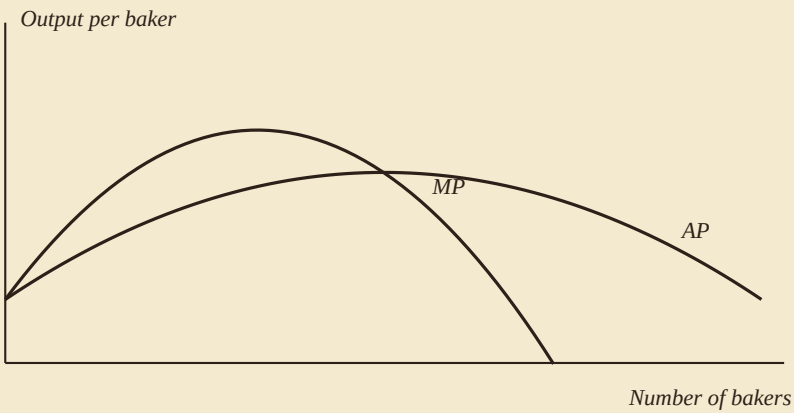


FIG. 9. Marginal product (MP) rises, then falls and cuts the average product (AP) curve at its highest point.

Why it happens.

The cause is simple. The oven is fixed. At first, extra bakers share tasks and use the oven better, so MP rises. Past a point, bakers crowd the kitchen and wait for the oven, so each adds less. The law holds if the extra units are of equal skill, if technology is unchanged and if at least one input is fixed. Output is in three stages: rising MP, falling but positive MP (where a sensible firm operates), and negative MP.

RETURNS TO SCALE

Returns to scale the change in output when all inputs are changed in the same proportion.

The long-run cousin of diminishing returns.

Returns to scale means the change in output when every input is changed in the same proportion. It is a long-run idea, since all inputs vary. Diminishing returns is a short-run idea, since one input is fixed. Do not confuse them. If output rises by a larger percentage than inputs, there are **increasing returns to scale**. If it rises by the same percentage, they are **constant**. If it rises by a smaller percentage, they are **decreasing**.

WORKED EXAMPLE. Types of returns to scale

GIVEN	Units of all inputs: 1, 2, 4, 8. Output: 100, 250, 500, 700.
FIND	Whether returns are increasing, constant or decreasing at each step.
FORMUL A	%change in output compared with %change in inputs.
STEPS	Step 1 to 2: inputs +100%; output 100 to 250 is +150%. Step 2 to 4: inputs +100%; output 250 to 500 is +100%. Step 4 to 8: inputs +100%; output 500 to 700 is +40%.
ANSWER	Increasing, then constant, then decreasing returns.
CHECK	Check: +150% exceeds +100%, +100% equals +100%, and +40% is below +100%. Each label fits.

Why returns to scale change.

At first, a larger firm gains economies: workers specialise, big machines work well, bulk buying lowers costs. Later, size brings strain: managers lose touch, orders move slowly and costs rise. Smith’s pin factory (1776) shows the first effect. Ford’s Model T line shows it too: the price fell from \$850 in 1908 to under \$300 by the mid-1920s, as assembly-line output grew.

CASE FILE: STAFFING, 1 SEPT 2024

The bakery sets four bakers, since MP is still 60 and AP is at its highest (60). A fifth baker would add only 40 loaves, which is less than the average. The owner leaves the number at four.

SUMMARY. The production function links inputs to output. In the short run, extra units of a variable input eventually add less. In the long run, doubling all inputs may more than double, exactly double or less than double output.

THINK IT THROUGH. Using the table, find MP of the fifth baker, and AP with six bakers. Why does the owner stop at four, and not seven?

WHERE PEOPLE GO WRONG. “Diminishing returns means total output falls.” Not at first. Total product still rises while MP is positive.

IN HISTORY. In 1767 the French minister Turgot saw that adding more labour to a fixed plot of land gave smaller and smaller extra harvests. Ricardo used the same idea to explain rent.

REVIEW QUESTIONS

1. Define the production function.
2. Distinguish diminishing returns from returns to scale.
3. Inputs double and output rises 60%. What returns?



CHAPTER THE NINTH

THE COSTS OF PRODUCTION

What does a loaf really cost?

The chapter begins with a baker who thinks her own time is free. It separates explicit and implicit costs, builds the cost curves from a table, compares short and long run, and finds the cheapest mix of inputs.

The owner of Rupali Bakery adds up her bills for flour, rent, wages and gas, and calls the total her cost. She has left out her own work and the money she put into the shop. Is her real cost larger than the bills? Economists say yes, and the difference changes the answer to the question “Am I making a profit?”

Start with a question you can almost answer.

OPPORTUNITY COST AND KINDS OF COST

Recall from Chapter 1 that **opportunity cost** is the value of the best alternative given up. To measure it, ask what the same resources would earn in their next best use. In the bakery, one cake needs the oven for as long as five loaves. So the opportunity cost of one cake is 5 loaves, and of one loaf it is 1/5 of a cake, that is, 0.2 cake. The cost is stated in goods, not only in money.

Opportunity cost, measured.

An **explicit cost** is a payment to an outsider, such as for flour or rent. An **implicit cost** is the value of the owner’s own resources that are used in the business. If the owner could earn Tk 1,000 a day in another job, that sum is an implicit cost of her time. If her savings could earn Tk 500 a day elsewhere, that is an implicit cost of her capital. Two examples of implicit cost are therefore the owner’s time and the owner’s own money.

Explicit cost a cost that needs a payment to someone outside the firm.

Implicit cost the value of resources the owner supplies, for which no payment is made.

Two kinds of cost: paid and unpaid.

Costs also differ by how they react to output. A **fixed cost** (FC) does not change with output in the short run, such as rent and the oven. A **variable cost** (VC) changes with output, such as flour and gas. **Total cost** (TC) is fixed plus variable cost. Each cost can also be stated per unit: **average cost** (AC) is TC divided by output, and **marginal cost** (MC) is the extra cost of one more unit.

Fixed cost a cost that does not change with output in the short run.

Variable cost a cost that changes as output changes.

Fixed does not move; variable does.

COSTS OF THE BAKERY, IN TAKA PER DAY (FC = 1,500)

Loaves	VC	TC	AC	AVC	MC
100	2,400	3,900	39.0	24.0	24
200	4,200	5,700	28.5	21.0	18
300	5,700	7,200	24.0	19.0	15
400	7,600	9,100	22.8	19.0	19
500	10,500	12,000	24.0	21.0	29

WORKED EXAMPLE. Cost per loaf

GIVEN	At 300 loaves, FC = Tk 1,500 and VC = Tk 5,700. Going from 300 to 400 loaves raises VC to Tk 7,600.
FIND	AC at 300 loaves, and MC of the step from 300 to 400.
FORMUL	$AC = TC / Q$. $MC = \text{change in } TC / \text{change in } Q$.
A	
STEPS	$TC = 1,500 + 5,700 = 7,200$. $AC = 7,200 / 300 = 24$. At 400: $TC = 1,500 + 7,600 = 9,100$. Change in $TC = 9,100 - 7,200 = 1,900$. $MC = 1,900 / 100 = 19$.
ANSWER	$AC = \text{Tk } 24$ a loaf; $MC = \text{Tk } 19$ a loaf.
CHECK	Check: MC (19) is below AC (24), so AC should be falling as output rises to 400. In the table it falls to 22.8.

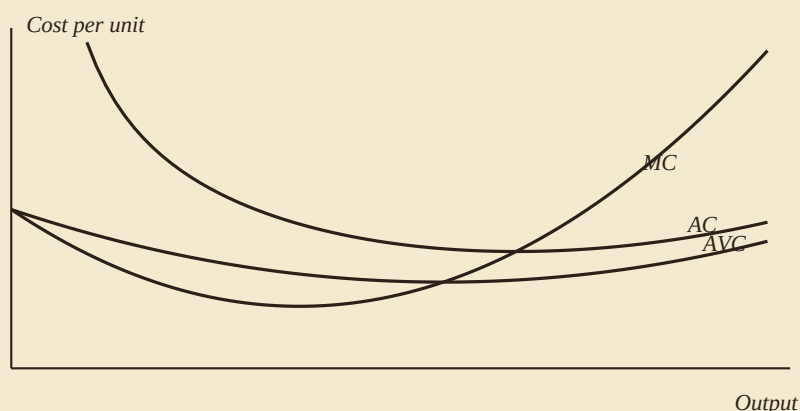


FIG. 10. Short-run cost curves. MC cuts both AVC and AC at their lowest points.

The pattern.

Look at the curves. AC and AVC fall at first and then rise, so each is U-shaped. AC falls at first because fixed cost is spread over more loaves. It rises later because of diminishing returns (Chapter 8). MC rises faster than the others and cuts them at their lowest points. A good rule: if MC is below the average, the average falls; if MC is above, the average rises. A marginal figure pulls the average in its direction.

SHORT RUN AND LONG RUN

Long-run average cost curve the lowest cost of each output when all inputs can change.

In the long run, no cost is fixed.

In the short run, the bakery has one oven, so it chooses a cost curve for that size. In the long run, it can choose a larger or smaller oven. The **long-run average cost curve** (LAC) joins the lowest points available from all the short-run curves, so it touches each of them from below. It is U-shaped as well: it falls with economies of scale and rises with diseconomies of scale (Chapter 8).

LEAST-COST COMBINATION

Least-cost combination the mix of inputs that produces a given output at the lowest total cost.

Spend each taka where it buys most.

A firm can often make the same output with different mixes of labour and machines. The **least-cost combination** is the mix that gives a target output at the lowest cost. The rule is: the extra output per taka should be equal for every input. In words, marginal product of labour divided by the wage must equal marginal product of capital divided by its hire price.

WORKED EXAMPLE. Least-cost mix

GIVEN	A baker's wage is Tk 800 a day; a machine hire is Tk 400 a day. Extra output: baker 40 loaves; machine 24 loaves.
FIND	Which input is the better buy.
FORMUL A	MP of labour / wage compared with MP of capital / hire price.
STEPS	Baker: $40 / 800 = 0.05$ loaf per taka. Machine: $24 / 400 = 0.06$ loaf per taka.
ANSWER	Machines give more per taka, so hire more machines and fewer bakers.
CHECK	Check: as machines increase, their MP falls (diminishing returns), until the two ratios are equal. Then no further change pays.

CASE FILE: COSTS, 1 OCT 2024

At 500 loaves a day: total cost Tk 12,000 (explicit). Implicit costs: owner's time Tk 1,000, own capital Tk 500. Full economic cost: Tk 13,500.

SUMMARY. Opportunity cost is the best alternative given up. Costs may be explicit or implicit, fixed or variable. AC and MC are U-shaped; MC cuts AC at its minimum. The long-run curve allows all inputs to change. The cheapest mix equalises extra output per taka.

THINK IT THROUGH. If FC is Tk 1,500 and VC at 200 loaves is Tk 4,200, find AFC, AVC and AC. Which cost is the owner's own time?

WHERE PEOPLE GO WRONG. "Sunk and fixed costs should change the price." In the short run, fixed costs do not change with output, so they do not affect the choice of how much to bake.

IN HISTORY. The term "opportunity cost" was first used by Friedrich von Wieser in 1914. Before him, Ricardo had already compared the cost of a good with the other goods that the same labour could make.

REVIEW QUESTIONS

1. What is opportunity cost?
2. Name two implicit costs.
3. $AC = 24$, $MC = 29$. Does AC rise or fall?



CHAPTER THE TENTH

MARKET STRUCTURES AND THE FIRM

Who decides the price: the seller or the market?

The chapter begins with a baker who must accept the price and a monopolist who may choose it. It defines market structure, derives equilibrium for the competitive firm and industry, and compares monopoly and monopolistic competition.

The bakery cannot sell a loaf at Tk 50 when the shop next door sells at Tk 30, because buyers walk away. The only water company in a small town can raise its price and still sell. What separates the two sellers? Not their wish for profit, but the number of rivals they face and how alike their goods are.

Start with a question you can almost answer.

MARKET STRUCTURE

Market structure means the features of a market that decide how firms behave: the number of sellers, whether the goods are alike, and how easily new firms can enter. Four main types exist. Price-setting power rises as we move down the table.

Market structure the features of a market that decide how firms behave, mainly the number of sellers, the kind of product and the ease of entry.

Competition decides who sets the price.

FOUR MARKET STRUCTURES

Type	Sellers	Product	Entry	Control of price
Perfect competition	Very many	Identical	Free	None
Monopolistic competition	Many	Differentiated	Free	Some
Oligopoly	Few	Alike or differentiated	Hard	Considerable
Monopoly	One	No close substitute	Blocked	Large

PERFECT COMPETITION

Perfect competition means a market with very many small buyers and sellers, identical goods, free entry and exit, and full knowledge of prices. **Pure competition** is a narrower idea: it needs only the first three, with no element of monopoly, and it does not need full knowledge or free movement of resources. In perfect competition, each firm is a **price taker**. It accepts the market price, so its extra revenue from one more unit (marginal revenue, MR) equals the price.

Perfect competition a market with very many small buyers and sellers, identical goods, free entry and full information.

Price is set by the market.

Equilibrium of the firm.

A firm that wishes to earn the most profit expands output as long as the extra unit adds more to revenue than to cost. It stops where MR equals MC, with MC rising. For the competitive firm, MR equals price, so the rule is $P = MC$. In the short run the firm may earn a profit, break even or make a loss. It continues to produce in the short run if price covers at least average variable cost.

WORKED EXAMPLE. Short-run profit of the firm

GIVEN	Market price Tk 30. Costs from Chapter 9: at 500 loaves, $TC = \text{Tk } 12,000$ and $MC = \text{Tk } 29$.
FIND	The output that maximises profit, and the profit.
FORMUL A	Rule: $P = MC$. Profit = $(P - AC) \times Q$.
STEPS	At 400 loaves MC is Tk 19, below the price; the firm gains by baking more. At 500 loaves MC is Tk 29, just below Tk 30, so 500 is the best output on the table. $AC = 12,000 / 500 = 24$. Profit = $(30 - 24) \times 500 = 6 \times 500 = 3,000$.
ANSWER	The firm bakes 500 loaves and earns Tk 3,000 a day.
CHECK	Check: revenue $30 \times 500 = 15,000$; cost 12,000; the difference is 3,000. It matches.

The industry in the short and long run.

The **industry** means all firms that make the same good. In the short run, the industry is in equilibrium when market demand equals the sum of the supplies of all firms, which fixes the price each firm accepts. In the long run, profit attracts new firms. Their entry raises supply and lowers price. Entry continues until price equals the lowest AC, and each firm earns only **normal profit**, the minimum reward needed to keep it in business. A loss has the opposite effect: firms leave, supply falls, and price rises. The industry reaches long-run equilibrium when no firm wishes to enter or leave.

MONOPOLY

Monopoly a market with a single seller of a good that has no close substitute.

One seller sets the price.

Monopoly means a market with only one seller and no close substitute. A **pure monopoly** is the extreme case. A **bilateral monopoly** has one seller and also only one buyer, such as a single union facing a single employer. The monopolist faces the whole market demand, so to sell more it must lower the price on all units. Its marginal revenue is therefore below price. It maximises profit where MR equals MC, then charges the highest price buyers will pay for that output, which is above MC.

MONOPOLISTIC COMPETITION

Monopolistic competition a market with many sellers of similar but not identical goods, and free entry.

Many sellers, each with a small monopoly.

Short and long run.

Monopolistic competition means a market with many sellers whose goods are similar but not identical, with free entry. Chamberlin and Robinson described it in 1933. **Product differentiation** is its typical feature. It means that each firm makes a version of the good that buyers see as different, through brand, quality, taste or location. Rupali’s loaf, with its own recipe and shop, is one example. Differentiation gives each firm a small monopoly and a downward-sloping demand curve.

In the short run, the firm acts like a monopolist: it sets MR equal to MC and may earn a profit. In the long run, profit invites new firms with close substitutes. The firm’s demand shifts left and becomes flatter, until the demand curve just touches the AC curve. Price equals AC, and the firm earns only normal profit. Yet the output is below the point of lowest cost. This is called **excess capacity**.

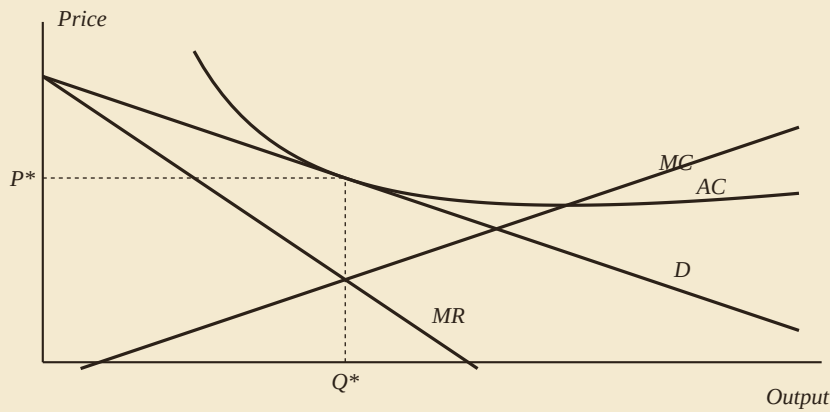


FIG. 11. Long-run equilibrium under monopolistic competition. The demand curve (D) touches AC at one point, and MR meets MC at the same output.

MARKET STRUCTURE AND PRICING

Structure decides whether a firm takes the price or makes it. A perfect competitor takes it. A monopolist and a monopolistic competitor choose it, within the limit of demand. An **oligopolist**, one of a few sellers, must also guess how rivals will answer a price change, so prices tend to stay stable. Many firms aim for goals other than profit, such as sales, market share or growth. Profit is still the most useful single guide, but it is not the only goal.

The role of structure in pricing.

CASE FILE: MARKET POSITION, 1 JAN 2025

The bakery sells a loaf with its own recipe, and the shop two streets away sells a similar one. Rupali Bakery therefore faces monopolistic competition. It has some control over price, and any profit above normal draws in new bakeries.

SUMMARY. Market structure depends on number of sellers, kind of product and entry. A competitive firm sets $P = MC$; entry drives profit to normal in the long run. A monopolist sets $MR = MC$ and charges a price above MC . Monopolistic competition has differentiation and, in the long run, normal profit and excess capacity.

THINK IT THROUGH. At Tk 30, would the firm of Chapter 9 bake 300 loaves? Compare price with MC at 300 and say whether more or less output would raise profit.

WHERE PEOPLE GO WRONG. “Perfect and pure competition are the same.” They are not. Perfect competition adds full knowledge and perfect mobility to the conditions of pure competition.

IN HISTORY. In 1933, two books appeared in the same year, by Edward Chamberlin in the United States and Joan Robinson in England. Each showed that most real firms are neither perfect competitors nor pure monopolists.

REVIEW QUESTIONS

1. Define market structure.
2. State the profit-maximising rule of a competitive firm.
3. What is product differentiation?



CHAPTER THE ELEVENTH

RENT, WAGES, INTEREST AND PROFIT

Who gets what the bakery earns?

The chapter begins with the bakery's daily takings and asks who receives them. It explains how rent, wages, interest and profit are fixed, using Ricardo's theory, the marginal productivity idea and the main theories of profit.

On a good day Rupali Bakery takes Tk 15,000. Part goes to the landlord, part to the bakers, part to the lender who financed the oven, and part is left for the owner. What decides each share? Economists call this question **distribution**: the sharing of the product among the owners of land, labour, capital and enterprise.

Start with a question you can almost answer.

RENT

Economic rent means the payment to a factor whose supply is fixed, in excess of what is needed to keep it in use. Land is the main example, since the nation's land cannot be increased by a higher price. Ricardo (1817) built a theory on it. Rent is different from the everyday word, which means a payment to a landlord for a house or shop.

Economic rent the payment to a factor in fixed supply, above what is needed to keep it in use.

Rent is the reward for what is in fixed supply.

WORKED EXAMPLE. Ricardo's theory

GIVEN	Three plots of equal size, each worked with the same cost. Wheat yields: plot A 40, plot B 30, plot C 20 units.
FIND	The rent of each plot.
FORMUL A	Rent = yield of the plot minus yield of the marginal (worst) plot used.
STEPS	Plot C is the marginal plot: it only covers its costs, so its rent is 0. Rent of B = $30 - 20 = 10$. Rent of A = $40 - 20 = 20$.
ANSWER	Rents: A 20, B 10, C 0 units of wheat.
CHECK	Check: rent measures the extra advantage over the worst land. A has the largest advantage (20), so it has the largest rent.

Ricardo said that rent is a **differential surplus**: it arises from differences in fertility and location. On the marginal land there is no rent, so rent does not enter the price of grain. Critics reply that land has alternative uses, so for the economy as a whole it does have a cost. The modern view agrees with Ricardo for a single use, and adds that rent can be viewed as a cost for a user who must outbid others. **Quasi-rent** is a temporary surplus on a man-made asset, such as a machine, until supply can be increased. **Artificial rent** is a surplus created by restrictions, such as a licence, and it differs from the real rent of land.

Differential surplus the extra produce of better land over the worst land in use.

Rent is a surplus, not a cost on the margin.

Nominal wage the wage paid in money.

Real wage what the money wage can buy.

Real wages count what buys.

Why wages vary.

How wages are determined.

Interest the payment for the use of borrowed money over a period of time.

Interest is the price of waiting.

Economic profit total revenue minus all costs, explicit and implicit.

Accounting profit total revenue minus explicit costs only.

Two ways to count profit.

WAGES

The **nominal wage** is the wage in money. The **real wage** is what that money can buy, found by correcting for prices. The formula is: real wage = nominal wage divided by the price index, times 100.

WORKED EXAMPLE. Nominal and real wage

GIVEN	A baker's wage rises from Tk 800 to Tk 880. The price index rises from 100 to 120.
FIND	The new real wage, and whether the worker gained.
FORMULA	Real wage = nominal wage / price index x 100.
ANSWERS	Real wage = $880 / 120 \times 100 = 733.3$ (in base-year taka). Before: $800 / 100 \times 100 = 800$.
ANSWER	The real wage fell from Tk 800 to Tk 733.
CHECK	Check: wages rose 10% but prices rose 20%, so buying power must have fallen. It did.

Wages differ for reasons that are simple to state. Skilled work needs training, which costs time and money. Dangerous or unpleasant work needs extra pay. Some jobs are steady and others are not. Trade unions and law may raise some wages. Wages differ between regions because of different costs of living and different supply of labour. A job that gives status may pay less, because people want it for other reasons.

In a competitive labour market, wages are set by demand and supply. The employer's demand rests on **marginal productivity**: a worker is hired if the value of the extra output is at least the wage. In the bakery, each loaf adds Tk 15 of value after materials. The fourth baker adds $60 \times 15 = \text{Tk } 900$, above the wage of 800, so is hired. The fifth adds $40 \times 15 = \text{Tk } 600$, below 800, so is not. Labour supply and bargaining also matter, so the result may differ.

INTEREST

Interest means the payment for the use of borrowed money for a period. Lenders give up spending now and take risk, and interest rewards them. In the **loanable funds** view, the interest rate is set where the supply of savings meets the demand for loans. Keynes (1936) added that people hold money for safety and convenience, so interest is the price of giving up that liquidity. A loan of Tk 100,000 at 10 per cent costs Tk 10,000 a year. If prices rise 6 per cent, the **real interest rate** is about $10 - 6 = 4$ per cent.

PROFIT

Accounting profit is revenue minus explicit costs. **Economic profit** is revenue minus all costs, including implicit ones. Using the bakery's figures at 500 loaves: revenue Tk 15,000 less explicit costs Tk 12,000 gives an accounting profit of Tk 3,000. Subtract the implicit costs of Tk 1,500 and the economic profit is Tk 1,500. **Normal profit** is the reward needed to keep the owner in business; it is already counted among the implicit costs.

Four main theories answer. Knight (1921) said profit rewards the bearing of **risk**, the uncertainty that cannot be insured. The marginal productivity theory treats the entrepreneur as one more factor, paid for the extra output she brings. Schumpeter (1911) said profit rewards **innovation**, the new method or product, and falls away as rivals copy it. A fourth view is that monopoly power or luck may also give surplus profit, which does not reward effort.

Why profit occurs.

CASE FILE: THE DAY'S TAKINGS, 1 OCT 2024

Takings Tk 15,000: materials, gas and other running costs; four bakers at Tk 800 = 3,200; rent; interest on the oven loan; and the rest to the owner. The owner's economic profit is Tk 1,500, as computed above.

SUMMARY. Rent rewards a factor in fixed supply and is a differential surplus on land. Wages depend on marginal productivity, supply and bargaining; real wages correct for prices. Interest rewards waiting and is set by loanable funds and liquidity. Profit rewards risk, innovation and enterprise, and economic profit is smaller than accounting profit.

THINK IT THROUGH. Plot A yields 50, B 35 and C 25, all at the same cost. Find the rent of A and B by Ricardo's method.

WHERE PEOPLE GO WRONG. "A higher nominal wage means a higher real wage." Not if prices rise faster. Always compare the two percentage changes.

IN HISTORY. In 1815 Ricardo wrote about the Corn Laws, the duties on imported grain in Britain, and argued that they raised rents of landowners at the cost of the rest. He used the theory of rent to make the case.

REVIEW QUESTIONS

1. Define economic rent.
2. Distinguish nominal wage from real wage.
3. Revenue 10,000, explicit costs 7,000, implicit costs 1,200. Find both profits.

ANSWERS TO THE REVIEW QUESTIONS

CHAPTER I

1. Because every person and every country has limited resources compared with wants, whether rich or poor.
2. Micro: why does a loaf cost Tk 30? Macro: why are prices across the nation rising?
3. Normative. It says what ought to be and rests on values; facts alone cannot settle it.

CHAPTER II

1. The extra satisfaction a person gets from one more unit of a good.
2. 60 minus 50 = 10 utils.
3. Units of the same kind and reasonable size; no change in taste, income or prices.

CHAPTER III

1. When the price of a good rises, quantity demanded falls, other things equal.
2. Any two of: income, tastes, prices of related goods, expectations, number of buyers.
3. A basic good for which demand rises when its price rises.

CHAPTER IV

1. The percentage change in quantity demanded divided by the percentage change in price.
2. %change Q = -25%; %change P = 10%; Ed = 2.5, elastic.
3. The two goods are complements.

CHAPTER V

1. The difference between what a buyer would pay and what is paid.
2. They slope downward and are convex to the origin; they never cross.
3. 10 buns.

CHAPTER VI

1. When price rises, quantity supplied rises, other things equal.
2. Cheaper inputs, better technology, a subsidy, more sellers.
3. Exactly 1 at every point.

CHAPTER VII

1. The price at which quantity demanded equals quantity supplied.
2. A surplus; price falls.
3. $100 - 2P = 3P - 20$, so $120 = 5P$ and $P = 24$.

CHAPTER VIII

1. The relationship between inputs and the greatest output they can produce.
2. Diminishing returns: one input varies, others fixed (short run). Returns to scale: all inputs vary together (long run).
3. Decreasing returns to scale.

CHAPTER IX

1. The value of the best alternative given up.
2. The owner's time and the owner's own capital.
3. Rise, because MC is above AC.

CHAPTER X

1. The features of a market, such as number of sellers, product type and entry, that decide how firms behave.
2. Produce where price equals marginal cost, with MC rising.
3. The making of goods that are similar but seen by buyers as different.

CHAPTER XI

1. A payment to a factor in fixed supply, above what is needed to keep it in use.
2. Nominal wage is in money; real wage is what that money can buy.
3. Accounting profit 3,000; economic profit 1,800.

GLOSSARY AND INDEX

Each entry gives the term, its meaning, and the chapter and page of first use.

Accounting profit: total revenue minus explicit costs only. *Ch. 11, p. 46.*

Budget line: all combinations of two goods that the consumer's income can buy at given prices. *Ch. 5, p. 22.*

Complementary goods: goods used together, such as bread and butter. *Ch. 3, p. 14.*

Consumer's surplus: the difference between what a buyer is willing to pay and what the buyer actually pays. *Ch. 5, p. 21.*

Consumption: the use of goods and services to satisfy wants. *Ch. 5, p. 22.*

Cross elasticity of demand: the percentage change in quantity of one good divided by the percentage change in the price of another good. *Ch. 4, p. 18.*

Demand: the quantity of a good that buyers are willing and able to buy at each price in a given period. *Ch. 3, p. 13.*

Differential surplus: the extra produce of better land over the worst land in use. *Ch. 11, p. 45.*

Economic law: a statement of the usual relation between economic events, true when other things stay equal. *Ch. 1, p. vi.*

Economic profit: total revenue minus all costs, explicit and implicit. *Ch. 11, p. 46.*

Economic rent: the payment to a factor in fixed supply, above what is needed to keep it in use. *Ch. 11, p. 45.*

Economic system: the way a society organises the choices of what, how and for whom to produce. *Ch. 1, p. v.*

Economics: the study of how people choose to use limited resources to satisfy wants. *Ch. 1, p. iv.*

Elastic demand: quantity changes by a larger percentage than price. *Ch. 4, p. 17.*

Equilibrium: a state in which quantity demanded equals quantity supplied, so there is no pressure for change. *Ch. 7, p. 29.*

Exceptional supply curve: a supply curve that does not slope upward in the usual way. *Ch. 6, p. 26.*

Explicit cost: a cost that needs a payment to someone outside the firm. *Ch. 9, p. 37.*

Fixed cost: a cost that does not change with output in the short run. *Ch. 9, p. 37.*

Giffen good: a very basic good for which demand rises when its price rises. *Ch. 3, p. 14.*

Implicit cost: the value of resources the owner supplies, for which no payment is made. *Ch. 9, p. 37.*

Income elasticity of demand: the percentage change in quantity demanded divided by the percentage change in income. *Ch. 4, p. 18.*

Indifference curve: a line showing all combinations of two goods that give a consumer equal satisfaction. *Ch. 5, p. 22.*

Inelastic demand: quantity changes by a smaller percentage than price. *Ch. 4, p. 17.*

Interest: the payment for the use of borrowed money over a period of time. *Ch. 11, p. 46.*

Law of demand: when the price of a good rises, the quantity demanded falls, other things equal. *Ch. 3, p. 13.*

Law of diminishing marginal utility: as a person consumes more units, the extra satisfaction from each further unit falls. *Ch. 2, p. ix.*

Law of diminishing returns: when one input is added to fixed inputs, the extra output from each further unit eventually falls. *Ch. 8, p. 34.*

Law of supply: when the price of a good rises, the quantity supplied rises, other things equal. *Ch. 6, p. 25.*

Least-cost combination: the mix of inputs that produces a given output at the lowest total cost. *Ch. 9, p. 38.*

Long-run average cost curve: the lowest cost of each output when all inputs can change. *Ch. 9, p. 38.*

Macroeconomics: the study of the whole economy, such as total output, prices and jobs. *Ch. 1, p. v.*

Marginal utility: the extra satisfaction from one more unit. *Ch. 2, p. viii.*

Market: any arrangement that brings buyers and sellers of a good together to trade. *Ch. 7, p. 29.*

- Market structure:** the features of a market that decide how firms behave, mainly the number of sellers, the kind of product and the ease of entry. *Ch. 10, p. 41.*
- Microeconomics:** the study of the choices of single households, firms and markets. *Ch. 1, p. v.*
- Monopolistic competition:** a market with many sellers of similar but not identical goods, and free entry. *Ch. 10, p. 42.*
- Monopoly:** a market with a single seller of a good that has no close substitute. *Ch. 10, p. 42.*
- Nominal wage:** the wage paid in money. *Ch. 11, p. 46.*
- Opportunity cost:** the value of the best alternative given up when a choice is made. *Ch. 1, p. v.*
- Perfect competition:** a market with very many small buyers and sellers, identical goods, free entry and full information. *Ch. 10, p. 41.*
- Positive statement:** a statement about what is, which facts can test. *Ch. 1, p. vi.*
- Price elasticity of demand:** the percentage change in quantity demanded divided by the percentage change in price. *Ch. 4, p. 17.*
- Price elasticity of supply:** the percentage change in quantity supplied divided by the percentage change in price. *Ch. 6, p. 26.*
- Producer's surplus:** the difference between the price a seller receives and the lowest price the seller would accept. *Ch. 5, p. 22.*
- Production:** the creation of goods and services that satisfy wants. *Ch. 8, p. 33.*
- Production function:** the relation between the inputs used and the maximum output obtained. *Ch. 8, p. 33.*
- Real wage:** what the money wage can buy. *Ch. 11, p. 46.*
- Returns to scale:** the change in output when all inputs are changed in the same proportion. *Ch. 8, p. 34.*
- Scarcity:** resources are limited while wants are not. *Ch. 1, p. iv.*
- Substitute goods:** goods that can replace each other, such as tea and coffee. *Ch. 3, p. 14.*
- Supply:** the quantity of a good that sellers are willing and able to offer at each price in a given period. *Ch. 6, p. 25.*
- Total utility:** the whole satisfaction from all the units consumed. *Ch. 2, p. viii.*
- Utility:** the satisfaction a person gets from using a good. *Ch. 2, p. viii.*
- Variable cost:** a cost that changes as output changes. *Ch. 9, p. 37.*

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MICROECONOMY

BIJOY?!

PRINCIPLES TO FINANCE



*The Finance of the Firm:
An Introduction for First-Year Students*

A TEXTBOOK FOR FIRST-YEAR STUDENTS

BIJOY?!



PRINCIPLES TO FINANCE



*The Finance of the Firm:
An Introduction for First-Year Students*

BIJOY ? !

TO THE READER

This book teaches how a business finds, uses and controls its money. It assumes no earlier knowledge of business. If you can do arithmetic and basic algebra, you can follow every step. A word you have not met will be explained when it first appears.

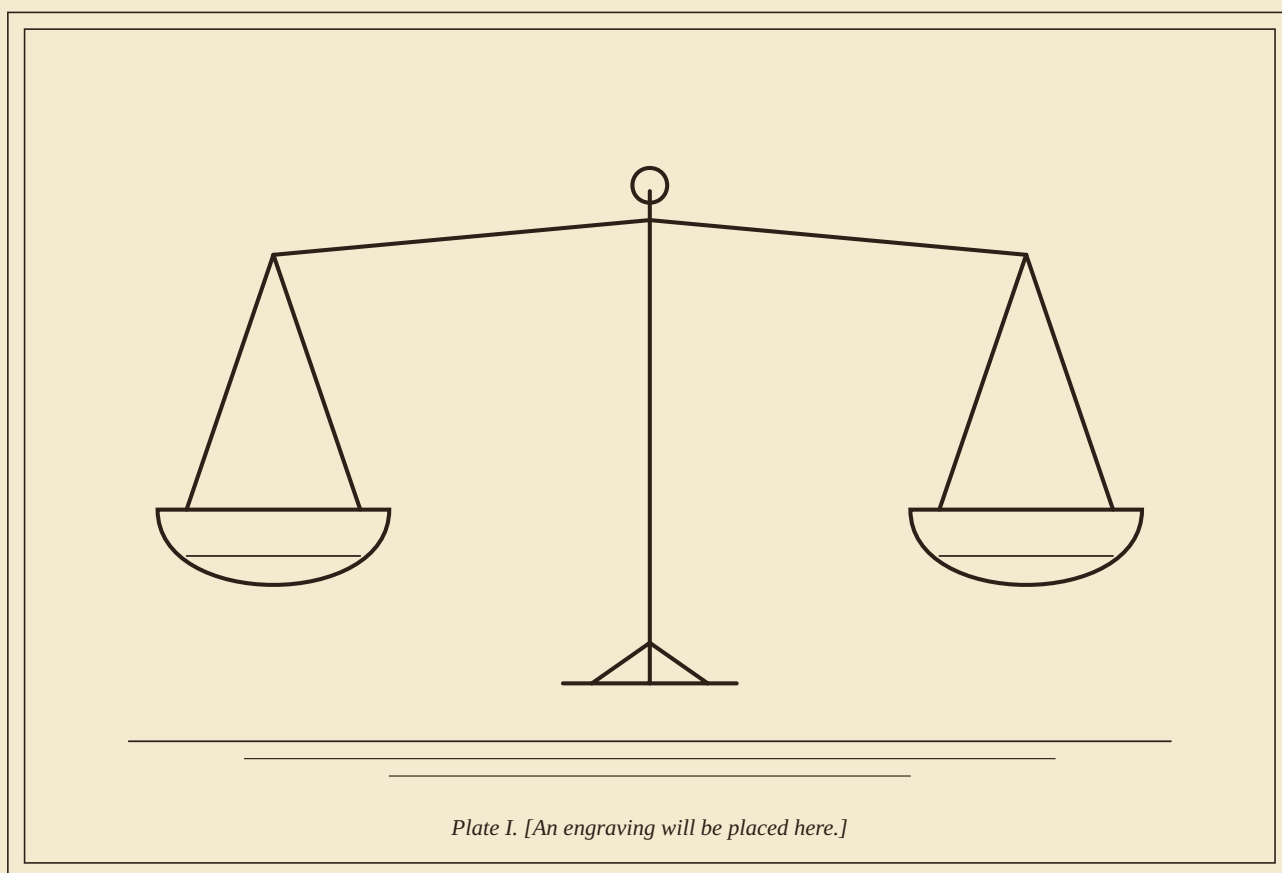
Each page has a wide column of short notes on the left. An italic note repeats the idea of the paragraph beside it in one line. A note in upright type, with a bold word at its start, gives the meaning of a new term. A word in bold in the main text is a new term, set in bold once only. Small raised numbers point to footnotes at the foot of the page. All terms are collected again in the glossary at the back.

Money is written in Bangladeshi taka, shown as Tk. Large sums use the South Asian grouping: Tk. 1,00,000 means one lakh, which is one hundred thousand, and Tk. 10,00,000 means ten lakh, which is one million. Laws, taxes and rates change; where a figure is an assumption it is said to be one. The Bakery in the case files is an invented business, used for illustration. It is not a real firm.

Each chapter begins with a puzzle, builds an idea in steps, and ends with a summary, a problem, common mistakes, a true story from history, and review questions. Short answers are printed at the back. Work each calculation with a pencil before you read the answer.

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CHAPTER I

The Purpose of Finance

Money is easy to count and hard to use well.



This chapter explains what finance is, what a financial manager decides, why the aim of a firm is wealth and not only profit, and why owners and managers may want different things.

A business can earn a profit and still run short of money.

Finance is about finding money and using it well.

Finance. The way a business finds, uses and controls its money.

Accounting looks back; finance looks forward.

Financial manager. The person who decides how a firm finds and uses its money.

Four kinds of decision cover the whole task.

Investment decision. A choice about which assets, such as machines or shops, the firm should buy.

Financing decision. A choice about how to raise money, from owners or from lenders.

Dividend. A share of profit paid out to the owners of a company.

A PUZZLE ABOUT MONEY

A baker sells bread for a month. At the end, her records show a profit of Tk. 20,000. On Friday the flour supplier asks to be paid, and the bank balance is Tk. 4,000. How can a business make a profit and still be unable to pay its bills? The answer lies in the way money moves through a business. The study of that movement is the subject of this book.

WHAT FINANCE MEANS

Finance means the way a business finds money, uses money and controls money. A business needs money to begin, for an oven, flour and rent. It needs more money to continue, for wages and repairs. Finance asks three plain questions. Where will the money come from? Where will it be placed? How will we know it was placed well?

Finance is close to two other subjects. Economics explains how people and markets make choices when resources are limited. Accounting records what has happened in a business. Finance uses both: it takes the records of the past and the ideas of economics to decide what the business should do next.

The person who makes these choices is the **financial manager**. In a small firm the owner does this work. In a large firm a team does it. The financial manager is also a manager in the ordinary sense, so finance and management cannot be separated: every plan of the firm, whether to hire, to expand or to buy, needs money, and the financial manager says whether the money can be found and what it will cost.¹

Before moving on. Finance means finding, using and controlling money. It looks forward, where accounting looks back. The financial manager makes the choices about money.

THE DECISIONS OF THE FINANCIAL MANAGER

The first decision concerns what to buy. An **investment decision** means a choice about which assets the firm should buy, such as a machine, a shop or a delivery van. An asset is anything of value that the firm owns. The second decision concerns where the money comes from. A **financing decision** means a choice about how to raise money: from the owners, from lenders, or from the profits the firm has kept.

The third decision concerns daily money. The working-capital decision means a choice about how much cash, stock and credit the firm needs from day to day to keep running. The fourth is the dividend decision. A **dividend** is a share of profit paid to the owners. The firm must decide how much profit to pay out and how much to keep for growth.

THE FOUR DECISIONS

Decision	The question it answers	Example
Investment	What should we buy?	A second oven
Financing	Where does the money come from?	A bank loan or owners' savings
Working capital	How do we keep daily money flowing?	How much flour to hold in stock
Dividend	How much profit do we pay out?	Pay owners now or keep the money for growth

Good finance keeps the business running and growing.

The firm needs one clear aim to judge its decisions.

Profit maximisation. Aiming for the largest possible profit.

Go slowly here.

Wealth is the value of the owners' investment.

Wealth maximisation. Aiming to make the owners' investment as valuable as possible.

When owners hire managers, their interests may differ.

Shareholders. The people who own shares, and so own part of a company.

Principal. A person who asks another person to act for them.

Agent. A person who acts on behalf of another.

Agency problem. The risk that agents act for themselves instead of for the owners.

Agency costs. The costs of watching agents and of the losses when

Read each row from left to right: the decision, the question behind it, and a case from a bakery. Later chapters teach how to make each one well.

Finance affects how well a business runs and whether it survives. A firm with the right money at the right time can buy materials, pay workers and keep machines busy. A firm that lacks money may stop production even when customers want its goods. A firm that borrows too much may be unable to pay its lenders. A firm that keeps idle cash earns nothing from it. In each case the quality of financial decisions shows in the firm's efficiency and in its success.

THE GOAL OF THE FIRM

Decisions need a standard. Without one, we cannot say whether a choice was good. The oldest standard is **profit maximisation**, which means aiming for the largest profit. It sounds sensible, but it has three weaknesses.

Picture two market stalls. Stall A will earn Tk. 100 profit this year and Tk. 100 next year. Stall B will earn Tk. 50 this year and Tk. 200 next year. Which has the larger profit? Stall B, in total. Yet the owner who needs money now may prefer A. The first weakness is that profit maximisation ignores *when* the money arrives. Chapter II explains why money received earlier is worth more.

The second weakness is risk. A profit that is almost certain and a profit that depends on luck count as equal, though they are not. The third is that profit, as accountants measure it, is not the same as cash, as the puzzle showed. A firm can also raise this year's profit by cutting maintenance or training, and so harm the years after.

A better standard is **wealth maximisation**. It means aiming to make the owners' investment as valuable as possible. The value of a firm depends on the cash it will give to its owners in future, the timing of that cash, and the risk of receiving it. This one idea includes the three things that profit leaves out. For this reason, profit is important but is not the ultimate goal.

You should now be able to say: *a firm should choose the plan that adds most to the value of the owners' investment, taking account of time and risk, and not simply the plan with the largest profit this year.*

Pause and check. Why might Stall B be worth more than Stall A, even though the owner who needs cash now prefers A?

OWNERS AND MANAGERS

In a large company the owners cannot run the firm. The owners are called **shareholders**. They hire managers to run it for them. In finance the owner is called the **principal** and the manager is called the **agent**. The principal wants the value of the firm to grow. The agent may want a high salary, a comfortable life or a larger firm, even when these do not raise the owners' wealth.

This gap is the **agency problem**. It has a cost. The owners may spend money to watch the managers, for example by hiring auditors, and some bad decisions may still be made. These costs together are called **agency costs**. Owners reduce the problem in two main ways: by linking the manager's pay to the firm's value, for instance through bonuses paid in shares, and by monitoring, through audits, reports and a board of directors.

Exceptions matter. In a business run by its single owner there is no agency problem, since principal and agent are the same person. The problem also exists

they act against the owners.

between lenders and owners, because owners may take risks with money that lenders supplied.



WORKED CASE: THE BAKERY

WORKED EXAMPLE: WHY DID CASH FALL WHEN THE BAKERY MADE A PROFIT?

SITUATION. *Case file, illustrative.* In January 2025 the owners put Tk. 5,00,000 into the Bakery. Oven and fittings cost Tk. 3,50,000, leaving cash of Tk. 1,50,000. In its first month the Bakery sold goods worth Tk. 1,50,000, of which Tk. 60,000 was sold to a café on 30 days' credit. Costs of Tk. 1,30,000 were paid in cash.

QUESTION. What was the profit, and what happened to cash?

REASONING. Profit is sales minus costs: $1,50,000 - 1,30,000 = \text{Tk. } 20,000$. Cash received was $1,50,000 - 60,000 = \text{Tk. } 90,000$. Cash paid was Tk. 1,30,000. Cash change is $90,000 - 1,30,000 = \text{minus Tk. } 40,000$.

CONCLUSION. The Bakery made a profit of Tk. 20,000, yet its cash fell by Tk. 40,000, from 1,50,000 to Tk. 1,10,000. The Tk. 60,000 owed by the café is profit that has not yet become cash.

CHECK. Profit 20,000 less the unpaid 60,000 gives minus 40,000, which matches the cash change.

CLOSING THE CHAPTER

SUMMARY

Finance is the way a firm finds, uses and controls money. The financial manager makes investment, financing, working-capital and dividend decisions. The sound goal is wealth maximisation, because it counts time and risk. Where owners and managers are different people, agency costs arise and must be controlled.

THINK IT THROUGH

A manager can raise this year's profit by Tk. 50,000 by stopping all repairs to the ovens. Explain why this may reduce the owners' wealth.

WHERE PEOPLE GO WRONG

Many students think profit and cash are the same; the case above shows they are not. Others think wealth maximisation means ignoring profit; it does not, since profit is one source of future cash.

IN HISTORY

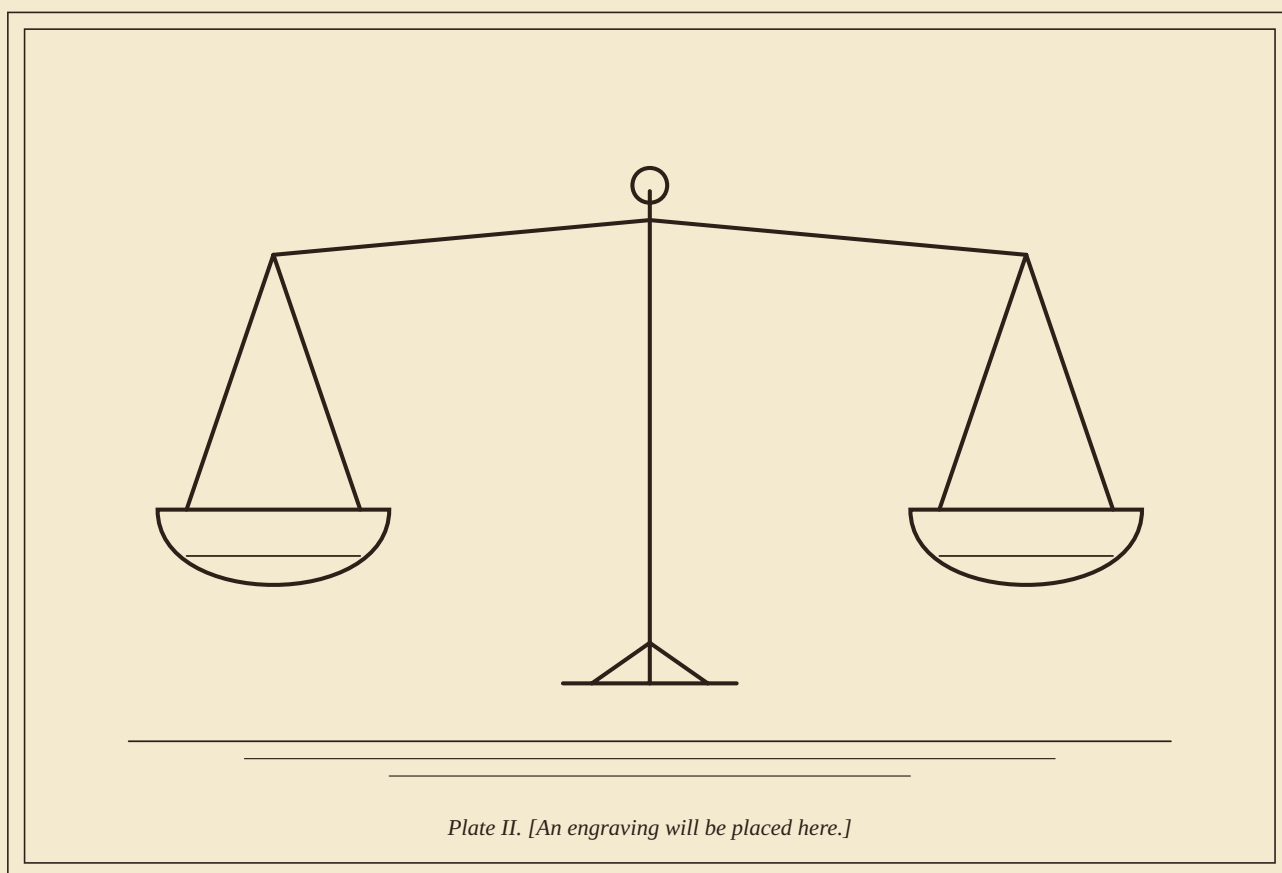
In 1602 the Dutch East India Company offered shares to the public, and ordinary people became part-owners of a business run by others far away. In 1776 Adam Smith wrote that directors of such companies, handling other people's money, could hardly be expected to watch it as anxiously as owners watch their own.²

REVIEW QUESTIONS

1. What is meant by finance?
2. Name the four decisions of a financial manager and say what each answers.
3. Why is profit maximisation not the ultimate goal of a firm?

Profit and cash are different things.

¹ Management is the wider task of organising people and work to reach a goal. Finance is the part of that task that deals with money.



CHAPTER II

Money Across Time

A sum in your hand is worth more than the same sum promised later.



This chapter explains why money has a time value, and how to move a sum forward to its future value or back to its present value, for a single sum, for equal payments, and for payments that differ.

² Smith, *The Wealth of Nations*, Book V. The point was formalised by Jensen and Meckling in 1976.

Would you take Tk. 1,000 today or Tk. 1,000 in a year?

Money today can earn; money later cannot.

Time value of money. The idea that a sum of money is worth more now than the same sum later.

Go slowly here.

Interest. The payment a borrower makes for the use of money, shown as a percentage of the sum.

Compound interest. Interest calculated on the original sum plus the interest already earned.

Future value pushes money forward; present value pulls it back.

Future value. What a sum of money today will be worth at a later date, after earning interest.

Present value. What a sum due in the future is worth today.

Discount rate. The yearly rate used to reduce a future sum to its present value.

Common slips with single sums.

A PUZZLE ABOUT WAITING

A friend offers you a choice: Tk. 1,000 today, or Tk. 1,000 in one year. Most people take the money today. Now the offer changes: Tk. 1,000 today, or Tk. 1,100 in one year. The choice is harder. Somewhere between 1,000 and 1,100 lies a point where you are indifferent. That point is the key to this chapter.

THE TIME VALUE OF MONEY

The **time value of money** means that a sum of money is worth more now than the same sum later. There are three reasons. Money held today can be placed in a bank or a business and earn a return. Prices usually rise, so later money buys less. And a promise of future money may not be kept, so waiting carries risk.

Suppose you place Tk. 1,000 in a bank that pays 10% a year. **Interest** means the payment for the use of money. After one year you hold $1,000 + 100 = \text{Tk. } 1,100$. In the second year the bank pays 10% on 1,100, not on 1,000. You hold $1,100 + 110 = \text{Tk. } 1,210$. Interest earned on earlier interest is called **compound interest**.

You should now be able to say: *compound interest grows a sum faster than simple interest, because each year's interest earns interest too.*

MOVING A SUM FORWARD AND BACK

The **future value** (FV) is what a sum today will be worth later. The **present value** (PV) is what a future sum is worth today. The **discount rate** is the rate used to move a future sum back to today. We write r for the rate as a decimal and n for the number of years.

Rule: $\text{future value} = \text{present value} \times (1 + r)^n$. $\text{Present value} = \text{future value} \div (1 + r)^n$. The two are opposites.

WORKED EXAMPLE: SINGLE SUM

GIVEN. $\text{PV} = \text{Tk. } 1,000$; $r = 10\%$ a year; $n = 2$ years.

FIND. the future value, and then the present value of that future value.

FORMULA. $\text{FV} = \text{PV} \times (1 + r)^n$; $\text{PV} = \text{FV} / (1 + r)^n$.

STEPS. $\text{FV} = 1,000 \times 1.10^2 = 1,000 \times 1.21 = \text{Tk. } 1,210$. $\text{PV} = 1,210 / 1.21 = \text{Tk. } 1,000$.

ANSWER. Tk. 1,000 today equals Tk. 1,210 in two years at 10%.

CHECK. This matches the year-by-year result above: 1,100, then 1,210. Moving forward and back returns the starting sum.

WORKED EXAMPLE: A MESSIER CASE: TK. 25,000 FOR 5 YEARS AT 12%

GIVEN. $\text{PV} = \text{Tk. } 25,000$; $r = 12\%$; $n = 5$.

FIND. FV.

FORMULA. $\text{FV} = \text{PV} \times (1 + r)^n$.

STEPS. $1.12^5 = 1.7623$. $\text{FV} = 25,000 \times 1.7623 = \text{Tk. } 44,059$.

ANSWER. Tk. 25,000 grows to about Tk. 44,059 in five years.

CHECK. Simple interest would give $25,000 + 5 \times 3,000 = \text{Tk. } 40,000$. Compound interest must be larger, and it is.

Common slips: using 10 instead of 0.10 in the formula; forgetting that n counts years, so 6 months is $n = 0.5$; and dividing when the question asks you to multiply.

Before moving on. Multiply by $(1 + r)^n$ to go forward; divide to go back. Compound interest earns interest on interest.

EQUAL PAYMENTS: ANNUITIES

An annuity is a series of equal payments at equal gaps.

Annuity. A series of equal payments made at equal intervals.

Ordinary annuity. An annuity whose payments are made at the end of each period.

Annuity due. An annuity whose payments are made at the start of each period.

An **annuity** is a series of equal payments made at equal intervals. Rent, loan instalments and regular savings are annuities. In an **ordinary annuity** each payment falls at the end of its period. In an **annuity due** each payment falls at the start. A third kind, the perpetuity, never ends; its present value is the payment divided by the rate.

The formulas are: FV of an ordinary annuity = payment $\times [(1 + r)^n - 1] / r$. PV of an ordinary annuity = payment $\times [1 - (1 + r)^{-n}] / r$. For an annuity due, multiply the ordinary result by $(1 + r)$, because every payment comes one period earlier and so earns or loses one more period of interest.

WORKED EXAMPLE: SAVING Tk. 1,000 A YEAR FOR 3 YEARS AT 10%

GIVEN. payment Tk. 1,000 at the end of each year; $r = 10\%$; $n = 3$.

FIND. the future value, the present value, and the future value if payments start at the beginning of each year.

FORMULA. $FV = P \times [(1 + r)^n - 1] / r$; $PV = P \times [1 - (1 + r)^{-n}] / r$.

STEPS. Factor for FV = $(1.331 - 1) / 0.10 = 3.31$, so FV = Tk. 3,310. Factor for PV = 2.48685, so PV = Tk. 2,486.85. Annuity due: $3,310 \times 1.10 = \text{Tk. } 3,641$.

ANSWER. The ordinary annuity grows to Tk. 3,310. Its present value is about Tk. 2,487. The annuity due grows to Tk. 3,641.

CHECK. Add each payment by hand: $1,000 \times 1.21 + 1,000 \times 1.10 + 1,000 = 1,210 + 1,100 + 1,000 = \text{Tk. } 3,310$.

PAYMENTS THAT DIFFER: MIXED STREAMS

A mixed stream is handled one payment at a time.

Mixed stream. A series of payments that are not all equal.

A **mixed stream** is a series of payments that are not all equal. No short formula exists. We move each payment separately and add the results. For the present value we divide each payment by $(1 + r)^n$ for its own year. For the future value we multiply each by $(1 + r)$ raised to the years remaining.

WORKED EXAMPLE: PRESENT VALUE OF A MIXED STREAM

GIVEN. Year 1 Tk. 1,000; Year 2 Tk. 2,000; Year 3 Tk. 3,000; discount rate 10%.

FIND. the present value.

FORMULA. $PV = \text{sum of each payment} / (1 + r)^n$.

STEPS. Year 1: $1,000 \times 0.9091 = 909.09$. Year 2: $2,000 \times 0.8264 = 1,652.89$. Year 3: $3,000 \times 0.7513 = 2,253.94$.

ANSWER. The total present value is Tk. 4,815.93.

CHECK. The total is below the plain sum of 6,000, as it must be, because later money is worth less today.

Exceptions deserve a note. If interest is added more often than once a year, say every month, use r divided by the number of periods and n multiplied by it. When comparing two streams, discount both at the same rate; the stream with the higher present value is worth more today.

Case file, illustrative. The Bakery sets aside Tk. 1,00,000 at the end of each of three years, in an account paying 10%. The ordinary annuity gives $1,00,000 \times 3.31 = \text{Tk. } 3,31,000$, enough toward a second oven.

The case file: saving for a new oven.

CLOSING THE CHAPTER

SUMMARY

Money has a time value because it can earn, prices rise and promises carry risk. Future value moves money forward and present value moves it back. Annuities use short formulas; mixed streams are moved one payment at a time.

THINK IT THROUGH

Stream P pays Tk. 5,000 at the end of each of 3 years. Stream Q pays 2,000, 4,000 and 7,000 in those years. Find the present value of each at 10% and say which is worth more today.

WHERE PEOPLE GO WRONG

Students add payments from different years as if they were equal. Others use an ordinary annuity formula for payments made at the start of the period.

IN HISTORY

In 1202 the Italian mathematician Leonardo of Pisa, called Fibonacci, published *Liber Abaci*, which taught merchants of Europe the Hindu-Arabic numerals. It included problems on interest and on the value of money at different times, tools for the traders of his day.

REVIEW QUESTIONS

1. What is meant by the time value of money?
2. What is an annuity, and what are its kinds?
3. Find the present values of streams P and Q at 10%. Which is better?
4. Tk. 1,00,000 is saved at the end of each year for 5 years at 8%. What is the future value?

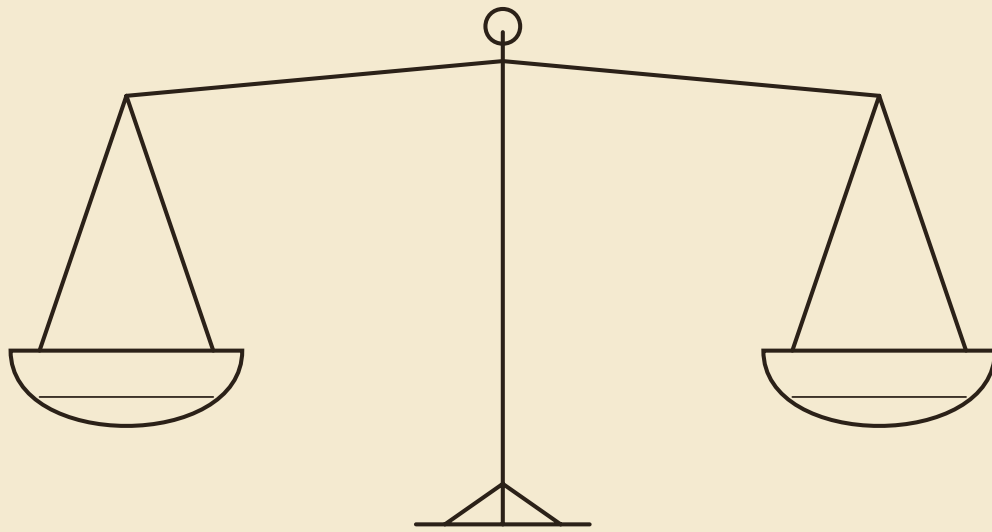


Plate III. [An engraving will be placed here.]

CHAPTER III

Judging Projects

A machine costs money now and pays back slowly; is it worth buying?



This chapter explains capital budgeting: why it matters, the steps of the process, which cash flows count, and how to judge projects by payback and by net present value.

Two machines cost the same but pay back differently.

Large, long-lasting spending needs careful judgement.

Capital budgeting. Planning and judging large spending whose benefits last for several years.

Capital budgeting follows six steps in order.

Only extra cash caused by the project counts.

Cash flow. Money actually received or paid by the firm.

Incremental. Extra; the change in cash caused by choosing the project.

Sunk cost. Money already spent that cannot be recovered whatever is decided.

Payback asks how soon the money returns.

Payback period. The time a project takes to return its initial cost in cash.

NPV measures the wealth a project adds.

Net present value. The present value of a project's cash inflows minus its initial cost.

A PUZZLE ABOUT TWO MACHINES

Machine A and Machine B each cost Tk. 10,000. A returns Tk. 5,000 a year for three years. B returns Tk. 2,000, then 4,000, then 10,000. Both return Tk. 15,000 and Tk. 16,000 in total. Which should the owner buy? A simple count of total money received favours B, but A returns money sooner. We need a method that treats both the size and the timing of money fairly.

WHAT CAPITAL BUDGETING IS

Capital budgeting means planning and judging large spending whose benefits last for several years, such as a new oven, a shop or a factory. It is necessary for four reasons. The sums are large. The benefits last a long time. Such decisions are hard to reverse, since a machine cannot be returned once installed. And they shape what the firm can do in future, so a poor choice weakens the firm for years.

THE STEPS OF THE PROCESS

First, the firm generates proposals, with ideas coming from managers, workers and customers. Second, it estimates the cash flows each proposal will cause. Third, it evaluates each proposal using a technique. Fourth, it selects and approves the best. Fifth, it carries out the project. Sixth, it reviews the results later and compares them with the estimates, so that future estimates improve.

CASH FLOWS THAT COUNT

A **cash flow** is money actually received or paid. The project's cash flows come in three groups: the initial outlay, which is the cost of starting; the operating cash inflows, which are the extra yearly receipts after tax; and the terminal cash flow, such as the sale of the machine at the end.

Only **incremental** cash flows count, which means the change caused by the project. A **sunk cost** is money already spent, such as a survey paid for last year; it is ignored, since it cannot be recovered. A use of an owned resource counts at what it could earn elsewhere. Depreciation is not a cash payment, so it is not counted as a cash flow, though it affects the tax paid.

TWO TECHNIQUES

The **payback period** (PBP) is the time a project takes to return its initial cost. The rule is to accept a project if its payback is shorter than a limit the firm has set, and to prefer the shorter payback. Its strength is simplicity and its attention to how soon the money returns. Its weakness is serious: it ignores the time value of money and it ignores all cash after the payback date.

The **net present value** (NPV) is the present value of the cash inflows minus the initial cost. The rule is to accept a project if NPV is greater than zero, and to prefer the project with the highest NPV. A positive NPV means the project earns more than the discount rate, and so adds to the owners' wealth. This links back to Chapter I: NPV is wealth maximisation made measurable.

Other techniques exist. The internal rate of return finds the discount rate at which NPV is zero. The profitability index divides the present value of inflows by the cost. They are met in later courses.

WORKED EXAMPLE: MACHINES A AND B AT A DISCOUNT RATE OF 10%

GIVEN. cost Tk. 10,000 each. A: 5,000, 5,000, 5,000. B: 2,000, 4,000, 10,000. Rate 10%.

FIND. payback period and NPV of each; which to prefer.

FORMULA. Payback = full years + (cost still unpaid / cash flow of the next year).

NPV = sum of PV of inflows - initial cost.

STEPS. A pays back in $10,000 / 5,000 = 2.0$ years. B returns 2,000 and 6,000 by the end of years 1 and 2, so 4,000 is unpaid; $4,000 / 10,000 = 0.4$, so payback is 2.4 years. NPV of A = $5,000 \times 2.48685 - 10,000 = 12,434.26 - 10,000 = \text{Tk. } 2,434.26$. NPV of B = $2,000 \times 0.9091 + 4,000 \times 0.8264 + 10,000 \times 0.7513 - 10,000 = 12,637.11 - 10,000 = \text{Tk. } 2,637.11$.

ANSWER. A has the shorter payback (2.0 against 2.4 years), but B has the higher NPV. The two methods disagree. NPV counts timing and all cash flows, so B is preferred as the better addition to wealth.

CHECK. B's large final payment of 10,000 is worth Tk. 7,513.15 today, which more than makes up for its slower start.

WORKED EXAMPLE: THE BAKERY'S SECOND OVEN AT 12%

GIVEN. cost Tk. 5,00,000. Inflows: Year 1 1,80,000; Year 2 2,00,000; Year 3 2,20,000; Year 4 1,50,000. Rate 12%.

FIND. NPV and the decision.

FORMULA. NPV = sum of inflow / $(1 + r)^n$ - cost.

STEPS. $1,80,000 \times 0.8929 = 1,60,714$; $2,00,000 \times 0.7972 = 1,59,439$; $2,20,000 \times 0.7118 = 1,56,592$; $1,50,000 \times 0.6355 = 95,328$. Total = 5,72,072. NPV = total - 5,00,000.

ANSWER. NPV is about Tk. 72,072. It is positive, so the project is acceptable.

CHECK. The plain sum of inflows is 7,50,000, and the discounted total is lower, as expected.

A second case, with messier numbers.

Exceptions and slips. When two projects exclude each other, as when only one oven fits the shop, choose by the higher NPV. When money is limited, rank by profitability. The cost belongs at time zero and is not discounted. Students often use accounting profit instead of cash flow, or forget the cost, or discount at the wrong rate. The discount rate should reflect the risk of the project.

CLOSING THE CHAPTER**SUMMARY**

Capital budgeting judges large, long-lived spending. Its steps run from proposals to review. Only incremental cash flows count. Payback is simple but ignores timing and later cash; NPV counts both and measures added wealth.

THINK IT THROUGH

A project costs Tk. 20,000 and returns Tk. 8,000 a year for 3 years. At 10%, find its payback and its NPV. Would you accept it?

WHERE PEOPLE GO WRONG

Students treat sunk costs as relevant and count depreciation as cash. Others prefer the quicker payback although the other project adds more wealth.

IN HISTORY

New York State began the Erie Canal in 1817 and opened it in 1825. It cost about seven million dollars, a vast sum for the state, and was paid for by bor-

rowing. Tolls repaid the debt early, and the canal lowered the cost of moving goods between the Atlantic coast and the Great Lakes.

REVIEW QUESTIONS

1. What is capital budgeting, and why is it necessary?
2. List the steps of capital budgeting.
3. Solve the Think it through problem.

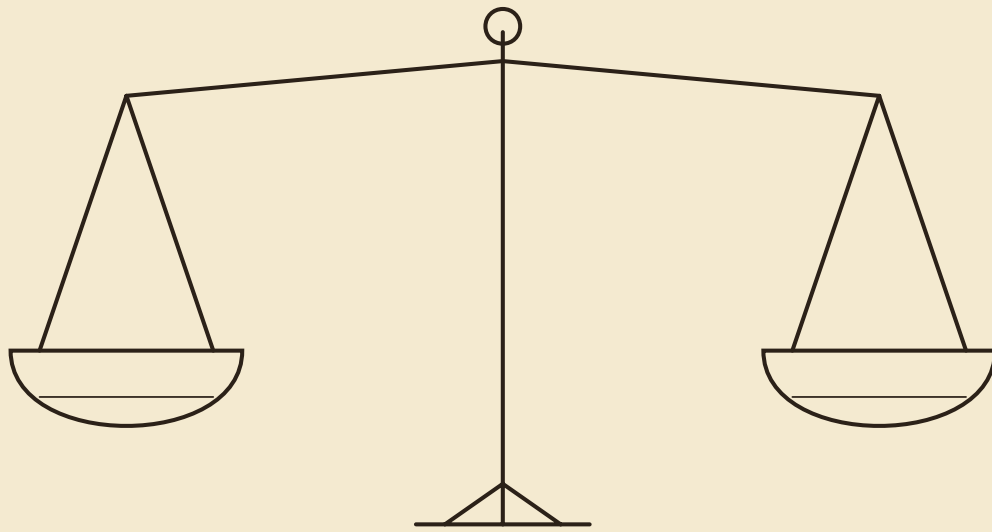


Plate IV. [An engraving will be placed here.]

CHAPTER IV

Risk and Return

Two roads reach the same place; one is safer.



This chapter explains return and risk, the kinds of risk a firm faces, how to measure risk with expected value, standard deviation and the coefficient of variation, and how risk sets the return that investors demand.

Same average, different spread.

Return is the gain; risk is doubt about the gain.

Return. The gain from an investment, shown as a percentage of the amount invested.

Risk. The chance that the actual return differs from the expected return.

Business risk comes from operations; financial risk from borrowing.

Business risk. The risk that comes from the firm's operations, such as changes in sales and costs.

Financial risk. The extra risk that comes from using borrowed money.

Risk is measured by how far returns spread from their average.

Go slowly here.

Probability distribution. A list of possible returns, each with the chance that it occurs.

Expected value. The average of possible returns, each weighted by its probability.

Standard deviation. A measure of how widely returns spread around the expected value.

Coefficient of variation. The standard deviation divided by the expected value; risk per unit of return.

A PUZZLE ABOUT TWO ROUTES

You travel to the city by two routes. Route X takes 40, 45 or 50 minutes. Route Y takes 20, 45 or 70 minutes. Both average 45 minutes. Yet if you must reach an examination on time, you choose X. Y is riskier because its results are more spread out. Finance measures this spread in the same way.

RETURN AND RISK

Return means the gain from an investment as a percentage of the amount invested. Rate of return = (selling price - buying price + cash received) / buying price. If you buy a share for Tk. 100, receive a dividend of Tk. 4 and sell for Tk. 108, the return is $(108 - 100 + 4) / 100 = 12\%$.

Risk means the chance that the actual return will differ from the one expected. Both bad and good surprises are included in the idea, though people worry about the bad.

KINDS OF RISK

Business risk means the risk that comes from the firm's operations: sales may fall, costs may rise, a rival may arrive. It exists even if the firm has no debt. **Financial risk** means the extra risk that comes from borrowing. Interest must be paid whether sales are good or poor, so debt makes the owners' return less certain. Chapter V studies this link.

Risks can also be sorted by whether they can be removed. Risk that belongs to one firm, such as a strike or a failed product, can be reduced by holding shares in many firms. Risk that affects all firms, such as a recession or a rise in interest rates, cannot. Investors are paid mainly for bearing the second kind.

MEASURING RISK

The measure uses four steps. A **probability distribution** lists the possible returns and the chance of each. The **expected value** is the average, with each return weighted by its chance. The **standard deviation** measures the spread: take each return's distance from the expected value, square it, weight it by its chance, add, and take the square root. The **coefficient of variation** divides the standard deviation by the expected value, so it shows risk for each unit of return and allows fair comparison.

WORKED EXAMPLE: INVESTMENTS X AND Y

GIVEN. three possible conditions with probabilities 0.25, 0.50, 0.25. Returns of X: 8%, 12%, 16%. Returns of Y: 4%, 12%, 20%.

FIND. expected value, standard deviation and coefficient of variation of each.

FORMULA. Expected value = sum of (probability x return). Standard deviation = square root of sum of [probability x (return - expected value)²]. CV = standard deviation / expected value.

STEPS. X: expected value = $0.25 \times 8 + 0.50 \times 12 + 0.25 \times 16 = 2 + 6 + 4 = 12\%$. Squared distances: $(8-12)^2 = 16$, 0, 16. Weighted: $0.25 \times 16 + 0 + 0.25 \times 16 = 8$. Standard deviation = square root of 8 = 2.83%. Y: expected value = $1 + 6 + 5 = 12\%$. Squared distances: 64, 0, 64. Weighted sum = 32. Standard deviation = 5.66%.

ANSWER. Both average 12%. X has standard deviation 2.83% and CV 0.24. Y has standard deviation 5.66% and CV 0.47. Y is twice as risky for the same return.

CHECK. Y's returns lie twice as far from the average as X's, and its standard deviation is twice as large.

You should now be able to say: *standard deviation tells me how widely returns spread; a larger value means more risk.*

Exceptions. Standard deviation compares well only when expected values are equal; when they differ, use the coefficient of variation. Standard deviation treats gains and losses alike. Past results do not guarantee future ones.

RISK AND THE RETURN INVESTORS DEMAND

More risk must be paid for with more return.

Risk premium. The extra return investors require for taking risk instead of a safe investment.

Required return. The minimum return an investor needs to accept an investment.

Investors dislike risk, so they require extra reward for bearing it. The **risk premium** is that extra return. The **required return** is the minimum return needed to accept an investment: $\text{required return} = \text{risk-free return} + \text{risk premium}$. The risk-free return is what a safe government security pays.

The Capital Asset Pricing Model (CAPM) sets the premium by the risk that cannot be removed. $\text{Required return} = \text{risk-free return} + \beta \times (\text{market return} - \text{risk-free return})$. Beta shows how strongly an investment moves with the whole market. With a risk-free return of 5%, a market return of 11% and a beta of 1.5, the required return is $5 + 1.5 \times (11 - 5) = 14\%$. A beta above 1 means more risk than the market as a whole.

MARKETS AND INFORMATION

Prices in an efficient market already reflect information.

Efficient market. A market in which prices already reflect the available information.

An **efficient market** is one in which prices already reflect the available information. Three forms are named. In the weak form, prices reflect all past prices, so studying old price charts brings no gain. In the semi-strong form, prices also reflect all public information, such as published accounts. In the strong form, prices reflect all information, even private. Most evidence supports the weak and, to a large degree, the semi-strong form.

CLOSING THE CHAPTER

SUMMARY

Return is the gain on an investment; risk is the chance that the gain differs from what is expected. Business and financial risk are the two main kinds in a firm. Expected value, standard deviation and the coefficient of variation measure risk. Investors demand a premium for risk.

THINK IT THROUGH

Asset M has expected return 10% and standard deviation 4%. Asset N has expected return 20% and standard deviation 6%. Which is less risky for each unit of return?

WHERE PEOPLE GO WRONG

Students forget to take the square root, or compare standard deviations when expected values differ. Others think a high return is a sign of low risk.

IN HISTORY

In 1688, according to the records, Edward Lloyd's coffee house in London was already a meeting place for shipowners, captains and merchants. Insurers gathered there to share the risks of sea voyages, and from this habit grew the market called Lloyd's.

REVIEW QUESTIONS

1. What is risk?
2. Distinguish financial risk from business risk.
3. Solve the Think it through problem.
4. A share costs Tk. 50, pays Tk. 2 and is sold for Tk. 54. Find the return.

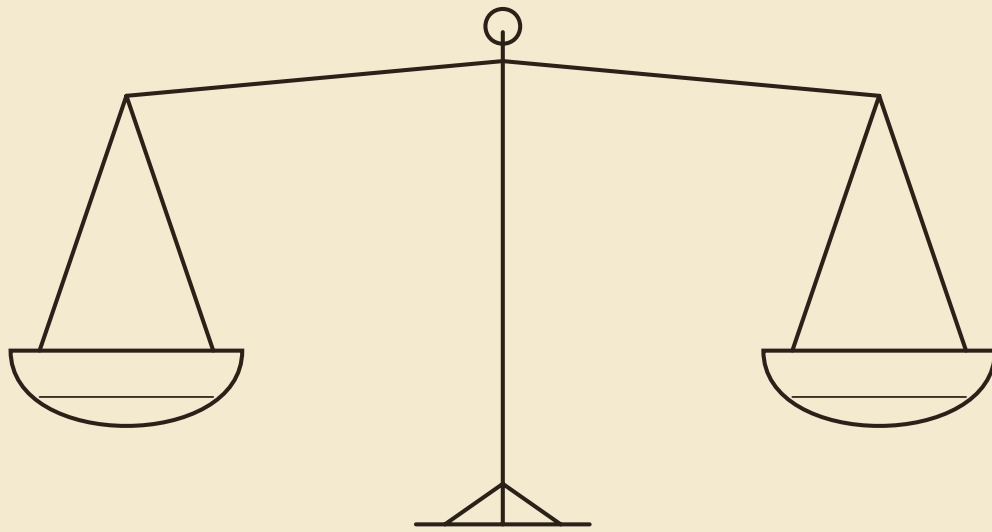


Plate V. [An engraving will be placed here.]

CHAPTER V

Leverage

A small push on one end lifts a large weight on the other.



This chapter explains operating, financial and total leverage, shows how to measure each one, and discusses why leverage can magnify both profit and trouble.

The same lever that lifts also drops.

Leverage means fixed costs magnify results.

Leverage. The use of fixed costs to magnify the effect of a change in sales on profit.

Fixed cost. A cost that stays the same whatever the level of sales.

Operating leverage. The effect of fixed operating costs on how profit before interest and tax changes with sales.

Financial leverage. The effect of fixed financing costs, such as interest, on how profit after interest responds to EBIT.

Total leverage. The combined effect of operating and financial leverage on profit after interest as sales change.

The degree of leverage is a multiplier.

Degree of operating leverage. The percentage change in EBIT caused by a one per cent change in sales.

Degree of financial leverage. The percentage change in profit after interest caused by a one per cent change in EBIT.

Degree of total leverage. The percentage change in profit after interest caused by a one per cent change in sales.

Break-even is where profit before interest and tax is zero.

Leverage raises both reward and risk.

A PUZZLE ABOUT A LEVER

A shopkeeper rents a shop for a fixed Tk. 40,000 a month and pays little for goods. Another buys goods at a high price but pays a low rent. In a month when sales rise by 10%, which shopkeeper's profit rises more? The first, since the rent does not rise with sales. In a month when sales fall by 10%, the same shopkeeper's profit falls more. A fixed cost works like a lever: it magnifies the effect of sales on profit, upward and downward.

WHAT LEVERAGE IS

Leverage means the use of fixed costs, which are costs that do not change when sales change, to magnify the effect of a change in one quantity on another. A **fixed cost** stays the same whatever the level of sales, as rent does. A variable cost rises and falls with sales, as flour does. The word comes from the lever of Archimedes.

Two kinds of fixed cost give two kinds of leverage. **Operating leverage** comes from fixed operating costs, such as rent and machines, and shows how profit before interest and tax (EBIT) responds to sales. **Financial leverage** comes from fixed financing costs, mainly interest, and shows how profit after interest (EBT) responds to EBIT. **Total leverage** combines both.

MEASURING LEVERAGE

The **degree of operating leverage** (DOL) is the percentage change in EBIT for each 1% change in sales. At a given sales level, $DOL = \text{contribution} / \text{EBIT}$, where $\text{contribution} = \text{sales} - \text{variable costs}$. The **degree of financial leverage** (DFL) = EBIT / EBT . The **degree of total leverage** (DTL) = $\text{contribution} / \text{EBT}$, which equals $DOL \times DFL$.

WORKED EXAMPLE: LEVERAGE OF A SMALL FIRM

GIVEN. price Tk. 50 per unit; variable cost Tk. 30 per unit; fixed operating cost Tk. 40,000; interest Tk. 10,000; sales 4,000 units.

FIND. DOL, DFL and DTL, and the effect of a 10% rise in sales.

FORMULA. $DOL = \text{contribution} / \text{EBIT}$; $DFL = \text{EBIT} / \text{EBT}$; $DTL = DOL \times DFL$.

STEPS. Contribution per unit = $50 - 30 = 20$. Contribution = $4,000 \times 20 = 80,000$. EBIT = $80,000 - 40,000 = 40,000$. EBT = $40,000 - 10,000 = 30,000$. $DOL = 80,000 / 40,000 = 2.00$. $DFL = 40,000 / 30,000 = 1.33$. $DTL = 80,000 / 30,000 = 2.67$, which equals 2.00×1.33 .

ANSWER. $DOL = 2.00$, $DFL = 1.33$, $DTL = 2.67$. A 10% rise in sales should raise EBT by about 26.7%.

CHECK. At 4,400 units, contribution is 88,000, EBIT is 48,000 (up 20%), EBT is 38,000 (up 26.7%). It matches.

The break-even quantity is fixed operating cost divided by contribution per unit: $40,000 / 20 = 2,000$ units. Below it, the firm makes a loss before interest.

WHY LEVERAGE MATTERS

Leverage raises the reward of good times and the damage of bad times. A firm with high operating leverage, such as one with a costly factory, is hurt badly when sales fall. A firm with high financial leverage must pay interest even in a poor year, so it may fail to pay and lose its assets. From the owner's side, debt

can raise the return on the owner's money when business is good. From the lender's side, high debt means a higher chance of not being repaid. From the manager's side, the task is to choose a level that gains from leverage without inviting failure.

Exceptions. DOL is different at each sales level, and it cannot be found at break-even, where EBIT is zero. Leverage figures assume that price, variable cost and fixed cost stay unchanged.

CLOSING THE CHAPTER

SUMMARY

Fixed costs act as a lever. Operating leverage comes from fixed operating costs; financial leverage from fixed financing costs; total leverage combines them as $DTL = DOL \times DFL$. Leverage magnifies gains and losses and therefore adds to risk.

THINK IT THROUGH

A firm sells 2,000 units at Tk. 60, with variable cost Tk. 40 a unit, fixed operating cost Tk. 20,000 and interest Tk. 5,000. Find DOL, DFL and DTL.

WHERE PEOPLE GO WRONG

Students use sales instead of contribution in DOL. Others forget that DTL is the product, not the sum, of DOL and DFL.

IN HISTORY

The Greek writer Pappus, who lived in the fourth century, reported that Archimedes had claimed he could move the Earth if given a place to stand. Whatever the truth of the saying, it made the lever a lasting picture of small force producing large effect, and financiers later borrowed the word.

REVIEW QUESTIONS

1. Define operating, financial and total leverage.
2. Why is leverage significant for a firm?
3. Solve the Think it through problem.

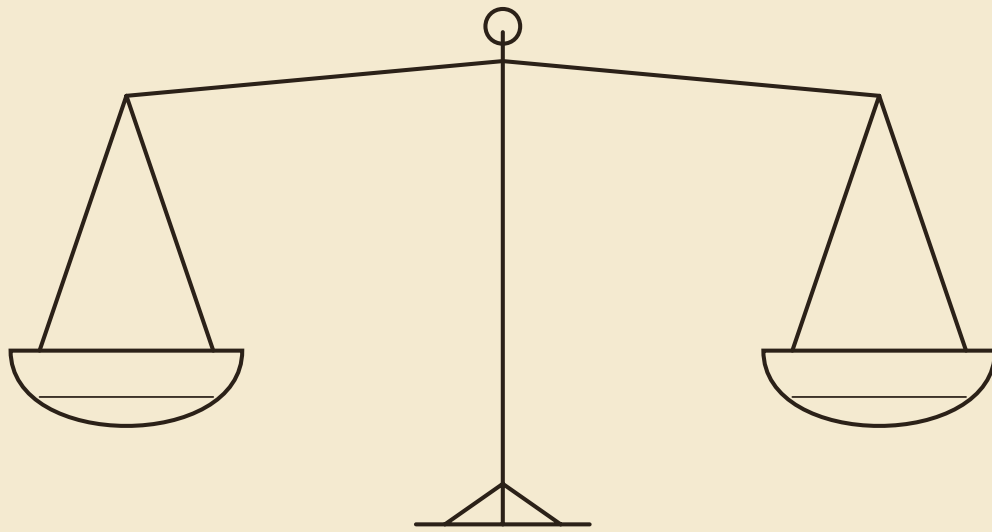


Plate VI. [An engraving will be placed here.]

CHAPTER VI

Valuing Bonds and Shares

What is a promise of future money worth today?



This chapter explains how securities are valued, how a bond is priced from its features, what its status and yield mean, how a bond differs from a share, and how a share is valued.

A paper that promises money has a price.

Value is the present value of expected cash at the required return.

Security. A paper or record that gives its holder a claim on money from a borrower or a company.

Intrinsic value. What a security is worth, found from the present value of the cash it is expected to pay.

A bond is a loan split into small, tradable pieces.

Bond. A long-term loan to a firm or government, with fixed payments, that can be bought and sold.

Face value. The sum printed on the bond and repaid at maturity; also called par value.

Coupon rate. The yearly interest rate on the face value, set when the bond is issued.

Maturity. The date on which the bond is repaid.

Required rate of return. The yield investors need from a bond of this risk, set by the market.

The bond sells at a discount, par or premium.

Bond status. Whether a bond sells below, at or above its face value.

Yield to maturity. The yearly return an investor earns by buying a bond at its price and holding it to maturity.

A PUZZLE ABOUT A PROMISE

A friend offers to sell you a paper. It promises Tk. 100 each year for three years and Tk. 1,000 at the end. How much should you pay? Not Tk. 1,300, the total promised, since later money is worth less today. The right price is the present value of everything the paper promises. This chapter applies Chapter II to real securities.

THE IDEA OF VALUE

A **security** means a paper or record that gives its holder a claim on money. **Intrinsic value** means what a security is worth, found from the present value of the cash it is expected to pay, discounted at the return investors require. If the market price is below this value, the security looks cheap; if above, it looks costly.

BONDS

A **bond** means a long-term loan to a firm or government, divided into pieces that can be traded. Its main features are these. The **face value** is the sum repaid at the end, often Tk. 1,000. The **coupon rate** is the yearly interest as a percentage of face value. The **maturity** is the date of repayment. Other features include the claim on assets if the issuer fails, and whether it can be redeemed early.

The price of a bond follows the rule of Chapter II. Price = present value of the interest payments (an annuity) + present value of the face value (a single sum), both at the required return. The **required rate of return** is set by the market for bonds of that risk.

WORKED EXAMPLE: VALUE OF A THREE-YEAR BOND

GIVEN. face value Tk. 1,000; coupon rate 10% paid yearly; 3 years; required return 12%.

FIND. the bond's value.

FORMULA. Value = interest x annuity factor + face value x single-sum factor.

STEPS. Interest = $1,000 \times 10\% = \text{Tk. } 100$. Annuity factor at 12% for 3 years = 2.40183. Single-sum factor = 0.71178. Value = $100 \times 2.40183 + 1,000 \times 0.71178 = 240.18 + 711.78$.

ANSWER. The value is Tk. 951.96.

CHECK. The coupon (10%) is below the required return (12%), so the bond must be worth less than its face value of 1,000. It is.

BOND STATUS AND YIELD

The **bond status** tells whether the bond sells at a discount (below face value), at par (equal) or at a premium (above). Compare the coupon rate with the required return. If the coupon is lower, the bond sells at a discount, because buyers must be compensated. If equal, at par. If higher, at a premium. When market rates rise, bond prices fall, and when they fall, prices rise.

The **yield to maturity** (YTM) is the yearly return earned by buying at the market price and holding to maturity. It is the discount rate that makes the present value of all payments equal to the price. In the example, a price of Tk. 951.96 gives a YTM of 12%. At a required return of 8% the same bond is worth Tk. 1051.54, a premium. At 10% it is worth Tk. 1,000, which is par.

WORKED EXAMPLE: A BOND PAYING INTEREST TWICE A YEAR

GIVEN. face value Tk. 1,000; coupon 6.5% a year paid half-yearly; 2 years; required return 8% a year.

FIND. the value.

FORMULA. Use half-yearly periods: halve the coupon and the rate, and double the years.

STEPS. Interest each half-year = $1,000 \times 6.5\% / 2 = \text{Tk. } 32.50$. Periods = 4. Rate per period = 4%. Annuity factor = 3.62990; single-sum factor = 0.85480. Value = $32.50 \times 3.62990 + 1,000 \times 0.85480 = 117.97 + 854.80$.

ANSWER. The value is Tk. 972.78.

CHECK. The coupon (6.5%) is below the required return (8%), so the value is below 1,000. It is.

SHARES

A share is a piece of ownership, with no fixed repayment.

Share. A unit of ownership in a company; also called common stock.

A **share** means a unit of ownership in a company, also called common stock. The holder receives dividends if the firm pays them and gains if the share price rises. Unlike a bond, a share has no maturity and no promised payment.

BOND AND SHARE COMPARED

Feature	Bond	Share
Holder is	A lender	An owner
Payment	Fixed interest	Dividend, not promised
Repayment	Face value at maturity	None; sell to another buyer
Voting rights	Usually none	Usually yes
Claim if firm fails	Paid before owners	Paid last
Risk to holder	Lower	Higher

Read each row across: it states how a bond and a share differ on one point. Because a share is riskier, its required return is usually higher.

The value of a share is again a present value. If the next dividend is D , the required return is r , and dividends grow forever at a steady rate g , then value = $D / (r - g)$. With $D = \text{Tk. } 5$, $r = 12\%$ and $g = 4\%$, value = $5 / (0.12 - 0.04) = \text{Tk. } 62.50$. If the dividend does not grow, value = D / r . The formula works only when r is greater than g , and it is only as good as the forecasts that go into it.

CLOSING THE CHAPTER**SUMMARY**

A security is worth the present value of the cash it is expected to pay, at the return investors require. A bond is priced from its coupon, face value, maturity and required return. Its status depends on coupon against required return. A share has no fixed payment and is valued from its expected dividends.

THINK IT THROUGH

A bond has face value Tk. 1,000, coupon 8% paid yearly, 3 years to maturity. Find its value when the required return is 10%, and state its status.

WHERE PEOPLE GO WRONG

Students use the total of payments as the price, or use the annual rate for half-yearly interest. Others think a rise in market rates raises bond prices.

IN HISTORY

In Holland in 1636 and 1637, contemporaries reported that prices for tulip bulbs rose sharply and then collapsed. Historians still debate how large and how harmful the episode was, but it remains a warning about prices that float far above any present value of expected cash.

REVIEW QUESTIONS

1. What is a bond? Name its key features.
2. How does the required return decide bond status?
3. Solve the Think it through problem.
4. How does a bond differ from a share?

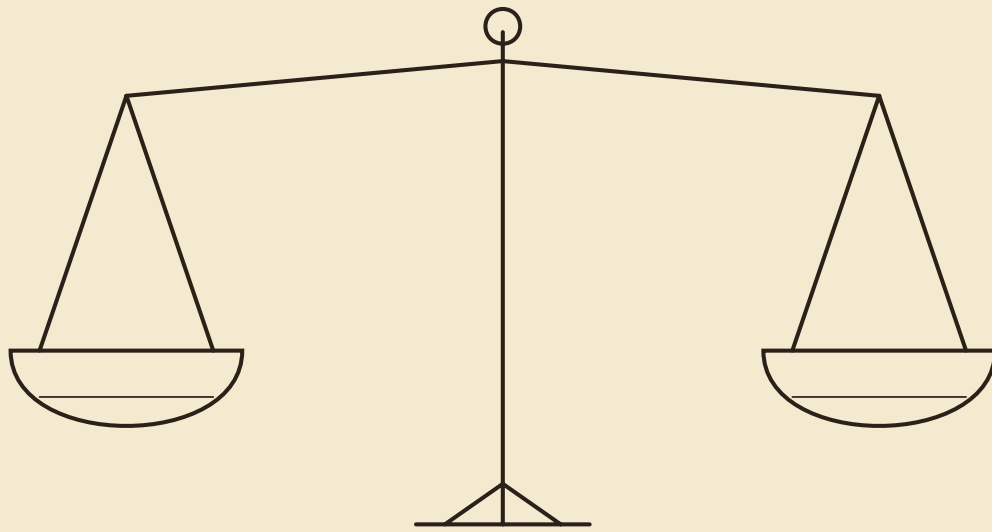


Plate VII. [An engraving will be placed here.]

CHAPTER VII

Short-Term Money

A firm can fail with a full order book and an empty till.



This chapter explains working capital, the length of financing, the matching principle, the cost of trade credit, the delay called float, and how to estimate the working capital a firm needs.

Orders do not pay wages; cash does.

Working capital is the money tied up in daily operations.

Working capital. The money a firm needs for daily operations, held in cash, stock and money owed by customers.

Liquidity. How quickly and easily an asset can be turned into cash without loss.

Financing is named by how long the money is borrowed.

Trade credit. Goods bought now and paid for later, which is a free or cheap loan from the supplier.

Match the life of the money to the life of the asset.

Matching principle. Financing an asset for about as long as the asset will be used.

A discount for early payment has a hidden interest rate.

Cash discount. A reduction in price offered for payment within a short time.

A PUZZLE ABOUT A FULL ORDER BOOK

The Bakery receives a large order from a hotel. To bake it, the Bakery must buy flour on Monday and pay wages on Friday. The hotel will pay in forty days. For forty days the Bakery spends money and receives none. If it cannot find that money, the best order of the year turns into a crisis. Short-term money is about surviving this gap.

WORKING CAPITAL

Working capital means the money a firm needs for daily operations. It is held in current assets: cash, stock of goods and the money customers owe. **Liquidity** means how quickly an asset can be turned into cash. Too little working capital risks failure to pay bills. Too much leaves money idle that could earn a return elsewhere.

SHORT, MEDIUM AND LONG FINANCING

Financing is grouped by maturity. Short-term financing is repaid within one year. Medium-term, or intermediate, financing runs from one year to about five years. Long-term financing runs for more than five years. The limits vary between books and countries, so the grouping is a guide. The next chapter studies long-term sources.

Common sources of short-term finance are these. **Trade credit** means goods bought now and paid for later. Bank overdrafts and short bank loans give quick money at a stated interest. Accrued expenses, such as wages owed but not yet paid, are a form of free credit. Firms may also borrow against money their customers owe.

THE MATCHING PRINCIPLE

The **matching principle** says to finance an asset for about as long as it will be used. A delivery van used for five years should be paid for with money repaid over about five years. Seasonal stock should be financed with short loans. If a firm finances a long-lived machine with an overdraft, the lender may demand repayment before the machine has earned the money. If it finances stock with long-term debt, it pays interest on money it does not need all year.

THE COST OF TRADE CREDIT

A supplier may say "2/10 net 30". It means: take 2% off if you pay within 10 days; otherwise pay the full sum by day 30. The **cash discount** is a reward for paying early. Not taking it is like borrowing from the supplier at a cost. Cost = $[\text{discount} / (100 - \text{discount})] \times [360 / (\text{final day} - \text{discount day})]$.

WORKED EXAMPLE: THE COST OF GIVING UP A DISCOUNT

GIVEN. terms 2/10 net 30. A bank loan costs 14% a year.

FIND. the yearly cost of trade credit, and whether to take the discount.

FORMULA. Cost = $[d / (100 - d)] \times [360 / (N - D)]$, where d is the discount per cent, D the discount day and N the final day.

STEPS. $d / (100 - d) = 2 / 98 = 0.0204$. $N - D = 30 - 10 = 20$ days, so $360 / 20 = 18$. Cost = $0.0204 \times 18 = 36.73\%$.

ANSWER. Not taking the discount costs about 36.7% a year, more than the bank's 14%. Take the discount, and borrow from the bank if cash is needed.

CHECK. For 20 days the cost is $2 / 98 = 2.04\%$; there are 18 such periods in a year, and 2.04×18 is about 36.7.

With terms 1/10 net 40 the cost is $(1 / 99) \times (360 / 30) = 12.12\%$, below 14%, so the firm should pay on day 40. Exceptions: the formula uses simple interest; with compounding the true cost is higher. Paying late costs less in the short run but may harm the firm's name with suppliers.

FLOAT AND THE CASH CYCLE

Float is the delay between paying and the money moving.

Float. The time between a payment being sent and the money becoming usable by the receiver.

Cash conversion cycle. The days between paying for materials and receiving cash from customers.

Float means the time between sending a payment and the receiver having usable funds. It has three parts: mail time, while the cheque travels; processing time, while the receiver records and deposits it; and clearing time, while the banks move the money. A firm wants short float on money it receives and, within honest limits, longer float on money it pays.

The **cash conversion cycle** shows how long money is tied up: inventory days + receivables days - payables days. If the Bakery holds stock for 40 days, waits 30 days for customers and pays suppliers after 20 days, the cycle is $40 + 30 - 20 = 50$ days.

WORKED EXAMPLE: ESTIMATING WORKING CAPITAL

GIVEN. yearly cost of sales Tk. 36,00,000; cash conversion cycle 50 days; 360 days in a year.

FIND. the working capital needed.

FORMULA. Working capital = daily cost $\times$ length of cycle in days.

STEPS. Daily cost = $36,00,000 / 360 = \text{Tk. } 10,000$. Working capital = $10,000 \times 50 = \text{Tk. } 5,00,000$.

ANSWER. The Bakery needs about Tk. 5,00,000 to run its daily operations.

CHECK. A cycle of 50 days is about $50/360$ of a year, and $50/360$ of 36,00,000 is Tk. 5,00,000.

CLOSING THE CHAPTER

SUMMARY

Working capital keeps daily operations running. Financing is short, medium or long term, and should match the life of the asset. Trade credit has a cost that can be calculated. Float delays money, and the cash conversion cycle shows how much working capital is needed.

THINK IT THROUGH

Terms are 3/10 net 45 and the bank charges 18%. Find the cost of trade credit and decide whether to take the discount.

WHERE PEOPLE GO WRONG

Students forget to divide the discount by 100 minus the discount. Others use the full credit period instead of the days after the discount period.

IN HISTORY

Francesco Datini, a merchant of Prato in Italy who died in 1410, left behind account books and a very large number of letters. They show how medieval traders used credit and bills of exchange to trade without carrying coins, an early lesson in managing the gap between paying and being paid.

REVIEW QUESTIONS

1. Define short-term, medium-term and long-term financing.
2. Find the cost of trade credit for 2/15 net 45.
3. Solve the Think it through problem.
4. What is float and what are its parts?

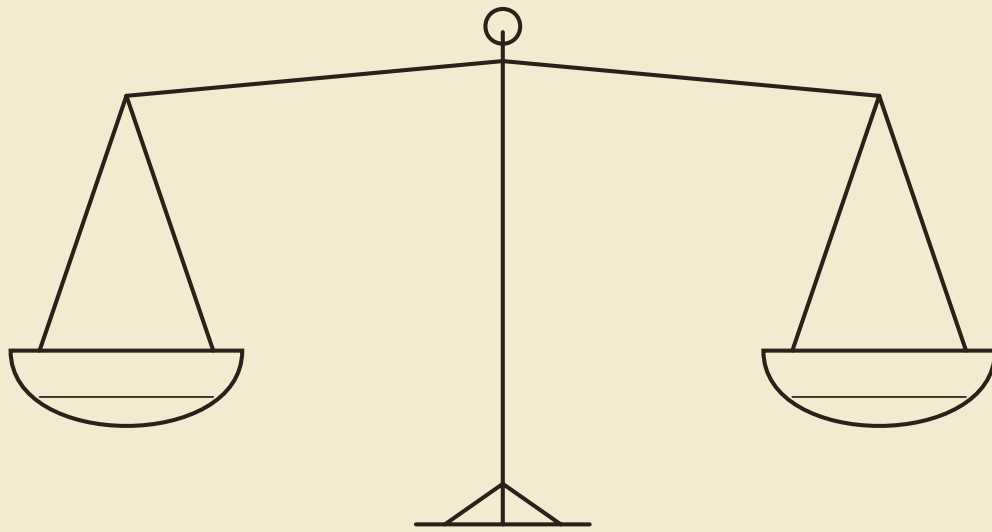


Plate VIII. [An engraving will be placed here.]

CHAPTER VIII

Long-Term Funds

A firm that plans to last must find money that lasts.



This chapter explains the long-term sources of money open to a firm: term loans, bonds, preferred shares, ordinary shares and retained profits. It shows how funds are raised in the capital market and which institutions supply them in Bangladesh.

New firms have little to offer lenders.

Six features let us compare sources.

Debt is repaid with interest on fixed dates.

Term loan. A loan from a bank or other lender for a fixed period, repaid in instalments.

Debenture. A long-term bond issued by a company, usually paying fixed interest.

Owners accept the highest risk and expect the highest return.

Preferred stock. Shares that pay a fixed dividend and rank before ordinary shares, usually without a vote.

Common stock. The ordinary shares of a company; holders vote and share in the profit left after others are paid.

Retained earnings. Profit that the firm keeps and reinvests instead of paying out as dividends.

A PUZZLE ABOUT STARTING OUT

A new firm has no profits, no record and few assets. Yet it needs a building and machines that will serve for ten years. Lenders hesitate and owners have limited savings. Where does the money come from, and what does each source ask in return? Long-term finance is the answer, and it is especially important for a new business, since a start-up's assets earn slowly and cannot be paid for from short loans.

FEATURES OF ANY LONG-TERM SOURCE

Each source can be compared on six points. Its claim on income: who is paid first. Its claim on assets if the firm fails. Its control: whether the supplier gets a vote. Its cost: the return the supplier requires. Its maturity: when repayment is due. Its risk to the firm: what happens if payment is missed. Debt is cheaper and has no vote but must be paid. Ownership money costs more but has no fixed payment.

BORROWED MONEY

A **term loan** means a loan for a fixed period, usually from one to ten years, repaid in instalments. It is common for buying machines. The lender often asks for security, which is an asset that the lender may take if the loan is not repaid. A **debenture** is a long-term bond issued by a company, which is how a company borrows from many lenders instead of one. The methods of debt thus include bank term loans, bonds and debentures.

Interest on debt is usually a cost that reduces taxable profit, so the real cost of debt is lower than the interest rate. After-tax cost of debt = interest rate $\times$ (1 - tax rate). If a firm pays 12% and the tax rate is 30% (an assumption for illustration; check the current rate), the after-tax cost is $12 \times 0.70 = 8.4\%$.

OWNERSHIP MONEY

Preferred stock means shares that pay a fixed dividend and are paid before ordinary shareholders. They are a mixture: like debt in their fixed payment, and like ownership in that a missed dividend does not bankrupt the firm.

Common stock, also called ordinary shares, means the shares of the true owners. Holders vote, receive what is left after others are paid, and bear the most risk. Their required return is therefore the highest.

Retained earnings means profit that the firm keeps instead of paying as dividends. It is the cheapest source to obtain, with no fees and no new owners, but it is not free: the owners could have used the money elsewhere, so they expect it to earn at least their required return.

THE SOURCES COMPARED

Source	Paid	Vote	Cost to firm	If payment missed
Debt	Fixed interest, first	No	Lowest after tax	Lender may take assets
Preferred stock	Fixed dividend, second	Usually no	Middle	Dividend is owed, firm survives
Common stock	What remains, last	Yes	Highest	No legal default
	Not applicable			Not applicable

Source	Paid	Vote	Cost to firm	If payment missed
Retained earnings		Existing owners	Same as common stock	

Read the table across: each row describes one source on the six features above.

RAISING MONEY IN THE CAPITAL MARKET

Large firms sell securities to many investors.

Capital market. The market where long-term funds are raised by selling shares and bonds.

Initial public offering. The first sale of a company's shares to the public.

The **capital market** is the market where long-term funds are raised and traded. A company can sell shares to the public for the first time in an **initial public offering** (IPO). It can offer new shares to its present owners in a rights issue. It can sell shares to a few large investors in a private placement. It can issue bonds or debentures. Each method needs approval from the regulator, a document describing the firm, and fees to the people who arrange the sale.

In Bangladesh, shares are traded on the Dhaka Stock Exchange and the Chittagong Stock Exchange, and the Bangladesh Securities and Exchange Commission regulates the market. Long-term finance is also supplied by commercial banks, by development finance institutions such as the Bangladesh Development Bank Limited and the Investment Corporation of Bangladesh, and by lenders for infrastructure such as IDCOL. Names and rules change, so check the current position before relying on any of this.

The **cost of capital** is the average return a firm must earn to satisfy all who supplied its long-term money. A firm should accept an investment only if it earns more than this cost. When we judge a project, we assume the project does not change the firm's business risk or financial risk; otherwise the cost of capital would change too.

Cost of capital is the average price of the firm's money.

Cost of capital. The average return the firm must earn to satisfy all those who supplied its long-term money.

CLOSING THE CHAPTER

SUMMARY

Long-term sources are debt, preferred stock, common stock and retained earnings. They differ in claim, control, cost and risk. Firms raise them in the capital market through shares, bonds and loans. The cost of capital is the average price of all of them.

THINK IT THROUGH

A firm can borrow at 15% with a tax rate of 30% or raise equity at 18%. Which is cheaper after tax, and why might the firm still not borrow without limit?

WHERE PEOPLE GO WRONG

Students say retained earnings cost nothing. Others forget that interest lowers tax and so compare before-tax debt with equity.

IN HISTORY

In 1694 the English government needed money for war with France. A group of subscribers lent it 1,200,000 pounds and in return was set up as the Bank of England. It was an early case of the public being invited to supply long-term money to a borrower.

REVIEW QUESTIONS

1. Why is long-term financing important for a new business?
2. Name the main sources of long-term money.
3. Solve the Think it through problem.

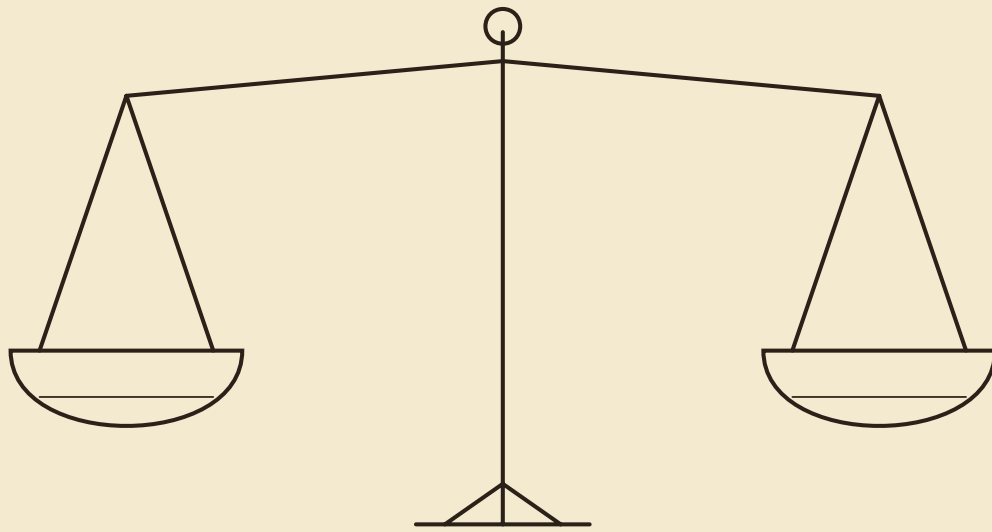


Plate IX. [An engraving will be placed here.]

CHAPTER IX

Leasing

To use a thing is not always to own it.



This chapter explains leasing, the two main kinds of lease, and the steps for deciding whether a firm should lease or buy an asset.

The Bakery needs a van but not the cost of one.

A lease is rent for the use of an asset.

Lease. A contract in which the owner of an asset lets another person use it for a fee over a set period.

Lessor. The owner of the asset who lets it to another.

Lessee. The person who uses the asset and pays the fee.

An operating lease is a short rental; a financial lease is a long commitment.

Operating lease. A lease for a short part of the asset's life, which the user can cancel, and where the owner keeps the risks of ownership.

Financial lease. A long lease that covers most of the asset's life and cannot be cancelled, so it works like a loan.

Compare the present value of the after-tax outflows.

A PUZZLE ABOUT A VAN

The Bakery needs a delivery van. Buying costs Tk. 1,00,000 at once. A dealer offers instead to let the Bakery use the van for four years for a yearly fee. Which is better? The answer depends on cost, tax and what remains at the end. To compare fairly, we must put all the cash flows on the same footing.

WHAT A LEASE IS

A **lease** means a contract in which the owner lets another person use an asset for a fee over a set period. The owner is the **lessor**. The user is the **lessee**. Leasing is a form of financing, since the lessee gets an asset without paying its full price now.

TWO KINDS OF LEASE

An **operating lease** lasts for part of the life of the asset, can usually be cancelled, and leaves the lessor with the costs of repairs and with the asset at the end. A **financial lease** covers most of the life of the asset, usually cannot be cancelled, and the lessee bears repairs and similar costs. It works like a loan used to buy the asset.

OPERATING AND FINANCIAL LEASES

Point	Operating lease	Financial lease
Length	Part of asset's life	Most of asset's life
Cancellation	Usually allowed	Usually not allowed
Repairs	Lessor pays	Lessee pays
Total payments	Less than the asset's cost	About the cost plus interest
Best for	Short or changing needs	Long, steady needs

Read each row across: it compares the two leases on one point. The reasons to lease include less money paid at the start, flexibility, and sometimes tax gain. The drawbacks include a higher total cost and, at the end, no ownership.

THE LEASE OR BUY DECISION

The decision has four steps. Step one: find the after-tax cash outflows of leasing. Step two: find those of buying. Step three: discount both at the after-tax cost of debt, because both are in effect ways of borrowing. Step four: choose the plan with the lower present value of outflows. Lease payments reduce taxable profit; so do the interest and depreciation of a purchase. Tax rules differ between countries, so state the rules used.

WORKED EXAMPLE: LEASE OR BUY A VAN

GIVEN. price Tk. 1,00,000; 4 years. Lease payment Tk. 30,000 at each year end. Purchase by a 10% loan with instalments of Tk. 31,500 a year. Depreciation Tk. 25,000 a year. Interest: Year 1 10,000, Year 2 7,500, Year 3 5,000, Year 4 2,500. Tax rate 30% (assumed).

FIND. which plan has the lower present value of after-tax outflows.

FORMULA. After-tax lease cost = payment $\times$ (1 - tax rate). After-tax purchase cost = instalment - tax rate $\times$ (depreciation + interest). Discount rate = 10% $\times$ (1 - 0.30) = 7%.

STEPS. Lease: $30,000 \times 0.70 = 21,000$ each year. $PV = 21,000 \times 3.38721 = 71,131$.
Purchase outflows: Year 1: $31,500 - 0.30 \times 35,000 = 21,000$; Year 2: $31,500 - 0.30 \times 32,500 = 21,750$; Year 3: $31,500 - 0.30 \times 30,000 = 22,500$; Year 4: $31,500 - 0.30 \times 27,500 = 23,250$. $PV = 74,727$.

ANSWER. Leasing has the lower present value, Tk. 71,131 against Tk. 74,727 for buying, so lease the van.

CHECK. Each yearly lease outflow (21,000) is below or equal to each purchase outflow, so leasing must have the lower present value. This test is not always possible; here it is.

Exceptions: this comparison leaves out the value of the van at the end of four years, which a buyer keeps. If that value is large, buying may win. The residual value must be estimated and added. Maintenance, insurance and risk of the asset becoming old also matter.

CLOSING THE CHAPTER

SUMMARY

A lease lets the lessee use an asset owned by the lessor. Operating leases are short and cancellable; financial leases are long and work like loans. The lease-or-buy decision compares the present value of after-tax outflows at the after-tax cost of debt.

THINK IT THROUGH

Explain why a firm needing a computer for six months would prefer an operating lease to a financial lease.

WHERE PEOPLE GO WRONG

Students discount at the before-tax rate, or forget the tax saving on depreciation and interest. Others ignore the value of the asset at the end.

IN HISTORY

In 1952 a firm in San Francisco, the United States Leasing Corporation, began to lease equipment to businesses that wanted to use machines without buying them. Leasing then spread as a standard way of financing equipment.

REVIEW QUESTIONS

1. What is leasing? Distinguish operating from financial leases.
2. State the four steps of the lease-or-buy decision.
3. Why is the after-tax cost of debt used as the discount rate?

Answers to Review Questions

CHAPTER I

1. Finance is the way a business finds, uses and controls money; it asks where money comes from, where it is placed and how its use is judged.
2. Investment (what to buy), financing (where money comes from), working capital (daily money), dividend (how much profit to pay out).
3. It ignores timing, risk and the difference between profit and cash, and may harm the future; wealth maximisation counts all three.

CHAPTER II

1. A sum today is worth more than the same sum later, because money can earn, prices rise and the future is uncertain.
2. A series of equal payments at equal intervals; ordinary (end of period), due (start of period), and perpetuity (never ends).
3. $P = \text{Tk. } 12,434.26$; $Q = \text{Tk. } 10,383.17$. P is worth more today.
4. $1,00,000 \times 5.8666 = \text{Tk. } 5,86,660$.

CHAPTER III

1. Planning and judging large, long-lived spending; necessary because sums are large, benefits long, decisions hard to reverse and the future of the firm depends on them.
2. Generate proposals; estimate cash flows; evaluate; select; implement; review.
3. $\text{Payback} = 20,000 / 8,000 = 2.5$ years. $\text{NPV} = 8,000 \times 2.48685 - 20,000 = \text{Tk. } -105.18$. It is negative, so reject.

CHAPTER IV

1. The chance that the actual return differs from the expected return.
2. Business risk comes from operations and exists without debt; financial risk is the extra risk from borrowing.
3. $\text{CV of } M = 4/10 = 0.40$; $\text{CV of } N = 6/20 = 0.30$. N is less risky for each unit of return.
4. $(54 - 50 + 2) / 50 = 12\%$.

CHAPTER V

1. Operating: fixed operating costs magnify the effect of sales on EBIT. Financial: fixed financing costs magnify the effect of EBIT on EBT. Total: the two combined, from sales to EBT.
2. It magnifies both profit and loss, so it raises risk; managers must balance the gain against the danger of failing to meet fixed costs.
3. $\text{Contribution} = 2,000 \times 20 = 40,000$; $\text{EBIT} = 20,000$; $\text{EBT} = 15,000$. $\text{DOL} = 2.00$; $\text{DFL} = 1.33$; $\text{DTL} = 2.67$.

CHAPTER VI

1. A long-term loan split into tradable pieces; face value, coupon rate and maturity are its key features.
2. Coupon below required return: discount. Equal: par. Above: premium.

3. $80 \times 2.48685 + 1,000 \times 0.75131 = \text{Tk. } 950.26$; it sells at a discount.

4. A bond is a loan with fixed payments and a repayment date; a share is ownership with no promised payment and no maturity.

CHAPTER VII

1. Repaid within one year; one to about five years; more than five years.

2. $(2/98) \times (360/30) = 24.49\%$.

3. $(3/97) \times (360/35) = 31.81\%$. This exceeds 18%, so take the discount.

4. The delay between sending a payment and the money becoming usable; mail, processing and clearing time.

CHAPTER VIII

1. A new firm needs lasting assets that earn slowly; short loans would fall due before the assets repay them.

2. Term loans, bonds or debentures, preferred stock, common stock, retained earnings.

3. Debt: $15 \times 0.70 = 10.5\%$, cheaper than equity at 18%. Heavy debt raises financial risk, so lenders and owners demand more.

CHAPTER IX

1. Leasing lets one person use an asset owned by another for a fee. Operating leases are short, cancellable, and the lessor bears costs; financial leases are long, not cancellable, and work like loans.

2. After-tax outflows of lease; after-tax outflows of purchase; discount both at the after-tax cost of debt; choose the lower present value.

3. Both plans are in effect borrowing, and interest saves tax, so the after-tax cost of borrowing is the fair rate.

Glossary

Agency costs. The costs of watching agents and of the losses when they act against the owners. (*Chapter I*)

Agency problem. The risk that agents act for themselves instead of for the owners. (*Chapter I*)

Agent. A person who acts on behalf of another. (*Chapter I*)

Annuity. A series of equal payments made at equal intervals. (*Chapter II*)

Annuity due. An annuity whose payments are made at the start of each period. (*Chapter II*)

Bond. A long-term loan to a firm or government, with fixed payments, that can be bought and sold. (*Chapter VI*)

Bond status. Whether a bond sells below, at or above its face value. (*Chapter VI*)

Business risk. The risk that comes from the firm's operations, such as changes in sales and costs. (*Chapter IV*)

Capital budgeting. Planning and judging large spending whose benefits last for several years. (*Chapter III*)

Capital market. The market where long-term funds are raised by selling shares and bonds. (*Chapter VIII*)

Cash conversion cycle. The days between paying for materials and receiving cash from customers. (*Chapter VII*)

Cash discount. A reduction in price offered for payment within a short time. (*Chapter VII*)

Cash flow. Money actually received or paid by the firm. (*Chapter III*)

Coefficient of variation. The standard deviation divided by the expected value; risk per unit of return. (*Chapter IV*)

Common stock. The ordinary shares of a company; holders vote and share in the profit left after others are paid. (*Chapter VIII*)

Compound interest. Interest calculated on the original sum plus the interest already earned. (*Chapter II*)

Cost of capital. The average return the firm must earn to satisfy all those who supplied its long-term money. (*Chapter VIII*)

Coupon rate. The yearly interest rate on the face value, set when the bond is issued. (*Chapter VI*)

Debenture. A long-term bond issued by a company, usually paying fixed interest. (*Chapter VIII*)

Degree of financial leverage. The percentage change in profit after interest caused by a one per cent change in EBIT. (*Chapter V*)

Degree of operating leverage. The percentage change in EBIT caused by a one per cent change in sales. (*Chapter V*)

Degree of total leverage. The percentage change in profit after interest caused by a one per cent change in sales. (*Chapter V*)

Discount rate. The yearly rate used to reduce a future sum to its present value. (*Chapter II*)

Dividend. A share of profit paid out to the owners of a company. (*Chapter I*)

Efficient market. A market in which prices already reflect the available information. (*Chapter IV*)

Expected value. The average of possible returns, each weighted by its probability. (*Chapter IV*)

- Face value.** The sum printed on the bond and repaid at maturity; also called par value. *(Chapter VI)*
- Finance.** The way a business finds, uses and controls its money. *(Chapter I)*
- Financial lease.** A long lease that covers most of the asset's life and cannot be cancelled, so it works like a loan. *(Chapter IX)*
- Financial leverage.** The effect of fixed financing costs, such as interest, on how profit after interest responds to EBIT. *(Chapter V)*
- Financial manager.** The person who decides how a firm finds and uses its money. *(Chapter I)*
- Financial risk.** The extra risk that comes from using borrowed money. *(Chapter IV)*
- Financing decision.** A choice about how to raise money, from owners or from lenders. *(Chapter I)*
- Fixed cost.** A cost that stays the same whatever the level of sales. *(Chapter V)*
- Float.** The time between a payment being sent and the money becoming usable by the receiver. *(Chapter VII)*
- Future value.** What a sum of money today will be worth at a later date, after earning interest. *(Chapter II)*
- Incremental.** Extra; the change in cash caused by choosing the project. *(Chapter III)*
- Initial public offering.** The first sale of a company's shares to the public. *(Chapter VIII)*
- Interest.** The payment a borrower makes for the use of money, shown as a percentage of the sum. *(Chapter II)*
- Intrinsic value.** What a security is worth, found from the present value of the cash it is expected to pay. *(Chapter VI)*
- Investment decision.** A choice about which assets, such as machines or shops, the firm should buy. *(Chapter I)*
- Lease.** A contract in which the owner of an asset lets another person use it for a fee over a set period. *(Chapter IX)*
- Lessee.** The person who uses the asset and pays the fee. *(Chapter IX)*
- Lessor.** The owner of the asset who lets it to another. *(Chapter IX)*
- Leverage.** The use of fixed costs to magnify the effect of a change in sales on profit. *(Chapter V)*
- Liquidity.** How quickly and easily an asset can be turned into cash without loss. *(Chapter VII)*
- Matching principle.** Financing an asset for about as long as the asset will be used. *(Chapter VII)*
- Maturity.** The date on which the bond is repaid. *(Chapter VI)*
- Mixed stream.** A series of payments that are not all equal. *(Chapter II)*
- Net present value.** The present value of a project's cash inflows minus its initial cost. *(Chapter III)*
- Operating lease.** A lease for a short part of the asset's life, which the user can cancel, and where the owner keeps the risks of ownership. *(Chapter IX)*
- Operating leverage.** The effect of fixed operating costs on how profit before interest and tax changes with sales. *(Chapter V)*
- Ordinary annuity.** An annuity whose payments are made at the end of each period. *(Chapter II)*
- Payback period.** The time a project takes to return its initial cost in cash. *(Chapter III)*
- Preferred stock.** Shares that pay a fixed dividend and rank before ordinary shares, usually without a vote. *(Chapter VIII)*
- Present value.** What a sum due in the future is worth today. *(Chapter II)*
- Principal.** A person who asks another person to act for them. *(Chapter I)*
- Probability distribution.** A list of possible returns, each with the chance that it occurs. *(Chapter IV)*

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- Profit maximisation.** Aiming for the largest possible profit. (*Chapter I*)
- Required rate of return.** The yield investors need from a bond of this risk, set by the market. (*Chapter VI*)
- Required return.** The minimum return an investor needs to accept an investment. (*Chapter IV*)
- Retained earnings.** Profit that the firm keeps and reinvests instead of paying out as dividends. (*Chapter VIII*)
- Return.** The gain from an investment, shown as a percentage of the amount invested. (*Chapter IV*)
- Risk.** The chance that the actual return differs from the expected return. (*Chapter IV*)
- Risk premium.** The extra return investors require for taking risk instead of a safe investment. (*Chapter IV*)
- Security.** A paper or record that gives its holder a claim on money from a borrower or a company. (*Chapter VI*)
- Share.** A unit of ownership in a company; also called common stock. (*Chapter VI*)
- Shareholders.** The people who own shares, and so own part of a company. (*Chapter I*)
- Standard deviation.** A measure of how widely returns spread around the expected value. (*Chapter IV*)
- Sunk cost.** Money already spent that cannot be recovered whatever is decided. (*Chapter III*)
- Term loan.** A loan from a bank or other lender for a fixed period, repaid in instalments. (*Chapter VIII*)
- Time value of money.** The idea that a sum of money is worth more now than the same sum later. (*Chapter II*)
- Total leverage.** The combined effect of operating and financial leverage on profit after interest as sales change. (*Chapter V*)
- Trade credit.** Goods bought now and paid for later, which is a free or cheap loan from the supplier. (*Chapter VII*)
- Wealth maximisation.** Aiming to make the owners' investment as valuable as possible. (*Chapter I*)
- Working capital.** The money a firm needs for daily operations, held in cash, stock and money owed by customers. (*Chapter VII*)
- Yield to maturity.** The yearly return an investor earns by buying a bond at its price and holding it to maturity. (*Chapter VI*)

Further Reading

Brealey, R. and Myers, S. *Principles of Corporate Finance*. A standard text on investment, risk and valuation.

Ross, S., Westerfield, R. and Jordan, B. *Fundamentals of Corporate Finance*. A clear text for beginners.

Van Horne, J. and Wachowicz, J. *Fundamentals of Financial Management*. Strong on working capital and leverage.

Gitman, L. *Principles of Managerial Finance*. Many worked examples.

Jensen, M. and Meckling, W. (1976). Theory of the firm: managerial behavior, agency costs and ownership structure. *Journal of Financial Economics*. The founding paper of agency theory.

Smith, A. (1776). *An Inquiry into the Nature and Causes of the Wealth of Nations*.

Maths You Will Use

PERCENTAGES

A percentage is a number out of 100. To find 12% of Tk. 500, multiply: $500 \times 0.12 = \text{Tk. } 60$. To turn 12% into a decimal, divide by 100.

PERCENTAGE CHANGE

Percentage change = $(\text{new value} - \text{old value}) / \text{old value} \times 100$. A rise from 40,000 to 48,000 is $8,000 / 40,000 = 0.20$, which is 20%.

RATIOS AND AVERAGES

A ratio compares two numbers by division. An average is the sum of the values divided by how many values there are.

REARRANGING AN EQUATION

Whatever you do to one side, do to the other. From $FV = PV \times 1.1$, divide both sides by 1.1 to get $PV = FV / 1.1$.

POWERS

1.1 to the power 3 means $1.1 \times 1.1 \times 1.1 = 1.331$. A negative power means division: 1.1 to the power -3 is $1 / 1.331 = 0.7513$.

ROUNDING

Keep at least four decimal places in factors, and round money only in the last step. Small differences in the last digit are normal.

READING A GRAPH

The horizontal axis shows one quantity, such as time. The vertical axis shows another, such as money. Each point shows the two values together.



PRINCIPLES TO FINANCE

BIJOY?!